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Chapter 1

Special Relativity and Flat Spacetime

1.1 Prelude

Core Concept of General Relativity (GR)

General relativity is Einstein's theory of space, time, and gravitation. The essential idea is that gravity is not a force represented by fields in spacetime but is instead an inherent property of spacetime itself. The phenomenon experienced as "gravity" is a manifestation of the curvature of spacetime. The objective is to understand spacetime, curvature, and the mechanism by which curvature becomes gravity.

Newtonian Gravity: A Review

Newtonian gravity is defined by two fundamental components: an equation for the gravitational field as influenced by matter and an equation for the response of matter to this field. This can be formulated in two equivalent ways.

Force-Based Formulation

Field Creation: The force between two objects of masses M and m is described by the inverse-square law:

$$F = -\frac{GMm}{r^2}\hat{e}_r \tag{1.1}$$

Matter Response: This force imparts an acceleration to a particle of mass m according to Newton's second law:

$$F = ma \tag{1.2}$$

Potential-Based Formulation

From Newton's 2nd law of gravity and the gravitational force law (1.1), (1.2) we can derive the gravitational potential Φ and rewrite the two components of Newtonian gravity.

$$a = -\frac{GM}{r^2}\hat{e_r} \tag{1.3}$$

By getting the flux of the gravity for a body:

$$a.dA = (-G\frac{M}{r^2}\hat{r}).(dA\hat{r}) = -G\frac{M}{r^2}\ .\ dA$$

$$Flux = \oint_s a.dA = -G\frac{M}{r^2}\oint_s dA = -4\pi G M_{enclosed}$$

Using the enclosed mass relation:

$$M_{enclosed} = \int_v \rho dV$$

now we can write:

$$\oint_s a.dA = -4\pi G\int_v \rho dV$$

By using the divergence theorem:

$$\int_v (\nabla.a) dV = -4\pi G \int_v \rho dV$$

On the final step we can use $a = -\nabla\Phi$ to get The gravitational potential Φ is related to the mass density ρ by Poisson's equation:

$$\nabla^2 \Phi = 4\pi G \rho \tag{1.4}$$

Matter Response: The acceleration of a particle is given by the gradient of the potential:

$$a = -\nabla\Phi \tag{1.5}$$

Generic Proof To Get Poisson's Equation

We could start by assuming field a(r) which represent the acceleration at some point (r), but fist we have to define the newtownian gravity equation for a small mass element (dm) at some point (r'):

$$da = -G\frac{dm}{|r-r'|^2}\hat{e}_{r-r'} \tag{1.6}$$

where we also have to define the dM:

$$dM = \rho(\mathbf{r'})dV' \tag{1.7}$$

and finally the gravitational flux Φ through some closed surface (S):

$$\Phi = \oint_S \mathbf{a} \cdot d\mathbf{A} \tag{1.8}$$

Now the next step is to get the total acceleration at point (r) by integrating over the whole source volume (V_{source}):

$$\begin{aligned} a(r) &= \int_{V_{\text{source}}} da \\ &= \int_{V_{\text{source}}} -G\frac{dM}{|\mathbf{r}-\mathbf{r'}|^2}\hat{\mathbf{e}}_{(\mathbf{r}-\mathbf{r'})} \\ &= \int_{V_{\text{source}}} -G\frac{\rho(\mathbf{r'})}{|\mathbf{r}-\mathbf{r'}|^2}\hat{\mathbf{e}}_{(\mathbf{r}-\mathbf{r'})}dV' \end{aligned}$$

Now to get the flux (Φ) it's equal to:

$$\begin{aligned} \Phi = \oint_S \mathbf{a}\cdot d\mathbf{A} &= \oint_S \left(\int_{V_{\text{source}}} -G\frac{\rho(\mathbf{r'})}{|\mathbf{r}-\mathbf{r'}|^2}\hat{\mathbf{e}}_{(\mathbf{r}-\mathbf{r'})}dV'\right)\cdot d\mathbf{A} \\ &= \int_{V_{\text{source}}} -G\ \rho(\mathbf{r'})\ \left(\oint_S \frac{\hat{\mathbf{e}}_{(\mathbf{r}-\mathbf{r'})}}{|\mathbf{r}-\mathbf{r'}|^2}\cdot d\mathbf{A}\right) dV' \end{aligned}$$

Now to claculate the surface integral we have to consider two cases:

- If the source point (r') is outside the surface (S), then the surface integral is zero:

$$\oint_S \frac{\hat{\mathbf{e}}_{(\mathbf{r}-\mathbf{r'})}}{|\mathbf{r}-\mathbf{r'}|^2}\cdot d\mathbf{A} = 0$$

- If the source point (r') is inside the surface (S), then the surface integral is:

$$\oint_S \frac{\hat{\mathbf{e}}_{(\mathbf{r}-\mathbf{r'})}}{|\mathbf{r}-\mathbf{r'}|^2}\cdot d\mathbf{A} = 4\pi$$

So we don't care about the case where the source isn't inside so:

$$\Phi = \int_{V_{\text{source}}} -4\pi G \; \rho(\mathbf{r}')dV' = -4\pi G \int_{V_{\text{source}}} \rho(\mathbf{r}')dV' \tag{1.9}$$

We now get from (1.8), (1.9), (1.5):

$$\begin{aligned}
\oint_S \mathbf{a}\cdot d\mathbf{A} &= -4\pi G \int_{V_{\text{source}}} \rho(\mathbf{r}')dV' \\
\int_V (\nabla\cdot\mathbf{a})dV &= -4\pi G \int_{V_{\text{source}}} \rho(\mathbf{r}')dV' \\
\nabla\cdot\mathbf{a} &= -4\pi G \; \rho(\mathbf{r}') \\
\nabla^2\Phi &= 4\pi G \; \rho(\mathbf{r}')
\end{aligned}$$

General Relativity: A Preview

General Relativity replaces the two components of Newtonian theory with statements about spacetime curvature.

How Energy-Momentum Curves Spacetime

The response of spacetime curvature to the presence of matter and energy is governed by Einstein's equation:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{1.10}$$

How Matter Responds to Spacetime Curvature

Free particles move along paths of "shortest possible distance," known as geodesics. In a curved spacetime, these are the closest equivalent to straight lines. The parameterized paths $x^\mu(\lambda)$ of these particles obey the geodesic equation:

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0 \tag{1.11}$$

The Metric Tensor and the Learning Path

The metric tensor $g_{\mu\nu}$ is the fundamental concept that encodes the geometry of a space. Characterizes curvature by expressing deviations from the flat-space Pythagorean theorem. From the metric, one can derive the Riemann curvature tensor (used in Einstein's equation) and the geodesic equation.

1.2 Space and Time, Separately and Together

Special relativity (SR) is a theory describing the fundamental structure of spacetime—the four-dimensional arena in which all physical processes occur. It replaces Newtonian mechanics, which also describes motion in space and time, but treats them as fundamentally separate entities.

Spacetime and Worldlines

Spacetime: A four-dimensional set of points (events), each labeled by three spatial coordinates and one time coordinate.
Worldline: The one-dimensional curve traced by a particle as it moves through spacetime. The key difference between the Newtonian and relativistic views lies in how they treat the relationship between space and time.

Newtonian Spacetime: Absolute Simultaneity

In the Newtonian picture, time flows uniformly and independently of space. Spacetime is imagined as a stack of "time slices," each representing all of space at one instant of universal time. Simultaneity is absolute—observers everywhere agree on which events occur at the same moment. Particle trajectories move forward through these slices, unconstrained by any upper limit on their velocity.

Relativistic Spacetime: The Light Cone Structure

Special relativity eliminates absolute time and simultaneity. At each event, the invariant structure is instead the *light cone*—the set of all possible directions that light rays through that event can take. Events inside the future light cone can be influenced by the event, and events inside the past light cone can influence it. Physical trajectories (worldlines of particles) must always remain within the light cone—no signal or matter can travel faster than light.
Conceptual contrast: Newtonian spacetime has absolute simultaneity; relativistic spacetime has causal cones.

The Invariant Spacetime Interval

The unification of space and time into a single four-dimensional entity is supported by the existence of a quantity invariant under all changes of inertial frames.
Analogy with Euclidean Geometry: In a two-dimensional plane,

$$(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2$$

is invariant under rotation. The projections Δx and Δy change, but the distance Δs does not.

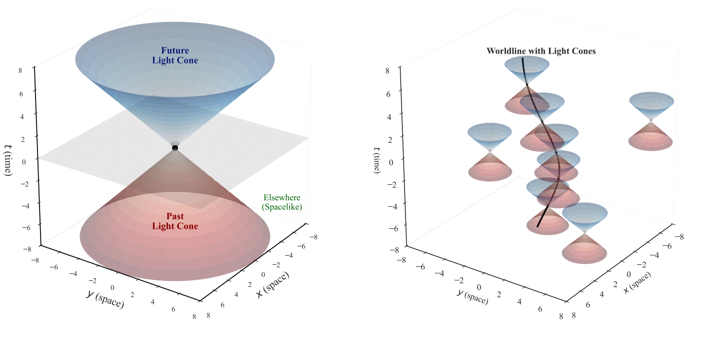


Figure 1.1: Left: Newtonian spacetime slices. Right: Relativistic light cone structure. (See source notebook).

Likewise, in spacetime the invariant quantity (the *interval*) is

$$(\Delta s)^2 = -(c\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$$

and remains the same for all inertial observers:

$$(\Delta s)^2 = -(c\Delta t')^2 + (\Delta x')^2 + (\Delta y')^2 + (\Delta z')^2$$

This invariance defines the geometry of flat spacetime, known as **Minkowski space**.

Minkowski Space: Coordinates and Metric

We define spacetime coordinates x^μ ($\mu = 0, 1, 2, 3$):

$$x^0 = ct, \quad x^1 = x, \quad x^2 = y, \quad x^3 = z$$

and often use units where $c = 1$. The metric tensor of flat spacetime is

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{1.12}$$

so that

$$(\Delta s)^2 = \eta_{\mu\nu}\, \Delta x^\mu \Delta x^\nu$$

Causal Structure and Proper Time

Intervals in spacetime fall into three classes:

- **Timelike:** $(\Delta s)^2 < 0$ — events can influence one another (connected by slower-than-light motion).
- **Spacelike:** $(\Delta s)^2 > 0$ — no causal connection possible.
- **Null:** $(\Delta s)^2 = 0$ — connected by light.

For timelike separations, the *proper time* is defined as

$$(\Delta\tau)^2 = -(\Delta s)^2 = -\eta_{\mu\nu}\,\Delta x^\mu \Delta x^\nu \tag{1.13}$$

It is the time measured by a clock moving along the worldline between two events.

Inertial Frames and Clock Synchronization

Inertial frames are constructed from unaccelerated observers with synchronized clocks. Clocks are synchronized by light signals according to

$$t_2 = \frac{1}{2}(t_1 + t_1')$$

ensuring that light travel time is equal in both directions.

The Twin Paradox

An inertial observer's proper time between two events A and C is simply $\Delta\tau = \Delta t$. A traveling twin moving out and back with velocity v experiences a total proper time

$$\Delta\tau_{\text{moving}} = \sqrt{(\Delta t)^2 - (\Delta x)^2} = \Delta t\sqrt{1 - v^2} < \Delta t$$

Hence, the accelerated twin ages less. This is not a contradiction but a manifestation of the geometry of spacetime: the straight (inertial) worldline maximizes proper time.

General Trajectories and Proper Time Integral

The infinitesimal spacetime line element is

$$ds^2 = \eta_{\mu\nu}\,dx^\mu dx^\nu$$

and for any timelike trajectory $x^\mu(\lambda)$, the total proper time is

$$\Delta\tau = \int \sqrt{-\eta_{\mu\nu}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda}}\,d\lambda$$

and for any spacelike trajectory $x^\mu(\lambda)$, the total distance is

$$\Delta s = \int \sqrt{-\eta_{\mu\nu}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda}}\,d\lambda$$

This definition applies even to accelerated motion in flat spacetime.

1.3 Lorentz Transformations

This section provides a formal description of the coordinate transformations between different inertial frames. These are defined as the transformations that leave the spacetime interval, $(\Delta s)^2$, invariant.

Types of Spacetime Transformations

The transformations that preserve the interval consist of translations and linear transformations.

- **Translations:** These are simple shifts of the coordinate origin in space or time.

$$x^{\mu'} = \delta^{\mu'}_{\mu}(x^{\mu} + a^{\mu}) \tag{1.14}$$

  Here, a^{μ} is a constant four-vector representing the shift. Translations leave coordinate differences (Δx^{μ}) unchanged, so the invariance of the interval under translations is trivial.

- **Linear Transformations:** These include spatial rotations and boosts (changes to a frame with a constant velocity). They are described by multiplying the coordinate vector by a spacetime-independent 4×4 matrix, $\Lambda^{\mu'}{}_{\nu}$.

$$x^{\mu'} = \Lambda^{\mu'}_{\nu} x^{\nu} \tag{1.15}$$

  In matrix notation, this is written as $x' = \Lambda x$.

The Condition for a Lorentz Transformation

To find the properties of the matrix Λ, we impose the condition that the interval must be invariant.
Derivation

1. The spacetime interval is $(\Delta s)^2 = \eta_{\mu\nu}\Delta x^{\mu}\Delta x^{\nu}$. In matrix notation, this is $(\Delta s)^2 = (\Delta x)^T \eta (\Delta x)$.
2. In the new coordinate system, the interval is $(\Delta s')^2 = (\Delta x')^T \eta (\Delta x')$.
3. The coordinate differences transform as $\Delta x' = \Lambda \Delta x$. Substituting this into the expression for $(\Delta s')^2$ gives:
$$(\Delta s')^2 = (\Lambda \Delta x)^T \eta (\Lambda \Delta x) = (\Delta x)^T \Lambda^T \eta \Lambda (\Delta x)$$
4. For the interval to be invariant, we must have $(\Delta s)^2 = (\Delta s')^2$ for any arbitrary Δx. This requires that the matrices themselves be equal.

This leads to the fundamental condition that a matrix Λ must satisfy to be a Lorentz transformation.

$$\eta = \Lambda^T \eta \Lambda \quad \text{(Matrix Form)} \tag{1.16}$$

$$\eta_{\rho\sigma} = \Lambda^{\mu'}_{\rho} \Lambda^{\nu'}_{\sigma} \eta_{\mu'\nu'} \quad \text{(Index Form)} \tag{1.17}$$

The Lorentz Group

The set of all matrices Λ that satisfy the condition $\eta = \Lambda^T \eta \Lambda$ forms a mathematical group under matrix multiplication, known as the **Lorentz group**.

- **Analogy to the Rotation Group O(3):** There is a close analogy between the Lorentz group and the group of rotations in 3D Euclidean space, O(3). A 3D rotation matrix R preserves the Euclidean distance, which means it satisfies the condition $R^T R = I$, where I is the 3×3 identity matrix. The identity matrix is the metric of flat 3D space. The condition can be written as $I = R^T I R$ to make the analogy clear. The Lorentz group, often denoted **O(3,1)**, is thus a generalization of the rotation group where the Euclidean metric (I) is replaced by the Minkowski metric (η). A metric like η, which has one negative and three positive eigenvalues, is called **Lorentzian**.

- **Subgroups of the Lorentz Group:** The full group O(3,1) includes not only continuous rotations and boosts but also discrete transformations: **parity reversal** (spatial inversion, e.g., $x \to -x$) and **time reversal** ($t \to -t$). We can restrict to transformations with determinant $|\Lambda| = 1$. This is the **proper Lorentz group, SO(3,1)**. However, SO(3,1) still includes transformations that reverse the direction of time (e.g., a combined parity and time reversal).

To isolate the transformations continuously connected to the identity, we add the condition that the time-time component must be positive, $\Lambda^0{}_0 \geq 1$. This defines the **proper orthochronous** or **restricted Lorentz group**, which describes all physical transformations between inertial frames without reversing space or time orientation.

- **The Poincaré Group:** The set of both Lorentz transformations and spacetime translations forms the 10-parameter **Poincaré group**, which represents the full set of spacetime symmetries in special relativity.

Explicit Transformation Matrices

Spatial Rotation: A rotation by an angle θ in the $x - y$ plane has the form:

$$\Lambda^{\mu'}{}_{\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{1.18}$$

Boost: A boost can be thought of as a "rotation" between a time and a space direction. A boost along the x-direction is parameterized by a quantity ϕ called the **rapidity** or **boost parameter**.

$$\Lambda^{\mu'}{}_{\nu} = \begin{pmatrix} \cosh\phi & -\sinh\phi & 0 & 0 \\ -\sinh\phi & \cosh\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{1.19}$$

The rapidity ϕ is related to the relative velocity v between the frames.

Derivation of Velocity-Rapidity Relation

1. The transformed coordinates are $t' = t\cosh\phi - x\sinh\phi$ and $x' = -t\sinh\phi + x\cosh\phi$.
2. The origin of the primed frame is defined by the condition $x' = 0$.
3. Setting $x' = 0$ implies $-t\sinh\phi + x\cosh\phi = 0$.
4. The velocity of this origin as seen from the unprimed frame is $v = x/t$. Solving for this gives:

$$v = \frac{x}{t} = \frac{\sinh\phi}{\cosh\phi} = \tanh\phi$$

Using the identities $\cosh\phi = \gamma$ and $\sinh\phi = v\gamma$ (where $\gamma = 1/\sqrt{1 - v^2}$), the boost transformation can be written in the more familiar form:

$$t' = \gamma(t - vx) \tag{1.20}$$
$$x' = \gamma(x - vt) \tag{1.21}$$

Geometric Interpretation of Boosts

On a spacetime diagram, a Lorentz boost appears as a transformation of the coordinate axes.

- **Transformation of Axes:** The axes of the boosted frame are rotated towards each other, appearing to "scissor" together. The new time axis (t'-axis, defined by $x' = 0$) is the line $t = x/v$. The new space axis (x'-axis, defined by $t' = 0$) is the line $t = vx$.
- **Invariance of the Light Cone:** The paths of light rays, $x = \pm t$, are identical to the paths $x' = \pm t'$. The light cone itself is unchanged by the boost. This is the geometric expression of the postulate that the speed of light is the same in all inertial frames.
- **Relativity of Simultaneity:** Events that are simultaneous in the (t, x) frame (i.e., lie on a line of constant t) are not simultaneous in the (t', x') frame (they do not lie on a line of constant t').

1.4 Vectors

To probe the structure of Minkowski space in more detail, it is necessary to introduce the concepts of vectors and tensors. In spacetime, vectors are four-dimensional and are often referred to as **four-vectors**.

Vectors and the Tangent Space

Beyond dimensionality, the most important concept is that each vector is located at a **given point in spacetime**. The notion of "free" vectors that can be moved around is not useful in the context of curved spaces.

- At each point p in spacetime, we associate the set of all possible vectors located at that point. This set is a vector space known as the **tangent space** at p, or T_p.
- **Geometric Intuition:** The name is inspired by imagining the set of vectors at a point on a curved two-dimensional surface as comprising a plane tangent to that point. It is important to think of these vectors as existing only at that single point.

A vector space is a collection of objects (vectors) that can be added together and multiplied by real numbers in a linear way. For any two vectors V and W and real numbers a and b, we have:

$$(a+b)(V+W) = aV + bV + aW + bW \tag{1.22}$$

Basis Vectors and Components

While a vector is a well-defined geometric object, it is often useful to decompose it into components with respect to a set of **basis vectors**. A basis is a set of vectors that is linearly independent and spans the vector space. For the four-dimensional tangent space at a point in Minkowski space, a basis consists of four vectors.

Let us consider a basis of four vectors $\hat{e}_{(\mu)}$, where $\mu \in \{0, 1, 2, 3\}$. Any abstract vector A can be written as a linear combination of these basis vectors:

$$A = A^\mu \hat{e}_{(\mu)} \tag{1.23}$$

The coefficients A^μ are the **components** of the vector A. It is crucial to distinguish between the abstract geometric entity (the vector A) and its components (A^μ), which are merely numerical coefficients in a chosen basis. The parentheses around the index on the basis vectors, $\hat{e}_{(\mu)}$, serve as a reminder that this is a collection of vectors, not the components of a single vector.

Example: Tangent Vector to a Curve

A standard example of a vector in spacetime is the tangent vector to a curve. A parameterized path through spacetime is specified by the coordinates as a function of a parameter, $x^\mu(\lambda)$. The tangent vector $V(\lambda)$ to this curve has components:

$$V^\mu = \frac{dx^\mu}{d\lambda} \tag{1.24}$$

Transformation Properties of Vectors

Under a Lorentz transformation, the coordinates x^μ change, and therefore the components of the tangent vector must also change. The components transform according to the Lorentz transformation matrix Λ:

$$V^{\mu'} = \Lambda^{\mu'}{}_\nu V^\nu \tag{1.25}$$

However, the abstract vector V itself is a geometric object and must be invariant under a change of coordinates. We can use this fact to derive the transformation properties of the basis vectors.

Derivation of Basis Vector Transformation

1. The vector V is invariant, so it can be expressed in either the old basis $\{\hat{e}_{(\mu)}\}$ or the new basis $\{\hat{e}_{(\nu')}\}$:
$$V = V^\mu \hat{e}_{(\mu)} = V^{\nu'} \hat{e}_{(\nu')}$$

2. Substitute the transformation rule for the components, $V^{\nu'} = \Lambda^{\nu'}{}_\mu V^\mu$, into the equation:
$$V = (\Lambda^{\nu'}{}_\mu V^\mu) \hat{e}_{(\nu')}$$

3. Comparing this with the original expression $V = V^\mu \hat{e}_{(\mu)}$, and noting that this must hold for any vector (i.e., for any set of components V^μ), we can equate the basis vectors:
$$\hat{e}_{(\mu)} = \Lambda^{\nu'}{}_\mu \hat{e}_{(\nu')}$$

4. To find how the new basis vectors are expressed in terms of the old ones, we multiply by the inverse of the Lorentz transformation. We denote the inverse matrix by $\Lambda^\mu{}_{\nu'}$, such that $\Lambda^\mu{}_{\nu'} \Lambda^{\nu'}{}_\rho = \delta^\mu_\rho$. This gives the transformation rule for the basis vectors:
$$\hat{e}_{(\nu')} = \Lambda^\mu{}_{\nu'} \hat{e}_{(\mu)} \tag{1.26}$$

This "contravariant" transformation behavior (one transforms oppositely to the other) ensures that the full geometric object, $V = V^\mu \hat{e}_{(\mu)}$, is invariant under coordinate changes. This is the foundation of tensor notation.

1.5 Dual Vectors (One-Forms)

Once a vector space is established, one can define an associated vector space of equal dimension known as the **dual vector space**. The dual space to the tangent space T_p is called the **cotangent space** and is denoted T_p^*. Its elements are called **dual vectors**, **covariant vectors**, or **one-forms**.

Definition of a Dual Vector

A dual vector $\omega \in T_p^*$ is a **linear map** from the original vector space T_p to the real numbers $\mathbb{R}$. This means that for any vectors $V, W \in T_p$ and any real numbers a, b:
$$\omega(aV + bW) = a\omega(V) + b\omega(W) \in \mathbb{R} \tag{1.27}$$

The set of these maps forms a vector space itself. If ω and η are dual vectors, then their linear combination acts on a vector V as:
$$(a\omega + b\eta)(V) = a\omega(V) + b\eta(V) \tag{1.28}$$

Basis and Components of Dual Vectors

To make this construction more concrete, we can introduce a set of **basis dual vectors** $\hat{\theta}^{(\nu)}$ by demanding a specific relationship with the basis vectors $\hat{e}_{(\mu)}$ of the original vector space:
$$\hat{\theta}^{(\nu)}(\hat{e}_{(\mu)}) = \delta^\nu_\mu \tag{1.29}$$

where δ^ν_μ is the Kronecker delta. Every dual vector can then be written as a linear combination of this basis. The components are labeled with lower indices:
$$\omega = \omega_\mu \hat{\theta}^{(\mu)} \tag{1.30}$$

The action of a dual vector on a vector can then be expressed simply in terms of their components.

Derivation of the Component Action

$$\begin{aligned}
\omega(V) &= (\omega_\mu \hat{\theta}^{(\mu)})(V^\nu \hat{e}_{(\nu)}) \\
&= \omega_\mu V^\nu \hat{\theta}^{(\mu)}(\hat{e}_{(\nu)}) \quad \text{(by linearity)} \\
&= \omega_\mu V^\nu \delta^\mu_\nu \\
&= \omega_\mu V^\mu
\end{aligned}$$

This yields the simple and powerful result:

$$\omega(V) = \omega_\mu V^\mu \in \mathbb{R} \tag{1.31}$$

This also implies that vectors can be seen as linear maps on dual vectors, $V(\omega) \equiv \omega(V)$, meaning the dual of the dual space is the original vector space itself.

Dual Vector Fields and Transformation Properties

In spacetime, we are interested in fields of vectors and dual vectors. The action of a dual vector field on a vector field is not a single number, but a **scalar** (a function) on spacetime. A scalar is a quantity without indices that is unchanged by Lorentz transformations.

The components of a dual vector must transform in such a way that the scalar product $\omega_\mu V^\mu$ is invariant. Since we know $V^{\mu'} = \Lambda^{\mu'}{}_\nu V^\nu$, the components of a dual vector must transform with the inverse matrix:

$$\omega_{\mu'} = \Lambda^\nu{}_{\mu'} \omega_\nu \tag{1.32}$$

Correspondingly, the basis dual vectors transform with the original Lorentz matrix:

$$\hat{\theta}^{(\rho')} = \Lambda^{\rho'}{}_\sigma \hat{\theta}^{(\sigma)} \tag{1.33}$$

This is consistent with the general rule of index placement: upper indices transform with Λ, lower indices transform with the inverse.

Example: The Gradient of a Scalar Function

The simplest and most important physical example of a dual vector is the **gradient** of a scalar function ϕ. The gradient $d\phi$ is the dual vector whose components are the partial derivatives of ϕ:

$$d\phi = \frac{\partial \phi}{\partial x^\mu} \hat{\theta}^{(\mu)} \tag{1.34}$$

The transformation of these components under a Lorentz transformation follows directly from the chain rule for partial derivatives:

$$\frac{\partial \phi}{\partial x^{\mu'}} = \frac{\partial x^\nu}{\partial x^{\mu'}} \frac{\partial \phi}{\partial x^\nu} = \Lambda^\nu{}_{\mu'} \frac{\partial \phi}{\partial x^\nu} \tag{1.35}$$

This is precisely the transformation law for the components of a dual vector. Common shorthand notations for the partial derivative are:

$$\frac{\partial \phi}{\partial x^\mu} = \partial_\mu \phi = \phi_{,\mu} \tag{1.36}$$

The natural action of the gradient (a dual vector) on the tangent vector to a curve $x^\mu(\lambda)$ (a vector) is the ordinary derivative of the function along that curve:

$$\partial_\mu \phi \frac{dx^\mu}{d\lambda} = \frac{d\phi}{d\lambda} \tag{1.37}$$

1.6 Tensors

A straightforward generalization of vectors and dual vectors is the notion of a tensor. Just as a dual vector is a linear map from vectors to $\mathbb{R}$, a tensor is a multilinear map from a collection of dual vectors and vectors to $\mathbb{R}$.

Definition of a Tensor

A **tensor** T of type (or rank) (k, l) is a multilinear map from k copies of the cotangent space T_p^* and l copies of the tangent space T_p to the real numbers:

$$T : \underbrace{T_p^* \times \cdots \times T_p^*}_{k \text{ times}} \times \underbrace{T_p \times \cdots \times T_p}_{l \text{ times}} \to \mathbb{R} \tag{1.38}$$

Multilinearity means that the tensor acts linearly in each of its arguments. For a tensor of type (1, 1), we have:

$$T(a\omega + b\eta, cV + dW) = acT(\omega, V) + adT(\omega, W) + bcT(\eta, V) + bdT(\eta, W) \tag{1.39}$$

From this point of view: a scalar is a type (0,0) tensor, a vector is a type (1,0) tensor, and a dual vector is a type (0,1) tensor.

Tensor Products and Basis

To construct a basis for the space of all (k, l) tensors, we define the **tensor product** $\otimes$. If T is a (k, l) tensor and S is an (m, n) tensor, their tensor product $T \otimes S$ is a $(k + m, l + n)$ tensor defined by its action on a collection of dual vectors ($\omega^{(i)}$) and vectors ($V^{(j)}$):

$$(T \otimes S)(\omega^{(1)}, \ldots, V^{(1)}, \ldots) = T(\omega^{(1)}, \ldots, V^{(1)}, \ldots) \times S(\omega^{(k+1)}, \ldots, V^{(l+1)}, \ldots) \tag{1.40}$$

A basis for the space of (k, l) tensors is formed by taking all tensor products of the basis vectors and dual vectors:

$$\hat{e}_{(\mu_1)} \otimes \cdots \otimes \hat{e}_{(\mu_k)} \otimes \hat{\theta}^{(\nu_1)} \otimes \cdots \otimes \hat{\theta}^{(\nu_l)} \tag{1.41}$$

Tensor Components and Transformation

An arbitrary tensor T can be written as a linear combination of basis tensors, where the coefficients are its **components**:

$$T = T^{\mu_1 \ldots \mu_k}{}_{\nu_1 \ldots \nu_l} \hat{e}_{(\mu_1)} \otimes \cdots \otimes \hat{\theta}^{(\nu_l)} \tag{1.42}$$

The components can be found by acting the tensor on the basis vectors and dual vectors:

$$T^{\mu_1 \ldots \mu_k}{}_{\nu_1 \ldots \nu_l} = T(\hat{\theta}^{(\mu_1)}, \ldots, \hat{\theta}^{(\mu_k)}, \hat{e}_{(\nu_1)}, \ldots, \hat{e}_{(\nu_l)}) \tag{1.43}$$

The transformation of tensor components under a Lorentz transformation $\Lambda^{\mu'}{}_{\mu}$ is a direct generalization of the vector and dual vector cases: each upper index gets transformed like a vector, and each lower index gets transformed like a dual vector.

$$T^{\mu'_1 \ldots \mu'_k}{}_{\nu'_1 \ldots \nu'_l} = \Lambda^{\mu'_1}{}_{\mu_1} \ldots \Lambda^{\mu'_k}{}_{\mu_k} \Lambda^{\nu_1}{}_{\nu'_1} \ldots \Lambda^{\nu_l}{}_{\nu'_l} T^{\mu_1 \ldots \mu_k}{}_{\nu_1 \ldots \nu_l} \tag{1.44}$$

Tensors can also be thought of as maps between other tensors. For example, a (1,1) tensor can map a vector to another vector: $T^\mu{}_\nu : V^\nu \to T^\mu{}_\nu V^\nu$. This viewpoint emphasizes that tensors are geometric objects with a meaning independent of any coordinate system.

Examples of Tensors in Spacetime

- **The Metric Tensor, $\eta_{\mu\nu}$:** A type (0,2) tensor. Its action on two vectors defines the **inner product** (or scalar product):

$$\eta(V, W) = \eta_{\mu\nu} V^\mu W^\nu = V \cdot W \tag{1.45}$$

  The norm of a vector is $V \cdot V$. Unlike in Euclidean space, this is not positive definite. A vector V is called:

  - **timelike** if $V \cdot V < 0$.
  - **lightlike** or **null** if $V \cdot V = 0$.
  - **spacelike** if $V \cdot V > 0$.

- **The Inverse Metric, $\eta^{\mu\nu}$:** A type (2,0) tensor defined by the relation:

$$\eta^{\mu\rho}\eta_{\rho\nu} = \delta^\mu_\nu \tag{1.46}$$

  In flat spacetime with Cartesian coordinates, the components of $\eta^{\mu\nu}$ are numerically identical to those of $\eta_{\mu\nu}$.

- **The Levi-Civita Symbol, $\tilde{\epsilon}_{\mu\nu\rho\sigma}$:** A type (0,4) tensor defined as:

$$\tilde{\epsilon}_{\mu\nu\rho\sigma} = \begin{cases} +1 & \text{if } \mu\nu\rho\sigma \text{ is an even permutation of } 0123 \\ -1 & \text{if } \mu\nu\rho\sigma \text{ is an odd permutation of } 0123 \\ 0 & \text{otherwise} \end{cases} \tag{1.47}$$

  Note: This object is technically a "tensor density" and will be treated more carefully in the context of general relativity.

- **The Electromagnetic Field Strength Tensor, $F_{\mu\nu}$:** A type (0,2) antisymmetric tensor that unifies the electric and magnetic fields.

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix} = -F_{\nu\mu} \tag{1.48}$$

  This tensor structure elegantly shows how electric and magnetic fields transform into one another under Lorentz boosts.

1.7 Manipulating Tensors

With the definition of tensors established, we can now systematize some of their important properties and operations.

Contraction

Contraction is an operation that turns a tensor of type (k, l) into a tensor of type $(k-1, l-1)$. It is performed by summing over one upper and one lower index. For example, if we have a tensor $T^{\mu\nu\rho}{}_{\sigma\lambda}$, we can contract the second upper index (ν) with the second lower index (λ) to form a new tensor:

$$S^{\mu\rho}{}_\sigma = T^{\mu\nu\rho}{}_{\sigma\nu} \tag{1.49}$$

It is only permissible to contract an upper index with a lower index; contracting two indices of the same type does not result in a well-defined tensor. Note also that the order of the remaining indices matters, so that in general:

$$T^{\mu\nu\rho}{}_{\sigma\nu} \neq T^{\mu\rho\nu}{}_{\sigma\nu} \tag{1.50}$$

Raising and Lowering Indices

The metric $\eta_{\mu\nu}$ and its inverse $\eta^{\mu\nu}$ can be used to raise and lower the indices on any tensor. This effectively changes a tensor of one type to another. Given a tensor $T^{\alpha\beta}{}_{\gamma\delta}$, we can define new tensors:

$$T^{\alpha\beta\mu}{}_{\delta} = \eta^{\mu\gamma} T^{\alpha\beta}{}_{\gamma\delta} \quad \text{(raising the } \gamma \text{ index)} \tag{1.51}$$

$$T_{\mu}{}^{\beta}{}_{\gamma\delta} = \eta_{\mu\alpha} T^{\alpha\beta}{}_{\gamma\delta} \quad \text{(lowering the } \alpha \text{ index)} \tag{1.52}$$

$$T_{\mu\nu}{}^{\rho\sigma} = \eta_{\mu\alpha}\eta_{\nu\beta}\eta^{\rho\gamma}\eta^{\sigma\delta} T^{\alpha\beta}{}_{\gamma\delta} \tag{1.53}$$

This operation does not change the horizontal position of an index relative to others. For example, vectors and dual vectors can be converted into one another:

$$V_\mu = \eta_{\mu\nu} V^\nu, \quad \omega^\mu = \eta^{\mu\nu}\omega_\nu \tag{1.54}$$

Because the metric and its inverse are truly inverses, a pair of indices being contracted over can be raised and lowered simultaneously without changing the result:

$$A^\lambda B_\lambda = \eta^{\lambda\rho} A_\rho \eta_{\lambda\sigma} B^\sigma = \delta^\rho_\sigma A_\rho B^\sigma = A_\sigma B^\sigma \tag{1.55}$$

In a Lorentzian spacetime, the components of a vector V^μ are not numerically equal to the components of its dual vector V_μ:

$$V^\mu = (V^0, V^1, V^2, V^3) \quad \Longrightarrow \quad V_\mu = (-V^0, V^1, V^2, V^3) \tag{1.56}$$

This distinction becomes even more critical in curved spacetimes, where the metric components are more complicated.

Symmetry and Antisymmetry

A tensor is **symmetric** in a set of its indices if it is unchanged upon their exchange. For example, $S_{\mu\nu\rho}$ is symmetric in its first two indices if:

$$S_{\mu\nu\rho} = S_{\nu\mu\rho} \tag{1.57}$$

A tensor is **antisymmetric** (or skew-symmetric) in a set of its indices if it changes sign upon their exchange. For example, $A_{\mu\nu\rho}$ is antisymmetric in its first and third indices if:

$$A_{\mu\nu\rho} = -A_{\rho\nu\mu} \tag{1.58}$$

The metric $\eta_{\mu\nu}$ is symmetric, while the electromagnetic field strength tensor $F_{\mu\nu}$ is antisymmetric.

Symmetrization and Antisymmetrization

Any tensor can be symmetrized or antisymmetrized over any number of its upper or lower indices.

- **Symmetrization** (denoted by round brackets) is the average over all permutations of the indices:

$$T_{(\mu_1 \dots \mu_n)} = \frac{1}{n!}(\text{sum of } T \text{ with indices } \mu_1 \dots \mu_n \text{ permuted}) \tag{1.59}$$

- **Antisymmetrization** (denoted by square brackets) is the alternating sum over all permutations:

$$T_{[\mu_1 \dots \mu_n]} = \frac{1}{n!}(\text{alternating sum of } T \text{ with indices } \mu_1 \dots \mu_n \text{ permuted}) \tag{1.60}$$

  For example, for three indices:

$$T_{[\mu\nu\rho]} = \frac{1}{6}(T_{\mu\nu\rho} - T_{\mu\rho\nu} + T_{\rho\mu\nu} - T_{\nu\mu\rho} + T_{\nu\rho\mu} - T_{\rho\nu\mu}) \tag{1.61}$$

For any two indices, a tensor can be decomposed into its symmetric and antisymmetric parts: $T_{\mu\nu} = T_{(\mu\nu)} + T_{[\mu\nu]}$. This simple decomposition does not hold for three or more indices due to the existence of more complex symmetry properties.

The Trace

The **trace** of a (1,1) tensor $X^\mu{}_\nu$ is its contraction, which is a scalar:

$$X = X^\lambda{}_\lambda \tag{1.62}$$

For a (0,2) tensor $Y_{\mu\nu}$, the trace is defined as raising one index and then contracting:

$$Y = Y^\lambda{}_\lambda = \eta^{\mu\nu} Y_{\mu\nu} \tag{1.63}$$

Note that this is not the sum of the diagonal components of $Y_{\mu\nu}$. For example, the trace of the metric itself is:

$$\eta^{\mu\nu}\eta_{\mu\nu} = \delta^\mu_\mu = 4 \tag{1.64}$$

A Note on Partial Derivatives

In the special case of **flat spacetime with inertial coordinates**, the partial derivative of a (k, l) tensor is a well-defined $(k, l+1)$ tensor. For example, if $T^\mu{}_\nu$ is a tensor, then so is:

$$S_\alpha{}^\mu{}_\nu = \partial_\alpha T^\mu{}_\nu \tag{1.65}$$

Important: This property will **fail** in more general (curved) spacetimes or non-inertial coordinate systems. We will later introduce a **covariant derivative** to replace the partial derivative in such cases. The one exception is the gradient of a scalar, $\partial_\alpha \phi$, which is always a well-defined (0,1) tensor. In flat spacetime, partial derivatives commute:

$$\partial_\mu \partial_\nu(\dots) = \partial_\nu \partial_\mu(\dots) \tag{1.66}$$

1.8 Maxwell's Equations

With the necessary tensor know-how accumulated, we can now illustrate these concepts using Maxwell's equations of electrodynamics. This exercise demonstrates the power and economy of the tensor formalism.

Maxwell's Equations in Traditional Notation

In 19th-century 3-vector notation, Maxwell's equations are:

$$\nabla \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{J} \tag{1.67}$$
$$\nabla \cdot \mathbf{E} = \rho \tag{1.68}$$
$$\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0 \tag{1.69}$$
$$\nabla \cdot \mathbf{B} = 0 \tag{1.70}$$

Here, $\mathbf{E}$ and $\mathbf{B}$ are the electric and magnetic field 3-vectors, $\mathbf{J}$ is the current 3-vector, and ρ is the charge density. These equations are, in fact, invariant under Lorentz transformations, but they do not look obviously so. Tensor notation makes this invariance manifest.

From 3-Vector to Tensor Formulation

The first step is to rewrite the equations in component notation and then combine them into a single tensor object. We begin by defining the current 4-vector $J^\mu = (\rho, J^x, J^y, J^z)$ and recalling the definition of the antisymmetric field strength tensor $F_{\mu\nu}$.

The goal is to show that the four equations above are equivalent to the following two tensor equations.

1. The Inhomogeneous Equations (with sources)

Derivation:

1. The first two of Maxwell's equations, $\nabla \cdot \mathbf{E} = \rho$ and $\nabla \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}$, involve the sources of the fields, ρ and $\mathbf{J}$.

2. In component notation (with $x^0 = t$), these are:

$$\partial_i E^i = J^0 \quad \text{and} \quad \tilde{\epsilon}^{ijk} \partial_j B_k - \partial_0 E^i = J^i$$

   where $\tilde{\epsilon}^{ijk}$ is the three-dimensional Levi-Civita symbol.

3. We can express the components of the field strength tensor with *upper* indices, $F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta}$, in terms of the electric and magnetic fields (1.48) :

$$F^{0i} = E^i, \quad F^{ij} = \tilde{\epsilon}^{ijk} B_k \tag{1.71}$$

4. Substituting these into the component equations gives:

$$\partial_i F^{0i} = J^0 \quad \text{and} \quad \partial_j F^{ji} - \partial_0 F^{0i} = J^i \tag{1.72}$$

5. Using the antisymmetry of the field strength tensor (i.e., $F^{\mu\nu} = -F^{\nu\mu}$, which implies $F^{00} = 0$ and $\partial_0 F^{00} = 0$), these two distinct equations can be combined into a single, compact 4-vector equation.

The result is the first of the covariant Maxwell equations:

$$\partial_\mu F^{\nu\mu} = J^\nu \tag{1.73}$$

2. The Homogeneous Equations (sourceless)
A similar line of reasoning reveals that the third and fourth of Maxwell's equations, $\nabla \cdot \mathbf{B} = 0$ and $\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$, can also be written in a single tensor form. The result is:

$$\partial_{[\mu} F_{\nu\lambda]} = 0 \tag{1.74}$$

Due to the antisymmetry of $F_{\mu\nu}$, this is equivalent to the expanded form:

$$\partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} = 0 \tag{1.75}$$

Significance of the Covariant Formulation

The four traditional Maxwell equations are replaced by two compact tensor equations, demonstrating the economy of the notation. More importantly, both sides of equations (1.73) and (1.74) are well-defined tensors. An equation that equates two tensors of the same type is said to be **manifestly covariant**. If such an equation is true in one inertial frame, it must be true in any Lorentz-transformed frame. This is why tensors are the natural language for relativity: they allow physical laws to be expressed in a frame-independent way.

- The tensor form is referred to as the **covariant form** of Maxwell's equations.
- The original 3-vector form is referred to as **noncovariant**.

1.9 Energy and Momentum

This section reviews the physics of energy and momentum in Minkowski spacetime, starting with single particles and extending to continuous media.

Particle Kinematics

The worldline of a massive particle is a timelike curve. Since the proper time τ is the time measured by a clock traveling along such a worldline, it is the natural parameter to use for the path, $x^\mu(\tau)$.

Four-Velocity and Four-Momentum

- **Four-Velocity:** The tangent vector to a particle's worldline parameterized by proper time is the **four-velocity**, U^μ.
$$U^\mu = \frac{dx^\mu}{d\tau} \tag{1.76}$$
From the definition of proper time, $d\tau^2 = -\eta_{\mu\nu}dx^\mu dx^\nu$, the four-velocity is automatically normalized:
$$\eta_{\mu\nu}U^\mu U^\nu = -1 \tag{1.77}$$
This reflects the fact that a massive particle always travels through spacetime at a constant "rate." In the particle's rest frame, its four-velocity has components $U^\mu = (1, 0, 0, 0)$.

- **Four-Momentum:** The **momentum four-vector** is defined as the rest mass m (an invariant scalar) times the four-velocity.
$$p^\mu = mU^\mu \tag{1.78}$$
The components of the four-momentum are the energy and the three-momentum: $p^\mu = (E, p^x, p^y, p^z)$.
  - In the particle's rest frame, $p^\mu = (m, 0, 0, 0)$, giving $E = m$ (or $E = mc^2$ in standard units).
  - For a particle moving with three-velocity $v = dx/dt$ along the x-axis, the components are:
$$p^\mu = (\gamma m, v\gamma m, 0, 0) \tag{1.79}$$
where $\gamma = 1/\sqrt{1 - v^2}$.
  - The norm of the four-momentum is an invariant:
$$p_\mu p^\mu = \eta_{\mu\nu}p^\mu p^\nu = -m^2 \tag{1.80}$$
This gives the relativistic energy-momentum relation: $E^2 - p^2 = m^2$, or
$$E = \sqrt{m^2 + p^2} \tag{1.81}$$

Four-Force

The relativistic analogue of Newton's Second Law introduces the **force four-vector** f^μ:
$$f^\mu = m\frac{d^2x^\mu}{d\tau^2} = \frac{dp^\mu}{d\tau} \tag{1.82}$$
For electromagnetism, the Lorentz force law has the unique tensorial generalization:
$$f^\mu = qF^\mu{}_\lambda U^\lambda \tag{1.83}$$
where q is the particle's charge and F^μ_λ is the electromagnetic field strength tensor.

The Energy-Momentum Tensor

For extended systems like fluids or fields, we use the **energy-momentum tensor** (or stress-energy tensor), $T^{\mu\nu}$.

- **Physical Meaning:** $T^{\mu\nu}$ is the flux of the μ-component of four-momentum (p^μ) across a surface of constant x^ν.

- **Components in the Rest Frame:**
  - $T^{00} = \rho$ (Energy density)
  - $T^{0i} = T^{i0}$ (Momentum density / Energy flux)
  - T^{ij} (Stress, i.e., flux of i-momentum in the j-direction). The diagonal terms T^{ii} (no sum) represent pressure:
$$p_i = T^{ii} \tag{1.84}$$

Examples of Energy-Momentum Tensors

- **Dust:** A collection of non-interacting particles all at rest with respect to each other. It is characterized by a single four-velocity field U^μ and a rest-frame number density n. The **number-flux four-vector** is:

$$N^\mu = nU^\mu \tag{1.85}$$

If each particle has mass m, the rest-frame energy density is $\rho = mn$. The energy-momentum tensor for dust is:

$$T^{\mu\nu}_{\text{dust}} = \rho U^\mu U^\nu \tag{1.86}$$

Dust has zero pressure. here we assumed rest frame so now we are going to see how would the N^μ and the $T^{\mu\nu}$ look like in an moving frame where we use $U^{\mu'}$. on the inertial frame we have $U^\mu = (1, 0, 0, 0)$ which give

$$N^\mu = nU^\mu = (n, 0, 0, 0)$$

and also we have the energy tensor:

$$T^{\mu\nu} = \rho U^\mu U^\nu = \rho \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{1.87}$$

Now if we move to a transformed frame where the particles are moving with some velocity $U^{\mu'}$ the four-velocity transforms as:

$$U^{\mu'} = \Lambda^{\mu'}{}_\mu U^\mu \tag{1.88}$$

so the number flux and the energy tensor will transform as:

$$N^{\mu'} = \Lambda^{\mu'}{}_\mu N^\mu$$
$$T^{\mu'\nu'} = \Lambda^{\mu'}{}_\mu \, \Lambda^{\nu'}{}_\nu \, T^{\mu\nu}$$

where $\Lambda^{\mu'}{}_\mu$ is the transformation matrix corresponding to the new frame. if we start with lorentz rotation around arbitrary direction:

$$\Lambda^{\mu'}{}_\nu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & R_{11} & R_{12} & R_{13} \\ 0 & R_{21} & R_{22} & R_{23} \\ 0 & R_{31} & R_{32} & R_{33} \end{pmatrix} \tag{1.89}$$

where R_{ij} are the components of the 3D rotation matrix. applying this transformation to the four-velocity in the rest frame will give us the same four-velocity in the rest frame since the time component will remain unchanged and the spatial components will be zero.

$$U^{\mu'} = \Lambda^{\mu'}{}_\mu U^\mu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & R_{11} & R_{12} & R_{13} \\ 0 & R_{21} & R_{22} & R_{23} \\ 0 & R_{31} & R_{32} & R_{33} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = U^\mu \tag{1.90}$$

we can clearly see that the U^μ will remain the same after the transformation and so that the energy-tensor and the number density will be the same. So now let's consider a boost in an arbitrary direction with velocity components (v_x, v_y, v_z):

$$\Lambda^{\mu'}{}_\mu = \begin{pmatrix} \gamma & -\gamma v_x & -\gamma v_y & -\gamma v_z \\ -\gamma v_x & 1 + (\gamma - 1)\frac{v_x^2}{v^2} & (\gamma - 1)\frac{v_x v_y}{v^2} & (\gamma - 1)\frac{v_x v_z}{v^2} \\ -\gamma v_y & (\gamma - 1)\frac{v_y v_x}{v^2} & 1 + (\gamma - 1)\frac{v_y^2}{v^2} & (\gamma - 1)\frac{v_y v_z}{v^2} \\ -\gamma v_z & (\gamma - 1)\frac{v_z v_x}{v^2} & (\gamma - 1)\frac{v_z v_y}{v^2} & 1 + (\gamma - 1)\frac{v_z^2}{v^2} \end{pmatrix} \tag{1.91}$$

$$
\begin{aligned}
U^{\mu'} &= \Lambda^{\mu'}_{\mu} U^{\mu} \\
&= \begin{pmatrix} \gamma & -\gamma v_x & -\gamma v_y & -\gamma v_z \\ -\gamma v_x & 1+(\gamma-1)\frac{v_x^2}{v^2} & (\gamma-1)\frac{v_x v_y}{v^2} & (\gamma-1)\frac{v_x v_z}{v^2} \\ -\gamma v_y & (\gamma-1)\frac{v_y v_x}{v^2} & 1+(\gamma-1)\frac{v_y^2}{v^2} & (\gamma-1)\frac{v_y v_z}{v^2} \\ -\gamma v_z & (\gamma-1)\frac{v_z v_x}{v^2} & (\gamma-1)\frac{v_z v_y}{v^2} & 1+(\gamma-1)\frac{v_z^2}{v^2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\
&= \gamma \begin{pmatrix} 1 \\ -v_x \\ -v_y \\ -v_z \end{pmatrix}
\end{aligned}
$$

so now we can calculate the number-flux and the energy tensor four-vector in this frame:

$$
N^{\mu'} = nU^{\mu'} = n\gamma \begin{pmatrix} 1 \\ -v_x \\ -v_y \\ -v_z \end{pmatrix} \tag{1.92}
$$

now for the energy tensor we have:

$$
T^{\mu'\nu'} = \rho U^{\mu'} U^{\nu'} = \rho\gamma^2 \begin{pmatrix} 1 & -v_x & -v_y & -v_z \\ -v_x & v_x^2 & v_x v_y & v_x v_z \\ -v_y & v_y v_x & v_y^2 & v_y v_z \\ -v_z & v_z v_x & v_z v_y & v_z^2 \end{pmatrix} \tag{1.93}
$$

- **Perfect Fluid:** A fluid that is completely characterized by its rest-frame energy density ρ and an isotropic rest-frame pressure p. In its rest frame, its energy-momentum tensor is:

$$
T^{\mu\nu}_{\text{rest}} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix} \tag{1.94}
$$

The general, covariant expression that is valid in any inertial frame is:

$$
T^{\mu\nu} = (\rho + p)U^{\mu}U^{\nu} + p\eta^{\mu\nu} \tag{1.95}
$$

The physics of a specific perfect fluid is determined by an **equation of state**, $p = p(\rho)$. An important example is **vacuum energy**, for which $p_{\text{vac}} = -\rho_{\text{vac}}$, leading to $T^{\mu\nu}_{\text{vac}} = -\rho_{\text{vac}}\eta^{\mu\nu}$. Again, the four-velocity will be the same after transforming under rotation on arbitrary direction, and for the boost in arbitrary direction we will have:

$$
T^{\mu'\nu'} = (\rho + p)U^{\mu'}U^{\nu'} + p\eta^{\mu\nu}
$$

$$
= (\rho + p) \begin{pmatrix} \gamma \\ -\gamma v_x \\ -\gamma v_y \\ -\gamma v_z \end{pmatrix} \begin{pmatrix} \gamma & -\gamma v_x & -\gamma v_y & -\gamma v_z \end{pmatrix} + p \; dig(-1, 1, 1, 1)
$$

$$
= \begin{pmatrix} \gamma^2(\rho+p) - p & -\gamma^2 v_x(\rho+p) & -\gamma^2 v_y(\rho+p) & -\gamma^2 v_z(\rho+p) \\ -\gamma^2 v_x(\rho+p) & \gamma^2 v_x^2(\rho+p) + p & \gamma^2 v_x v_y(\rho+p) & \gamma^2 v_x v_z(\rho+p) \\ -\gamma^2 v_y(\rho+p) & \gamma^2 v_y v_x(\rho+p) & \gamma^2 v_y^2(\rho+p) + p & \gamma^2 v_y v_z(\rho+p) \\ -\gamma^2 v_z(\rho+p) & \gamma^2 v_z v_x(\rho+p) & \gamma^2 v_z v_y(\rho+p) & \gamma^2 v_z^2(\rho+p) + p \end{pmatrix}
$$

Conservation of Energy-Momentum

The energy-momentum tensor for a closed system is conserved, which is expressed as the vanishing of its four-divergence:

$$\begin{aligned}&\partial_\mu T^{\mu\nu} = \partial_\mu\left((\rho+p)U^\mu U^\nu\right) + \partial^\nu p = 0\\ &\implies 0 = \partial_\mu(\rho+p)U^\mu U^\nu + (\rho+p)U^\mu\partial_\mu U^\nu + (\rho+p)U^\nu\partial_\mu U^\mu + \partial^\nu p\end{aligned} \tag{1.96}$$

This is a set of four equations. The $\nu = 0$ component expresses conservation of energy, while the $\nu = k$ components express conservation of momentum.

Analysis for a Perfect Fluid: Applying the conservation law to the perfect fluid tensor yields:

$$\partial_\mu((\rho+p)U^\mu U^\nu) + \partial^\nu p = 0$$

This equation can be projected into parts parallel and perpendicular to the fluid's four-velocity U^μ.

Projection Parallel to U^μ (Energy Conservation): Contracting with U_ν (multipliying the equation (1.96)) and setting the result to zero gives the relativistic equation for energy conservation. using the fact that $U_\nu U^\nu = -1$ is constant along the flow, we have:

$$U_\nu\partial_\mu U^\nu = \frac{1}{2}\partial_\mu(U_\nu U^\nu) = 0 \tag{1.97}$$

Which can be shown as follows:

$$\begin{aligned}\partial_\mu(U_\nu U^\nu) &= U_\nu\partial_\mu U^\nu + \partial_\mu U_\nu U^\nu\\ &= U_\nu\partial_\mu U^\nu + \partial_\mu(g_{\nu\lambda}U^\lambda)U^\nu\\ &= U_\nu\partial_\mu U^\nu + \partial_\mu U^\lambda U_\lambda\\ &= 2U_\nu\partial_\mu U^\nu\end{aligned} \tag{1.98}$$

By Changing the dummy index λ to ν in the last term and by the fact that they commute, we see that it is identical to the first term. Thus, $U_\nu\partial_\mu U^\nu = 0$.

$$\begin{aligned}U_\nu\partial_\mu T^{\mu\nu} &= \partial_\mu(\rho+p)U^\mu U_\nu U^\nu + (\rho+p)U^\mu U_\nu\partial_\mu U^\nu + (\rho+p)U_\nu U^\nu\partial_\mu U^\mu + U_\nu\partial^\nu p\\ &= -\partial_\mu(\rho+p)U^\mu - (\rho+p)\partial_\mu U^\mu - U^\nu\partial_\nu p\end{aligned} \tag{1.99}$$

In the non-relativistic limit ($|v^i| \ll 1, p \ll \rho$, $\gamma \approx 1$), this becomes the familiar **continuity equation**: by applying the limits we get:

$$\begin{aligned}U_\nu\partial_\mu T^{\mu\nu} &\approx \partial_\mu(\rho)U^\mu + \rho\partial_\mu U^\mu\\ &= \partial_\mu(\rho U^\mu)\\ &= \partial_0(\rho U^0) + \partial_i(\rho U^i)\\ &= \partial_0(\rho\gamma) + \partial_i(\rho\gamma v^i)\\ &= \partial_t\rho + \nabla\cdot(\rho\mathbf{v})\end{aligned} \tag{1.100}$$

Projection Perpendicular to U^μ (Momentum Conservation): Projecting the conservation law onto the space orthogonal to U^μ (using the projection tensor $P^\sigma{}_\nu = \delta^\sigma_\nu + U^\sigma U_\nu$) gives the equations for momentum conservation. In the non-relativistic limit, this yields the **Euler equation** for fluid dynamics:

$$\rho[\partial_t\mathbf{v} + (\mathbf{v}\cdot\nabla)\mathbf{v}] = -\nabla p \tag{1.101}$$

1.10 Classical Field Theory

When we transition to general relativity, the metric $\eta_{\mu\nu}$ is promoted to a dynamical tensor field, $g_{\mu\nu}(x)$. GR is thus a particular example of a classical field theory. We can build intuition by first considering classical fields on the fixed background of flat spacetime.

From Mechanics to Field Theory

The dynamics of a system can be derived from the **principle of least action**.

- **Point-Particle Mechanics:** For a single particle with coordinate $q(t)$, the action S is the time integral of the Lagrangian, $L(q, \dot{q})$.
$$S = \int L(q, \dot{q})\, dt \tag{1.102}$$
The Lagrangian is typically the kinetic energy minus the potential energy, $L = K - V$. The trajectory that a particle follows is one that makes the action stationary ($\delta S = 0$), which leads to the **Euler-Lagrange equations**:
$$\frac{\partial L}{\partial q} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) = 0 \tag{1.103}$$
- **Classical Field Theory:** We replace the single coordinate $q(t)$ with a set of spacetime-dependent fields, $\Phi^i(x^\mu)$. The action becomes an integral of the **Lagrange density**, $\mathcal{L}$, over all of spacetime.
$$S = \int \mathcal{L}(\Phi^i, \partial_\mu \Phi^i)\, d^4x \tag{1.104}$$
The Lagrange density $\mathcal{L}$ must be a Lorentz scalar.

The Euler-Lagrange Equations for Fields

Derivation:

1. We require the action S to be stationary under an infinitesimal variation of the fields, $\Phi^i \to \Phi^i + \delta\Phi^i$.
2. The change in the action, δS, is found by expanding the Lagrangian:
$$\delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \Phi^i}\delta\Phi^i + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^i)}\partial_\mu(\delta\Phi^i)\right]$$
3. We integrate the second term by parts, $\int u\, dv = uv - \int v\, du$. Let $u = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^i)}$ and $dv = \partial_\mu(\delta\Phi^i)$.
$$\int d^4x \frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^i)}\partial_\mu(\delta\Phi^i) = -\int d^4x \left[\partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^i)}\right)\right]\delta\Phi^i + \text{surface term}$$
4. Assuming the variations vanish at the boundary, the surface term is zero. Substituting this back gives:
$$\delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \Phi^i} - \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^i)}\right)\right]\delta\Phi^i$$
5. For $\delta S = 0$ for any arbitrary variation $\delta\Phi^i$, the expression in the brackets must vanish.

This gives the Euler-Lagrange equations for a field theory:

$$\frac{\partial \mathcal{L}}{\partial \Phi^i} - \partial_\mu\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \Phi^i)}\right) = 0 \tag{1.105}$$

Example 1: Real Scalar Field

Consider a single real scalar field $\phi(x^\mu)$. The Lagrangian density is constructed by generalizing "kinetic minus potential energy."

$$\mathcal{L} = -\frac{1}{2}\eta^{\mu\nu}(\partial_\mu\phi)(\partial_\nu\phi) - V(\phi) \tag{1.106}$$

Here, the first term is the Lorentz-invariant combination of kinetic ($\frac{1}{2}\dot{\phi}^2$) and gradient ($-\frac{1}{2}(\nabla\phi)^2$) energy density. Applying the Euler-Lagrange equations:

- $\frac{\partial\mathcal{L}}{\partial\phi} = -\frac{dV}{d\phi}$
- $\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = -\eta^{\mu\nu}\partial_\nu\phi$

The equation of motion is:

$$\eta^{\mu\nu}\partial_\mu\partial_\nu\phi - \frac{dV}{d\phi} = 0 \quad \text{or} \quad \Box\phi - \frac{dV}{d\phi} = 0 \tag{1.107}$$

where $\Box \equiv \eta^{\mu\nu}\partial_\mu\partial_\nu$ is the d'Alembertian operator. For a simple harmonic oscillator potential, $V(\phi) = \frac{1}{2}m^2\phi^2$, this becomes the famous **Klein-Gordon equation**:

$$\Box\phi + m^2\phi = 0 \tag{1.108}$$

Example 2: Electromagnetism

The dynamical field for electromagnetism is the vector potential A_μ. Physical observables are expressed in terms of the gauge-invariant field strength tensor:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{1.109}$$

The Lagrangian density for electromagnetism coupled to a 4-current source J^μ is:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + A_\mu J^\mu \tag{1.110}$$

Applying the Euler-Lagrange equations for the field A_ν:

- $\frac{\partial\mathcal{L}}{\partial A_\nu} = J^\nu$
- A careful calculation shows $\frac{\partial\mathcal{L}}{\partial(\partial_\mu A_\nu)} = -F^{\mu\nu}$

Plugging these into the Euler-Lagrange equation gives:

$$J^\nu - \partial_\mu(-F^{\mu\nu}) = 0$$

Using the antisymmetry of $F^{\mu\nu}$, this becomes the inhomogeneous Maxwell's equation:

$$\partial_\mu F^{\nu\mu} = J^\nu \tag{1.111}$$

The other Maxwell equation, $\partial_{[\mu}F_{\nu\lambda]} = 0$, is automatically satisfied by the definition of $F_{\mu\nu}$ in terms of A_μ, since partial derivatives commute.

Energy-Momentum Tensors from the Action

The action principle provides a direct procedure for deriving the energy-momentum tensor for any field theory. For the examples above, the results are:

- **Scalar Field:**
$$T^{\mu\nu}_{\text{scalar}} = (\partial^\mu\phi)(\partial^\nu\phi) - \eta^{\mu\nu}\left[\frac{1}{2}(\partial_\lambda\phi)(\partial^\lambda\phi) + V(\phi)\right] \tag{1.112}$$
- **Electromagnetism:**
$$T^{\mu\nu}_{\text{EM}} = F^{\mu\lambda}F^\nu{}_\lambda - \frac{1}{4}\eta^{\mu\nu}F^{\lambda\sigma}F_{\lambda\sigma} \tag{1.113}$$

Using the equations of motion for each theory, these energy-momentum tensors can be shown to be conserved ($\partial_\mu T^{\mu\nu} = 0$).

Chapter 2

Manifolds

2.1 *Gravity as Geometry*

The foundational insight of general relativity is that gravity is not a force that propagates *through* spacetime, but is instead a manifestation of spacetime's own geometry. The physical concept that leads to this radical conclusion is the **Principle of Equivalence**, which formalizes the observed universality of the gravitational interaction. This principle can be formulated in several distinct, progressively stronger ways.

The Weak Equivalence Principle (WEP)

The WEP states that the **inertial mass** (m_i) and **gravitational mass** (m_g) of any object are equal.

- **Inertial Mass** relates an applied force to the resultant acceleration via Newton's Second Law:

$$F = m_i a \tag{2.1}$$

- **Gravitational Mass** acts as the "gravitational charge," determining the force experienced by an object in a gravitational potential Φ:

$$F_g = -m_g \nabla\Phi \tag{2.2}$$

The WEP sets $m_i = m_g$. The immediate and profound consequence is that the acceleration of any test particle in a gravitational field is *independent* of its own mass or composition:

$$a = -\nabla\Phi \tag{2.3}$$

This universality is unique to gravity. In contrast, in an electric field, particles with different charge-to-mass ratios follow different trajectories.

The "Sealed Box" Analogy

An alternative formulation of the WEP is that an observer in a small, sealed box, by observing the motion of freely-falling particles, cannot distinguish between being in a uniform gravitational field and being in a uniformly accelerating reference frame.

This equivalence is necessarily *local*. In a large enough box, an observer *could* detect inhomogeneities in the gravitational field. For example, particles dropped at opposite ends of the box would accelerate towards the center of the Earth, and their paths would not be perfectly parallel. These differential forces are known as **tidal forces**. A uniformly accelerating frame would, by contrast, have an apparent "force" that is perfectly uniform and parallel everywhere in the box.

The Einstein Equivalence Principle (EEP)

The EEP is a much stronger and more comprehensive statement. It postulates that *no* local experiment (whether using test particles, light, or electromagnetism) can distinguish uniform acceleration from a gravitational field.

- **Formal Statement:** In small enough regions of spacetime, the laws of physics reduce to those of special relativity.
- **Implication:** The WEP implies gravity couples universally to rest mass. The EEP extends this, implying that gravity must also couple universally to *all* forms of energy, such as the electromagnetic binding energy holding an atom together.

The Strong Equivalence Principle (SEP)

The SEP is the most stringent form, stating that the EEP applies to *all* laws of physics, including the laws of gravity itself. A hypothetical violation of the SEP would mean that an object with significant gravitational self-binding-energy (like a planet or a black hole) might fall differently in an external gravitational field than a small test particle.

From Equivalence to Geometry

The EEP is the crucial conceptual link to a geometric theory of gravity.

- Gravity is inescapable; there are no "gravitationally neutral" objects with which to compare accelerated motion.
- This makes it impossible to operationally define acceleration *due to* gravity in an unambiguous way.
- **Fundamental Redefinition:** We are forced to redefine an "unaccelerated" path. We now define an **unaccelerated** observer as one who is **freely falling**.
- **Gravity is Not a Force:** A force is what causes acceleration (a deviation from an unaccelerated, or "inertial," path). Since a freely-falling path *is* the new definition of an inertial path, gravity is not a force in this framework.

This redefinition has a profound consequence: we must abandon the idea of *global* inertial frames. In a gravitational field, two nearby freely-falling particles will still accelerate *relative* to each other (tidal forces). It is therefore impossible to construct a single, rigid, unaccelerated reference frame that covers all of spacetime.

We can only define **locally inertial frames**—freely-falling frames that are valid only in a "small enough" region of spacetime where tidal forces are negligible. In these local frames, the laws of physics take their special relativity form (this is the EEP).

This mathematical structure—a space that "looks locally like" flat Minkowski space but may have a different, non-trivial global structure—is precisely a **differentiable manifold**.

Prediction: Gravitational Redshift

The EEP alone, without the full machinery of GR, makes a powerful and testable prediction: the gravitational redshift.

Derivation from Accelerating Frames

First, consider a "thought experiment" in flat spacetime, far from any gravity. Two rockets accelerate uniformly with acceleration a, maintaining a constant separation z.

1. The rear rocket emits a photon of wavelength λ_0 at time t_0.
2. The photon must travel the distance z to reach the front rocket. This takes a time $\Delta t \approx z/c$. (We work to first order, assuming $v \ll c$).
3. In this travel time Δt, both rockets have increased their velocity (relative to their starting inertial frame) by $\Delta v = a\Delta t = az/c$.
4. The front rocket is therefore moving *away* from the photon's emission point with a velocity Δv (relative to the emitter's velocity at the moment of emission).
5. This relative velocity causes a standard first-order Doppler shift. The observed wavelength λ_1 is longer (redshifted):
$$\frac{\Delta\lambda}{\lambda_0} = \frac{\lambda_1 - \lambda_0}{\lambda_0} \approx \frac{\Delta v}{c} \tag{2.4}$$

6. Substituting our expression for Δv:

$$\frac{\Delta\lambda}{\lambda_0} = \frac{(az/c)}{c} = \frac{az}{c^2} \tag{2.5}$$

Applying the EEP

By the EEP, an observer in a sealed box cannot distinguish this acceleration from a uniform gravitational field a_g.

Therefore, a photon emitted from the ground (height $z = 0$) with wavelength λ_0 and received at the top of a tower of height z in a uniform gravitational field a_g must be redshifted by the exact same amount:

$$\frac{\Delta\lambda}{\lambda_0} = \frac{a_g z}{c^2} \tag{2.6}$$

In the weak-field limit, the change in the Newtonian potential is $\Delta\Phi = \Phi(z) - \Phi(0)$. Since $F_g = -m\nabla\Phi$ and $F_g \approx ma_g$ (in the "down" direction), we have $a_g \approx \Delta\Phi/z$. (Note: Carroll's $a_g = \nabla\Phi$ uses a sign convention for the acceleration of the *frame*, not the particle).

This gives the famous relation for gravitational redshift in terms of potential (setting $c = 1$):

$$\frac{\Delta\lambda}{\lambda_0} \approx \Delta\Phi \tag{2.7}$$

A photon "climbing" out of a potential well (moving to a higher Φ) loses energy and is redshifted.

Implication for Spacetime Geometry

This redshift has a profound geometric implication.

- An emitter at z_0 sends two consecutive wave crests. The time between them, as measured by a clock at z_0, is the period $\Delta t_0 = \lambda_0/c$.
- A receiver at z_1 observes these same two crests. The time between them, as measured by a clock at z_1, is $\Delta t_1 = \lambda_1/c$.
- The observed gravitational redshift means $\lambda_1 > \lambda_0$, which directly implies $\Delta t_1 > \Delta t_0$.
- **Conclusion:** Clocks run at different rates at different gravitational potentials. The clock at the top of the tower (z_1, higher potential) runs *faster* than the clock on the ground (z_0).

In a static (unchanging) gravitational field, the spacetime path of the second wave crest must be identical to the first, just shifted in time. In the "simple" geometry of flat spacetime, these congruent paths would imply $\Delta t_1 = \Delta t_0$. The fact that $\Delta t_1 \neq \Delta t_0$ is direct, experimental evidence that the geometry of spacetime is not flat—it is curved.

2.2 What Is a Manifold?

The EEP implies that spacetime is *locally* indistinguishable from Minkowski space. This suggests we should model spacetime as a mathematical object that looks like flat $\mathbb{R}^n$ (or in our case, $\mathbb{R}^{3,1}$) in any small neighborhood, even if its global structure is complex or "curved". This is the essential idea of a **differentiable manifold**.

Examples of Manifolds

- **$\mathbb{R}^n$:** The n-dimensional Euclidean space is the trivial example of an n-manifold.
- **The n-sphere, S^n:** The set of points a fixed distance from the origin in $\mathbb{R}^{n+1}$. S^1 is a circle, and S^2 is the familiar two-dimensional sphere. It is important to note that the embedding in a higher-dimensional space is just a convenient way to define it; the manifold exists as its own entity.

- **The n-torus, T^n:** This manifold is formed by taking an n-dimensional cube and "gluing" (identifying) opposite faces. The 2-torus, T^2, is the surface of a doughnut.
- **Riemann Surfaces:** A 2-torus with g "holes" is a Riemann surface of genus g. S^2 is genus 0, and T^2 is genus 1.
- **Lie Groups:** Manifolds that also have a continuous group structure, such as $SO(2)$ (the group of 2D rotations), which is topologically identical to S^1.
- **Direct Products:** The product of an n-manifold M and an m-manifold N is an $(n+m)$-manifold $M \times N$.

Examples of Non-Manifolds

Spaces that do not "look locally like $\mathbb{R}^n$" everywhere are not manifolds. Common examples are spaces with points where the dimension changes abruptly.

More subtle cases exist:

- **A single cone:** This *is* a smooth manifold. While the curvature is singular at the vertex, any local patch (even one containing the vertex) can be "unrolled" into a flat $\mathbb{R}^2$ plane, satisfying the definition.
- **A line segment [a, b]:** The endpoints a and b do not have neighborhoods that look like $\mathbb{R}$. This is an example of a "manifold with boundary."

Formal Definition

To formalize the intuitive idea of "smoothly sewing together flat patches," we need some preliminary definitions.

- **Map:** A rule $\phi : M \to N$ that assigns each element in set M to exactly one element in set N.
- **Map Properties:**
  - **One-to-one (injective):** Each element in N is mapped to by *at most* one element in M. (e.g., $\phi(x) = e^x$).
  - **Onto (surjective):** Each element in N is mapped to by *at least* one element in M. (e.g., $\phi(x) = x^3 - x$).
  - **Invertible (bijective):** Both one-to-one and onto. (e.g., $\phi(x) = x^3$).
- **Differentiability:** A map $\phi : \mathbb{R}^m \to \mathbb{R}^n$ is C^p if its p-th partial derivatives exist and are continuous. A C^∞ map is called **smooth**.
- **Diffeomorphism:** A smooth (C^∞) map $\phi : M \to N$ that has a smooth (C^∞) inverse ϕ^{-1}. This is the rigorous way of saying two manifolds are "the same".

The Definition of a Manifold

1. **Chart (or Coordinate System):** A pair (U, ϕ), where U is a subset of our set M, and $\phi : U \to \mathbb{R}^n$ is a one-to-one map from U to an *open set* $\phi(U)$ in $\mathbb{R}^n$.
2. C^∞ **Atlas:** A collection of charts $\{(U_\alpha, \phi_\alpha)\}$ such that:
   - The U_α completely cover M: $\cup_\alpha U_\alpha = M$.
   - The charts are **smoothly compatible**. On any region where two charts overlap ($U_\alpha \cap U_\beta \neq \emptyset$), the **transition map** $\phi_\alpha \circ \phi_\beta^{-1}$ (which maps from an open set in $\mathbb{R}^n$ to another open set in $\mathbb{R}^n$) is C^∞.
3. C^∞ n-**Manifold:** A set M equipped with a *maximal* C^∞ atlas. (The "maximal" condition is a technicality that just means we include all possible charts that are compatible with our atlas).

Atlases and Coordinate Systems

It is crucial that our definition does not require the manifold to be embedded in a higher-dimensional space. Our 4D spacetime does not need to "live inside" a 5D space to be a manifold.

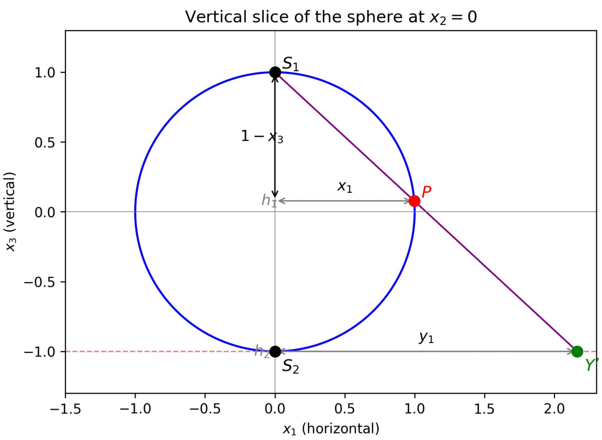
A single chart is often not enough to cover an entire manifold.

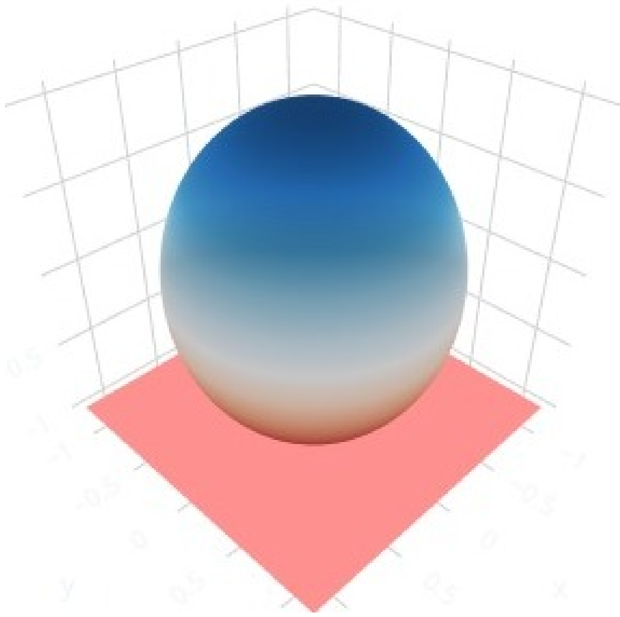
- **Example: S^1 (the circle).** A single coordinate θ is not a valid chart, because the map $\theta : S^1 \to \mathbb{R}$ would have an image like $[0, 2\pi)$, which is not an open set in $\mathbb{R}$. We need at least two charts (e.g., $\theta \in (0, 2\pi)$ and $\theta \in (-\pi, \pi)$) to cover the circle.

- **Example: S^2 (the sphere).** No single map can cover the 2-sphere without singularities (e.g., a Mercator projection misses the poles). We can, however, cover S^2 with an atlas of just two charts using **stereographic projection**.

Stereographic Projection on S^2

Let S^2 be the set $(x^1)^2 + (x^2)^2 + (x^3)^2 = 1$ in $\mathbb{R}^3$.





- **Chart 1 (U_N, ϕ_N):** U_N is the entire sphere *except* the North Pole $N = (0, 0, 1)$. The map ϕ_N projects any point $P = (x^1, x^2, x^3)$ from N onto the equatorial plane $x^3 = -1$. The coordinates are $y^i = (y^1, y^2)$. The map ϕ_N projects from N to the plane $x^3 = -1$. The coordinates are (y^1, y^2). By similar triangles, the map is:

$$(y^1, y^2) = \phi_N(x^1, x^2, x^3) = \left(\frac{2x^1}{1 - x^3}, \frac{2x^2}{1 - x^3} \right) \tag{2.8}$$

- **Chart 2 (U_S, ϕ_S):** U_S is the entire sphere *except* the South Pole $S = (0, 0, -1)$. The map ϕ_S projects from S onto the plane $x^3 = +1$. The coordinates are (z^1, z^2).

$$(z^1, z^2) = \phi_S(x^1, x^2, x^3) = \left(\frac{2x^1}{1 + x^3}, \frac{2x^2}{1 + x^3} \right) \tag{2.9}$$

These two charts co ver the entire sphere. On their overlap (the sphere minus both poles), we must check that the transition map is C^∞.

Derivation: Transition Map for S^2

We need to find the map $\phi_S \circ \phi_N^{-1}$, which takes coordinates (y^1, y^2) from the first chart and gives the corresponding coordinates (z^1, z^2) in the second chart.

1. **Invert ϕ_N:** We must find (x^1, x^2, x^3) in terms of (y^1, y^2). Let $Y^2 = (y^1)^2 + (y^2)^2$.

$$Y^2 = \frac{4(x^1)^2 + 4(x^2)^2}{(1-x^3)^2}$$

   Using $(x^1)^2 + (x^2)^2 = 1 - (x^3)^2 = (1-x^3)(1+x^3)$:

$$Y^2 = \frac{4(1-x^3)(1+x^3)}{(1-x^3)^2} = \frac{4(1+x^3)}{1-x^3}$$

   We can solve this for x^3:

$$Y^2(1-x^3) = 4(1+x^3) \implies Y^2 - Y^2x^3 = 4 + 4x^3$$

$$Y^2 - 4 = (Y^2+4)x^3 \implies x^3 = \frac{Y^2-4}{Y^2+4}$$

   Now we find x^1 and x^2. First, find the term $(1 - x^3)$:

$$1 - x^3 = 1 - \frac{Y^2-4}{Y^2+4} = \frac{(Y^2+4)-(Y^2-4)}{Y^2+4} = \frac{8}{Y^2+4}$$

   From the definition $y^1 = 2x^1/(1-x^3)$:

$$x^1 = \frac{y^1(1-x^3)}{2} = \frac{y^1}{2}\left(\frac{8}{Y^2+4}\right) = \frac{4y^1}{Y^2+4}$$

   Similarly, $x^2 = \frac{4y^2}{Y^2+4}$. This completes the inverse map $\phi_N^{-1} : (y^1, y^2) \mapsto (x^1, x^2, x^3)$.

2. **Apply ϕ_S:** Now we apply the second chart's map to these expressions.

$$(z^1, z^2) = \left(\frac{2x^1}{1+x^3}, \frac{2x^2}{1+x^3}\right)$$

   We need the term $(1 + x^3)$:

$$1 + x^3 = 1 + \frac{Y^2-4}{Y^2+4} = \frac{(Y^2+4)+(Y^2-4)}{Y^2+4} = \frac{2Y^2}{Y^2+4}$$

   Now substitute:

$$z^1 = \frac{2\left(\frac{4y^1}{Y^2+4}\right)}{\left(\frac{2Y^2}{Y^2+4}\right)} = \frac{8y^1}{2Y^2} = \frac{4y^1}{(y^1)^2+(y^2)^2}$$

$$z^2 = \frac{2\left(\frac{4y^2}{Y^2+4}\right)}{\left(\frac{2Y^2}{Y^2+4}\right)} = \frac{8y^2}{2Y^2} = \frac{4y^2}{(y^1)^2+(y^2)^2}$$

The transition map is:

$$(z^1, z^2) = \left(\frac{4y^1}{(y^1)^2+(y^2)^2}, \frac{4y^2}{(y^1)^2+(y^2)^2}\right) \tag{2.10}$$

This map is C^∞ (smooth) everywhere except at $(y^1, y^2) = (0, 0)$. This point corresponds to the South Pole, which is explicitly *excluded* from the domain of ϕ_N. Therefore, the transition map is smooth on the entire overlap region, and our two charts form a valid C^∞ atlas for S^2.

The Chain Rule

We will frequently use the chain rule for maps between coordinate patches. If we have maps $f : \mathbb{R}^m \to \mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}^l$, with coordinates $x^a \in \mathbb{R}^m$, $y^b \in \mathbb{R}^n$, and $z^c \in \mathbb{R}^l$, the chain rule relates the partial derivatives:

$$\frac{\partial (g \circ f)^c}{\partial x^a} = \sum_{b=1}^{n} \frac{\partial f^b}{\partial x^a} \frac{\partial g^c}{\partial y^b} \tag{2.11}$$

A common shorthand for this operation on a function is:

$$\frac{\partial}{\partial x^a} = \sum_{b=1}^{n} \frac{\partial y^b}{\partial x^a} \frac{\partial}{\partial y^b} \tag{2.12}$$

2.3 Vectors Again

Having established the concept of a manifold, we must redefine our physical objects (vectors, tensors) to live on this new structure. In Chapter 1, we introduced the **tangent space** T_p as the set of all vectors at a single point p. This was to break the (false) intuition of vectors stretching between different points. Now, we must provide a rigorous, intrinsic definition of T_p that doesn't rely on embedding the manifold in a higher-dimensional space.

A natural first guess is to define T_p as the set of all "tangent vectors" to all possible curves $\gamma(\lambda)$ that pass through p. The problem is that this is circular; we haven't yet defined what a "tangent vector" is. In a coordinate system x^μ, the components $dx^\mu/d\lambda$ are coordinate-dependent and don't represent the invariant geometric object we're after.

The solution is to notice that any curve $\gamma(\lambda)$ through p defines a **directional derivative operator**, $d/d\lambda$, which acts on any smooth scalar function $f : M \to \mathbb{R}$ (a space we call $\mathcal{F}$) to produce a real number:

$$\frac{d}{d\lambda}(f) \in \mathbb{R}$$

We will therefore *define* the tangent space T_p as the set of all such directional derivative operators at p.

Vectors as Derivative Operators

To show this is a valid definition, we must first establish that this set of operators forms a vector space.

1. Linearity: We can add two such operators and scale them by real numbers a, b:

$$X = a\frac{d}{d\lambda_1} + b\frac{d}{d\lambda_2}$$

This new operator X is clearly linear in its action on functions.

2. Leibniz Rule: A "good" derivative operator must also obey the product rule (Leibniz's rule). We can check this:

$$\begin{aligned}
X(fg) &= \left(a\frac{d}{d\lambda_1} + b\frac{d}{d\lambda_2}\right)(fg) \\
&= a\left(f\frac{dg}{d\lambda_1} + g\frac{df}{d\lambda_1}\right) + b\left(f\frac{dg}{d\lambda_2} + g\frac{df}{d\lambda_2}\right) \\
&= f\left(a\frac{dg}{d\lambda_1} + b\frac{dg}{d\lambda_2}\right) + g\left(a\frac{df}{d\lambda_1} + b\frac{df}{d\lambda_2}\right) \\
&= f(X(g)) + g(X(f))
\end{aligned}$$

The product rule is satisfied. Thus, the set of all such operators at p forms a vector space, which we identify as the tangent space T_p.

The Coordinate Basis

Now that T_p is defined, we can find a basis for it. Given a coordinate chart with coordinates x^μ (where $\mu = 0, \ldots, n-1$), there is a natural set of n directional derivatives: the **partial derivative operators** $\{\partial_\mu\} \equiv \{\frac{\partial}{\partial x^\mu}\}$. These operators form a basis for T_p. To prove this, we show that any arbitrary directional derivative $d/d\lambda$ (representing any vector V) can be written as a unique linear combination of these partials.

Derivation: Vector Components Let's assume $\phi : M \to \mathbb{R}^n$ is our coordinate chart, mapping point p to coordinates $x^\mu(p)$. A curve $\gamma : \mathbb{R} \to M$ and function $f : M \to \mathbb{R}$.

$$\begin{aligned}\frac{d}{d\lambda}(f) &= \frac{d}{d\lambda}(f \circ \gamma) \\ &= \frac{d}{d\lambda}[(f \circ \phi^{-1}) \circ (\phi \circ \gamma)] \\ &= \frac{d(\phi \circ \gamma)^\mu}{d\lambda} \frac{\partial (f \circ \phi^{-1})}{\partial x^\mu} \\ &= \frac{dx^\mu}{d\lambda}\frac{\partial f}{\partial x^\mu}\end{aligned} \tag{2.13}$$

so finally we have:

$$\frac{d}{d\lambda} = \frac{dx^\mu}{d\lambda}\partial_\mu$$

Physical Meaning: This is a profound result. It shows that any vector V (represented by $d/d\lambda$) can be written in the basis $V = V^\mu \hat{e}_{(\mu)}$ where the basis vectors are the partial derivatives $\hat{e}_{(\mu)} = \partial_\mu$ and the components V^μ are precisely the familiar $dx^\mu/d\lambda$.

This basis $\{\partial_\mu\}$ is called the **coordinate basis**. It is simple and natural, but it is crucial to remember that it is generally *not* an orthonormal basis.

Transformation Properties

This abstract definition makes deriving transformation laws trivial.

- **Basis Vector Transformation:** How does the basis $\{\partial_\mu\}$ change when we change coordinates from x^μ to $x^{\mu'}$? The new basis vectors $\partial_{\mu'} \equiv \frac{\partial}{\partial x^{\mu'}}$ are related to the old ones by the chain rule:

$$\partial_{\mu'} = \frac{\partial}{\partial x^{\mu'}} = \frac{\partial x^\mu}{\partial x^{\mu'}}\frac{\partial}{\partial x^\mu} = \frac{\partial x^\mu}{\partial x^{\mu'}}\partial_\mu \tag{2.14}$$

  This is the transformation law for the basis vectors.

- **Vector Component Transformation:** The abstract vector V itself is invariant to coordinate choice. Its expansion in either basis must be equal:

$$V = V^{\mu'}\partial_{\mu'} = V^\mu \partial_\mu$$

  Substituting the basis transformation:

$$V = V^{\mu'}\left(\frac{\partial x^\mu}{\partial x^{\mu'}}\partial_\mu\right)$$

  By comparing this to $V = V^\mu\partial_\mu$, and since the $\{\partial_\mu\}$ are a basis, the coefficients must be equal:

$$V^\mu = V^{\mu'}\frac{\partial x^\mu}{\partial x^{\mu'}}$$

  To find the transformation for $V^{\mu'}$, we just invert this relationship using the inverse Jacobian matrix:

$$V^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\mu}V^\mu \tag{2.15}$$

  This is the familiar transformation law for a contravariant vector, but now derived for *any* smooth coordinate change, not just the linear Lorentz transformations of flat space.

The Commutator (Lie Bracket)

A vector field $X = X^\mu(x)\partial_\mu$ is an operator that maps scalar fields to scalar fields ($X : \mathcal{F} \to \mathcal{F}$). This allows us to define the **commutator** (or **Lie bracket**) of two vector fields, X and Y, by its action on an arbitrary scalar field f:

$$[X, Y](f) \equiv X(Y(f)) - Y(X(f)) \tag{2.16}$$

Crucially, the resulting object $[X, Y]$ is *also* a vector field, meaning it is a linear, first-order differential operator that obeys the Leibniz rule.

Derivation: Component Form of the Commutator We can find the components of the vector field $Z = [X, Y]$ in a coordinate basis. The ν-th component Z^ν can be found by acting the operator on the coordinate function x^ν (since $Z(x^\nu) = Z^\mu\partial_\mu x^\nu = Z^\mu\delta^\nu_\mu = Z^\nu$).

$$\begin{aligned}
[X, Y]^\nu &= [X, Y](x^\nu) \\
&= X(Y(x^\nu)) - Y(X(x^\nu)) \\
&= X(Y^\mu\partial_\mu x^\nu) - Y(X^\lambda\partial_\lambda x^\nu) \\
&= X(Y^\mu\delta^\nu_\mu) - Y(X^\lambda\delta^\nu_\lambda) \\
&= X(Y^\nu) - Y(X^\nu) \\
&= (X^\lambda\partial_\lambda)Y^\nu - (Y^\mu\partial_\mu)X^\nu
\end{aligned}$$

By renaming the dummy index μ to λ in the second term, we get the standard expression:

$$[X, Y]^\nu = X^\lambda\partial_\lambda Y^\nu - Y^\lambda\partial_\lambda X^\nu \tag{2.17}$$

It may seem surprising that this is a well-defined tensor, since it involves partial derivatives of vector components, which are not themselves tensors. However, the non-tensorial "extra pieces" that arise in a coordinate transformation of the first term ($\partial_\lambda Y^\nu$) are precisely canceled by those from the second term ($\partial_\lambda X^\nu$), leaving a valid tensor transformation law.

A simple, important example is the commutator of two coordinate basis vectors:

$$[\partial_\mu, \partial_\nu](f) = \partial_\mu(\partial_\nu f) - \partial_\nu(\partial_\mu f) = 0$$

This is a direct consequence of the fact that partial derivatives commute.

2.4 Tensors Again

With a rigorous definition of vectors and the tangent space T_p, we can retrace our steps from Chapter 1 to define dual vectors (one-forms) and, more generally, tensors on a manifold.

Dual Vectors (One-Forms)

The **cotangent space** T^*_p is defined as the dual vector space to T_p. That is, T^*_p is the set of all linear maps $\omega : T_p \to \mathbb{R}$.

The canonical example of a one-form is the **gradient** of a scalar function f, denoted df. Its "action" on a vector V is defined to be the directional derivative of f along V:

$$df(V) \equiv V(f)$$

If we use our definition of a vector V as the tangent vector to a curve $x^\mu(\lambda)$, $V = d/d\lambda$, then its action is just the ordinary derivative:

$$df\left(\frac{d}{d\lambda}\right) = \frac{df}{d\lambda} \tag{2.18}$$

Basis and Components

Just as the partial derivatives $\{\partial_\mu\}$ form a natural basis for the tangent space T_p, the gradients of the coordinate functions $\{dx^\mu\}$ form a natural basis for the cotangent space T_p^*.

We can prove this by showing that $\{dx^\mu\}$ is the **dual basis** to $\{\partial_\nu\}$, which means their "action" on each other is the Kronecker delta.

1. Start with the definition of the action of the one-form dx^μ on the vector ∂_ν:

$$dx^\mu(\partial_\nu) \equiv \partial_\nu(x^\mu)$$

2. The partial derivative of the coordinate function x^μ with respect to the coordinate x^ν is:

$$\partial_\nu(x^\mu) = \frac{\partial x^\mu}{\partial x^\nu} = \delta^\mu_\nu$$

Thus, we have our duality relationship:

$$dx^\mu(\partial_\nu) = \delta^\mu_\nu \tag{2.19}$$

Any arbitrary one-form ω can be expanded in this basis using its components ω_μ:

$$\omega = \omega_\mu dx^\mu$$

Transformation Properties

The transformation laws follow directly from the chain rule.

- **Basis One-Forms:** The new basis one-forms $dx^{\mu'}$ are related to the old dx^μ by the standard rule for total differentials:

$$dx^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\mu} dx^\mu \tag{2.20}$$

- **Components:** The components ω_μ must transform in the "opposite" (inverse) way to keep the abstract one-form ω invariant.

$$\omega = \omega_\mu dx^\mu = \omega_{\mu'} dx^{\mu'}$$

Substituting the basis transformation:

$$\omega = \omega_{\mu'} \left(\frac{\partial x^{\mu'}}{\partial x^\mu} dx^\mu \right)$$

By matching the dx^μ components, we find $\omega_\mu = \omega_{\mu'} \frac{\partial x^{\mu'}}{\partial x^\mu}$. We can invert this by multiplying by $\frac{\partial x^\mu}{\partial x^{\nu'}}$ to get the standard form:

$$\omega_{\nu'} = \frac{\partial x^\mu}{\partial x^{\nu'}} \omega_\mu \tag{2.21}$$

General Tensors

A tensor T of type (or rank) (k, l) is a multilinear map from k copies of the cotangent space and l copies of the tangent space to the real numbers:

$$T : \underbrace{T_p^* \times \cdots \times T_p^*}_{k \text{ times}} \times \underbrace{T_p \times \cdots \times T_p}_{l \text{ times}} \to \mathbb{R}$$

In a coordinate basis, any tensor can be expanded using its components:

$$T = T^{\mu_1 \ldots \mu_k}{}_{\nu_1 \ldots \nu_l} \partial_{\mu_1} \otimes \cdots \otimes \partial_{\mu_k} \otimes dx^{\nu_1} \otimes \cdots \otimes dx^{\nu_l} \tag{2.22}$$

The components are found by acting the tensor on the basis vectors and one-forms:

$$T^{\mu_1 \ldots \mu_k}{}_{\nu_1 \ldots \nu_l} = T(dx^{\mu_1}, \ldots, dx^{\mu_k}, \partial_{\nu_1}, \ldots, \partial_{\nu_l}) \tag{2.23}$$

The general **tensor transformation law** replaces the Lorentz matrices from Chapter 1 with the more general Jacobian matrices of the coordinate transformation. Each index transforms according to its type (contravariant or covariant):

$$T^{\mu'_1 \ldots \mu'_k}{}_{\nu'_1 \ldots \nu'_l} = \underbrace{\frac{\partial x^{\mu'_1}}{\partial x^{\mu_1}} \cdots \frac{\partial x^{\mu'_k}}{\partial x^{\mu_k}}}_{\text{contravariant indices}} \underbrace{\frac{\partial x^{\nu_1}}{\partial x^{\nu'_1}} \cdots \frac{\partial x^{\nu_l}}{\partial x^{\nu'_l}}}_{\text{covariant indices}} T^{\mu_1 \ldots \mu_k}{}_{\nu_1 \ldots \nu_l} \tag{2.24}$$

Example: Transforming a (0,2) Tensor

Applying the full transformation law (2.30) is cumbersome. It is often much simpler to transform a tensor by transforming its basis vectors and one-forms directly.

Let's consider a symmetric (0,2) tensor S on a 2D manifold with coordinates $(x^1, x^2) = (x, y)$. In this frame, S has components:

$$S_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & x^2 \end{pmatrix}$$

As an abstract tensor, this is written (suppressing the $\otimes$ symbol):

$$S = S_{\mu\nu} dx^\mu dx^\nu = 1\,(dx)^2 + x^2 (dy)^2 \tag{2.25}$$

Now, let's change to a new coordinate system (x', y') defined by:

$$x' = \frac{2x}{y}, \quad y' = \frac{y}{2}$$

Derivation:

1. **Invert the transformation** to express old coordinates in terms of new:

$$y = 2y'$$

$$x = \frac{x'y}{2} = \frac{x'(2y')}{2} = x'y'$$

2. **Find the old basis one-forms** in terms of the new ones by differentiation:

$$dx = \frac{\partial x}{\partial x'} dx' + \frac{\partial x}{\partial y'} dy' = y' dx' + x' dy'$$

$$dy = \frac{\partial y}{\partial x'} dx' + \frac{\partial y}{\partial y'} dy' = 0\, dx' + 2dy' = 2dy'$$

3. **Substitute** these expressions directly into the abstract tensor S:

$$S = (y' dx' + x' dy')^2 + (x'y')^2 (2dy')^2$$

We must be careful to treat $dx'dy'$ as $dx' \otimes dy'$:

$$S = (y' dx' + x' dy') \otimes (y' dx' + x' dy') + (x'y')^2 (2dy' \otimes 2dy')$$

4. **Expand** the tensor products:

$$S = (y')^2 (dx' \otimes dx') + x'y' (dx' \otimes dy') + x'y' (dy' \otimes dx') + (x')^2 (dy' \otimes dy') + 4(x'y')^2 (dy' \otimes dy')$$

5. **Combine** terms (and revert to the $dx'dy'$ shorthand, remembering it's symmetric):

$$S = (y')^2(dx')^2 + x'y'(dx'dy' + dy'dx') + \left[(x')^2 + 4(x'y')^2\right](dy')^2$$

By reading off the components $S = S_{\mu'\nu'}dx^{\mu'}dx^{\nu'}$, we find the new component matrix:

$$S_{\mu'\nu'} = \begin{pmatrix} (y')^2 & x'y' \\ x'y' & (x')^2 + 4(x'y')^2 \end{pmatrix} \tag{2.26}$$

This matches the result one would get from the full transformation law (2.30), but was much easier to derive.

$$S_{\mu'\nu'} = \frac{\partial x^{\mu}}{\partial x^{\mu'}}\frac{\partial x^{\nu}}{\partial x^{\nu'}}S_{\mu\nu} \tag{2.27}$$

First we calculate the partial derivatives:

$$\frac{\partial x}{\partial x'} = y', \quad \frac{\partial x}{\partial y'} = x', \quad \frac{\partial y}{\partial x'} = 0, \quad \frac{\partial y}{\partial y'} = 2$$

So now we can compute each component:

$$\begin{aligned}
S_{x'x'} &= \frac{\partial x}{\partial x'}\frac{\partial x}{\partial x'}S_{xx} + \frac{\partial x}{\partial x'}\frac{\partial y}{\partial x'}S_{xy} + \frac{\partial y}{\partial x'}\frac{\partial x}{\partial x'}S_{yx} + \frac{\partial y}{\partial x'}\frac{\partial y}{\partial x'}S_{yy} \\
&= (y')(y')(1) + (y')(0)(0) + (0)(y')(0) + (0)(0)(x^2) = (y')^2 \\
S_{x'y'} &= \frac{\partial x}{\partial x'}\frac{\partial x}{\partial y'}S_{xx} + \frac{\partial x}{\partial x'}\frac{\partial y}{\partial y'}S_{xy} + \frac{\partial y}{\partial x'}\frac{\partial x}{\partial y'}S_{yx} + \frac{\partial y}{\partial x'}\frac{\partial y}{\partial y'}S_{yy} \\
&= (y')(x')(1) + (y')(2)(0) + (0)(x')(0) + (0)(2)(x^2) = x'y' \\
S_{y'y'} &= \frac{\partial x}{\partial y'}\frac{\partial x}{\partial y'}S_{xx} + \frac{\partial x}{\partial y'}\frac{\partial y}{\partial y'}S_{xy} + \frac{\partial y}{\partial y'}\frac{\partial x}{\partial y'}S_{yx} + \frac{\partial y}{\partial y'}\frac{\partial y}{\partial y'}S_{yy} \\
&= (x')(x')(1) + (x')(2)(0) + (2)(x')(0) + (2)(2)(x^2) = (x')^2 + 4(x'y')^2
\end{aligned}$$

which give us the equation (2.26) again.

The Failure of Partial Derivatives

In Chapter 1, we noted that in flat space with *inertial* coordinates, the partial derivative of a tensor was also a tensor. This property **fails** in a general manifold (or even in flat space with curvilinear coordinates).

The partial derivative of a scalar $\partial_\mu \phi$ is a valid (0,1) tensor (a one-form). However, the partial derivative of a vector $\partial_\mu V^\nu$ or a one-form $\partial_\mu W_\nu$ is **not a tensor**.

Proof (for a one-form): Let's see what happens when we try to transform the components of $\partial_\mu W_\nu$ to the new x' frame. The object we are transforming is $K_{\mu\nu} = \partial_\mu W_\nu$. The transformed components would be $K_{\mu'\nu'} = \partial_{\mu'} W_{\nu'}$.

1. Start with the transformed object $\partial_{\mu'} W_{\nu'}$.

$$\partial_{\mu'} W_{\nu'} = \frac{\partial}{\partial x^{\mu'}}(W_{\nu'})$$

2. Substitute the transformation law for the one-form component $W_{\nu'}$:

$$\partial_{\mu'} W_{\nu'} = \frac{\partial}{\partial x^{\mu'}}\left(\frac{\partial x^{\nu}}{\partial x^{\nu'}}W_\nu\right)$$

3. Apply the product rule (since both $\frac{\partial x^\nu}{\partial x^{\nu'}}$ and W_ν are functions of position):

$$\partial_{\mu'} W_{\nu'} = \left(\frac{\partial}{\partial x^{\mu'}}\frac{\partial x^{\nu}}{\partial x^{\nu'}}\right)W_\nu + \frac{\partial x^{\nu}}{\partial x^{\nu'}}\left(\frac{\partial}{\partial x^{\mu'}}W_\nu\right)$$

4. Apply the chain rule to both terms:

$$\partial_{\mu'} W_{\nu'} = \underbrace{\left(\frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial^2 x^\nu}{\partial x^\mu \partial x^{\nu'}} \right) W_\nu}_{\text{Term 1}} + \underbrace{\frac{\partial x^\nu}{\partial x^{\nu'}} \left(\frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial}{\partial x^\mu} W_\nu \right)}_{\text{Term 2}}$$

Rearranging the terms, we get:

$$\partial_{\mu'} W_{\nu'} = \underbrace{\frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} (\partial_\mu W_\nu)}_{\text{This is the correct tensor transformation...}} + \underbrace{W_\nu \frac{\partial^2 x^\nu}{\partial x^{\mu'} \partial x^{\nu'}}}_{\text{...but this extra "bad" term is not zero!}} \tag{2.28}$$

This "bad" second term arises from the derivative of the transformation matrix itself. It is zero *only if* the transformation is linear (all second derivatives $\frac{\partial^2 x}{\partial x' \partial x'}$ are zero), as is the case for Lorentz transformations. For a general coordinate change, this term is non-zero, and $\partial_\mu W_\nu$ fails to transform as a (0,2) tensor.

This fundamental problem is the primary motivation for introducing a new, more powerful type of derivative that *is* a tensor: the **covariant derivative**.

2.5 The Metric

The metric tensor, now denoted $g_{\mu\nu}$, is the central object in general relativity. It is a symmetric, (0,2) tensor field that endows the manifold with a notion of distance, time, and causal structure.

We will almost always assume the metric is **nondegenerate**, meaning its determinant $g = \det(g_{\mu\nu})$ is non-zero. This ensures that an inverse metric $g^{\mu\nu}$ exists, which is defined by the relation:

$$g^{\mu\nu} g_{\nu\sigma} = \delta^\mu_\sigma \tag{2.29}$$

The metric and its inverse are the tools we use to raise and lower indices on all other tensors (e.g., $V_\mu = g_{\mu\nu} V^\nu$).

The metric $g_{\mu\nu}$ plays numerous roles, including:

- Defining the causal structure (past, future, light cones).
- Allowing the computation of proper time and path lengths.
- Determining the "shortest paths" (geodesics) on which test particles move.
- Replacing the single Newtonian potential Φ.
- Defining locally inertial frames and thus a notion of "no rotation."
- Determining the speed of light and causality.
- Replacing the 3D Euclidean dot product.

The Line Element

As in flat space, we write the metric in the form of the **line element**, ds^2. This is a convenient shorthand for the abstract (0,2) tensor g:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \tag{2.30}$$

Here, the dx^μ are formally understood as the basis one-forms. The inner product of two vectors V and W is $g(V, W) = g_{\mu\nu} V^\mu W^\nu$.

Example: Flat Space in Spherical Coordinates

The metric components $g_{\mu\nu}$ depend on the coordinate system. Even flat Euclidean space has non-trivial metric components in curvilinear coordinates. The Cartesian line element for flat 3D space is $ds^2 = (dx)^2 + (dy)^2 + (dz)^2$. The transformation to spherical coordinates (r, θ, ϕ) is:

$$\begin{aligned} x &= r \sin\theta \cos\phi \\ y &= r \sin\theta \sin\phi \\ z &= r \cos\theta \end{aligned}$$

Derivation: We find the new metric by calculating the differentials dx, dy, and dz and substituting them into the Cartesian line element.

1. **Calculate the differentials** using the multivariable chain rule:

$$\begin{aligned} dx &= (\sin\theta\cos\phi)dr + (r\cos\theta\cos\phi)d\theta - (r\sin\theta\sin\phi)d\phi \\ dy &= (\sin\theta\sin\phi)dr + (r\cos\theta\sin\phi)d\theta + (r\sin\theta\cos\phi)d\phi \\ dz &= (\cos\theta)dr - (r\sin\theta)d\theta \end{aligned}$$

2. **Square the differentials.** We must compute $(dx)^2 + (dy)^2 + (dz)^2$. This is tedious, but many cross-terms will cancel. Let's first compute $(dx)^2 + (dy)^2$:

$$\begin{aligned} (dx)^2 = {} & (\sin^2\theta\cos^2\phi)dr^2 + (r^2\cos^2\theta\cos^2\phi)d\theta^2 + (r^2\sin^2\theta\sin^2\phi)d\phi^2 \\ & + 2(r\sin\theta\cos\theta\cos^2\phi)drd\theta - 2(r\sin^2\theta\sin\phi\cos\phi)drd\phi \\ & - 2(r^2\sin\theta\cos\theta\sin\phi\cos\phi)d\theta d\phi \end{aligned}$$

$$\begin{aligned} (dy)^2 = {} & (\sin^2\theta\sin^2\phi)dr^2 + (r^2\cos^2\theta\sin^2\phi)d\theta^2 + (r^2\sin^2\theta\cos^2\phi)d\phi^2 \\ & + 2(r\sin\theta\cos\theta\sin^2\phi)drd\theta + 2(r\sin^2\theta\sin\phi\cos\phi)drd\phi \\ & + 2(r^2\sin\theta\cos\theta\sin\phi\cos\phi)d\theta d\phi \end{aligned}$$

3. **Sum** $(dx)^2 + (dy)^2$ by collecting terms and using $\cos^2\phi + \sin^2\phi = 1$:
   - dr^2 term: $(\sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi)dr^2 = \sin^2\theta dr^2$
   - $d\theta^2$ term: $(r^2\cos^2\theta\cos^2\phi + r^2\cos^2\theta\sin^2\phi)d\theta^2 = r^2\cos^2\theta d\theta^2$
   - $d\phi^2$ term: $(r^2\sin^2\theta\sin^2\phi + r^2\sin^2\theta\cos^2\phi)d\phi^2 = r^2\sin^2\theta d\phi^2$
   - $drd\theta$ term: $2(r\sin\theta\cos\theta(\cos^2\phi + \sin^2\phi))drd\theta = 2r\sin\theta\cos\theta drd\theta$
   - $drd\phi$ term: The cross terms cancel to 0.
   - $d\theta d\phi$ term: The cross terms cancel to 0.

   So, $(dx)^2 + (dy)^2 = \sin^2\theta dr^2 + r^2\cos^2\theta d\theta^2 + r^2\sin^2\theta d\phi^2 + 2r\sin\theta\cos\theta drd\theta$.

4. **Now add $(dz)^2$:**

$$(dz)^2 = (\cos\theta dr - r\sin\theta d\theta)^2 = \cos^2\theta dr^2 - 2r\sin\theta\cos\theta drd\theta + r^2\sin^2\theta d\theta^2$$

5. **Sum all three parts:**

$$\begin{aligned} ds^2 &= \left[(dx)^2 + (dy)^2\right] + (dz)^2 \\ &= \left[\sin^2\theta dr^2 + r^2\cos^2\theta d\theta^2 + r^2\sin^2\theta d\phi^2 + 2r\sin\theta\cos\theta drd\theta\right] \\ &\quad + \left[\cos^2\theta dr^2 - 2r\sin\theta\cos\theta drd\theta + r^2\sin^2\theta d\theta^2\right] \end{aligned}$$

6. **Collect final terms:**

- dr^2 term: $(\sin^2\theta + \cos^2\theta)dr^2 = dr^2$
- $d\theta^2$ term: $(r^2\cos^2\theta + r^2\sin^2\theta)d\theta^2 = r^2 d\theta^2$
- $d\phi^2$ term: $r^2\sin^2\theta d\phi^2$
- $drd\theta$ term: $(2r\sin\theta\cos\theta - 2r\sin\theta\cos\theta)drd\theta = 0$

The final result is the metric for flat 3D space in spherical coordinates:

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2\sin^2\theta d\phi^2 \tag{2.31}$$

From this, we can read off the non-zero metric components: $g_{rr} = 1$, $g_{\theta\theta} = r^2$, and $g_{\phi\phi} = r^2\sin^2\theta$. This metric is for a *flat* space, even though its components depend on the coordinates r and θ. Curvature is a more subtle concept.

Example: The 2-Sphere S^2

We can find the metric of a 2-dimensional sphere of radius R by taking the 3D spherical metric, setting $r = R$ (a constant), and thus setting $dr = 0$.

$$ds^2 = R^2 d\theta^2 + R^2\sin^2\theta d\phi^2 \tag{2.32}$$

For a unit sphere ($R = 1$), this is $ds^2 = d\theta^2 + \sin^2\theta d\phi^2$. This is a *curved* space.

Locally Inertial Coordinates

The EEP states that at any point p in spacetime, we can construct a **locally inertial coordinate system** (or **local Lorentz frame**) in which the laws of physics take their special relativity form.

In this frame (with hatted indices, $x^{\hat{\mu}}$), the metric at p must look like the flat Minkowski metric $\eta_{\hat{\mu}\hat{\nu}}$, and any non-gravitational "forces" must vanish. This means the first derivatives of the metric must also vanish at p. A coordinate system $x^{\hat{\mu}}$ is a **locally inertial coordinate (LIC)** system at p if:

$$g_{\hat{\mu}\hat{\nu}}(p) = \eta_{\hat{\mu}\hat{\nu}} \tag{2.33}$$

$$\partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}}(p) = 0 \tag{2.34}$$

This is the rigorous, mathematical statement of the Equivalence Principle. We can always find coordinates where gravity "vanishes" to first order at a point.

However, we *cannot* in general make the second derivatives $\partial_{\hat{\rho}}\partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}}(p)$ vanish. These components describe the failure of nearby locally inertial frames to be parallel; they encode the **curvature** of spacetime (tidal forces).

Derivation (Counting Degrees of Freedom): We can prove this is possible by counting the number of "free parameters" we have in a coordinate transformation. In 4D ($n = 4$):

1. **Zeroth Order (Setting $g_{\hat{\mu}\hat{\nu}} = \eta_{\hat{\mu}\hat{\nu}}$):**
   - **Components to fix:** $g_{\hat{\mu}\hat{\nu}}$ is symmetric, so it has $\frac{1}{2}n(n+1) = \frac{1}{2}(4)(5) = \mathbf{10}$ independent components.
   - **Parameters we control:** We can choose the $n \times n$ Jacobian matrix $\frac{\partial x^{\mu}}{\partial x^{\hat{\nu}}}$, which has $n^2 = \mathbf{16}$ independent parameters.
   - **Conclusion:** We have 16 parameters to fix 10 components. This is easily possible. (The $16 - 10 = 6$ remaining parameters are the 3 rotations and 3 boosts of the Lorentz group, which leave $\eta_{\hat{\mu}\hat{\nu}}$ unchanged).
2. **First Order (Setting $\partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}} = 0$):**
   - **Components to fix:** There are $n = 4$ choices for the derivative index $\hat{\sigma}$, and 10 components for $g_{\hat{\mu}\hat{\nu}}$. Total $= 4 \times 10 = \mathbf{40}$ components.

- **Parameters we control:** We can now choose the n^3 second derivatives $\frac{\partial^2 x^\mu}{\partial x^{\hat{\nu}} \partial x^{\hat{\sigma}}}$. This object is symmetric in its two lower indices. The number of components is $n \times \frac{1}{2}n(n+1) = 4 \times 10 = \mathbf{40}$ parameters.
- **Conclusion:** We have exactly 40 parameters to fix 40 components. It is always possible to set the first derivatives of the metric to zero.

3. **Second Order (Setting $\partial_{\hat{\rho}}\partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}} = 0$):**

   - **Components to fix:** The object $\partial_{\hat{\rho}}\partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}}$ is symmetric in $\hat{\mu}, \hat{\nu}$ (10 components) and in $\hat{\rho}, \hat{\sigma}$ (10 components). Total $= 10 \times 10 = \mathbf{100}$ components.
   - **Parameters we control:** We can now choose the third derivatives $\frac{\partial^3 x^\mu}{\partial x^{\hat{\nu}} \partial x^{\hat{\sigma}} \partial x^{\hat{\rho}}}$. This is symmetric in its three lower indices.
   - The number of independent components for 3 symmetric indices in $n = 4$ dimensions is $\frac{n(n+1)(n+2)}{3!} = \frac{4 \cdot 5 \cdot 6}{6} = 20$.
   - The total number of parameters we control is n (for the μ index) $\times\ 20 = 4 \times 20 = \mathbf{80}$ parameters.
   - **Conclusion:** We have 80 parameters of freedom but need to fix 100 components. We are short by **20** components.

These 20 components that we *cannot* set to zero by a clever choice of coordinates are the 20 independent components of the **Riemann curvature tensor**, $R_{\hat{\rho}\hat{\sigma}\hat{\mu}\hat{\nu}}$.

Utility of Locally Inertial Coordinates

The existence of LICs is an incredibly powerful tool. We can solve complex problems by:

1. Transforming to a LIC at the point of interest p.
2. In this frame, $g_{\hat{\mu}\hat{\nu}} = \eta_{\hat{\mu}\hat{\nu}}$ and all physics reduces to Special Relativity.
3. Solving the problem using SR.
4. Writing the answer as a general tensor equation.

Since a tensor equation that is true in one coordinate system is true in all, this "SR" answer is the correct, general answer for any curved spacetime.

Example: Relative Velocity What is the 3-velocity v of a rocket (with 4-velocity V^μ) as measured by an observer (with 4-velocity U^μ)?

1. Go to the observer's LIC. In this frame, the observer is at rest, so $U^{\hat{\mu}} = (1, 0, 0, 0)$.
2. The rocket's 4-velocity in this frame is $V^{\hat{\mu}} = (\gamma, \gamma v^1, \gamma v^2, \gamma v^3)$, where $v^2 = (v^1)^2 + (v^2)^2 + (v^3)^2$ and $\gamma = (1 - v^2)^{-1/2}$.
3. In this SR frame, let's compute the scalar product $g_{\hat{\mu}\hat{\nu}} U^{\hat{\mu}} V^{\hat{\nu}} = \eta_{\hat{\mu}\hat{\nu}} U^{\hat{\mu}} V^{\hat{\nu}}$:
$$\eta_{\hat{\mu}\hat{\nu}} U^{\hat{\mu}} V^{\hat{\nu}} = \eta_{00} U^{\hat{0}} V^{\hat{0}} = (-1)(1)(\gamma) = -\gamma$$
4. We have the SR result $\gamma = -(\eta_{\hat{\mu}\hat{\nu}} U^{\hat{\mu}} V^{\hat{\nu}})$.
5. We know $\gamma = (1 - v^2)^{-1/2}$, so $\gamma^{-2} = 1 - v^2$, which gives $v = \sqrt{1 - \gamma^{-2}}$.
6. Substitute our SR result for γ:
$$v = \sqrt{1 - (-\eta_{\hat{\mu}\hat{\nu}} U^{\hat{\mu}} V^{\hat{\nu}})^{-2}} = \sqrt{1 - (\eta_{\hat{\mu}\hat{\nu}} U^{\hat{\mu}} V^{\hat{\nu}})^{-2}}$$
7. This is a tensor equation. It must be true in *any* coordinate system.

The general, covariant expression for the relative 3-velocity is:
$$v = \sqrt{1 - (g_{\mu\nu} U^\mu V^\nu)^{-2}} \tag{2.35}$$

2.6 *An Expanding Universe*

A simple yet non-trivial example of a Lorentzian geometry is the flat **Robertson-Walker (RW) metric**. This metric describes a homogeneous and isotropic universe where the spatial sections (at constant time) are flat Euclidean 3-space, but the distance between points is scaled by a time-dependent **scale factor**, $a(t)$.

The line element is:

$$ds^2 = -dt^2 + a^2(t)[(dx)^2 + (dy)^2 + (dz)^2] \tag{2.36}$$

Observers who remain at constant spatial coordinates (x, y, z) are called **comoving** observers. The physical distance between two comoving observers grows over time as $a(t)$.

A common set of solutions for $a(t)$ are power laws (e.g., from a Big Bang singularity at $t = 0$):

$$a(t) = t^q, \quad \text{for } 0 < q < 1 \tag{2.37}$$

(For example, a universe dominated by radiation has $q = 1/2$, and one dominated by matter has $q = 2/3$.)

Light Cones and Causal Structure

The causal structure is defined by null paths ($ds^2 = 0$). Let's find the light cone for an observer at the origin, considering only a path in the $x - t$ plane ($dy = dz = 0$). For $a(t) = t^q$, the null condition $ds^2 = 0$ becomes:

$$0 = -dt^2 + a(t)^2(dx)^2 = -dt^2 + t^{2q}(dx)^2$$

This implies:

$$\frac{dx}{dt} = \pm t^{-q} \tag{2.38}$$

Justifying the dx/dt Notation

The expression $\frac{dx}{dt}$ can seem "sloppy," as dx and dt are basis one-forms, not simple differentials. The procedure is justified by formally considering the action of the metric tensor g on the tangent vector $V = \frac{dx^\mu}{d\lambda}\partial_\mu$ to a null curve $x^\mu(\lambda)$.

1. The "null curve" condition is $g(V, V) = 0$.

$$\begin{aligned} g(V, V) &= g_{\mu\nu}V^\mu V^\nu \\ &= g_{\mu\nu}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} \\ &= g_{tt}\frac{dt}{d\lambda}\frac{dt}{d\lambda} + g_{xx}\frac{dx}{d\lambda}\frac{dx}{d\lambda} \quad (\text{since } dy = dz = 0) \\ &= -1\left(\frac{dt}{d\lambda}\right)^2 + a(t)^2\left(\frac{dx}{d\lambda}\right)^2 = 0 \end{aligned}$$

2. For $a(t) = t^q$, this is:

$$-\left(\frac{dt}{d\lambda}\right)^2 + t^{2q}\left(\frac{dx}{d\lambda}\right)^2 = 0 \tag{2.39}$$

3. We can rearrange this to:

$$\frac{\left(\frac{dx}{d\lambda}\right)^2}{\left(\frac{dt}{d\lambda}\right)^2} = \frac{1}{t^{2q}} = t^{-2q}$$

4. By the standard 1D chain rule, $\frac{dx}{dt} = \frac{dx/d\lambda}{dt/d\lambda}$. Therefore, we have:

$$\left(\frac{dx}{dt}\right)^2 = t^{-2q} \implies \frac{dx}{dt} = \pm t^{-q}$$

This confirms the "sloppy" notation gives the correct result for the coordinate velocity of a null path.

The Particle Horizon

We can solve this differential equation by separation of variables to find the path of a light ray $x(t)$:

$$\int_{x_0}^{x(t)} dx' = \pm \int_{t_0}^{t} (t')^{-q} dt'$$

$$x(t) - x_0 = \pm \left[\frac{(t')^{1-q}}{1-q} \right]_{t_0}^{t}$$

If we consider a light ray emitted from the singularity at $t_0 = 0$ (and $x_0 = 0$), the total comoving distance it can travel by time t is:

$$x(t) = \pm \frac{t^{1-q}}{1-q}$$

Since $0 < q < 1$, the exponent $1 - q$ is positive, so this distance is finite. This means that at any given time t, an observer at the origin can only see events within a finite comoving distance. This boundary is the **particle horizon**.

The light cones are tangent to the $t = 0$ singularity. A key feature of this geometry is that the past light cones of two spatially separated, comoving observers (at the same time t) may not overlap. This means there are regions of the universe that have never been in causal contact, which leads to the "horizon problem" in cosmology.

2.7 Causality

In physics, we often pose an **initial-value problem**: given the state of a system on a "slice" of time, can we predict its state at a later time? In General Relativity, the dynamical nature of spacetime itself makes this a complex question. The study of how information can (or cannot) propagate is the study of **causal structure**.

Fundamental Definitions

We begin with a set of definitions that formalize how points in spacetime are causally related.

- **Causal Curve:** A curve that is everywhere timelike or null. This represents a path information or an observer can travel.
- **Causual Future $J^+(S)$:** For a set of points S, $J^+(S)$ is the set of all points that can be reached from S by a future-directed *causal* (timelike or null) curve.
- **Chronological Future $I^+(S)$:** The set of all points that can be reached from S by a future-directed, strictly *timelike* curve.
- **Causal/Chronological Past $(J^-(S), I^-(S))$:** The same definitions, but for past-directed curves.
- **Achronal Set:** A set S where no two points in S are connected by a timelike curve (e.g., any spacelike surface).

Domains of Dependence and Cauchy Surfaces

The most important concept for the initial-value problem is the domain of dependence.

- **Domain of Dependence $D(S)$:** For an achronal set S, its **future domain of dependence**, $D^+(S)$, is the set of all points p such that *every* past-directed, inextendible causal curve through p must intersect S.

- **Physical Intuition:** $D(S)$ is the set of all points in spacetime for which the state of the universe is completely and uniquely determined by the initial data specified on S. Any event outside of $D(S)$ could be influenced by information that *never* passed through S.
- **Cauchy Horizon $H(S)$:** The boundary of the domain of dependence, $H(S) = \text{boundary}(D(S))$. The future Cauchy horizon is $H^+(S) = \text{boundary}(D^+(S))$.
- **Cauchy Surface (or Cauchy Slice):** An achronal set Σ whose domain of dependence is the *entire* manifold, $D(\Sigma) = M$.
- **Globally Hyperbolic:** A spacetime that possesses a Cauchy surface is called **globally hyperbolic**. These are the "nicest" spacetimes for which the initial-value problem is well-posed.

How Causal Structure Can Fail

A spacetime may fail to be globally hyperbolic (i.e., no Cauchy surface exists) for several reasons. This means predictability breaks down.

1. Poor Choice of Hypersurface

In a simple spacetime like Minkowski space, we can choose a "bad" initial slice Σ that is not a Cauchy surface. For example, a spacelike surface that is bounded in the future by some point p's light cone. Information from p can never reach Σ, so p is not in $D(\Sigma)$. This is a fault of the *slice*, not the spacetime.

2. Closed Timelike Curves (CTCs)

A more severe problem is when the spacetime geometry itself is pathological. The light cones can be "tilted" by curvature so much that they wrap around, allowing a future-directed timelike path to return to its own past. This is a **closed timelike curve (CTC)**.

An example is a 2D cylindrical spacetime $R \times S^1$ (with (t, x) coordinates where $x \equiv x + 1$) with the metric:

$$ds^2 = -\cos(\lambda)dt^2 - \sin(\lambda)(dtdx + dxdt) + \cos(\lambda)dx^2 \tag{2.40}$$

where $\lambda = \cot^{-1}(t)$.

- As $t \to -\infty$, $\lambda \to 0$ and $ds^2 \to -dt^2 + dx^2$. This is normal.
- As $t \to +\infty$, $\lambda \to \pi$ and $ds^2 \to +dt^2 - dx^2$. The x-direction has become timelike.

A curve of constant t (for $t > 0$) is a path $\gamma(\lambda) = (t_0, \lambda)$ which is a closed loop in the x-direction. Its tangent vector is $V = \partial_x$, and its "length" is $g(V, V) = g_{xx} = \cos(\lambda)$. When $t > 0$, $\lambda \in (0, \pi/2)$ so $\cos(\lambda) > 0$. Whoops, wait.

Let's re-state that:

- As $t \to -\infty$, $\lambda \to 0$, $ds^2 \to -dt^2 + dx^2$. t is timelike, x is spacelike.
- As $t \to +\infty$, $\lambda \to \pi$, $ds^2 \to +dt^2 - dx^2$. t is spacelike, x is timelike.

At some point, the light cones "tip over." The book states this happens for $t > 0$. A closed curve at constant $t_0 > 0$ (like $x \in [0, 1]$) is a CTC because the x-direction has become timelike ($g_{xx} = \cos(\lambda) < 0$). If we specify initial data on a surface Σ at $t < 0$, we can only predict the future up to the point where the CTCs form. This "boundary" is a Cauchy horizon.

3. Singularities

A **singularity** is a point that is not part of the manifold, but can be reached by a causal curve in a finite parameter time (it's an "edge" of spacetime). If a curve can "run into" a singularity, it does not go on to intersect the initial slice Σ. Therefore, any point p whose past contains a singularity may not be in the domain of dependence $D(\Sigma)$. Singularities create Cauchy horizons.

In GR, singularities are almost unavoidable (due to the attractive nature of gravity), which suggests that classical GR is not a complete theory of physics at all scales.

2.8 Tensor Densities

We sometimes encounter objects that are not tensors, but are "close." A **tensor density** is an object that transforms like a tensor, but with an additional factor of the Jacobian determinant of the coordinate transformation, $|J| = \left|\frac{\partial x^{\mu'}}{\partial x^{\mu}}\right|$, raised to some power. This power is called the **weight** W.

The Levi-Civita Symbol

The most common example is the **Levi-Civita symbol**, $\tilde{\epsilon}_{\mu_1\mu_2\ldots\mu_n}$. We define its components to be the same in *any* (right-handed) coordinate system:

$$\tilde{\epsilon}_{\mu_1\mu_2\ldots\mu_n} = \begin{cases} +1 & \text{if } \mu_1\ldots\mu_n \text{ is an even permutation of } 0\ldots(n-1) \\ -1 & \text{if } \mu_1\ldots\mu_n \text{ is an odd permutation of } 0\ldots(n-1) \\ 0 & \text{otherwise} \end{cases} \tag{2.41}$$

Because its components are fixed by definition, it does not transform as a tensor. Its transformation law can be found from the general formula for the determinant of a matrix $M^{\mu}{}_{\mu'}$:

$$\tilde{\epsilon}_{\mu'_1\ldots\mu'_n}\det(M) = \tilde{\epsilon}_{\mu_1\ldots\mu_n} M^{\mu_1}{}_{\mu'_1}\ldots M^{\mu_n}{}_{\mu'_n}$$

By setting $M^{\mu}{}_{\mu'} = \frac{\partial x^{\mu}}{\partial x^{\mu'}}$, we find $\det(M) = \left|\frac{\partial x^{\mu}}{\partial x^{\mu'}}\right| = |J|^{-1}$. This gives:

$$\tilde{\epsilon}_{\mu'_1\ldots\mu'_n}|J|^{-1} = \tilde{\epsilon}_{\mu_1\ldots\mu_n}\frac{\partial x^{\mu_1}}{\partial x^{\mu'_1}}\cdots\frac{\partial x^{\mu_n}}{\partial x^{\mu'_n}}$$

Multiplying by $|J|$, we get the transformation law:

$$\tilde{\epsilon}_{\mu'_1\ldots\mu'_n} = |J|\left(\tilde{\epsilon}_{\mu_1\ldots\mu_n}\frac{\partial x^{\mu_1}}{\partial x^{\mu'_1}}\cdots\frac{\partial x^{\mu_n}}{\partial x^{\mu'_n}}\right) \tag{2.42}$$

This is the standard transformation for a (0,n) tensor, but multiplied by $|J|^1$. Thus, the Levi-Civita symbol is a **tensor density of weight** $W = +1$.

The Determinant of the Metric

Another important density is the determinant of the metric, $g = \det(g_{\mu\nu})$. We can derive its transformation law from the tensor transformation of $g_{\mu\nu}$:

$$g_{\mu'\nu'} = \frac{\partial x^{\mu}}{\partial x^{\mu'}}\frac{\partial x^{\nu}}{\partial x^{\nu'}}g_{\mu\nu}$$

In matrix notation, this is $g' = A^T g A$, where A is the matrix $A^{\mu}{}_{\mu'} = \frac{\partial x^{\mu}}{\partial x^{\mu'}}$. Taking the determinant of both sides:

$$\begin{aligned} \det(g') &= \det(A^T g A) = \det(A^T)\det(g)\det(A) \\ g' &= (\det(A))^2 g \end{aligned}$$

Since $\det(A) = \left|\frac{\partial x^{\mu}}{\partial x^{\mu'}}\right| = |J|^{-1}$, we have:

$$g' = (|J|^{-1})^2 g \quad\implies\quad g' = |J|^{-2} g \tag{2.43}$$

Thus, g is a **scalar density of weight** $W = -2$.

From Densities to Tensors

We can construct a true tensor from a density of weight W by multiplying it by $\sqrt{|g|}^{-W}$. (We use $|g|$ because g is negative for a Lorentzian metric).

The Levi-Civita Tensor: We can define a true (0,n) tensor, the **Levi-Civita tensor** (or volume element form), $\epsilon_{\mu_1 \ldots \mu_n}$, by absorbing the correct factor of $\sqrt{|g|}$. The symbol $\tilde{\epsilon}$ has $W = +1$. We must multiply by $\sqrt{|g|}^{-(+1)} = 1/\sqrt{|g|}$. The Levi-Civita tensor is defined as:

$$\epsilon_{\mu_1 \mu_2 \ldots \mu_n} = \sqrt{|g|}\tilde{\epsilon}_{\mu_1 \mu_2 \ldots \mu_n} \tag{2.44}$$

As shown above, the weight $W = -1$ of $\sqrt{|g|}$ (since $\sqrt{|g'|} = |J|^{-1}\sqrt{|g|}$) precisely cancels the weight $W = +1$ of $\tilde{\epsilon}$, resulting in a true tensor (weight $W = 0$).

Contravariant Tensors: We can also define the contravariant tensor $\epsilon^{\mu_1 \cdots \mu_n}$ by raising the indices on $\epsilon_{\mu_1 \ldots \mu_n}$ with $g^{\mu\nu}$:

$$\begin{aligned}
\epsilon^{\mu_1 \ldots \mu_n} &= g^{\alpha_1 \mu_1} \ldots g^{\alpha_n \mu_n} \; \epsilon_{\alpha_1 \ldots \alpha_n} \\
&= \sqrt{|g|} \; g^{\alpha_1 \mu_1} \ldots g^{\alpha_n \mu_n} \; \tilde{\epsilon}_{\alpha_1 \ldots \alpha_n} = \sqrt{|g_{\mu\alpha}|} \; \tilde{\epsilon}^{\mu_1 \ldots \mu_n} |g^{\mu\alpha}| \\
&= \sqrt{|g|} \; |g|^{-1} \; \tilde{\epsilon}^{\mu_1 \ldots \mu_n} = \frac{1}{\sqrt{|g|}} \; \tilde{\epsilon}^{\mu_1 \ldots \mu_n}
\end{aligned}$$

One can show that this is related to a contravariant *symbol* $\tilde{\epsilon}^{\mu_1 \cdots \mu_n}$ (whose components are $\mathrm{sgn}(g)\tilde{\epsilon}_{\mu_1 \ldots \mu_n}$) by:

$$\epsilon^{\mu_1 \mu_2 \ldots \mu_n} = \frac{1}{\sqrt{|g|}}\tilde{\epsilon}^{\mu_1 \mu_2 \ldots \mu_n} \tag{2.45}$$

Contraction Identity

A very useful identity relates the product of two Levi-Civita *tensors* (not symbols). When p indices are contracted in n dimensions, the result is an antisymmetrized product of Kronecker deltas:

$$\epsilon^{\mu_1 \ldots \mu_p \alpha_1 \ldots \alpha_{n-p}} \epsilon_{\mu_1 \ldots \mu_p \beta_1 \ldots \beta_{n-p}} = (-1)^s p!(n-p)! \delta^{[\alpha_1}_{\beta_1} \ldots \delta^{\alpha_{n-p}]}_{\beta_{n-p}} \tag{2.46}$$

where s is the number of negative eigenvalues of the metric (so $s = 1$ for our Lorentzian signature).

The most common case is $p = n - 1$:

$$\epsilon^{\mu_1 \ldots \mu_{n-1} \alpha} \epsilon_{\mu_1 \ldots \mu_{n-1} \beta} = (-1)^s (n-1)! \delta^\alpha_\beta \tag{2.47}$$

Proof of the Contraction Identity

The proof relies on relating the Levi-Civita *tensor* to the generalized Kronecker delta.

Step 1: Relating Tensor to Symbol

The Levi-Civita tensor ϵ is related to the Levi-Civita symbol $\tilde{\epsilon}$ (where components are $0, \pm 1$) by the metric determinant g:

$$\epsilon_{\mu_1 \ldots \mu_n} = \sqrt{|g|} \, \tilde{\epsilon}_{\mu_1 \ldots \mu_n}, \qquad \epsilon^{\mu_1 \ldots \mu_n} = \frac{\mathrm{sgn}(g)}{\sqrt{|g|}} \, \tilde{\epsilon}^{\mu_1 \ldots \mu_n} \tag{2.48}$$

The sign of the determinant is $\mathrm{sgn}(g) = (-1)^s$. When we multiply the two tensors, the factors of $\sqrt{|g|}$ cancel, leaving only the sign:

$$\epsilon^{\mu_1 \ldots \mu_n} \epsilon_{\nu_1 \ldots \nu_n} = (-1)^s \, \tilde{\epsilon}^{\mu_1 \ldots \mu_n} \tilde{\epsilon}_{\nu_1 \ldots \nu_n} \tag{2.49}$$

Step 2: The Generalized Kronecker Delta

The product of two Levi-Civita symbols is defined as the **Generalized Kronecker Delta**, which is the determinant of a matrix of individual Kronecker deltas:

$$\tilde{\epsilon}^{\mu_1 \dots \mu_n} \tilde{\epsilon}_{\nu_1 \dots \nu_n} = \delta^{\mu_1 \dots \mu_n}_{\nu_1 \dots \nu_n} = \det \begin{pmatrix} \delta^{\mu_1}_{\nu_1} & \dots & \delta^{\mu_1}_{\nu_n} \\ \vdots & & \vdots \\ \delta^{\mu_n}_{\nu_1} & \dots & \delta^{\mu_n}_{\nu_n} \end{pmatrix} \tag{2.50}$$

This determinant can be written compactly using antisymmetrization notation (where the brackets $[\dots]$ indicate a sum over permutations divided by $n!$):

$$\delta^{\mu_1 \dots \mu_n}_{\nu_1 \dots \nu_n} = n!\, \delta^{[\mu_1}_{\nu_1} \dots \delta^{\mu_n]}_{\nu_n} \tag{2.51}$$

we multiply by $n!$ because the antisymmetrization notation includes a $1/n!$ factor.

Step 3: Contraction

We now contract p pairs of indices. Without loss of generality, let the first p indices be the ones summed over ($\mu_1 = \nu_1, \dots, \mu_p = \nu_p$).

In the determinant expansion, summing over a pair of indices removes them from the determinant but introduces a factorial factor counting the permutations of those indices. Contracting p indices yields:

$$\delta^{\mu_1 \dots \mu_p \alpha_1 \dots \alpha_{n-p}}_{\mu_1 \dots \mu_p \beta_1 \dots \beta_{n-p}} = p!\, \delta^{\alpha_1 \dots \alpha_{n-p}}_{\beta_1 \dots \beta_{n-p}} \tag{2.52}$$

Finally, we expand the remaining generalized delta of rank $(n-p)$ using the antisymmetrization notation:

$$\delta^{\alpha_1 \dots \alpha_{n-p}}_{\beta_1 \dots \beta_{n-p}} = (n-p)!\, \delta^{[\alpha_1}_{\beta_1} \dots \delta^{\alpha_{n-p}]}_{\beta_{n-p}} \tag{2.53}$$

Combining these steps with the sign factor $(-1)^s$ from Step 1 gives the final identity:

$$\epsilon^{\mu_1 \dots \mu_p \alpha_1 \dots} \epsilon_{\mu_1 \dots \mu_p \beta_1 \dots} = (-1)^s p!(n-p)! \delta^{[\alpha_1}_{\beta_1} \dots \delta^{\alpha_{n-p}]}_{\beta_{n-p}} \tag{2.54}$$

2.9 Differential Forms

Differential p-forms are a special, and extremely useful, class of tensors. A p-form is defined as a **(0, p) tensor that is completely antisymmetric**.

- A 0-form is just a scalar function ϕ.
- A 1-form is a dual vector ω_μ.
- The Levi-Civita tensor $\epsilon_{\mu\nu\rho\sigma}$ is a 4-form (in 4D).

The space of p-forms at a point on an n-dimensional manifold has a dimensionality of $\binom{n}{p} = \frac{n!}{p!(n-p)!}$.

- **Why this dimension?** This is "n choose p". Since the tensor is totally antisymmetric, the order of indices doesn't matter (it just adds a sign), and indices cannot repeat. Thus, we are simply counting how many unique ways we can choose p distinct indices from n available dimensions.

The Wedge Product

Given a p-form A and a q-form B, we can define their **wedge product** $A \wedge B$, which is a $(p+q)$-form. It is defined as their antisymmetrized tensor product:

$$(A \wedge B)_{\mu_1 \dots \mu_{p+q}} = \frac{(p+q)!}{p!q!} A_{[\mu_1 \dots \mu_p} B_{\mu_{p+1} \dots \mu_{p+q}]} \tag{2.55}$$

Anatomy of the Wedge Factors

- **The Factor** $\frac{(p+q)!}{p!q!}$**:**
  - *Role:* Normalization to ensure basis unity.
  - *Origin:* The antisymmetrization bracket $[\dots]$ includes a factor of $\frac{1}{(p+q)!}$. Without the pre-factor, the wedge product of basis differentials (like $dx \wedge dy$) would equal $\frac{1}{2}(dx \otimes dy - dy \otimes dx)$. We want the basis to be normalized to 1, not 1/2.
  - The factor $\frac{(p+q)!}{p!q!}$ represents the number of "shuffles"—ways to interleave the indices of A and B while keeping the internal order of A and B fixed.
- **Commutation Property** $(-1)^{pq}$**:**

$$A \wedge B = (-1)^{pq} B \wedge A \tag{2.56}$$

Proof: To transform $A \wedge B$ into $B \wedge A$, we must swap every index of A past every index of B.

1. A has p indices. B has q indices.
2. Take the first index of A. To move it past all of B, you need q swaps.
3. You must do this for all p indices of A.
4. Total swaps $= p \times q$. Each swap contributes (-1). Total sign $= (-1)^{pq}$.

The Exterior Derivative

The **exterior derivative** d is an operator that turns a p-form field into a $(p+1)$-form field. It is defined as the antisymmetrized partial derivative:

$$(dA)_{\mu_1 \dots \mu_{p+1}} = (p+1)\partial_{[\mu_1} A_{\mu_2 \dots \mu_{p+1}]} \tag{2.57}$$

Anatomy of the Exterior Derivative

- **The Factor** $(p+1)$**:**
  - *Role:* Cancels the $1/(p+1)!$ in the antisymmetrizer to match the "curl" convention.
  - *Example:* For a 1-form A, $(dA)_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. If we just wrote $\partial_{[\mu} A_{\nu]}$, we would get $\frac{1}{2}(\dots)$. The factor $(1+1) = 2$ cancels the half.
- **The Graded Leibniz Rule (Product Rule):**

$$d(A \wedge B) = (dA) \wedge B + (-1)^p A \wedge (dB) \tag{2.58}$$

Derivation/Intuition: Consider the derivative ∂_λ acting on the product $A \dots B \dots$. By the standard chain rule:

$$\partial(AB) = (\partial A)B + A(\partial B)$$

However, d involves antisymmetrizing the derivative index λ with all other indices.

1. **First Term** $(dA) \wedge B$**:** The derivative index is naturally adjacent to A. No swapping needed.
2. **Second Term** $A \wedge (dB)$**:** The derivative index λ starts at the far left (by definition of the operation $d(A \wedge B)$). For ∂_λ to act on B, it must "jump" over all indices of A.
3. Since A is a p-form, there are p indices to jump over.
4. Moving one index (λ) past p indices costs $(-1)^p$.

Examples:

- $p = 0$ **(Gradient of a scalar):** $A = \phi$

$$(d\phi)_\mu = (0+1)\partial_{[\mu}\phi = \partial_\mu \phi$$

- $p = 1$ **(Curl of a 1-form):** $A = A_\lambda$

$$(dA)_{\mu\nu} = (1+1)\partial_{[\mu}A_{\nu]} = 2 \cdot \frac{1}{2}(\partial_\mu A_\nu - \partial_\nu A_\mu) = \partial_\mu A_\nu - \partial_\nu A_\mu$$

This is precisely the electromagnetic field strength tensor, $F_{\mu\nu} = (dA)_{\mu\nu}$.

Why d is a Tensor

In Section 2.4, we showed that the partial derivative $\partial_\mu W_\nu$ is *not* a tensor because its transformation law contains an "extra" non-tensorial piece:

$$(\partial_{\mu'} W_{\nu'})_{\text{bad}} = W_\nu \frac{\partial^2 x^\nu}{\partial x^{\mu'} \partial x^{\nu'}}$$

The exterior derivative dA *is* a tensor because this "bad" term is symmetric in the indices μ' and ν' (since $\partial_{\mu'}\partial_{\nu'} = \partial_{\nu'}\partial_{\mu'}$), but the exterior derivative antisymmetrizes over these indices.

Derivation (for $p = 1$): Let's find the "bad" non-tensorial part of the transformed $(dA)_{\mu'\nu'}$.

$$\begin{aligned}(dA)_{\mu'\nu'} &= \partial_{\mu'} A_{\nu'} - \partial_{\nu'} A_{\mu'} \\ \text{Bad part of } (dA)_{\mu'\nu'} &= \left[W_\nu \frac{\partial^2 x^\nu}{\partial x^{\mu'} \partial x^{\nu'}}\right] - \left[W_\mu \frac{\partial^2 x^\mu}{\partial x^{\nu'} \partial x^{\mu'}}\right]\end{aligned}$$

In the second term, we can rename the dummy index $\mu \to \nu$. Since the order of partial derivatives doesn't matter, $\partial_{\nu'}\partial_{\mu'} = \partial_{\mu'}\partial_{\nu'}$.

$$\text{Bad part} = W_\nu \frac{\partial^2 x^\nu}{\partial x^{\mu'} \partial x^{\nu'}} - W_\nu \frac{\partial^2 x^\nu}{\partial x^{\mu'} \partial x^{\nu'}} = 0$$

The non-tensorial terms exactly cancel, leaving only the parts that obey the correct tensor transformation law. This holds for any p-form.

Key Property: $d^2 = 0$

The exterior derivative of an exterior derivative is always zero:

$$d(dA) = 0 \quad (\text{or } d^2 = 0) \tag{2.59}$$

This is a direct consequence of the fact that partial derivatives commute. **Derivation:** Let's apply d to the 2-form $F = dA$:

This leads to two important definitions:

- A p-form A is **closed** if $dA = 0$.
- A p-form A is **exact** if $A = dB$ for some $(p-1)$-form B.

The $d^2 = 0$ property means that **all exact forms are automatically closed**. The converse is only true in "topologically trivial" spacetimes (this is **Poincaré's Lemma**).

The Double Dual Identity: $\star\star$

Applying the Hodge star twice returns the original form, up to a sign.

$$\star\star A = (-1)^{s+p(n-p)} A \tag{2.60}$$

Proof & Origin of Factors: Let's apply the star twice to a p-form A (resulting in indices ρ):

$$\begin{aligned}(\star\star A)_\rho &= \frac{1}{(n-p)!}\epsilon^{\sigma\dots}_{\rho\dots}(\star A)_{\sigma\dots} \\ &= \frac{1}{(n-p)!}\epsilon^{\sigma\dots}_{\rho\dots}\left(\frac{1}{p!}\epsilon^{\nu\dots}_{\sigma\dots}A_{\nu\dots}\right)\end{aligned}$$

Combining these gives us the product of two Levi-Civita tensors contracted over the internal indices σ. This generates three distinct factors:

- **1. The Normalization Factor:** $\frac{1}{p!(n-p)!}$
  - *Role:* Fixes overcounting.
  - *Origin:* The Einstein summation convention sums over *all* possible permutations of indices (e.g., $dx^1 \wedge dx^2$ and $dx^2 \wedge dx^1$). Since our forms are antisymmetric, these are mathematically identical. We divide by the factorial of the form's degree to count each physical component exactly once.
- **2. The Permutation Factor:** $(-1)^{p(n-p)}$
  - *Role:* Aligns the indices for contraction.
  - *Origin:* In the expression $\epsilon^{\sigma\dots}_{\rho\dots}\epsilon^{\nu\dots}_{\sigma\dots}$, the contracted indices (σ) are at the **front** of the first epsilon but at the **back** of the second.
  - To contract them properly (using $\delta^{\dots}_{\dots}$ identities), we must "move" the block of $(n-p)$ indices past the block of p indices.
  - This requires exactly $p \times (n-p)$ swaps. Each swap contributes a (-1).
- **3. The Metric Signature Factor:** $(-1)^s$
  - *Role:* Accounts for the geometry of spacetime (Time vs. Space).
  - *Origin:* We are contracting $\epsilon^{\dots}$ (upper) with $\epsilon_{\dots}$ (lower). Lowering indices uses the metric $g_{\mu\nu}$.
  - In Euclidean space, the determinant is positive. In Lorentzian space (Relativity), the determinant is negative ($g = -1$).
  - This "cost" of lowering the indices introduces a minus sign for each time dimension. s is the number of minus signs in the metric signature ($s = 1$ for standard GR).

Summary:

$$\underbrace{(-1)^s}_{\text{Metric}} \times \underbrace{(-1)^{p(n-p)}}_{\text{Swapping Indices}} \times A$$

This is exactly the i-th component of the 3D cross product, $\mathbf{U} \times \mathbf{V}$. This shows why the cross product only exists in 3D (as a map from two vectors to one vector).

Example: Maxwell's Equations

This formalism provides an incredibly elegant way to write Maxwell's equations.

- Define the **vector potential 1-form** A and the **current 1-form** J.
- Define the **Faraday 2-form** F as the exterior derivative of A:

$$F = dA \tag{2.61}$$

- **Homogeneous Equation:** The $d^2 = 0$ property automatically gives one of Maxwell's equations (the sourceless one):

$$dF = d(dA) = 0 \tag{2.62}$$

  In components, expanding the exterior derivative gives:

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0 \quad \Longrightarrow \quad \partial_{[\lambda} F_{\mu\nu]} = 0$$

  This corresponds to the identity $\nabla \cdot \mathbf{B} = 0$ and Faraday's law $\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$.

- **Inhomogeneous Equation:** The second (source) equation is written using the Hodge star:

$$d(\star F) = \star J \tag{2.63}$$

  Derivation of Tensor Form: To recover the standard tensor equation, apply the Hodge star $\star$ to both sides of the equation:

$$\star d(\star F) = \star(\star J)$$

  1. **RHS:** In 4D Lorentzian spacetime with signature $(- + ++)$, acting on a 1-form J, the double dual identity is $\star\star = 1$. Thus, $\star \star J = J$.
  2. **LHS:** The term $\star d \star$ is related to the codifferential (divergence). For a 2-form F, the identity is $(\star d \star F)^\nu = \nabla_\mu F^{\mu\nu}$.

  Substituting these back allows us to strip the forms and equate the tensor components:

$$\nabla_\mu F^{\mu\nu} = J^\nu \tag{2.64}$$

  This equation represents Gauss's law $\nabla \cdot \mathbf{E} = \rho$ and Ampere's law $\nabla \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}$.

2.10 Integration

In standard multivariable calculus, we learn that under a coordinate change, the volume element $d^n x$ transforms with the Jacobian determinant:

$$d^n x' = \left| \frac{\partial x^{\mu'}}{\partial x^\mu} \right| d^n x = |J| d^n x \tag{2.65}$$

This is a non-tensorial transformation law. From our work in the previous section, we can see this means the coordinate volume element $d^n x$ is a **tensor density of weight** $W = +1$.

The Volume Element as an n-Form

The proper way to understand integration on a manifold is to recognize that the integrand is an n-form. An integral over an n-dimensional region $\Sigma \subset M$ is a map from an n-form field ω to the real numbers:

$$\int_\Sigma : \omega \to \mathbb{R} \tag{2.66}$$

This is intuitive for a line integral ($n = 1$), where the integrand is $\omega = \omega(x)dx$, a 1-form.

For a general n-dimensional volume, the volume element $d\mu$ is an object that assigns a (real) number to an infinitesimal region. This region can be defined by n tangent vectors $(V_1, \ldots, V_n)$ that span it (like an infinitesimal parallelepiped).

Therefore, the volume element $d\mu$ is a $(0, n)$ tensor, i.e., an n-form. It is **totally antisymmetric** because swapping any two of the defining vectors (e.g., V_1, V_2) should flip the sign of the *oriented* volume.

The Invariant Volume Element

We are looking for a true n-form that is invariant under coordinate changes and can serve as our volume element. We already have two objects:

- The coordinate volume element $d^n x$, which is a density of weight $W = +1$.
- The scalar $\sqrt{|g|}$, which is a density of weight $W = -1$. (Recall $g' = |J|^{-2} g \implies \sqrt{|g'|} = |J|^{-1}\sqrt{|g|}$)

The product of these two objects will have weight $W = (+1) + (-1) = 0$, meaning it is a true scalar (and in this case, an n-form).

$$\sqrt{|g'|}d^n x' = \left(|J|^{-1}\sqrt{|g|}\right)(|J|d^n x) = \sqrt{|g|}d^n x$$

Thus, the **invariant volume element** is $\sqrt{|g|}d^n x$.

This object is, in fact, precisely the **Levi-Civita tensor** ϵ we defined in Section 2.8.

Derivation: We can show this by starting with the abstract definition of the ϵ tensor and its components.

1. The abstract Levi-Civita tensor ϵ is a (0,n) tensor.
2. We can expand it in a basis. As a (0,n) tensor, it is a linear combination of $\otimes dx^{\mu_i}$. Because it is also a totally antisymmetric n-form, we can write it more compactly using the wedge product:
$$\epsilon = \frac{1}{n!}\epsilon_{\mu_1 \ldots \mu_n} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_n}$$
3. Now, substitute the definition of the components, $\epsilon_{\mu_1 \ldots \mu_n} = \sqrt{|g|}\tilde{\epsilon}_{\mu_1 \ldots \mu_n}$:
$$\epsilon = \frac{1}{n!}\sqrt{|g|}\tilde{\epsilon}_{\mu_1 \ldots \mu_n} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_n}$$
4. The combination of the symbol and the wedge product has a simple identity:
$$\frac{1}{n!}\tilde{\epsilon}_{\mu_1 \ldots \mu_n} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_n} = dx^0 \wedge dx^1 \wedge \cdots \wedge dx^{n-1}$$
5. We define the shorthand $d^n x \equiv dx^0 \wedge dx^1 \wedge \cdots \wedge dx^{n-1}$.
6. Substituting this back, we find the identity:
$$\epsilon = \sqrt{|g|}d^n x \tag{2.67}$$

The invariant volume element is the Levi-Civita tensor.

Therefore, the invariant integral of a scalar function $\phi(x)$ over an n-dimensional manifold is written as:

$$S = \int \phi(x)\sqrt{|g|}d^n x \tag{2.68}$$

This is also written more abstractly, but equivalently, as:

$$S = \int \phi(x)\epsilon \tag{2.69}$$

Chapter 3

Curvature

3.1 Overview

Curvature and the Metric

We intuitively know that a surface like a sphere is curved, while a plane is flat. In General Relativity, we must formalize this using the metric tensor $g_{\mu\nu}$. Curvature depends on the metric, but the relationship is subtle: even a flat space can have a complicated-looking metric if we choose a weird coordinate system (like polar coordinates). Our goal is to construct a mathematical object—the curvature tensor—that tells us whether the space is genuinely curved, independent of our coordinate choice.

The Mathematical Roadmap

This chapter builds a hierarchy of structures, starting from the metric and ending with Einstein's equations. Here is the summary of the key objects we will derive.

1. The Connection (Christoffel Symbols)

To do calculus in curved space, we need a way to connect vectors lying in tangent spaces at different points. This structure is called a **connection**. In GR, we use a unique connection derived from the metric, characterized by the **Christoffel symbols** $\Gamma^{\lambda}_{\mu\nu}$:

$$\Gamma^{\lambda}_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma} (\partial_{\mu} g_{\nu\sigma} + \partial_{\nu} g_{\sigma\mu} - \partial_{\sigma} g_{\mu\nu}). \tag{3.1}$$

Note: Γ is not a tensor; it is a tool to correct for the "turning" of the coordinate grid.

2. The Covariant Derivative

The partial derivative ∂_{μ} is not a good operator in curved space because it doesn't transform as a tensor. We replace it with the **Covariant Derivative** ∇_{μ}, constructed using the connection:

$$\nabla_{\mu} V^{\nu} = \partial_{\mu} V^{\nu} + \Gamma^{\nu}_{\mu\sigma} V^{\sigma}. \tag{3.2}$$

Here, V^{ν} is a vector field. The extra term ΓV subtracts the changes that are purely due to the coordinate system curving, leaving behind the true geometric change of the vector.

3. Geodesics

A straight line is a path that does not accelerate. In curved space, this generalizes to a **geodesic**. A curve $x^{\mu}(\lambda)$ is a geodesic if it satisfies the geodesic equation:

$$\frac{d^2 x^{\mu}}{d\lambda^2} + \Gamma^{\mu}_{\rho\sigma} \frac{dx^{\rho}}{d\lambda} \frac{dx^{\sigma}}{d\lambda} = 0. \tag{3.3}$$

This equation says that the "acceleration" of the path (second derivative) is exactly canceled by the curvature of the coordinates (the Γ terms), meaning the particle is freely falling.

4. The Riemann Curvature Tensor

Finally, the definitive measure of curvature is the **Riemann tensor** $R^{\rho}_{\sigma\mu\nu}$. It is constructed from derivatives of the connection:

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_{\mu} \Gamma^{\rho}_{\nu\sigma} - \partial_{\nu} \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda} \Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda} \Gamma^{\lambda}_{\mu\sigma}. \tag{3.4}$$

Intuition: This tensor vanishes ($R = 0$) if and only if the space is perfectly flat. If any component is non-zero, the space is curved. Einstein's equation will eventually link parts of this tensor to the energy and momentum of matter in the universe.

3.2 Covariant Derivatives

The Problem with Partial Derivatives

In flat space with inertial coordinates, the partial derivative operator ∂_μ is sufficient to map tensors to tensors. However, on a general manifold, the partial derivative of a tensor does not transform as a tensor.

Let us verify this failure for a vector field V^ν. We transform coordinates from x^μ to $x^{\mu'}$. The transformation law for the vector itself is:

$$V^{\nu'} = \frac{\partial x^{\nu'}}{\partial x^\nu} V^\nu. \tag{3.5}$$

Taking the partial derivative $\partial_{\mu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \partial_\mu$ of this expression:

$$\partial_{\mu'} V^{\nu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \partial_\mu \left(\frac{\partial x^{\nu'}}{\partial x^\nu} V^\nu \right) \tag{3.6}$$

$$= \frac{\partial x^\mu}{\partial x^{\mu'}} \left[\left(\partial_\mu \frac{\partial x^{\nu'}}{\partial x^\nu} \right) V^\nu + \frac{\partial x^{\nu'}}{\partial x^\nu} (\partial_\mu V^\nu) \right] \tag{3.7}$$

$$= \underbrace{\frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} (\partial_\mu V^\nu)}_{\text{Tensor part}} + \underbrace{\frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial^2 x^{\nu'}}{\partial x^\mu \partial x^\nu} V^\nu}_{\text{Non-tensorial part}}. \tag{3.8}$$

The second term involves the second derivative of the coordinate transformation. Unless the transformation is linear (as in Special Relativity), this term is non-zero, meaning $\partial_{\mu'} V^{\nu'}$ is not a tensor.

Intuition: The partial derivative compares a vector at point x with a vector at $x + dx$. In curvilinear coordinates (or curved space), the coordinate basis vectors themselves change direction and length between these points. The "non-tensorial part" above essentially captures the changes in the coordinate grid, contaminating our measurement of how the vector field itself is changing.

Defining the Covariant Derivative

We seek a new operator ∇_μ (the covariant derivative) that fixes this by accounting for the background geometry. We postulate that ∇_μ is a linear map from (k, l) tensors to $(k, l + 1)$ tensors satisfying the Leibniz rule:

$$\nabla(T \otimes S) = (\nabla T) \otimes S + T \otimes (\nabla S). \tag{3.9}$$

For a vector V^ν, we define the covariant derivative as the partial derivative plus a linear correction term involving the connection coefficients $\Gamma^\nu_{\mu\lambda}$:

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\lambda} V^\lambda. \tag{3.10}$$

Note: $\Gamma^\nu_{\mu\lambda}$ is not a tensor. It is a set of n^3 functions (coefficients) specifically designed to cancel the non-tensorial trash generated by the partial derivative.

Transformation Law for Connection Coefficients

We can derive exactly how Γ must transform by demanding that $\nabla_\mu V^\nu$ acts like a tensor. The target transformation law is:

$$\nabla_{\mu'} V^{\nu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \nabla_\mu V^\nu. \tag{3.11}$$

Let us expand both sides. **LHS (Primed coordinates):**

$$\nabla_{\mu'} V^{\nu'} = \partial_{\mu'} V^{\nu'} + \Gamma^{\nu'}_{\mu'\lambda'} V^{\lambda'}. \tag{3.12}$$

Substituting the partial derivative result we found earlier:

$$\nabla_{\mu'} V^{\nu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \partial_\mu V^\nu + \frac{\partial x^\mu}{\partial x^{\mu'}} V^\nu \frac{\partial^2 x^{\nu'}}{\partial x^\mu \partial x^\nu} + \Gamma^{\nu'}_{\mu'\lambda'} V^{\lambda'}. \tag{3.13}$$

RHS (Transformed Unprimed coordinates):

$$\frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \left(\partial_\mu V^\nu + \Gamma^\nu_{\mu\lambda} V^\lambda \right) = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \partial_\mu V^\nu + \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \Gamma^\nu_{\mu\lambda} V^\lambda. \tag{3.14}$$

Equating LHS and RHS, the term with $\partial_\mu V^\nu$ cancels on both sides. We are left with:

$$\frac{\partial x^\mu}{\partial x^{\mu'}} V^\nu \frac{\partial^2 x^{\nu'}}{\partial x^\mu \partial x^\nu} + \Gamma^{\nu'}_{\mu'\lambda'} V^{\lambda'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \Gamma^\nu_{\mu\lambda} V^\lambda. \tag{3.15}$$

We want to isolate $\Gamma^{\nu'}_{\mu'\lambda'}$. First, rewrite dummy indices to factor out the vector V. On the LHS term $\Gamma^{\nu'}_{\mu'\lambda'} V^{\lambda'}$, rewrite $V^{\lambda'} = \frac{\partial x^\lambda}{\partial x^{\lambda'}} V^\lambda$. On the LHS second derivative term, change dummy index $\nu \to \lambda$ so it acts on V^λ.

$$\left(\Gamma^{\nu'}_{\mu'\lambda'} \frac{\partial x^\lambda}{\partial x^{\lambda'}} + \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial^2 x^{\nu'}}{\partial x^\mu \partial x^\lambda} \right) V^\lambda = \left(\frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \Gamma^\nu_{\mu\lambda} \right) V^\lambda. \tag{3.16}$$

Since this must hold for *any* vector V^λ, the terms in the parentheses must be equal:

$$\Gamma^{\nu'}_{\mu'\lambda'} \frac{\partial x^\lambda}{\partial x^{\lambda'}} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \Gamma^\nu_{\mu\lambda} - \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial^2 x^{\nu'}}{\partial x^\mu \partial x^\lambda}. \tag{3.17}$$

Multiply by $\frac{\partial x^{\lambda'}}{\partial x^\lambda}$ to isolate $\Gamma^{\nu'}_{\mu'\lambda'}$. Note that $\frac{\partial x^\lambda}{\partial x^{\lambda'}} \frac{\partial x^{\sigma'}}{\partial x^\lambda} = \delta^{\sigma'}_{\lambda'}$.

$$\Gamma^{\nu'}_{\mu'\lambda'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\lambda}{\partial x^{\lambda'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \Gamma^\nu_{\mu\lambda} - \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\lambda}{\partial x^{\lambda'}} \frac{\partial^2 x^{\nu'}}{\partial x^\mu \partial x^\lambda}. \tag{3.18}$$

This is the transformation law for the connection coefficients. The second term is the inhomogeneous part that makes Γ non-tensorial.

Covariant Derivative of One-Forms

To find the rule for one-forms ω_ν, we assume two additional properties:

1. Commutation with contraction: $\nabla_\mu (V^\lambda \omega_\lambda) = \partial_\mu (V^\lambda \omega_\lambda)$ (since $V^\lambda \omega_\lambda$ is a scalar).
2. Leibniz rule holds.

Let's expand the derivative of the scalar contraction:

$$\nabla_\mu (\omega_\lambda V^\lambda) = (\nabla_\mu \omega_\lambda) V^\lambda + \omega_\lambda (\nabla_\mu V^\lambda) \tag{3.19}$$

$$\partial_\mu (\omega_\lambda V^\lambda) = (\nabla_\mu \omega_\lambda) V^\lambda + \omega_\lambda (\partial_\mu V^\lambda + \Gamma^\lambda_{\mu\sigma} V^\sigma). \tag{3.20}$$

We also know directly that $\partial_\mu (\omega_\lambda V^\lambda) = (\partial_\mu \omega_\lambda) V^\lambda + \omega_\lambda (\partial_\mu V^\lambda)$. Equating the two lines:

$$(\partial_\mu \omega_\lambda) V^\lambda + \omega_\lambda \partial_\mu V^\lambda = (\nabla_\mu \omega_\lambda) V^\lambda + \omega_\lambda \partial_\mu V^\lambda + \omega_\lambda \Gamma^\lambda_{\mu\sigma} V^\sigma. \tag{3.21}$$

Cancel $\omega_\lambda \partial_\mu V^\lambda$ from both sides:

$$(\partial_\mu \omega_\lambda) V^\lambda = (\nabla_\mu \omega_\lambda) V^\lambda + \omega_\sigma \Gamma^\sigma_{\mu\lambda} V^\lambda. \tag{3.22}$$

Since V^λ is arbitrary:

$$\partial_\mu \omega_\lambda = \nabla_\mu \omega_\lambda + \Gamma^\sigma_{\mu\lambda}\omega_\sigma \quad \Longrightarrow \quad \nabla_\mu \omega_\lambda = \partial_\mu \omega_\lambda - \Gamma^\sigma_{\mu\lambda}\omega_\sigma. \tag{3.23}$$

Notice the sign change compared to vectors. This pattern generalizes to any tensor: **add** a Γ term for every upper index, and **subtract** a Γ term for every lower index.

The Christoffel Connection

There are infinitely many possible connections Γ on a manifold. However, general relativity uses a specific unique connection derived from the metric $g_{\mu\nu}$. This connection is defined by two constraints:

1. **Torsion-Free:** $\Gamma^\lambda_{\mu\nu} = \Gamma^\lambda_{\nu\mu}$. (The coefficients are symmetric in lower indices).
2. **Metric Compatibility:** $\nabla_\rho g_{\mu\nu} = 0$. (The metric is covariantly constant).

1. Vanishing Covariant Derivative of the Inverse Metric

We aim to show $\nabla_\rho g^{\mu\nu} = 0$. Start with the definition of the inverse metric:

$$g^{\mu\sigma} g_{\sigma\nu} = \delta^\mu_\nu \tag{3.24}$$

Apply the covariant derivative ∇_ρ to both sides. Recall that $\nabla_\rho \delta^\mu_\nu = 0$:

$$\nabla_\rho (g^{\mu\sigma} g_{\sigma\nu}) = 0 \tag{3.25}$$

Expand using the Leibniz (product) rule:

$$(\nabla_\rho g^{\mu\sigma}) g_{\sigma\nu} + g^{\mu\sigma}(\nabla_\rho g_{\sigma\nu}) = 0 \tag{3.26}$$

Assume metric compatibility ($\nabla_\rho g_{\sigma\nu} = 0$). The second term vanishes:

$$(\nabla_\rho g^{\mu\sigma}) g_{\sigma\nu} = 0 \tag{3.27}$$

Contract with the inverse metric $g^{\nu\lambda}$:

$$(\nabla_\rho g^{\mu\sigma}) g_{\sigma\nu} g^{\nu\lambda} = 0 \tag{3.28}$$

Using $g_{\sigma\nu} g^{\nu\lambda} = \delta^\lambda_\sigma$:

$$(\nabla_\rho g^{\mu\sigma}) \delta^\lambda_\sigma = \nabla_\rho g^{\mu\lambda} = 0 \quad \text{(Q.E.D.)} \tag{3.29}$$

2. Vanishing Covariant Derivative of the Levi-Civita Tensor

We aim to show $\nabla_\lambda \epsilon_{\mu\nu\rho\sigma} = 0$. The Levi-Civita tensor is defined using the permutation symbol $\tilde{\epsilon}_{\mu\nu\rho\sigma}$ (constants $0, \pm 1$) and the metric determinant g:

$$\epsilon_{\mu\nu\rho\sigma} = \sqrt{|g|}\, \tilde{\epsilon}_{\mu\nu\rho\sigma} \tag{3.30}$$

Consider a **Locally Inertial Coordinate System** (Riemann Normal Coordinates) at a point p. In these coordinates:

$$\Gamma^\alpha_{\beta\gamma}(p) = 0 \quad \text{and} \quad \partial_\lambda g_{\mu\nu}(p) = 0 \tag{3.31}$$

Consequently, the partial derivative of the determinant vanishes:

$$\partial_\lambda g = g g^{\mu\nu} \partial_\lambda g_{\mu\nu} = 0 \implies \partial_\lambda \sqrt{|g|} = 0 \tag{3.32}$$

In this frame, the covariant derivative reduces to the partial derivative:

$$\nabla_\lambda \epsilon_{\mu\nu\rho\sigma} = \partial_\lambda(\sqrt{|g|}\tilde{\epsilon}_{\mu\nu\rho\sigma}) = (\partial_\lambda \sqrt{|g|})\tilde{\epsilon}_{\mu\nu\rho\sigma} + \sqrt{|g|}(\partial_\lambda \tilde{\epsilon}_{\mu\nu\rho\sigma}) \tag{3.33}$$

Since $\tilde{\epsilon}$ are constants ($\partial\tilde{\epsilon} = 0$) and $\partial\sqrt{|g|} = 0$:

$$\nabla_\lambda \epsilon_{\mu\nu\rho\sigma} = 0 \tag{3.34}$$

Since $\nabla_\lambda \epsilon_{\mu\nu\rho\sigma}$ is a tensor, if it vanishes in one coordinate system, it vanishes in all coordinate systems.

$$\nabla_\lambda \epsilon_{\mu\nu\rho\sigma} = 0 \quad \text{(Q.E.D.)} \tag{3.35}$$

We can derive an explicit formula for this connection (called the **Christoffel symbols**) by permuting the indices of the metric compatibility equation.

First, write the definition of metric compatibility:

$$\nabla_\rho g_{\mu\nu} = \partial_\rho g_{\mu\nu} - \Gamma^\lambda_{\rho\mu} g_{\lambda\nu} - \Gamma^\lambda_{\rho\nu} g_{\mu\lambda} = 0. \tag{3.36}$$

Permute indices cyclically ($\rho \to \mu \to \nu \to \rho$):

$$\nabla_\mu g_{\nu\rho} = \partial_\mu g_{\nu\rho} - \Gamma^\lambda_{\mu\nu} g_{\lambda\rho} - \Gamma^\lambda_{\mu\rho} g_{\nu\lambda} = 0. \tag{3.37}$$

$$\nabla_\nu g_{\rho\mu} = \partial_\nu g_{\rho\mu} - \Gamma^\lambda_{\nu\rho} g_{\lambda\mu} - \Gamma^\lambda_{\nu\mu} g_{\rho\lambda} = 0. \tag{3.38}$$

We define a linear combination: (3.36) − (3.37) − (3.38) = 0.

$$\begin{aligned}
&\left(\partial_\rho g_{\mu\nu} - \Gamma^\lambda_{\rho\mu} g_{\lambda\nu} - \Gamma^\lambda_{\rho\nu} g_{\mu\lambda}\right) \\
-&\left(\partial_\mu g_{\nu\rho} - \Gamma^\lambda_{\mu\nu} g_{\lambda\rho} - \Gamma^\lambda_{\mu\rho} g_{\nu\lambda}\right) \\
-&\left(\partial_\nu g_{\rho\mu} - \Gamma^\lambda_{\nu\rho} g_{\lambda\mu} - \Gamma^\lambda_{\nu\mu} g_{\rho\lambda}\right) = 0.
\end{aligned}$$

Now we use the symmetries:

- Metric symmetry: $g_{\alpha\beta} = g_{\beta\alpha}$.
- Torsion-free connection: $\Gamma^\lambda_{\alpha\beta} = \Gamma^\lambda_{\beta\alpha}$.

Let's group the Γ terms.

- Terms with $g_{\nu\lambda}$: $-\Gamma^\lambda_{\rho\mu} g_{\lambda\nu}$ (from eq 1) cancels with $+\Gamma^\lambda_{\mu\rho} g_{\nu\lambda}$ (from eq 2).
- Terms with $g_{\mu\lambda}$: $-\Gamma^\lambda_{\rho\nu} g_{\mu\lambda}$ (from eq 1) cancels with $+\Gamma^\lambda_{\nu\rho} g_{\lambda\mu}$ (from eq 3).
- Terms with $g_{\rho\lambda}$: We are left with $+\Gamma^\lambda_{\mu\nu} g_{\lambda\rho}$ (from eq 2) and $+\Gamma^\lambda_{\nu\mu} g_{\rho\lambda}$ (from eq 3). These sum to $2\Gamma^\lambda_{\mu\nu} g_{\lambda\rho}$.

The equation simplifies to:

$$\partial_\rho g_{\mu\nu} - \partial_\mu g_{\nu\rho} - \partial_\nu g_{\rho\mu} + 2\Gamma^\lambda_{\mu\nu} g_{\lambda\rho} = 0. \tag{3.39}$$

Solving for the Γ term:

$$2\Gamma^\lambda_{\mu\nu} g_{\lambda\rho} = \partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}. \tag{3.40}$$

Multiply both sides by $\frac{1}{2}g^{\sigma\rho}$ (the inverse metric) to raise the index $\lambda \to \sigma$:

$$\Gamma^\sigma_{\mu\nu} = \frac{1}{2} g^{\sigma\rho}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}). \tag{3.41}$$

Intuition: This formula tells us that the "force" (connection) needed to keep a coordinate system straight depends entirely on the rates of change of the metric components. If the metric is constant (flat space Cartesian), all partials are zero, and $\Gamma = 0$.

Example: Polar Coordinates in Flat Space

Consider the plane $\mathbb{R}^2$ with polar coordinates (r, θ). The metric is:

$$ds^2 = dr^2 + r^2 d\theta^2 \implies g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}, \quad g^{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & r^{-2} \end{pmatrix}. \tag{3.42}$$

Let us calculate $\Gamma^\theta_{r\theta}$. Using the formula:

$$\Gamma^\theta_{r\theta} = \frac{1}{2} g^{\theta\lambda}(\partial_r g_{\theta\lambda} + \partial_\theta g_{\lambda r} - \partial_\lambda g_{r\theta}). \tag{3.43}$$

The sum over λ involves $\lambda = r$ and $\lambda = \theta$. Since $g^{\theta\lambda}$ is diagonal, only $\lambda = \theta$ contributes:

$$\Gamma^\theta_{r\theta} = \frac{1}{2} g^{\theta\theta}(\partial_r g_{\theta\theta} + \partial_\theta g_{\theta r} - \partial_\theta g_{r\theta}). \tag{3.44}$$

Substitute $g_{\theta\theta} = r^2$ and $g_{\theta r} = 0$:

$$\Gamma^\theta_{r\theta} = \frac{1}{2}\left(\frac{1}{r^2}\right)(\partial_r(r^2) + 0 - 0) = \frac{1}{2r^2}(2r) = \frac{1}{r}. \tag{3.45}$$

Similarly, for $\Gamma^r_{\theta\theta}$:

$$\Gamma^r_{\theta\theta} = \frac{1}{2} g^{rr}(\partial_\theta g_{\theta r} + \partial_\theta g_{r\theta} - \partial_r g_{\theta\theta}) \tag{3.46}$$

$$= \frac{1}{2}(1)(0 + 0 - \partial_r(r^2)) \tag{3.47}$$

$$= -r. \tag{3.48}$$

(Note: Γ vanishes for coordinates where the metric is constant, like Cartesian coordinates).

Divergence Formula

A useful identity relates the trace of the Christoffel symbol to the determinant of the metric $g = \det(g_{\mu\nu})$.

$$\Gamma^\mu_{\mu\lambda} = \frac{1}{\sqrt{|g|}}\partial_\lambda\sqrt{|g|} \tag{3.49}$$

where $g = \det(g_{\mu\nu})$.

Step 1: Expand the Definition

We start with the general definition of the Christoffel symbol:

$$\Gamma^\sigma_{\rho\lambda} = \frac{1}{2} g^{\sigma\nu}(\partial_\rho g_{\nu\lambda} + \partial_\lambda g_{\nu\rho} - \partial_\nu g_{\rho\lambda}) \tag{3.50}$$

Contracting the indices by setting $\sigma = \mu$ and $\rho = \mu$:

$$\Gamma^\mu_{\mu\lambda} = \frac{1}{2} g^{\mu\nu}(\partial_\mu g_{\nu\lambda} + \partial_\lambda g_{\mu\nu} - \partial_\nu g_{\mu\lambda}) \tag{3.51}$$

Since μ and ν are dummy summation indices and the metric is symmetric ($g_{\mu\nu} = g_{\nu\mu}$), the first term $g^{\mu\nu}\partial_\mu g_{\nu\lambda}$ cancels the third term $-g^{\mu\nu}\partial_\nu g_{\mu\lambda}$. We are left with:

$$\Gamma^\mu_{\mu\lambda} = \frac{1}{2} g^{\mu\nu}\partial_\lambda g_{\mu\nu} \tag{3.52}$$

Step 2: Derivative of the Determinant

Recall Cramer's rule for the inverse of a matrix. The inverse metric $g^{\mu\nu}$ is related to the cofactor of the metric element $g_{\mu\nu}$ by:

$$g^{\mu\nu} = \frac{1}{g}\text{adj}(g)_{\mu\nu} \tag{3.53}$$

Rearranging this, we find the partial derivative of the determinant with respect to the component:

$$\frac{\partial g}{\partial g_{\mu\nu}} = \frac{\partial(\text{adj}(g)_{\mu\nu}\, g_{\mu\nu})}{\partial g_{\mu\nu}} = \text{adj}(g)_{\mu\nu} \tag{3.54}$$

we now get

$$\frac{\partial g}{\partial g_{\mu\nu}} = gg^{\mu\nu} \tag{3.55}$$

Using the chain rule, the derivative of the determinant g with respect to the parameter λ is:

$$\partial_\lambda g = \frac{\partial g}{\partial g_{\mu\nu}}\partial_\lambda g_{\mu\nu} = (gg^{\mu\nu})\partial_\lambda g_{\mu\nu} \tag{3.56}$$

Dividing both sides by g, we obtain the identity:

$$g^{\mu\nu}\partial_\lambda g_{\mu\nu} = \frac{1}{g}\partial_\lambda g \tag{3.57}$$

Step 3: Substitution

Substituting the result from Eq. (3.57) back into Eq. (3.52):

$$\Gamma^\mu_{\mu\lambda} = \frac{1}{2}\left(\frac{1}{g}\partial_\lambda g\right) \tag{3.58}$$

Using the logarithmic derivative rule $\frac{1}{x}dx = d(\ln x)$:

$$\Gamma^\mu_{\mu\lambda} = \frac{1}{2}\partial_\lambda(\ln|g|) = \partial_\lambda(\ln|g|^{1/2}) \tag{3.59}$$

This yields the final result:

$$\Gamma^\mu_{\mu\lambda} = \frac{1}{\sqrt{|g|}}\partial_\lambda\sqrt{|g|} \tag{3.60}$$

we can now get the useful divergence formula:

$$\nabla_\mu V^\mu = \frac{1}{\sqrt{|g|}}\partial_\mu(\sqrt{|g|}V^\mu) \tag{3.61}$$

There is also other formuls on of them is stock's theorem in curved space:

$$\int_{\mathcal{V}} d^n x\sqrt{|g|}\nabla_\mu V^\mu = \int_{\partial\mathcal{V}} d^{n-1}x\sqrt{|\gamma|}n_\mu V^\mu \tag{3.62}$$

3.3 Parallel Transport and Geodesics

The Concept of Parallel Transport

In flat space, comparing vectors at different points is easy: we just slide the vector from point A to point B keeping its components constant. This "sliding" is called **parallel transport**. In curved space, however, the result of parallel transport depends on the path taken.

To formalize this, we look for a way to "keep the vector constant" relative to the local geometry as we move along a path $x^\mu(\lambda)$. In flat space, "constant" means $\frac{dV^\mu}{d\lambda} = 0$. In curved space, we replace the ordinary derivative with the covariant derivative projected along the path. We define the **directional covariant derivative** (or intrinsic derivative) $\frac{D}{d\lambda}$ as:

$$\frac{D}{d\lambda} = \frac{dx^\mu}{d\lambda}\nabla_\mu. \tag{3.63}$$

The condition for a vector V^μ to be parallel transported along a curve $x^\mu(\lambda)$ is that its directional covariant derivative vanishes:

$$\frac{DV^\mu}{d\lambda} \equiv \frac{dx^\nu}{d\lambda}\nabla_\nu V^\mu = 0. \tag{3.64}$$

Expanding the covariant derivative $\nabla_\nu V^\mu = \partial_\nu V^\mu + \Gamma^\mu_{\nu\sigma}V^\sigma$:

$$\frac{dx^\nu}{d\lambda}\left(\frac{\partial V^\mu}{\partial x^\nu} + \Gamma^\mu_{\nu\sigma}V^\sigma\right) = 0. \tag{3.65}$$

Using the chain rule $\frac{dx^\nu}{d\lambda}\partial_\nu = \frac{d}{d\lambda}$, we get the differential equation for parallel transport:

$$\frac{dV^\mu}{d\lambda} + \Gamma^\mu_{\sigma\rho}\frac{dx^\sigma}{d\lambda}V^\rho = 0. \tag{3.66}$$

Intuition: This equation tells us exactly how the components V^μ must change ($dV/d\lambda$) to compensate for the twisting of the coordinate system (Γ) so that the vector remains "geometrically constant."

Geodesics: Definition 1 (Parallel Transport)

A straight line in flat space is a curve that never turns; its tangent vector always points in the same direction. Generalizing this to curved space, a **geodesic** is a curve whose tangent vector $U^\mu = \frac{dx^\mu}{d\lambda}$ is parallel transported along itself.

Substituting $V^\mu = \frac{dx^\mu}{d\lambda}$ into the parallel transport equation (3.64):

$$\frac{D}{d\lambda}\left(\frac{dx^\mu}{d\lambda}\right) = 0. \tag{3.67}$$

Expanding this:

$$\frac{d}{d\lambda}\left(\frac{dx^\mu}{d\lambda}\right) + \Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0. \tag{3.68}$$

This yields the **Geodesic Equation**:

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0. \tag{3.69}$$

Note on Affine Parameters: This form of the equation is only valid if λ is an **affine parameter** (like proper time τ for massive particles). If we choose a weird parameter (like $\alpha = \lambda^3$), the RHS would not be zero, but proportional to $dx^\mu/d\alpha$.

Geodesics: Definition 2 (Variational Principle)

A second way to define a straight line is the "shortest path" between two points. In spacetime, for timelike paths, geodesics actually *maximize* the proper time τ.

We define the action as the proper time integral:

$$\tau = \int\sqrt{-g_{\mu\nu}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda}}d\lambda. \tag{3.70}$$

Varying the square root is messy. It is a standard "trick" to instead vary the "energy" integral I, which yields the same paths if we parameterize by proper time:

$$I = \frac{1}{2}\int g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu d\tau, \tag{3.71}$$

where $\dot{x}^\mu \equiv \frac{dx^\mu}{d\tau}$. We consider a variation $x^\mu(\tau) \to x^\mu(\tau) + \delta x^\mu(\tau)$ that vanishes at the endpoints. The variation of the integral is:

$$\delta I = \frac{1}{2} \int [\delta(g_{\mu\nu})\dot{x}^\mu \dot{x}^\nu + g_{\mu\nu}\delta(\dot{x}^\mu)\dot{x}^\nu + g_{\mu\nu}\dot{x}^\mu \delta(\dot{x}^\nu)]\, d\tau. \tag{3.72}$$

We use two facts: 1. The metric varies because it depends on position: $\delta g_{\mu\nu} = \partial_\sigma g_{\mu\nu} \delta x^\sigma$. 2. Variation commutes with differentiation: $\delta \dot{x}^\mu = \frac{d}{d\tau}(\delta x^\mu)$.

Substituting these into δI:

$$\delta I = \frac{1}{2} \int \left[(\partial_\sigma g_{\mu\nu})\dot{x}^\mu \dot{x}^\nu \delta x^\sigma + g_{\mu\nu} \frac{d(\delta x^\mu)}{d\tau} \dot{x}^\nu + g_{\mu\nu}\dot{x}^\mu \frac{d(\delta x^\nu)}{d\tau} \right] d\tau. \tag{3.73}$$

The last two terms are identical (by relabeling indices $\mu \leftrightarrow \nu$ and using symmetry of g). We combine them:

$$\delta I = \frac{1}{2} \int \left[(\partial_\sigma g_{\mu\nu})\dot{x}^\mu \dot{x}^\nu \delta x^\sigma + 2 g_{\mu\nu}\dot{x}^\nu \frac{d(\delta x^\mu)}{d\tau} \right] d\tau. \tag{3.74}$$

Now, integrate the second term by parts to move the derivative off δx^μ:

$$\int g_{\mu\nu}\dot{x}^\nu \frac{d(\delta x^\mu)}{d\tau} d\tau = \underbrace{[g_{\mu\nu}\dot{x}^\nu \delta x^\mu]_{\text{boundaries}}}_{=0} - \int \frac{d}{d\tau}(g_{\mu\nu}\dot{x}^\nu)\delta x^\mu d\tau. \tag{3.75}$$

Use the chain rule on the time derivative: $\frac{d}{d\tau}(g_{\mu\nu}) = \partial_\rho g_{\mu\nu} \dot{x}^\rho$.

$$\frac{d}{d\tau}(g_{\mu\nu}\dot{x}^\nu) = (\partial_\rho g_{\mu\nu}\dot{x}^\rho)\dot{x}^\nu + g_{\mu\nu}\ddot{x}^\nu. \tag{3.76}$$

Substitute this back into the expression for δI and factor out δx^σ. (We relabel dummy index $\mu \to \sigma$ in the second term to match the first term's δx^σ):

$$\delta I = \int \left[\frac{1}{2} \partial_\sigma g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu - (g_{\sigma\nu}\ddot{x}^\nu + \partial_\mu g_{\sigma\nu}\dot{x}^\mu \dot{x}^\nu) \right] \delta x^\sigma d\tau. \tag{3.77}$$

We want to group the $\dot{x}\dot{x}$ terms. In the term $\partial_\mu g_{\sigma\nu}\dot{x}^\mu \dot{x}^\nu$, we can split the sum since μ and ν are symmetric dummies: $\partial_\mu g_{\sigma\nu}\dot{x}^\mu \dot{x}^\nu = \frac{1}{2}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu})\dot{x}^\mu \dot{x}^\nu$. Now the term in the brackets is:

$$-g_{\sigma\nu}\ddot{x}^\nu + \frac{1}{2} \left(\partial_\sigma g_{\mu\nu} - \partial_\mu g_{\sigma\nu} - \partial_\nu g_{\sigma\mu} \right) \dot{x}^\mu \dot{x}^\nu. \tag{3.78}$$

For $\delta I = 0$ for all variations δx^σ, the bracket term must vanish.

$$g_{\sigma\nu}\ddot{x}^\nu + \frac{1}{2} \left(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu} \right) \dot{x}^\mu \dot{x}^\nu = 0. \tag{3.79}$$

Multiply by the inverse metric $g^{\lambda\sigma}$:

$$\ddot{x}^\lambda + \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})\dot{x}^\mu \dot{x}^\nu = 0. \tag{3.80}$$

We recognize the Christoffel symbol formula ($\Gamma^\lambda_{\mu\nu}$) in the second term! Thus, the shortest path equation is exactly the geodesic equation derived from parallel transport:

$$\frac{d^2 x^\lambda}{d\tau^2} + \Gamma^\lambda_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0. \tag{3.81}$$

Intuition: We have proven that "straightest" (parallel tangent) and "shortest" (extremal proper time) are identical definitions in curved spacetime, provided the connection is the metric-compatible (Christoffel) one.

Null Geodesics

For massless particles (photons), the proper time vanishes ($ds^2 = 0$), so τ cannot be used as a parameter. However, the path is still a geodesic. We simply use an affine parameter λ such that the tangent vector is the 4-momentum $p^\mu = \frac{dx^\mu}{d\lambda}$.

The equation of motion is still:

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0 \quad \text{or} \quad p^\nu \nabla_\nu p^\mu = 0. \tag{3.82}$$

An observer with 4-velocity U^μ measures the energy of this photon to be:

$$E = -p_\mu U^\mu. \tag{3.83}$$

3.4 Properties of Geodesics

Affine Parameters

When we derived the geodesic equation $\frac{D}{d\lambda}(\frac{dx^\mu}{d\lambda}) = 0$, we implicitly assumed a "good" parameterization. For massive particles, we used proper time τ. However, any linear transformation of the parameter works just as well.

If λ is a parameter such that the geodesic equation holds:

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0, \tag{3.84}$$

then λ is called an **affine parameter**. Consider a new parameter $\alpha(\lambda)$. By the chain rule:

$$\frac{dx^\mu}{d\lambda} = \frac{dx^\mu}{d\alpha}\frac{d\alpha}{d\lambda}. \tag{3.85}$$

Differentiating again with respect to λ:

$$\frac{d^2x^\mu}{d\lambda^2} = \frac{d}{d\lambda}\left(\frac{dx^\mu}{d\alpha}\frac{d\alpha}{d\lambda}\right) \tag{3.86}$$

$$= \frac{d}{d\lambda}\left(\frac{dx^\mu}{d\alpha}\right)\frac{d\alpha}{d\lambda} + \frac{dx^\mu}{d\alpha}\frac{d^2\alpha}{d\lambda^2} \tag{3.87}$$

$$= \left(\frac{d^2x^\mu}{d\alpha^2}\frac{d\alpha}{d\lambda}\right)\frac{d\alpha}{d\lambda} + \frac{dx^\mu}{d\alpha}\frac{d^2\alpha}{d\lambda^2} \tag{3.88}$$

$$= \frac{d^2x^\mu}{d\alpha^2}\left(\frac{d\alpha}{d\lambda}\right)^2 + \frac{dx^\mu}{d\alpha}\frac{d^2\alpha}{d\lambda^2}. \tag{3.89}$$

Substitute this into the geodesic equation:

$$\left[\frac{d^2x^\mu}{d\alpha^2}\left(\frac{d\alpha}{d\lambda}\right)^2 + \frac{dx^\mu}{d\alpha}\frac{d^2\alpha}{d\lambda^2}\right] + \Gamma^\mu_{\rho\sigma}\left(\frac{dx^\rho}{d\alpha}\frac{d\alpha}{d\lambda}\right)\left(\frac{dx^\sigma}{d\alpha}\frac{d\alpha}{d\lambda}\right) = 0. \tag{3.90}$$

Grouping terms by $(d\alpha/d\lambda)^2$:

$$\left(\frac{d\alpha}{d\lambda}\right)^2\left[\frac{d^2x^\mu}{d\alpha^2} + \Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\alpha}\frac{dx^\sigma}{d\alpha}\right] + \frac{dx^\mu}{d\alpha}\frac{d^2\alpha}{d\lambda^2} = 0. \tag{3.91}$$

If α is also an affine parameter, the term in the brackets must vanish (it's the geodesic equation for α). This requires the remaining term to vanish:

$$\frac{dx^\mu}{d\alpha}\frac{d^2\alpha}{d\lambda^2} = 0 \quad \Longrightarrow \quad \frac{d^2\alpha}{d\lambda^2} = 0. \tag{3.92}$$

The solution is a linear transformation:

$$\alpha = a\lambda + b, \tag{3.93}$$

where a and b are constants. Thus, affine parameters are unique up to linear transformations.

Intuition: Just like in Newtonian mechanics where Newton's laws look simplest if time flows uniformly ($F = ma$), in GR geodesics look simplest if the "clock" (parameter) ticks uniformly. If you use a weird clock (like logarithmic time), the equation acquires "fictitious force" terms (the $d^2\alpha/d\lambda^2$ part).

Four-Momentum and Energy

For timelike geodesics (massive particles), we define the four-momentum using proper time τ:

$$p^\mu = mU^\mu = m\frac{dx^\mu}{d\tau}. \tag{3.94}$$

The geodesic equation can be written as conservation of momentum along the path:

$$p^\lambda \nabla_\lambda p^\mu = 0. \tag{3.95}$$

For **null geodesics** (photons), $d\tau = 0$, so we cannot define velocity U^μ. Instead, we choose an affine parameter λ such that the tangent vector *is* the momentum:

$$p^\mu = \frac{dx^\mu}{d\lambda}. \tag{3.96}$$

This p^μ satisfies the null condition $p_\mu p^\mu = 0$ (or $g_{\mu\nu}p^\mu p^\nu = 0$).

How do we measure energy? In Special Relativity, $E = \gamma m$. In General Relativity, energy is frame-dependent. If an observer has 4-velocity U^μ_{obs} (where $U \cdot U = -1$), they measure the energy of a particle with momentum p^μ as the projection of p^μ onto their time axis:

$$E = -p_\mu U^\mu_{\text{obs}}. \tag{3.97}$$

This formula works for both massive and massless particles.

Intuition: Energy is just "the time component of momentum." But "time component" depends on whose time you are using. The dot product $-p \cdot U$ picks out the component of p parallel to the observer's worldline U.

Maximizing Proper Time

We previously stated that geodesics maximize proper time. Why maximize and not minimize? Consider two points A and B in spacetime connected by a timelike geodesic. We can always construct a path between them made of "jagged" light rays (null curves). Since $ds^2 = 0$ for light, the total proper time along this jagged path is $\tau = \int d\tau = 0$. Since $\tau_{\text{geodesic}} > 0$ and $\tau_{\text{jagged}} = 0$, the geodesic is clearly not a minimum! Thus, locally, timelike geodesics are curves of **maximal** proper time. This explains the Twin Paradox: the twin who stays home (follows a geodesic) experiences the most proper time (ages the most). The traveling twin accelerates (leaves the geodesic) and thus experiences less time.

The Exponential Map and Riemann Normal Coordinates

We can use geodesics to construct a special coordinate system that looks locally flat. This relies on the exponential map. At a point p, take a tangent vector $k \in T_p$. There is a unique geodesic $x^\mu(\lambda)$ starting at p with initial tangent k:

$$x^\mu(0) = x^\mu_p, \quad \left.\frac{dx^\mu}{d\lambda}\right|_0 = k^\mu. \tag{3.98}$$

We define the point $\exp_p(k)$ as the point located at $\lambda = 1$ along this geodesic.

$$\exp_p(k) = x^\mu(\lambda = 1). \tag{3.99}$$

This maps the flat tangent space T_p onto the curved manifold M near p.

We define **Riemann Normal Coordinates** (RNC) $x^{\hat{\mu}}$ by choosing an orthonormal basis $\hat{e}_{(\hat{\mu})}$ in T_p and assigning coordinates to a point q based on the vector k that maps to it:

$$x^{\hat{\mu}}(q) = k^{\hat{\mu}}. \tag{3.100}$$

In these coordinates, the geodesics through p are simply straight lines passing through the origin:

$$x^{\hat{\mu}}(\lambda) = \lambda k^{\hat{\mu}}. \tag{3.101}$$

Let us prove that the Christoffel symbols vanish at p in RNC. Plug the straight line solution $x^{\hat{\mu}}(\lambda) = \lambda k^{\hat{\mu}}$ into the geodesic equation at p (where $\lambda = 0$):

$$\underbrace{\frac{d^2 x^{\hat{\mu}}}{d\lambda^2}}_{=0} + \Gamma^{\hat{\mu}}_{\hat{\rho}\hat{\sigma}}(p) \underbrace{\frac{dx^{\hat{\rho}}}{d\lambda}}_{k^{\hat{\rho}}} \underbrace{\frac{dx^{\hat{\sigma}}}{d\lambda}}_{k^{\hat{\sigma}}} = 0. \tag{3.102}$$

Since the second derivative of a linear function is zero, we get:

$$\Gamma^{\hat{\mu}}_{\hat{\rho}\hat{\sigma}}(p) k^{\hat{\rho}} k^{\hat{\sigma}} = 0. \tag{3.103}$$

This must hold for *any* vector k. The only way a quadratic form $M_{\rho\sigma} k^\rho k^\sigma$ vanishes for all k is if the symmetric part of M is zero. Since the Christoffel connection is torsion-free ($\Gamma^{\hat{\mu}}_{\hat{\rho}\hat{\sigma}} = \Gamma^{\hat{\mu}}_{\hat{\sigma}\hat{\rho}}$), it is entirely symmetric. Therefore:

$$\Gamma^{\hat{\mu}}_{\hat{\rho}\hat{\sigma}}(p) = 0. \tag{3.104}$$

Finally, applying metric compatibility $\nabla_\rho g_{\mu\nu} = 0$ at p:

$$\nabla_\rho g_{\mu\nu} = \partial_\rho g_{\mu\nu} - \Gamma^\lambda_{\rho\mu} g_{\lambda\nu} - \Gamma^\lambda_{\rho\nu} g_{\mu\lambda} = 0. \tag{3.105}$$

Since $\Gamma = 0$ at p, this implies:

$$\partial_{\hat{\rho}} g_{\hat{\mu}\hat{\nu}}(p) = 0. \tag{3.106}$$

Intuition: Riemann Normal Coordinates are the rigorous realization of "locally inertial frames" (freely falling elevators). At the specific point p, gravity "disappears" (metric derivatives vanish), but second derivatives of the metric (curvature) cannot be made to vanish. Gravity is not a force field; it is tidal deviations between geodesics.

3.5 The Expanding Universe Revisited

Metric and Lagrangian

We consider the spatially flat expanding universe metric (Robertson-Walker metric with $k = 0$):

$$ds^2 = -dt^2 + a^2(t)\delta_{ij} dx^i dx^j. \tag{3.107}$$

Here, $a(t)$ is the scale factor describing the expansion. To find the Christoffel symbols, we use the variational method with the "energy" integral I:

$$I = \frac{1}{2}\int g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu d\tau = \frac{1}{2}\int \left[-(\dot{t})^2 + a^2(t)\delta_{ij}\dot{x}^i\dot{x}^j\right] d\tau, \tag{3.108}$$

where dots denote differentiation with respect to proper time τ (or an affine parameter).

Derivation of Connection Coefficients

1. Time Component Equation Vary the action with respect to t: $t(\tau) \to t(\tau) + \delta t(\tau)$. The scale factor depends on t, so $\delta(a^2) = 2a\frac{da}{dt}\delta t = 2a\dot{a}\delta t$. The variation is:

$$\delta I = \frac{1}{2}\int \left[-2\dot{t}\delta\dot{t} + 2a\dot{a}\delta_{ij}\dot{x}^i\dot{x}^j\delta t\right] d\tau. \tag{3.109}$$

Integrate the first term by parts ($\int \dot{t}\delta\dot{t} = -\int \ddot{t}\delta t$):

$$\delta I = \int \left[\ddot{t} + a\dot{a}\delta_{ij}\dot{x}^i\dot{x}^j\right] \delta t d\tau = 0. \tag{3.110}$$

This yields the equation of motion for t:

$$\ddot{t} + a\dot{a}\delta_{ij}\dot{x}^i\dot{x}^j = 0. \tag{3.111}$$

Compare this to the general geodesic equation for $\mu = 0$:

$$\ddot{t} + \Gamma^0_{\rho\sigma}\dot{x}^\rho\dot{x}^\sigma = 0. \tag{3.112}$$

The quadratic terms in the equation of motion are purely spatial ($\dot{x}^i\dot{x}^j$). There are no $\dot{t}^2$ or $\dot{t}\dot{x}^i$ terms. Thus:

$$\Gamma^0_{00} = 0, \tag{3.113}$$
$$\Gamma^0_{0i} = \Gamma^0_{i0} = 0, \tag{3.114}$$
$$\Gamma^0_{ij} = a\dot{a}\delta_{ij}. \tag{3.115}$$

2. Spatial Component Equation Vary with respect to a spatial coordinate x^k: $x^k \to x^k + \delta x^k$. The metric coefficients for x do not depend on x (spatial homogeneity), so the variation is only in the derivative terms:

$$\delta I = \frac{1}{2}\int \left[2a^2(t)\delta_{ij}\dot{x}^i\delta\dot{x}^j\right] d\tau = \int a^2\delta_{kj}\dot{x}^j\delta\dot{x}^k d\tau. \tag{3.116}$$

(We used the Kronecker delta to set $i = k$). Integrate by parts:

$$\delta I = -\int \frac{d}{d\tau}(a^2\dot{x}^k)\delta x^k d\tau = 0. \tag{3.117}$$

Expanding the time derivative (product rule):

$$\frac{d}{d\tau}(a^2\dot{x}^k) = a^2\ddot{x}^k + 2a\frac{da}{d\tau}\dot{x}^k = a^2\ddot{x}^k + 2a(\dot{a}\dot{t})\dot{x}^k = 0. \tag{3.118}$$

Dividing by a^2:

$$\ddot{x}^k + 2\frac{\dot{a}}{a}\dot{t}\dot{x}^k = 0. \tag{3.119}$$

Compare to the geodesic equation for $\mu = k$:

$$\ddot{x}^k + \Gamma^k_{\rho\sigma}\dot{x}^\rho\dot{x}^\sigma = 0. \tag{3.120}$$

The cross terms $\dot{x}^0\dot{x}^k$ correspond to Γ^k_{0k} and Γ^k_{k0}. Since the connection is symmetric:

$$2\Gamma^k_{0j}\dot{t}\dot{x}^j = 2\frac{\dot{a}}{a}\delta^k_j\dot{t}\dot{x}^j \implies \Gamma^k_{0j} = \frac{\dot{a}}{a}\delta^k_j. \tag{3.121}$$

The purely spatial terms Γ^k_{ij} vanish because there is no $\dot{x}^i\dot{x}^j$ term in the equation of motion.

$$\Gamma^k_{0j} = \frac{\dot{a}}{a}\delta^k_j, \tag{3.122}$$
$$\Gamma^k_{ij} = 0, \quad \Gamma^k_{00} = 0. \tag{3.123}$$

Intuition: The non-zero Γ^k_{0j} term acts like a "friction" or "drag" term in the equation of motion ($\ddot{x} \propto -\dot{a}/a\dot{x}$). As the universe expands ($\dot{a} > 0$), the peculiar velocity of a particle decays.

Cosmological Redshift

Consider a photon (null geodesic) moving in the x-direction. The null condition $ds^2 = 0$ implies:

$$-dt^2 + a^2 dx^2 = 0 \implies \frac{dx}{dt} = \frac{1}{a(t)}. \tag{3.124}$$

We need the behavior of the energy. We assume an affine parameter λ. The t-component of the geodesic equation is:

$$\frac{d^2t}{d\lambda^2} + \Gamma^0_{ij}\frac{dx^i}{d\lambda}\frac{dx^j}{d\lambda} = \frac{d^2t}{d\lambda^2} + a\dot{a}\left(\frac{dx}{d\lambda}\right)^2 = 0. \tag{3.125}$$

From the null condition, we can relate $dx/d\lambda$ to $dt/d\lambda$:

$$\left(\frac{dt}{d\lambda}\right)^2 = a^2\left(\frac{dx}{d\lambda}\right)^2 \implies \left(\frac{dx}{d\lambda}\right)^2 = \frac{1}{a^2}\left(\frac{dt}{d\lambda}\right)^2. \tag{3.126}$$

Substitute this back into the geodesic equation:

$$\frac{d^2t}{d\lambda^2} + a\dot{a}\left[\frac{1}{a^2}\left(\frac{dt}{d\lambda}\right)^2\right] = 0 \implies \frac{d^2t}{d\lambda^2} + \frac{\dot{a}}{a}\left(\frac{dt}{d\lambda}\right)^2 = 0. \tag{3.127}$$

This equation can be rewritten using the chain rule: $\frac{\dot{a}}{a}(\frac{dt}{d\lambda})^2 = \frac{1}{a}\frac{da}{dt}\frac{dt}{d\lambda}\frac{dt}{d\lambda} = \frac{1}{a}\frac{da}{d\lambda}\frac{dt}{d\lambda}$.

$$\frac{d}{d\lambda}\left(\frac{dt}{d\lambda}\right) + \frac{1}{a}\frac{da}{d\lambda}\left(\frac{dt}{d\lambda}\right) = 0. \tag{3.128}$$

This is equivalent to:

$$\frac{1}{dt/d\lambda}\frac{d}{d\lambda}\left(\frac{dt}{d\lambda}\right) = -\frac{1}{a}\frac{da}{d\lambda} \implies \frac{d}{d\lambda}\ln\left(\frac{dt}{d\lambda}\right) = -\frac{d}{d\lambda}\ln a. \tag{3.129}$$

Integrating gives:

$$\frac{dt}{d\lambda} = \frac{\omega_0}{a(t)}. \tag{3.130}$$

The energy measured by a comoving observer (velocity $U^\mu = (1,0,0,0)$) is:

$$E = -p_\mu U^\mu = -g_{00}p^0U^0 = -(-1)\frac{dt}{d\lambda}(1) = \frac{\omega_0}{a(t)}. \tag{3.131}$$

Thus, the energy (and frequency) of a photon is inversely proportional to the scale factor.

$$\frac{E_2}{E_1} = \frac{a_1}{a_2}. \tag{3.132}$$

Intuition: As the universe stretches by a factor of 2, the wavelength of light traveling through it also stretches by a factor of 2, reducing its energy by half. This is distinct from Doppler shift; it happens continuously along the path.

Conservation of Energy-Momentum

The conservation law in curved space is $\nabla_\mu T^{\mu\nu} = 0$. For a perfect fluid, the energy-momentum tensor is:

$$T^{\mu\nu} = (\rho + p)U^\mu U^\nu + pg^{\mu\nu}. \tag{3.133}$$

For a comoving fluid, $U^\mu = (1, 0, 0, 0)$. The tensor components are diagonal:

$$T^{00} = \rho, \quad T^{ij} = g^{ij} p = \frac{p}{a^2}\delta^{ij}. \tag{3.134}$$

Time Component ($\nu = 0$): Expand $\nabla_\mu T^{\mu 0} = \partial_\mu T^{\mu 0} + \Gamma^\mu_{\mu\lambda} T^{\lambda 0} + \Gamma^0_{\mu\lambda} T^{\mu\lambda} = 0$. 1. $\partial_\mu T^{\mu 0} = \partial_0 \rho = \dot\rho$. 2. $\Gamma^\mu_{\mu\lambda} T^{\lambda 0}$: Only $\lambda = 0$ contributes. $\Gamma^\mu_{\mu 0} = \Gamma^0_{00} + \Gamma^i_{i0} = 0 + \delta^i_i \frac{\dot a}{a} = 3\frac{\dot a}{a}$. Term is $3\frac{\dot a}{a}\rho$. 3. $\Gamma^0_{\mu\lambda} T^{\mu\lambda}$: Since T is diagonal, $\mu = \lambda$. $\Gamma^0_{00} T^{00} + \Gamma^0_{ij} T^{ij} = 0 + (a\dot a \delta_{ij})(\frac{p}{a^2}\delta^{ij}) = a\dot a \frac{p}{a^2}(3) = 3\frac{\dot a}{a} p$. Combining these:

$$\dot\rho + 3\frac{\dot a}{a}(\rho + p) = 0. \tag{3.135}$$

Spatial Component ($\nu = i$): A similar expansion shows $\partial_i p = 0$, meaning pressure is spatially uniform.

Fluid Evolution: Given an equation of state $p = w\rho$, the continuity equation becomes:

$$\frac{\dot\rho}{\rho} = -3(1 + w)\frac{\dot a}{a} \implies \rho \propto a^{-3(1+w)}. \tag{3.136}$$

- **Matter (Dust):** $w = 0 \implies \rho \propto a^{-3}$ (Volume expansion).
- **Radiation:** $w = 1/3 \implies \rho \propto a^{-4}$ (Volume expansion + Redshift).
- **Vacuum Energy:** $w = -1 \implies \rho \propto$ const.

Intuition: "Conservation of energy" $\nabla_\mu T^{\mu\nu} = 0$ does *not* imply total energy $\int \rho dV$ is constant. For radiation, total energy decreases as the universe expands because photons lose energy to the gravitational field (the expansion). For vacuum energy, total energy increases because the energy density is constant while the volume grows.

3.6 The Riemann Curvature Tensor

Curvature and Parallel Transport

We have seen that parallel transport depends on the path taken. To quantify this, consider moving a vector V^σ around an infinitesimal closed loop defined by two vectors A^μ and B^ν. The change in the vector δV^ρ after completing the loop is a linear transformation of the original vector, proportional to the area of the loop:

$$\delta V^\rho = R^\rho{}_{\sigma\mu\nu} V^\sigma A^\mu B^\nu. \tag{3.137}$$

Here, $R^\rho{}_{\sigma\mu\nu}$ is the Riemann Curvature Tensor. It measures the non-commutativity of parallel transport.

Derivation via Commutator of Covariant Derivatives

A more efficient way to calculate the Riemann tensor is to evaluate the commutator of two covariant derivatives acting on a vector field V^ρ. The failure of covariant derivatives to commute $[\nabla_\mu, \nabla_\nu] \neq 0$ is the hallmark of curvature.

We compute $[\nabla_\mu, \nabla_\nu]V^\rho = \nabla_\mu(\nabla_\nu V^\rho) - \nabla_\nu(\nabla_\mu V^\rho)$. First, consider the term $\nabla_\mu(\nabla_\nu V^\rho)$. Note that $\nabla_\nu V^\rho$ is a $(1, 1)$ tensor (one upper index ρ, one lower index ν). We apply the covariant derivative rule for such a tensor:

$$\nabla_\mu(\nabla_\nu V^\rho) = \partial_\mu(\nabla_\nu V^\rho) - \Gamma^\lambda_{\mu\nu}(\nabla_\lambda V^\rho) + \Gamma^\rho_{\mu\sigma}(\nabla_\nu V^\sigma). \tag{3.138}$$

Now, expand the inner covariant derivative $\nabla_\nu V^\rho = \partial_\nu V^\rho + \Gamma^\rho_{\nu\lambda} V^\lambda$:

$$\begin{aligned}\nabla_\mu(\nabla_\nu V^\rho) &= \partial_\mu(\partial_\nu V^\rho + \Gamma^\rho_{\nu\sigma} V^\sigma) - \Gamma^\lambda_{\mu\nu}\nabla_\lambda V^\rho + \Gamma^\rho_{\mu\lambda}(\partial_\nu V^\lambda + \Gamma^\lambda_{\nu\sigma} V^\sigma) \\ &= \partial_\mu\partial_\nu V^\rho + (\partial_\mu \Gamma^\rho_{\nu\sigma}) V^\sigma + \Gamma^\rho_{\nu\sigma}(\partial_\mu V^\sigma) \\ &\quad - \Gamma^\lambda_{\mu\nu}\nabla_\lambda V^\rho + \Gamma^\rho_{\mu\lambda}\partial_\nu V^\lambda + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} V^\sigma. \end{aligned} \tag{3.139}$$

(Note: We relabeled dummy indices to σ and λ strategically to factor out V^σ later).

Now, subtract the same expression with μ and ν interchanged. 1. The partial derivatives commute: $\partial_\mu\partial_\nu V^\rho - \partial_\nu\partial_\mu V^\rho = 0$. 2. The terms $\Gamma^\rho_{\nu\sigma}\partial_\mu V^\sigma$ (from the first part) and $\Gamma^\rho_{\nu\lambda}\partial_\mu V^\lambda$ (from the second part, relabeled) cancel. Specifically: In $\nabla_\mu\nabla_\nu$: Term is $+\Gamma^\rho_{\nu\sigma}\partial_\mu V^\sigma$. In $\nabla_\nu\nabla_\mu$: Term is $+\Gamma^\rho_{\mu\sigma}\partial_\nu V^\sigma$. Wait, looking at the indices carefully: $\Gamma^\rho_{\mu\lambda}\partial_\nu V^\lambda$ in the first expansion cancels with the $\Gamma^\rho_{\nu\sigma}\partial_\mu V^\sigma$ term from the subtracted expansion (after dummy index relabeling).

We are left with the terms proportional to V^σ and the connection term:

$$[\nabla_\mu, \nabla_\nu] V^\rho = \left(\partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}\right) V^\sigma - (\Gamma^\lambda_{\mu\nu} - \Gamma^\lambda_{\nu\mu})\nabla_\lambda V^\rho. \tag{3.140}$$

We identify the two key tensors here: 1. The Torsion Tensor: $T^\lambda_{\mu\nu} = \Gamma^\lambda_{\mu\nu} - \Gamma^\lambda_{\nu\mu} = 2\Gamma^\lambda_{[\mu\nu]}$. 2. The Riemann Curvature Tensor:

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}. \tag{3.141}$$

Thus, the commutator is:

$$[\nabla_\mu, \nabla_\nu] V^\rho = R^\rho{}_{\sigma\mu\nu} V^\sigma - T^\lambda_{\mu\nu}\nabla_\lambda V^\rho. \tag{3.142}$$

If we assume the connection is torsion-free (Christoffel connection), $T^\lambda_{\mu\nu} = 0$, and the expression simplifies to a linear map on the vector:

$$[\nabla_\mu, \nabla_\nu] V^\rho = R^\rho{}_{\sigma\mu\nu} V^\sigma. \tag{3.143}$$

So now we going to proof that it's the same one that appears in the parallel transport around a loop.

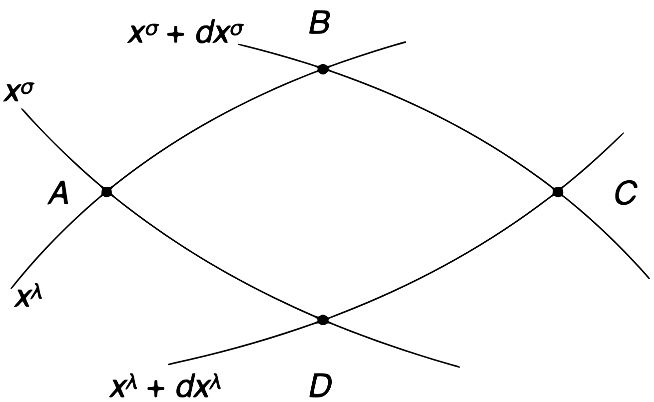


Figure 3.1: Parallel transport of a vector V^α around the closed loop $A \to B \to C \to D \to A$.

Derivation of the Riemann Curvature Tensor

We calculate the change in a vector V^α after parallel transporting it around a closed "diamond" loop defined by two coordinate displacements, dx^σ and dx^λ. The vertices of the loop are defined as follows:

- $A : (x^\sigma, x^\lambda)$
- $B : (x^\sigma + dx^\sigma, x^\lambda)$
- $C : (x^\sigma + dx^\sigma, x^\lambda + dx^\lambda)$

- $D : (x^\sigma, x^\lambda + dx^\lambda)$

The condition for parallel transport is $\nabla_{\vec{e}} V^\alpha = 0$, which leads to the differential equation:

$$\frac{\partial V^\alpha}{\partial x^\nu} = -\Gamma^\alpha{}_{\nu\mu} V^\mu \tag{3.144}$$

We integrate this equation along each leg of the loop:
Leg I ($A \to B$): Constant x^λ, increasing x^σ.

$$V^\alpha(B) = V^\alpha_{init} - \int_I \Gamma^\alpha{}_{\sigma\mu} V^\mu dx^\sigma \tag{3.145}$$

Leg II ($B \to C$): Constant $x^\sigma + dx^\sigma$, increasing x^λ.

$$V^\alpha(C) = V^\alpha(B) - \int_{II} \Gamma^\alpha{}_{\lambda\mu} V^\mu dx^\lambda \tag{3.146}$$

Leg III ($C \to D$): Constant $x^\lambda + dx^\lambda$, decreasing x^σ (note the sign change).

$$V^\alpha(D) = V^\alpha(C) + \int_{III} \Gamma^\alpha{}_{\sigma\mu} V^\mu dx^\sigma \tag{3.147}$$

Leg IV ($D \to A$): Constant x^σ, decreasing x^λ.

$$V^\alpha_{final} = V^\alpha(D) + \int_{IV} \Gamma^\alpha{}_{\lambda\mu} V^\mu dx^\lambda \tag{3.148}$$

The total change in the vector is $\delta V^\alpha = V^\alpha_{final} - V^\alpha_{init}$. Summing the contributions gives:

$$\delta V^\alpha = \int_{IV} \Gamma^\alpha{}_{\lambda\mu} V^\mu dx^\lambda - \int_{II} \Gamma^\alpha{}_{\lambda\mu} V^\mu dx^\lambda + \int_{III} \Gamma^\alpha{}_{\sigma\mu} V^\mu dx^\sigma - \int_I \Gamma^\alpha{}_{\sigma\mu} V^\mu dx^\sigma \tag{3.149}$$

We group terms by the direction of integration. Using the approximation that the difference between integrals along parallel paths is proportional to the derivative of the integrand:

$$\int_{III} - \int_I \approx \delta x^\lambda \int \frac{\partial}{\partial x^\lambda} (\Gamma^\alpha{}_{\sigma\mu} V^\mu) dx^\sigma \tag{3.150}$$

$$\int_{IV} - \int_{II} \approx -\delta x^\sigma \int \frac{\partial}{\partial x^\sigma} (\Gamma^\alpha{}_{\lambda\mu} V^\mu) dx^\lambda \tag{3.151}$$

Note the negative sign in the second term because Leg IV is at x^σ while Leg II is at $x^\sigma + dx^\sigma$. This allows us to write the total change as:

$$\delta V^\alpha \approx \delta x^\lambda \delta x^\sigma \left[\frac{\partial}{\partial x^\lambda} (\Gamma^\alpha{}_{\sigma\mu} V^\mu) - \frac{\partial}{\partial x^\sigma} (\Gamma^\alpha{}_{\lambda\mu} V^\mu) \right] \tag{3.152}$$

Expanding the derivatives using the product rule:

$$\delta V^\alpha \approx \delta x^\lambda \delta x^\sigma \left[(\partial_\lambda \Gamma^\alpha{}_{\sigma\mu}) V^\mu + \Gamma^\alpha{}_{\sigma\mu} (\partial_\lambda V^\mu) - (\partial_\sigma \Gamma^\alpha{}_{\lambda\mu}) V^\mu - \Gamma^\alpha{}_{\lambda\mu} (\partial_\sigma V^\mu) \right] \tag{3.153}$$

Since the vector is parallel transported, we substitute the derivatives of the vector components: $\partial_\nu V^\mu = -\Gamma^\mu{}_{\nu\rho} V^\rho$.

$$\delta V^\alpha \approx \delta x^\lambda \delta x^\sigma \left[(\partial_\lambda \Gamma^\alpha{}_{\sigma\mu} - \partial_\sigma \Gamma^\alpha{}_{\lambda\mu}) V^\mu + \Gamma^\alpha{}_{\sigma\mu} (-\Gamma^\mu{}_{\lambda\nu} V^\nu) - \Gamma^\alpha{}_{\lambda\mu} (-\Gamma^\mu{}_{\sigma\nu} V^\nu) \right] \tag{3.154}$$

Cleaning up the expression and relabeling dummy indices to factor out V^μ (swapping $\mu \leftrightarrow \nu$ in the last two terms):

$$\delta V^\alpha = \left[\partial_\lambda \Gamma^\alpha{}_{\sigma\mu} - \partial_\sigma \Gamma^\alpha{}_{\lambda\mu} + \Gamma^\alpha{}_{\lambda\nu} \Gamma^\nu{}_{\sigma\mu} - \Gamma^\alpha{}_{\sigma\nu} \Gamma^\nu{}_{\lambda\mu} \right] \delta x^\lambda \delta x^\sigma V^\mu \tag{3.155}$$

The term in the brackets is defined as the Riemann curvature tensor, $R^\alpha{}_{\mu\lambda\sigma}$:

$$R^\alpha{}_{\mu\lambda\sigma} = \partial_\lambda \Gamma^\alpha{}_{\sigma\mu} - \partial_\sigma \Gamma^\alpha{}_{\lambda\mu} + \Gamma^\alpha{}_{\lambda\nu} \Gamma^\nu{}_{\sigma\mu} - \Gamma^\alpha{}_{\sigma\nu} \Gamma^\nu{}_{\lambda\mu} \tag{3.156}$$

Thus, the change in the vector is given by:

$$\delta V^\alpha = R^\alpha{}_{\mu\lambda\sigma} V^\mu \delta x^\lambda \delta x^\sigma \tag{3.157}$$

Intuition: The Riemann tensor is composed of "derivatives of Γ" and "products of Γ". Since $\Gamma \sim \partial g$, the curvature involves second derivatives of the metric $\partial^2 g$. This matches our expectation that curvature describes the "acceleration" or "flexing" of the grid lines, not just their slope.

Theorem: Flatness and Constant Metrics

We wish to prove two statements regarding the relationship between the Riemann tensor and the metric:

1. If there exists a coordinate system where the metric components are constant everywhere ($\partial_\sigma g_{\mu\nu} = 0$), then the Riemann tensor vanishes ($R^\rho{}_{\sigma\mu\nu} = 0$).
2. Conversely, if the Riemann tensor vanishes everywhere, there exists a coordinate system in which the metric components are constant.

Part 1: Constant Metric $\implies$ Zero Curvature

This direction is straightforward. If we are in a coordinate system such that $\partial_\sigma g_{\mu\nu} = 0$ everywhere (not just at a point), then the Christoffel symbols vanish because they depend on derivatives of the metric:

$$\Gamma^\rho_{\mu\nu} = 0 \quad \text{and} \quad \partial_\sigma \Gamma^\rho_{\mu\nu} = 0. \tag{3.158}$$

By the definition of the Riemann tensor [Eq. (3.113)], if Γ and $\partial\Gamma$ are zero, then:

$$R^\rho{}_{\sigma\mu\nu} = 0. \tag{3.159}$$

Since $R^\rho{}_{\sigma\mu\nu} = 0$ is a tensor equation, if it is true in one coordinate system, it must be true in *any* coordinate system. Thus, a vanishing Riemann tensor is a necessary condition for finding coordinates where $g_{\mu\nu}$ is constant.

Part 2: Zero Curvature $\implies$ Constant Metric

We assume $R^\rho{}_{\sigma\mu\nu} = 0$ everywhere. We construct a one-form $\omega = \omega_\mu dx^\mu$ at a point p. For any path $x^\mu(\lambda)$ through p, we define a field along the path by parallel transport:

$$\frac{dx^\mu}{d\lambda} \nabla_\mu \omega_\nu = 0. \tag{3.160}$$

Because the Riemann tensor vanishes, parallel transport is path-independent. We can therefore define a unique one-form field ω_μ throughout the manifold. Since Eq. (3.160) must hold for arbitrary paths (arbitrary $dx^\mu/d\lambda$), the field must be covariantly constant:

$$\nabla_\mu \omega_\nu = 0. \tag{3.161}$$

The antisymmetric part of this equation is the exterior derivative:

$$\nabla_{[\mu}\omega_{\nu]} = \partial_{[\mu}\omega_{\nu]} = 0 \implies d\omega = 0. \tag{3.162}$$

Since ω is closed ($d\omega = 0$), and assuming a topologically simple region, it is also exact. Thus, there exists a scalar function α such that:

$$\omega_\mu = \partial_\mu \alpha. \tag{3.163}$$

We repeat this procedure to construct a set of n one-forms $\hat{\theta}^{(a)}$ (where $a \in \{1 \dots n\}$). We choose the initial one-forms at point p to form a normalized basis for the dual space T_p^*, such that the metric at p takes the canonical form:

$$ds^2(p) = \eta_{ab}\, \hat{\theta}^{(a)} \otimes \hat{\theta}^{(b)}. \tag{3.164}$$

Here, η_{ab} is a diagonal matrix with entries ± 1 (depending on the signature).

Now, parallel transport this entire set of basis forms all over the manifold. Since the Riemann tensor vanishes, the result is independent of the path. Since the metric is automatically parallel-transported (metric compatibility $\nabla g = 0$), the relationship between the metric and the basis remains unchanged everywhere:

$$ds^2(\text{anywhere}) = \eta_{ab}\,\hat{\theta}^{(a)} \otimes \hat{\theta}^{(b)}. \tag{3.165}$$

We have found a basis where the metric components are constant. However, this is just an abstract basis. To show that these correspond to coordinates, we recall that the one-forms are closed and therefore exact. Thus, there exist functions y^a such that:

$$\hat{\theta}^{(a)} = dy^a. \tag{3.166}$$

These n functions y^a are precisely the coordinates we sought. In this coordinate system, the metric becomes:

$$ds^2 = \eta_{ab}\,dy^a dy^b. \tag{3.167}$$

We have verified that if the Riemann tensor vanishes, the metric is effectively that of flat space, perhaps written in a "perverse" coordinate system.

3.7 Properties of the Riemann Tensor

Symmetries of the Riemann Tensor

The Riemann tensor has n^4 components in n dimensions (256 in 4D), but symmetries reduce this number significantly. To derive these, we evaluate the tensor in locally inertial coordinates (where $\Gamma = 0$ but $\partial\Gamma \neq 0$) at a point p. In these coordinates, the Riemann tensor with all lower indices is:

$$R_{\rho\sigma\mu\nu} = g_{\rho\lambda}R^{\lambda}{}_{\sigma\mu\nu} = \frac{1}{2}(\partial_\mu\partial_\sigma g_{\rho\nu} - \partial_\mu\partial_\rho g_{\nu\sigma} - \partial_\nu\partial_\sigma g_{\rho\mu} + \partial_\nu\partial_\rho g_{\mu\sigma}). \tag{3.168}$$

From this form, the symmetries are manifest: 1. Antisymmetry in the first pair:

$$R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu}. \tag{3.169}$$

2. Antisymmetry in the last pair:

$$R_{\rho\sigma\mu\nu} = -R_{\rho\sigma\nu\mu}. \tag{3.170}$$

3. Symmetry under exchange of pairs:

$$R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma}. \tag{3.171}$$

4. First Bianchi Identity (Algebraic): The sum of cyclic permutations of the last three indices vanishes. We define the sum of cyclic permutations of the last three indices (σ, μ, ν) as S. Using the expression for $R_{\rho\sigma\mu\nu}$ in locally inertial coordinates:

$$R_{\rho\sigma\mu\nu} = \frac{1}{2}\left(\partial_\mu\partial_\sigma g_{\rho\nu} - \partial_\mu\partial_\rho g_{\sigma\nu} - \partial_\nu\partial_\sigma g_{\rho\mu} + \partial_\nu\partial_\rho g_{\sigma\mu}\right) \tag{1}$$
$$R_{\rho\mu\nu\sigma} = \frac{1}{2}\left(\partial_\nu\partial_\mu g_{\rho\sigma} - \partial_\nu\partial_\rho g_{\mu\sigma} - \partial_\sigma\partial_\mu g_{\rho\nu} + \partial_\sigma\partial_\rho g_{\mu\nu}\right) \tag{2}$$
$$R_{\rho\nu\sigma\mu} = \frac{1}{2}\left(\partial_\sigma\partial_\nu g_{\rho\mu} - \partial_\sigma\partial_\rho g_{\nu\mu} - \partial_\mu\partial_\nu g_{\rho\sigma} + \partial_\mu\partial_\rho g_{\nu\sigma}\right) \tag{3}$$

Summing these three equations ($S = R_{\rho\sigma\mu\nu} + R_{\rho\mu\nu\sigma} + R_{\rho\nu\sigma\mu}$), we group terms to see the cancellations:

$$\begin{aligned}
2S = \quad & \partial_\mu\partial_\sigma g_{\rho\nu} - \partial_\sigma\partial_\mu g_{\rho\nu} && \text{(Term 1 of eq. 1 and Term 3 of eq. 2)} \\
& +\partial_\nu\partial_\mu g_{\rho\sigma} - \partial_\mu\partial_\nu g_{\rho\sigma} && \text{(Term 1 of eq. 2 and Term 3 of eq. 3)} \\
& +\partial_\sigma\partial_\nu g_{\rho\mu} - \partial_\nu\partial_\sigma g_{\rho\mu} && \text{(Term 1 of eq. 3 and Term 3 of eq. 1)} \\
& -\partial_\mu\partial_\rho g_{\sigma\nu} + \partial_\mu\partial_\rho g_{\nu\sigma} && \text{(Term 2 of eq. 1 and Term 4 of eq. 3)} \\
& -\partial_\nu\partial_\rho g_{\mu\sigma} + \partial_\nu\partial_\rho g_{\sigma\mu} && \text{(Term 2 of eq. 2 and Term 4 of eq. 1)} \\
& -\partial_\sigma\partial_\rho g_{\nu\mu} + \partial_\sigma\partial_\rho g_{\mu\nu} && \text{(Term 2 of eq. 3 and Term 4 of eq. 2)}
\end{aligned}$$

Because the metric is symmetric ($g_{\alpha\beta} = g_{\beta\alpha}$) and partial derivatives commute ($\partial_\alpha\partial_\beta = \partial_\beta\partial_\alpha$), every row above sums to zero:

$$R_{\rho\sigma\mu\nu} + R_{\rho\mu\nu\sigma} + R_{\rho\nu\sigma\mu} = 0$$

This is equivalent to the vanishing of the totally antisymmetric part: $R_{\rho[\sigma\mu\nu]} = 0$. The antisymmetrization of the last three indices of the Riemann tensor is defined as the sum over all permutations π of (σ, μ, ν), weighted by the sign of the permutation $\text{sgn}(\pi)$:

$$R_{\rho[\sigma\mu\nu]} = \frac{1}{3!}\sum_{\pi\in S_3} \text{sgn}(\pi) R_{\rho\pi(\sigma)\pi(\mu)\pi(\nu)}$$

Expanding this fully, we get:

$$\begin{aligned}
R_{\rho[\sigma\mu\nu]} = \frac{1}{6}\Big(& \underbrace{R_{\rho\sigma\mu\nu}}_{\text{even}} - \underbrace{R_{\rho\sigma\nu\mu}}_{\text{odd}} + \underbrace{R_{\rho\mu\nu\sigma}}_{\text{even}} \\
& - \underbrace{R_{\rho\mu\sigma\nu}}_{\text{odd}} + \underbrace{R_{\rho\nu\sigma\mu}}_{\text{even}} - \underbrace{R_{\rho\nu\mu\sigma}}_{\text{odd}}\Big) \qquad (1)
\end{aligned}$$

Now, we apply the antisymmetry of the last two indices from equation (3.130), which states $R_{\rho\alpha\beta\gamma} = -R_{\rho\alpha\gamma\beta}$. We use this to rewrite the "odd" terms in equation (1):

- $-R_{\rho\sigma\nu\mu} = +R_{\rho\sigma\mu\nu}$
- $-R_{\rho\mu\sigma\nu} = +R_{\rho\mu\nu\sigma}$
- $-R_{\rho\nu\mu\sigma} = +R_{\rho\nu\sigma\mu}$

Substituting these back into the expansion:

$$\begin{aligned}
R_{\rho[\sigma\mu\nu]} &= \frac{1}{6}\left(R_{\rho\sigma\mu\nu} + R_{\rho\sigma\mu\nu} + R_{\rho\mu\nu\sigma} + R_{\rho\mu\nu\sigma} + R_{\rho\nu\sigma\mu} + R_{\rho\nu\sigma\mu}\right) \\
&= \frac{1}{6}\left(2R_{\rho\sigma\mu\nu} + 2R_{\rho\mu\nu\sigma} + 2R_{\rho\nu\sigma\mu}\right) \\
&= \frac{1}{3}\left(R_{\rho\sigma\mu\nu} + R_{\rho\mu\nu\sigma} + R_{\rho\nu\sigma\mu}\right) \qquad (2)
\end{aligned}$$

From the First Bianchi Identity (equation 3.132), we know that the sum of the cyclic permutations is zero:

$$R_{\rho\sigma\mu\nu} + R_{\rho\mu\nu\sigma} + R_{\rho\nu\sigma\mu} = 0$$

Therefore:

$$R_{\rho[\sigma\mu\nu]} = \frac{1}{3}(0) = 0$$

This confirms that the vanishing of the cyclic sum is mathematically equivalent to the statement that the antisymmetric part of the last three indices is zero.

Intuition: These symmetries reduce the number of independent components from 256 to just 20 in 4D spacetime. These 20 numbers represent the "true" curvature degrees of freedom that cannot be removed by coordinate transformations (unlike the metric components themselves, which can be set to $\eta_{\mu\nu}$ to first order).

Step 1: Antisymmetry in Pairs

The Riemann tensor $R_{\rho\sigma\mu\nu}$ is antisymmetric in the first pair of indices ($\rho\sigma$) and the second pair ($\mu\nu$). For an n-dimensional space, the number of independent ways to choose a pair of distinct indices is equivalent to the number of strictly upper-triangular components in an $n \times n$ antisymmetric matrix. This is given by $N = \frac{1}{2}n(n-1)$. We can mathematically treat the first pair $[\rho\sigma]$ as a single composite index A, and the second pair $[\mu\nu]$ as another composite index B. Both A and B can take on N possible values.

Step 2: Symmetry Under Pair Exchange

The Riemann tensor is symmetric when you swap the first pair with the second pair entirely: $R_{AB} = R_{BA}$. This means we can view the Riemann tensor as a large $N \times N$ symmetric matrix. The number of independent components in any $N \times N$ symmetric matrix (including its diagonal) is $\frac{1}{2}N(N+1)$. Let us explicitly substitute $N = \frac{1}{2}n(n-1)$ into this symmetric matrix formula and expand it:

$$\begin{aligned}
\text{Components} &= \frac{1}{2}\left[\frac{1}{2}n(n-1)\right]\left[\frac{1}{2}n(n-1)+1\right] \\
&= \frac{1}{4}(n^2-n)\left[\frac{1}{2}(n^2-n)+\frac{2}{2}\right] \\
&= \frac{1}{4}(n^2-n)\left[\frac{1}{2}(n^2-n+2)\right] \\
&= \frac{1}{8}(n^2-n)(n^2-n+2)
\end{aligned} \tag{3.172}$$

We distribute the terms in equation (3.172) to get a single polynomial:

$$\begin{aligned}
\text{Components} &= \frac{1}{8}\left(n^2(n^2)+n^2(-n)+n^2(2)-n(n^2)-n(-n)-n(2)\right) \\
&= \frac{1}{8}(n^4-n^3+2n^2-n^3+n^2-2n) \\
&= \frac{1}{8}(n^4-2n^3+3n^2-2n)
\end{aligned} \tag{3.173}$$

Equation (3.173) perfectly confirms the first major polynomial provided in the text.

Step 3: The Algebraic Bianchi Identity

The tensor must also satisfy the algebraic Bianchi identity $R_{\rho[\sigma\mu\nu]} = 0$, which implies $R_{\rho\sigma\mu\nu} + R_{\rho\mu\nu\sigma} + R_{\rho\nu\sigma\mu} = 0$. This identity only provides a fundamentally new, independent constraint if all four indices (ρ, σ, μ, ν) are distinct. If any two indices are identical, the identity trivially reduces to $0 = 0$ due to the antisymmetry properties we already accounted for in Step 1. Therefore, the number of independent constraints is exactly the number of ways to choose 4 distinct indices from n dimensions. This is calculated using the binomial coefficient "n choose 4":

$$\text{Constraints} = \binom{n}{4} = \frac{n(n-1)(n-2)(n-3)}{4\cdot 3\cdot 2\cdot 1} = \frac{n^4-6n^3+11n^2-6n}{24} \tag{3.174}$$

Step 4: Subtracting Constraints for the Final Count

To find the true final number of independent components, we must subtract the number of Bianchi constraints (3.174) from our initial symmetric matrix count (3.173):

$$\text{Total} = \frac{1}{8}(n^4-2n^3+3n^2-2n) - \frac{1}{24}(n^4-6n^3+11n^2-6n) \tag{3.175}$$

To evaluate equation (3.175), we find a common denominator of 24 by multiplying the first term by 3/3:

$$\text{Total} = \frac{3(n^4-2n^3+3n^2-2n)}{24} - \frac{n^4-6n^3+11n^2-6n}{24} \tag{3.176}$$

We combine the numerators of equation (3.176) into a single fraction and distribute the negative sign:

$$\text{Total} = \frac{(3n^4 - 6n^3 + 9n^2 - 6n) - n^4 + 6n^3 - 11n^2 + 6n}{24} \tag{3.177}$$

Now, we group the like terms together in equation (3.177). Notice how the n^3 and n terms perfectly cancel out to zero:

$$\begin{aligned} \text{Total} &= \frac{(3n^4 - n^4) + (-6n^3 + 6n^3) + (9n^2 - 11n^2) + (-6n + 6n)}{24} \\ &= \frac{2n^4 + 0 - 2n^2 + 0}{24} \\ &= \frac{2n^4 - 2n^2}{24} \end{aligned} \tag{3.178}$$

Finally, we simplify equation (3.178) by factoring out $2n^2$ from the numerator:

$$\text{Total} = \frac{2n^2(n^2 - 1)}{24} = \frac{1}{12}n^2(n^2 - 1) \tag{3.179}$$

Equation (3.179) gives us the final, elegant formula. For a standard 4-dimensional spacetime ($n = 4$), plugging the value in yields $\frac{16(16-1)}{12} = \frac{16 \cdot 15}{12} = \frac{240}{12} = 20$ totally independent components.

The Second Bianchi Identity (Differential)

There is also a differential identity relating the derivatives of the Riemann tensor. By differentiating the expression for R in locally inertial coordinates (where $\partial g = 0$, so $\Gamma = 0$, but $\partial^2 g \neq 0$):

$$\nabla_\lambda R_{\rho\sigma\mu\nu} = \partial_\lambda R_{\rho\sigma\mu\nu} \quad (\text{at } p). \tag{3.180}$$

Summing cyclic permutations of the first three indices (the derivative index plus the first pair):

$$\nabla_\lambda R_{\rho\sigma\mu\nu} + \nabla_\rho R_{\sigma\lambda\mu\nu} + \nabla_\sigma R_{\lambda\rho\mu\nu} = 0. \tag{3.181}$$

Using antisymmetry, this is compactly written as:

$$\nabla_{[\lambda} R_{\rho\sigma]\mu\nu} = 0. \tag{3.182}$$

This is the Second Bianchi Identity. It is analogous to the Jacobi identity for commutators or the identity $d^2 = 0$ for exterior derivatives.

Ricci Tensor and Scalar

We can decompose the Riemann tensor into trace parts and trace-free parts. The Ricci Tensor $R_{\mu\nu}$ is defined by contracting the first and third indices:

$$R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}. \tag{3.183}$$

Properties:

- It is symmetric: $R_{\mu\nu} = R_{\nu\mu}$ (follows from the symmetries of Riemann).
- It encodes the change in volume of a shape as it moves along geodesics (as we will see later).

The Ricci Scalar (or Curvature Scalar) R is the trace of the Ricci tensor:

$$R = R^\mu{}_\mu = g^{\mu\nu} R_{\mu\nu}. \tag{3.184}$$

This is a single number at every point in spacetime describing the intrinsic curvature.

The Einstein Tensor

We contract the Second Bianchi Identity twice to find a conserved tensor. Start with:

$$\nabla_\lambda R_{\rho\sigma\mu\nu} + \nabla_\rho R_{\sigma\lambda\mu\nu} + \nabla_\sigma R_{\lambda\rho\mu\nu} = 0. \tag{3.185}$$

Contract with $g^{\nu\sigma}g^{\mu\lambda}$:

$$\nabla^\mu R_{\rho\mu} - \nabla_\rho R + \nabla^\nu R_{\rho\nu} = 0. \tag{3.186}$$

(Note: we used antisymmetry to rearrange indices). Combining terms gives $2\nabla^\mu R_{\rho\mu} - \nabla_\rho R = 0$, or:

$$\nabla^\mu \left(R_{\rho\mu} - \frac{1}{2} R g_{\rho\mu} \right) = 0. \tag{3.187}$$

The term in parentheses is so important it gets its own name, the Einstein Tensor $G_{\mu\nu}$:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}. \tag{3.188}$$

The identity guarantees that the Einstein tensor is divergence-free:

$$\nabla^\mu G_{\mu\nu} = 0. \tag{3.189}$$

Intuition: This geometric conservation law is crucial for General Relativity. Since energy-momentum is conserved ($\nabla^\mu T_{\mu\nu} = 0$), Einstein realized that $G_{\mu\nu}$ is the geometric object that must be proportional to $T_{\mu\nu}$.

The Weyl Tensor

The "trace-free" part of the Riemann tensor is called the Weyl Tensor $C_{\rho\sigma\mu\nu}$. It contains the information about tidal forces that preserve volume but distort shape (shearing). In n dimensions, it is defined by subtracting the traces (Ricci parts) from the Riemann tensor:

$$C_{\rho\sigma\mu\nu} = R_{\rho\sigma\mu\nu} - \frac{2}{n-2}(g_{\rho[\mu}R_{\nu]\sigma} - g_{\sigma[\mu}R_{\nu]\rho}) + \frac{2}{(n-1)(n-2)} g_{\rho[\mu}g_{\nu]\sigma}R. \tag{3.190}$$

Properties:

- It has the same symmetries as the Riemann tensor.
- It is totally traceless: $C^\lambda{}_{\sigma\lambda\nu} = 0$.
- It is conformally invariant: If we rescale the metric $g_{\mu\nu} \to \Omega^2(x) g_{\mu\nu}$, the Weyl tensor $C^\rho{}_{\sigma\mu\nu}$ (with one index up) is unchanged.
- It vanishes identically in $n = 3$ dimensions or fewer. In 3D, the Ricci tensor determines the full curvature. In 4D+, you can have curvature (gravity) even in vacuum ($R_{\mu\nu} = 0$) due to the Weyl tensor (e.g., gravitational waves).

We begin by decomposing the Riemann tensor $R_{\rho\sigma\mu\nu}$ into three parts:

1. The Weyl tensor $C_{\rho\sigma\mu\nu}$ (which is trace-free).
2. A part involving the Ricci tensor $R_{\mu\nu}$.
3. A part involving the Ricci scalar R.

We write this decomposition using unknown coefficients A and B. To maintain the symmetries of the Riemann tensor, the terms must be constructed as follows:

$$C_{\rho\sigma\mu\nu} = R_{\rho\sigma\mu\nu} + A\,\underbrace{(g_{\rho\mu}R_{\sigma\nu} - g_{\rho\nu}R_{\sigma\mu} - g_{\sigma\mu}R_{\rho\nu} + g_{\sigma\nu}R_{\rho\mu})}_{\text{Ricci Terms}} + B\,\underbrace{R(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu})}_{\text{Scalar Terms}} \tag{3.191}$$

The defining characteristic of the Weyl tensor is that it is **trace-free** on any pair of indices. We apply the contraction $g^{\rho\mu}$ to equation (3.304) and require the result to be zero:

$$g^{\rho\mu}C_{\rho\sigma\mu\nu} = 0 \tag{3.192}$$

Now, we compute the trace of each term on the right-hand side of (3.304) individually.

$$g^{\rho\mu}R_{\rho\sigma\mu\nu} = R_{\sigma\nu} \quad \text{(Definition of Ricci tensor)} \tag{3.193}$$

We contract $g^{\rho\mu}$ with the term $(g_{\rho\mu}R_{\sigma\nu} - g_{\rho\nu}R_{\sigma\mu} - g_{\sigma\mu}R_{\rho\nu} + g_{\sigma\nu}R_{\rho\mu})$:

$$\begin{aligned}
&\text{Term 1: } g^{\rho\mu}g_{\rho\mu}R_{\sigma\nu} = nR_{\sigma\nu} \quad (\text{since } g^{\rho\mu}g_{\rho\mu} = \delta^{\mu}_{\mu} = n)\\
&\text{Term 2: } -g^{\rho\mu}g_{\rho\nu}R_{\sigma\mu} = -\delta^{\mu}_{\nu}R_{\sigma\mu} = -R_{\sigma\nu}\\
&\text{Term 3: } -g^{\rho\mu}g_{\sigma\mu}R_{\rho\nu} = -\delta^{\rho}_{\sigma}R_{\rho\nu} = -R_{\sigma\nu}\\
&\text{Term 4: } g^{\rho\mu}g_{\sigma\nu}R_{\rho\mu} = g_{\sigma\nu}R \quad (\text{since } g^{\rho\mu}R_{\rho\mu} = R)
\end{aligned}$$

Summing these, the trace of the A term is:

$$A(nR_{\sigma\nu} - 2R_{\sigma\nu} + g_{\sigma\nu}R) = A[(n-2)R_{\sigma\nu} + g_{\sigma\nu}R] \tag{3.194}$$

We contract $g^{\rho\mu}$ with $R(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu})$:

$$\begin{aligned}
&\text{Term 1: } R(g^{\rho\mu}g_{\rho\mu})g_{\sigma\nu} = R(n)g_{\sigma\nu}\\
&\text{Term 2: } -R(g^{\rho\mu}g_{\rho\nu})g_{\sigma\mu} = -R(\delta^{\mu}_{\nu})g_{\sigma\mu} = -Rg_{\sigma\nu}
\end{aligned}$$

Summing these, the trace of the B term is:

$$BR(ng_{\sigma\nu} - g_{\sigma\nu}) = B(n-1)g_{\sigma\nu}R \tag{3.195}$$

We substitute the results from steps 2a, 2b, and 2c back into the trace equation (0 = Sum of traces):

$$0 = R_{\sigma\nu} + A[(n-2)R_{\sigma\nu} + g_{\sigma\nu}R] + B(n-1)g_{\sigma\nu}R \tag{3.196}$$

We group the terms by $R_{\sigma\nu}$ and $g_{\sigma\nu}R$:

$$0 = \underbrace{[1 + A(n-2)]}_{\text{Must be 0}}R_{\sigma\nu} + \underbrace{[A + B(n-1)]}_{\text{Must be 0}}g_{\sigma\nu}R \tag{3.197}$$

For this equation to hold for any geometry, the coefficients of $R_{\sigma\nu}$ and $g_{\sigma\nu}R$ must vanish independently.

Solving for A:

$$1 + A(n-2) = 0$$
$$A = -\frac{1}{n-2}$$

Solving for B:

$$A + B(n-1) = 0$$
$$-\frac{1}{n-2} + B(n-1) = 0$$
$$B = \frac{1}{(n-1)(n-2)}$$

We have found the coefficients:

$$C_{\rho\sigma\mu\nu} = R_{\rho\sigma\mu\nu} - \frac{1}{n-2}(\text{Ricci Terms}) + \frac{1}{(n-1)(n-2)}(\text{Scalar Terms})$$

However, your image uses the notation with a factor of **2**. This is due to the antisymmetrization bracket notation $T_{[\mu\nu]}$. The definition is:

$$T_{[\mu\nu]} = \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu}) \tag{3.198}$$

Let's look at the Ricci term in the image:

$$-\frac{2}{n-2}(g_{\rho[\mu}R_{\nu]\sigma} - g_{\sigma[\mu}R_{\nu]\rho}) \tag{3.199}$$

Expanding the first part $g_{\rho[\mu}R_{\nu]\sigma}$:

$$g_{\rho[\mu}R_{\nu]\sigma} = \frac{1}{2}(g_{\rho\mu}R_{\nu\sigma} - g_{\rho\nu}R_{\mu\sigma}) \tag{3.200}$$

Multiplying by $\frac{2}{n-2}$ cancels the $\frac{1}{2}$:

$$\frac{2}{n-2} \cdot \frac{1}{2}(g_{\rho\mu}R_{\nu\sigma} - g_{\rho\nu}R_{\mu\sigma}) = \frac{1}{n-2}(g_{\rho\mu}R_{\nu\sigma} - g_{\rho\nu}R_{\mu\sigma}) \tag{3.201}$$

This matches our derived coefficient $A = -\frac{1}{n-2}$ exactly. The same logic applies to the scalar term B.

3.8 Mathematical Interlude: Diffeomorphisms and Lie Derivatives

Active vs. Passive Transformations

In General Relativity, we often speak of "coordinate transformations." Traditionally, we view these as *passive*: the manifold M stays frozen, and we simply change the labels (charts) we use to describe points.

However, there is an equivalent *active* perspective called a **diffeomorphism**. A diffeomorphism is a smooth, invertible map $\phi : M \to M$ that moves points around on the manifold.

- *Passive:* Change the map, keep points fixed.
- *Active:* Keep the map fixed, move the points.

Diffeomorphisms allow us to compare tensors at different points by "pulling" a tensor from point $\phi(p)$ back to point p. This ability allows us to define a derivative operator that does not depend on a connection: the Lie Derivative.

The Lie Derivative

We want to measure how a tensor field changes as we move along the flow of a vector field V^μ. The integral curves of V^μ define a one-parameter family of diffeomorphisms ϕ_t. To take a derivative, we compare the tensor T at point p with the tensor at a nearby point $\phi_t(p)$.

Because tensors at different points live in different tangent spaces, we cannot subtract them directly. Instead, we use the diffeomorphism to **pull back** the tensor from $\phi_t(p)$ to p. The **Lie Derivative** is defined as:

$$\mathcal{L}_V T = \lim_{t\to 0} \frac{\phi_t^*[T(\phi_t(p))] - T(p)}{t}. \tag{3.202}$$

This operator satisfies the Leibniz rule and is linear. Crucially, it requires no connection (Γ) or metric (g), only the vector field V that defines the flow.

Calculating the Lie Derivative

To derive explicit formulas, it is convenient to choose a coordinate system adapted to the vector field V. We choose coordinates $\{x^1, x^2, \ldots, x^n\}$ such that V points entirely along the x^1 direction:

$$V^\mu = (1, 0, \ldots, 0) \implies V = \frac{\partial}{\partial x^1}. \tag{3.203}$$

In this coordinate system, the flow ϕ_t is simply a shift in the x^1 coordinate: $x^1 \to x^1 + t$. The Lie derivative reduces to a partial derivative:

$$\mathcal{L}_V T^{\mu\cdots}_{\nu\cdots} = \frac{\partial}{\partial x^1} T^{\mu\cdots}_{\nu\cdots}. \tag{3.204}$$

However, this expression is coordinate-dependent. We want a covariant formula.

1. Lie Derivative of a Vector Let U^μ be another vector field. In our adapted coordinates, $\mathcal{L}_V U^\mu = \partial_1 U^\mu$. Now consider the commutator (Lie Bracket) of the two vectors:

$$[V, U]^\mu = V^\nu \partial_\nu U^\mu - U^\nu \partial_\nu V^\mu. \tag{3.205}$$

In our adapted coordinates where $V^\nu = \delta^\nu_1$ (so $\partial_\nu V^\mu = 0$):

$$[V, U]^\mu = 1 \cdot \partial_1 U^\mu - 0 = \frac{\partial U^\mu}{\partial x^1}. \tag{3.206}$$

Since $\mathcal{L}_V U$ and $[V, U]$ have the same components in this coordinate system, and both are tensors, they must be equal in *all* coordinate systems.

$$\mathcal{L}_V U^\mu = [V, U]^\mu = V^\nu \partial_\nu U^\mu - U^\nu \partial_\nu V^\mu. \tag{3.207}$$

2. Lie Derivative of a One-Form Let ω_μ be a one-form. To find $\mathcal{L}_V \omega_\mu$, we look at the scalar contraction $S = \omega_\mu U^\mu$. Since S is a scalar function, its Lie derivative is just the directional derivative:

$$\mathcal{L}_V(\omega_\mu U^\mu) = V^\nu \partial_\nu(\omega_\mu U^\mu). \tag{3.208}$$

We can also apply the Leibniz rule to the Lie derivative:

$$\mathcal{L}_V(\omega_\mu U^\mu) = (\mathcal{L}_V \omega)_\mu U^\mu + \omega_\mu (\mathcal{L}_V U)^\mu. \tag{3.209}$$

Equating the two expressions:

$$V^\nu(\partial_\nu \omega_\mu) U^\mu + V^\nu \omega_\mu (\partial_\nu U^\mu) = (\mathcal{L}_V \omega)_\mu U^\mu + \omega_\mu (V^\nu \partial_\nu U^\mu - U^\nu \partial_\nu V^\mu). \tag{3.210}$$

The terms involving derivatives of U cancel out ($V^\nu \omega_\mu \partial_\nu U^\mu$ on both sides). We are left with terms proportional to U^μ:

$$V^\nu(\partial_\nu \omega_\mu) U^\mu = (\mathcal{L}_V \omega)_\mu U^\mu - \omega_\nu (\partial_\mu V^\nu) U^\mu. \tag{3.211}$$

(Note: we relabeled indices in the last term to factor out U^μ). Since this holds for arbitrary U^μ, we have:

$$\mathcal{L}_V \omega_\mu = V^\nu \partial_\nu \omega_\mu + \omega_\nu \partial_\mu V^\nu. \tag{3.212}$$

3. General Tensor Formula The pattern generalizes. For an arbitrary tensor, the Lie derivative consists of a partial derivative term plus a correction term for each index.

- Upper indices get a minus sign and derive the vector V: $-\partial V$.
- Lower indices get a plus sign and derive the vector V: $+\partial V$.

$$\mathcal{L}_V T^\mu{}_\nu = V^\lambda \partial_\lambda T^\mu{}_\nu - (\partial_\lambda V^\mu) T^\lambda{}_\nu + (\partial_\nu V^\lambda) T^\mu{}_\lambda. \tag{3.213}$$

Intuition: If we replace partial derivatives ∂ with covariant derivatives ∇ (assuming a torsion-free connection), all the connection coefficients Γ cancel out. Thus, we can write the formula manifestly covariantly:

$$\mathcal{L}_V T^\mu{}_\nu = V^\lambda \nabla_\lambda T^\mu{}_\nu - (\nabla_\lambda V^\mu) T^\lambda{}_\nu + (\nabla_\nu V^\lambda) T^\mu{}_\lambda. \tag{3.214}$$

Symmetries and the Metric

We can now rigorously define a symmetry. A diffeomorphism ϕ is a symmetry of a tensor T if pulling the tensor back leaves it unchanged: $\phi^* T = T$. For infinitesimal symmetries generated by a vector field K^μ, this corresponds to a vanishing Lie derivative:

$$\mathcal{L}_K T = 0. \tag{3.215}$$

The most important symmetries are Isometries, which are symmetries of the metric tensor $g_{\mu\nu}$.

$$\mathcal{L}_K g_{\mu\nu} = 0. \tag{3.216}$$

Let us expand this using the general formula for a $(0, 2)$ tensor:

$$\mathcal{L}_K g_{\mu\nu} = K^\sigma \nabla_\sigma g_{\mu\nu} + g_{\lambda\nu} \nabla_\mu K^\lambda + g_{\mu\lambda} \nabla_\nu K^\lambda. \tag{3.217}$$

If we use the metric-compatible connection (Christoffel), then $\nabla_\sigma g_{\mu\nu} = 0$. The first term vanishes. We lower the indices on the K terms:

$$\mathcal{L}_K g_{\mu\nu} = \nabla_\mu K_\nu + \nabla_\nu K_\mu. \tag{3.218}$$

Thus, the condition for K^μ to generate a symmetry is:

$$\nabla_\mu K_\nu + \nabla_\nu K_\mu = 0 \quad \Longrightarrow \quad \nabla_{(\mu} K_{\nu)} = 0. \tag{3.219}$$

This rigorously derives Killing's Equation, which we used in Section 3.8.

3.9 Symmetries and Killing Vectors

Isometries and Conservation Laws

Real-world metrics are often approximated by solutions with high degrees of symmetry (e.g., spherical symmetry for stars, isotropy for cosmology). A symmetry of the metric is formally called an **isometry**. Physically, if a metric $g_{\mu\nu}$ is independent of a specific coordinate x^{σ_*} (meaning $\partial_{\sigma_*} g_{\mu\nu} = 0$), the geometry looks the same as we translate along that coordinate.

This symmetry leads directly to a conservation law for particle motion. Recall the geodesic equation derived earlier for a particle with four-momentum p_μ:

$$m \frac{dp_\mu}{d\tau} = \frac{1}{2} (\partial_\mu g_{\nu\lambda}) p^\nu p^\lambda. \tag{3.220}$$

If the metric is independent of x^{σ_*}, then $\partial_{\sigma_*} g_{\nu\lambda} = 0$. Consequently, the RHS vanishes for the component $\mu = \sigma_*$:

$$\frac{dp_{\sigma_*}}{d\tau} = 0. \tag{3.221}$$

Thus, the momentum component p_{σ_*} is a constant of motion along the geodesic.

Intuition: This is the General Relativistic version of Noether's theorem. Independence of spatial position implies conservation of linear momentum; independence of time implies conservation of energy.

Killing Vectors

We want to express this symmetry in a coordinate-independent language. We define a vector field K^μ that "generates" the symmetry. If the metric is independent of x^{σ_*}, the generator is the coordinate basis vector $K = \partial_{\sigma_*}$, which has components $K^\mu = \delta^\mu_{\sigma_*}$.

In terms of this vector, the conserved quantity p_{σ_*} is the scalar product:

$$p_{\sigma_*} = K^\nu p_\nu. \tag{3.222}$$

We now ask: what is the general condition on a vector field K^μ such that $K^\nu p_\nu$ is conserved along *any* geodesic? We evaluate the derivative along the path:

$$\frac{d}{d\tau}(K_\nu p^\nu) = p^\mu \nabla_\mu (K_\nu p^\nu) = 0. \tag{3.223}$$

Expanding the covariant derivative via the Leibniz rule:

$$p^\mu \nabla_\mu (K_\nu p^\nu) = p^\mu (\nabla_\mu K_\nu) p^\nu + K_\nu (p^\mu \nabla_\mu p^\nu). \tag{3.224}$$

The second term vanishes because p^μ follows a geodesic ($p^\mu \nabla_\mu p^\nu = 0$). The first term involves the contraction of $\nabla_\mu K_\nu$ with $p^\mu p^\nu$. Since $p^\mu p^\nu$ is symmetric in μ and ν, it will only contract with the symmetric part of $\nabla_\mu K_\nu$.

$$p^\mu p^\nu \nabla_\mu K_\nu = p^\mu p^\nu \nabla_{(\mu} K_{\nu)}. \tag{3.225}$$

For this to vanish for *any* momentum p^μ, the symmetric part of the derivative must vanish. This yields **Killing's Equation**:

$$\nabla_{(\mu} K_{\nu)} = \frac{1}{2}(\nabla_\mu K_\nu + \nabla_\nu K_\mu) = 0. \tag{3.226}$$

Any vector field K^μ satisfying this equation is called a **Killing vector field**.

Intuition: The condition $\nabla_{(\mu} K_{\nu)} = 0$ geometrically means that if you flow along the vector field K, distances between neighboring points do not change. The metric is "dragged" into itself.

Properties of Killing Vectors

Killing vectors satisfy several useful identities related to curvature: We are given a coordinate system where the metric tensor $g_{\mu\nu}$ is independent of a specific coordinate x^σ. Mathematically, this condition is:

$$\partial_\sigma g_{\mu\nu} = 0 \tag{3.227}$$

We define the vector field K as the coordinate basis vector corresponding to x^σ:

$$K = \partial_\sigma \tag{3.228}$$

In component notation, the components of this vector K^ρ are given by the Kronecker delta (it has a 1 in the σ-position and 0 elsewhere):

$$K^\rho = \delta^\rho_\sigma \tag{3.229}$$

We aim to prove that K satisfies Killing's Equation:

$$\nabla_\mu K_\nu + \nabla_\nu K_\mu = 0 \tag{3.230}$$

Killing's equation is written in terms of the covariant components K_ν. We lower the index using the metric:

$$K_\nu = g_{\nu\rho} K^\rho = g_{\nu\rho} \delta^\rho_\sigma = g_{\nu\sigma} \tag{3.231}$$

So, the covariant components of our vector are simply the components of the metric column corresponding to σ.

We compute the term $\nabla_\mu K_\nu$ using the definition of the covariant derivative:

$$\nabla_\mu K_\nu = \partial_\mu K_\nu - \Gamma^\lambda_{\mu\nu} K_\lambda \tag{3.232}$$

Substitute $K_\nu = g_{\nu\sigma}$ and $K_\lambda = g_{\lambda\sigma}$:

$$\nabla_\mu K_\nu = \partial_\mu g_{\nu\sigma} - \Gamma^\lambda_{\mu\nu} g_{\lambda\sigma} \tag{3.233}$$

The term $\Gamma^{\lambda}_{\mu\nu} g_{\lambda\sigma}$ is actually the Christoffel symbol of the first kind, denoted $\Gamma_{\sigma\mu\nu}$. We expand it using the metric definition:

$$\Gamma^{\lambda}_{\mu\nu} g_{\lambda\sigma} = \frac{1}{2} g^{\lambda\rho} (\partial_\mu g_{\nu\rho} + \partial_\nu g_{\mu\rho} - \partial_\rho g_{\mu\nu}) g_{\lambda\sigma} \tag{3.234}$$

Since $g^{\lambda\rho} g_{\lambda\sigma} = \delta^{\rho}_{\sigma}$, the expression simplifies significantly by setting $\rho = \sigma$:

$$\Gamma^{\lambda}_{\mu\nu} g_{\lambda\sigma} = \frac{1}{2} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}) \tag{3.235}$$

Now substitute this back into our expression for $\nabla_\mu K_\nu$:

$$\begin{aligned}
\nabla_\mu K_\nu &= \partial_\mu g_{\nu\sigma} - \frac{1}{2} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}) \\
&= \partial_\mu g_{\nu\sigma} - \frac{1}{2} \partial_\mu g_{\nu\sigma} - \frac{1}{2} \partial_\nu g_{\mu\sigma} + \frac{1}{2} \partial_\sigma g_{\mu\nu} \\
&= \frac{1}{2} \partial_\mu g_{\nu\sigma} - \frac{1}{2} \partial_\nu g_{\mu\sigma} + \frac{1}{2} \partial_\sigma g_{\mu\nu} \qquad \text{(A)}
\end{aligned}$$

We now calculate the left-hand side of Killing's equation by writing equation (A) and adding the same equation with μ and ν swapped:

$$\begin{aligned}
\nabla_\mu K_\nu &= \frac{1}{2} \partial_\mu g_{\nu\sigma} - \frac{1}{2} \partial_\nu g_{\mu\sigma} + \frac{1}{2} \partial_\sigma g_{\mu\nu} \\
\nabla_\nu K_\mu &= \frac{1}{2} \partial_\nu g_{\mu\sigma} - \frac{1}{2} \partial_\mu g_{\nu\sigma} + \frac{1}{2} \partial_\sigma g_{\nu\mu}
\end{aligned}$$

Adding these two lines together:

$$\begin{aligned}
\nabla_\mu K_\nu + \nabla_\nu K_\mu = \quad & \left(\frac{1}{2} \partial_\mu g_{\nu\sigma} - \frac{1}{2} \partial_\mu g_{\nu\sigma} \right) \quad \text{(Cancels to 0)} \\
& + \left(-\frac{1}{2} \partial_\nu g_{\mu\sigma} + \frac{1}{2} \partial_\nu g_{\mu\sigma} \right) \quad \text{(Cancels to 0)} \\
& + \left(\frac{1}{2} \partial_\sigma g_{\mu\nu} + \frac{1}{2} \partial_\sigma g_{\nu\mu} \right)
\end{aligned}$$

Since the metric is symmetric ($g_{\mu\nu} = g_{\nu\mu}$), the last term becomes:

$$\nabla_\mu K_\nu + \nabla_\nu K_\mu = \partial_\sigma g_{\mu\nu} \tag{3.236}$$

We have derived that for the coordinate basis vector $K = \partial_\sigma$:

$$\nabla_{(\mu} K_{\nu)} = \frac{1}{2} \partial_\sigma g_{\mu\nu} \tag{3.237}$$

Therefore, Killing's equation $\nabla_{(\mu} K_{\nu)} = 0$ is satisfied **if and only if**:

$$\partial_\sigma g_{\mu\nu} = 0 \tag{3.238}$$

This confirms the statement in the text: if the metric is independent of the coordinate x^σ, then ∂_σ is a Killing vector.

1. **Second Derivative:** The second derivative of a Killing vector is determined by the Riemann tensor: we start with the defining property of a Killing vector field K_μ, which satisfies Killing's equation:

$$\nabla_\mu K_\nu + \nabla_\nu K_\mu = 0 \implies \nabla_\mu K_\nu = -\nabla_\nu K_\mu$$

We will also need the definition of the Riemann curvature tensor in terms of the commutator of covariant derivatives acting on a dual vector field (this is known as the Ricci identity). Using the convention from Carroll's text, this is:

$$[\nabla_\mu, \nabla_\nu] K_\rho = \nabla_\mu \nabla_\nu K_\rho - \nabla_\nu \nabla_\mu K_\rho = -R^{\sigma}{}_{\rho\mu\nu} K_\sigma$$

We begin by examining the second covariant derivative of the Killing vector, $\nabla_\mu \nabla_\nu K_\rho$, and we will systematically perform cyclic permutations of the indices using Killing's equation and the Ricci identity. First, apply Killing's equation to the inner derivative to swap ν and ρ:

$$\nabla_\mu \nabla_\nu K_\rho = -\nabla_\mu \nabla_\rho K_\nu$$

Next, apply the Ricci identity to the right-hand side to commute the covariant derivatives ∇_μ and ∇_ρ:

$$\nabla_\mu \nabla_\rho K_\nu = \nabla_\rho \nabla_\mu K_\nu - R^\sigma{}_{\nu\mu\rho} K_\sigma$$

Substituting this back into our expression gives:

$$\nabla_\mu \nabla_\nu K_\rho = -\left(\nabla_\rho \nabla_\mu K_\nu - R^\sigma{}_{\nu\mu\rho} K_\sigma\right) = -\nabla_\rho \nabla_\mu K_\nu + R^\sigma{}_{\nu\mu\rho} K_\sigma$$

Now, apply Killing's equation again to the inner derivative on the right-hand side ($\nabla_\mu K_\nu = -\nabla_\nu K_\mu$):

$$\nabla_\mu \nabla_\nu K_\rho = -\nabla_\rho(-\nabla_\nu K_\mu) + R^\sigma{}_{\nu\mu\rho} K_\sigma = \nabla_\rho \nabla_\nu K_\mu + R^\sigma{}_{\nu\mu\rho} K_\sigma$$

Apply the Ricci identity one more time to commute ∇_ρ and ∇_ν:

$$\nabla_\rho \nabla_\nu K_\mu = \nabla_\nu \nabla_\rho K_\mu - R^\sigma{}_{\mu\rho\nu} K_\sigma$$

Substitute this into the main equation:

$$\nabla_\mu \nabla_\nu K_\rho = \nabla_\nu \nabla_\rho K_\mu - R^\sigma{}_{\mu\rho\nu} K_\sigma + R^\sigma{}_{\nu\mu\rho} K_\sigma$$

Apply Killing's equation to the first term on the right-hand side ($\nabla_\rho K_\mu = -\nabla_\mu K_\rho$):

$$\nabla_\mu \nabla_\nu K_\rho = -\nabla_\nu \nabla_\mu K_\rho - R^\sigma{}_{\mu\rho\nu} K_\sigma + R^\sigma{}_{\nu\mu\rho} K_\sigma$$

Rearrange this equation to bring the covariant derivative terms to the left side:

$$\nabla_\mu \nabla_\nu K_\rho + \nabla_\nu \nabla_\mu K_\rho = -R^\sigma{}_{\mu\rho\nu} K_\sigma + R^\sigma{}_{\nu\mu\rho} K_\sigma$$

Now, recall the initial Ricci identity we stated at the beginning:

$$\nabla_\mu \nabla_\nu K_\rho - \nabla_\nu \nabla_\mu K_\rho = -R^\sigma{}_{\rho\mu\nu} K_\sigma$$

Add these last two equations together. The $\nabla_\nu \nabla_\mu K_\rho$ terms cancel out perfectly:

$$2\nabla_\mu \nabla_\nu K_\rho = -R^\sigma{}_{\mu\rho\nu} K_\sigma + R^\sigma{}_{\nu\mu\rho} K_\sigma - R^\sigma{}_{\rho\mu\nu} K_\sigma$$

To simplify the right-hand side, it is helpful to lower the upper index on the Riemann tensors using the metric ($R^\sigma{}_{\alpha\beta\gamma} K_\sigma = R_{\lambda\alpha\beta\gamma} g^{\lambda\sigma} K_\sigma = R_{\lambda\alpha\beta\gamma} K^\lambda$). Let's write the right side entirely with lowered Riemann indices contracted with K^λ:

$$2\nabla_\mu \nabla_\nu K_\rho = \left(-R_{\lambda\mu\rho\nu} + R_{\lambda\nu\mu\rho} - R_{\lambda\rho\mu\nu}\right) K^\lambda$$

We can now use the algebraic Bianchi identity, which states that the cyclic sum of the last three indices of the Riemann tensor vanishes:

$$R_{\lambda\mu\rho\nu} + R_{\lambda\rho\nu\mu} + R_{\lambda\nu\mu\rho} = 0 \implies R_{\lambda\nu\mu\rho} = -R_{\lambda\mu\rho\nu} - R_{\lambda\rho\nu\mu}$$

Substitute this expression for $R_{\lambda\nu\mu\rho}$ into our equation:

$$2\nabla_\mu \nabla_\nu K_\rho = \left(-R_{\lambda\mu\rho\nu} + \left(-R_{\lambda\mu\rho\nu} - R_{\lambda\rho\nu\mu}\right) - R_{\lambda\rho\mu\nu}\right) K^\lambda$$

$$2\nabla_\mu \nabla_\nu K_\rho = \left(-2R_{\lambda\mu\rho\nu} - R_{\lambda\rho\nu\mu} - R_{\lambda\rho\mu\nu}\right) K^\lambda$$

Next, apply the antisymmetry of the Riemann tensor on its last two indices ($R_{\lambda\rho\nu\mu} = -R_{\lambda\rho\mu\nu}$). This means the last two terms inside the parenthesis cancel out:

$$-R_{\lambda\rho\nu\mu} - R_{\lambda\rho\mu\nu} = R_{\lambda\rho\mu\nu} - R_{\lambda\rho\mu\nu} = 0$$

This leaves us with a much simpler expression:

$$2\nabla_\mu \nabla_\nu K_\rho = -2R_{\lambda\mu\rho\nu} K^\lambda \implies \nabla_\mu \nabla_\nu K_\rho = -R_{\lambda\mu\rho\nu} K^\lambda$$

To arrive at the exact form of equation (3.176), we need to raise the index ρ. Multiply both sides by the inverse metric $g^{\rho\alpha}$:

$$g^{\rho\alpha} \nabla_\mu \nabla_\nu K_\rho = -g^{\rho\alpha} R_{\lambda\mu\rho\nu} K^\lambda$$
$$\nabla_\mu \nabla_\nu K^\alpha = -R_{\lambda\mu}{}^{\alpha}{}_{\nu} K^\lambda$$

Finally, we use the symmetries of the Riemann tensor to match the textbook's index arrangement. First, swap the first pair of indices with the second pair, which leaves the tensor unchanged:

$$-R_{\lambda\mu}{}^{\alpha}{}_{\nu} = -g^{\rho\alpha} R_{\lambda\mu\rho\nu} = -g^{\rho\alpha} R_{\rho\nu\lambda\mu} = -R^{\alpha}{}_{\nu\lambda\mu}$$

Next, swap the last two indices, which introduces a minus sign:

$$-R^{\alpha}{}_{\nu\lambda\mu} = R^{\alpha}{}_{\nu\mu\lambda}$$

So we have:

$$\nabla_\mu \nabla_\nu K^\alpha = R^{\alpha}{}_{\nu\mu\lambda} K^\lambda$$

To make it match the equation exactly, we perform a simple change of dummy variables and free index labels. Let $\alpha \to \rho$, $\nu \to \sigma$, and the dummy summation index $\lambda \to \nu$:

$$\nabla_\mu \nabla_\sigma K^\rho = R^{\rho}{}_{\sigma\mu\nu} K^\nu$$

2. **Ricci Tensor:** Contracting the above equation gives: We begin with the relation for the second derivative of a Killing vector K^ρ, which is given in the text as equation (1.233):

$$\nabla_\mu \nabla_\sigma K^\rho = R^{\rho}{}_{\sigma\mu\nu} K^\nu \tag{3.239}$$

3. **Ricci Scalar:** The geometry does not change along the symmetry direction, so the directional derivative of the curvature scalar vanishes:

$$K^\lambda \nabla_\lambda R = 0. \tag{3.240}$$

To derive equation (3.253), we begin with the contracted Bianchi identity, which relates the divergence of the Ricci tensor to the gradient of the Ricci scalar R:

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2} \nabla_\nu R \tag{3.241}$$

Contracting both sides of equation (3.241) with the Killing vector field K^ν gives:

$$K^\nu \nabla^\mu R_{\mu\nu} = \frac{1}{2} K^\nu \nabla_\nu R \tag{3.242}$$

Next, we apply the product rule to the divergence of $R_{\mu\nu} K^\nu$:

$$\nabla^\mu (R_{\mu\nu} K^\nu) = (\nabla^\mu R_{\mu\nu}) K^\nu + R_{\mu\nu} \nabla^\mu K^\nu \tag{3.243}$$

Since K^ν is a Killing vector, it satisfies Killing's equation $\nabla^\mu K^\nu = -\nabla^\nu K^\mu$, making it antisymmetric. The Ricci tensor $R_{\mu\nu}$ is symmetric. The contraction of a symmetric and an antisymmetric tensor vanishes, so $R_{\mu\nu} \nabla^\mu K^\nu = 0$. Equation (3.243) simplifies to:

$$\nabla^\mu (R_{\mu\nu} K^\nu) = (\nabla^\mu R_{\mu\nu}) K^\nu \tag{3.244}$$

Substituting equation (3.244) into the left side of equation (3.242) yields:

$$\frac{1}{2}K^\nu \nabla_\nu R = \nabla^\mu (R_{\mu\nu} K^\nu) \tag{3.245}$$

From equation (3.177) in the text, we have $\nabla_\mu \nabla_\sigma K^\mu = R_{\sigma\nu} K^\nu$. Relabeling indices gives $R_{\mu\nu} K^\nu = \nabla^\alpha \nabla_\mu K_\alpha$. Substituting this into equation (3.245):

$$\frac{1}{2}K^\nu \nabla_\nu R = \nabla^\mu (\nabla^\alpha \nabla_\mu K_\alpha) \tag{3.246}$$

Let $A_{\mu\alpha} = \nabla_\mu K_\alpha$. By Killing's equation, $A_{\mu\alpha}$ is an antisymmetric tensor ($A_{\mu\alpha} = -A_{\alpha\mu}$). We can split the right side of equation (3.246) into two equal halves and swap the dummy indices μ and α on the second half. Because of the antisymmetry, swapping the indices introduces a minus sign:

$$\nabla^\mu \nabla^\alpha A_{\mu\alpha} = \frac{1}{2}\nabla^\mu \nabla^\alpha A_{\mu\alpha} - \frac{1}{2}\nabla^\alpha \nabla^\mu A_{\mu\alpha} = \frac{1}{2}[\nabla^\mu, \nabla^\alpha] A_{\mu\alpha} \tag{3.247}$$

To evaluate this commutator rigorously, we use the Ricci identity for a rank-2 covariant tensor $A_{\mu\alpha}$, which introduces the Riemann curvature tensor:

$$[\nabla_\rho, \nabla_\sigma] A_{\mu\alpha} = -R^\lambda{}_{\mu\rho\sigma} A_{\lambda\alpha} - R^\lambda{}_{\alpha\rho\sigma} A_{\mu\lambda} \tag{3.248}$$

To match our expression, we raise the derivative indices to μ and α by contracting with inverse metric tensors $g^{\mu\rho} g^{\alpha\sigma}$:

$$[\nabla^\mu, \nabla^\alpha] A_{\mu\alpha} = -g^{\alpha\sigma}(g^{\mu\rho} R^\lambda{}_{\mu\rho\sigma}) A_{\lambda\alpha} - g^{\mu\rho}(g^{\alpha\sigma} R^\lambda{}_{\alpha\rho\sigma}) A_{\mu\lambda} \tag{3.249}$$

We simplify the Riemann tensor contractions using the standard definition of the Ricci tensor, $R_{\beta\sigma} = R^\lambda{}_{\beta\lambda\sigma}$. For the first term, we use the antisymmetry of the Riemann tensor in its last two indices ($R^\lambda{}_{\mu\rho\sigma} = -R^\lambda{}_{\mu\sigma\rho}$) to obtain $g^{\mu\rho} R^\lambda{}_{\mu\rho\sigma} = -R^\lambda{}_\sigma$. For the second term, the contraction directly matches the Ricci trace definition: $g^{\alpha\sigma} R^\lambda{}_{\alpha\rho\sigma} = R^\lambda{}_\rho$. Substituting these contracted forms yields:

$$[\nabla^\mu, \nabla^\alpha] A_{\mu\alpha} = -g^{\alpha\sigma}(-R^\lambda{}_\sigma) A_{\lambda\alpha} - g^{\mu\rho}(R^\lambda{}_\rho) A_{\mu\lambda} = R^{\lambda\alpha} A_{\lambda\alpha} - R^{\lambda\mu} A_{\mu\lambda} \tag{3.250}$$

Now we apply the antisymmetry of A ($A_{\mu\lambda} = -A_{\lambda\mu}$) to the second term, which flips its sign. Additionally, because the Ricci tensor is symmetric ($R^{\lambda\mu} = R^{\mu\lambda}$), and α and μ in the first term are simply dummy indices being summed over, we can relabel them to show that both terms are identical:

$$[\nabla^\mu, \nabla^\alpha] A_{\mu\alpha} = R^{\mu\lambda} A_{\mu\lambda} + R^{\mu\lambda} A_{\mu\lambda} = 2R^{\mu\lambda} A_{\mu\lambda} \tag{3.251}$$

Dividing both sides by 2 produces the relation for the commutator acting on our tensor:

$$\frac{1}{2}[\nabla^\mu, \nabla^\alpha] A_{\mu\alpha} = R^{\mu\lambda} A_{\mu\lambda} \tag{3.252}$$

Here, we are contracting the Ricci tensor $R^{\mu\lambda}$, which is perfectly symmetric ($R^{\mu\lambda} = R^{\lambda\mu}$), with $A_{\mu\lambda}$, which is perfectly antisymmetric ($A_{\mu\lambda} = -A_{\lambda\mu}$). The contraction of any symmetric tensor with an antisymmetric tensor is identically zero. Therefore, the right side of equation (3.246) evaluates exactly to zero:

$$\frac{1}{2}K^\nu \nabla_\nu R = 0 \implies K^\lambda \nabla_\lambda R = 0 \tag{3.253}$$

This perfectly recovers equation (3.253).

Now, for equation (3.257), we must show that the current $J_T^\mu = K_\nu T^{\mu\nu}$ is conserved, which means its covariant divergence vanishes:

$$\nabla_\mu J_T^\mu = \nabla_\mu (K_\nu T^{\mu\nu}) \tag{3.254}$$

Applying the product rule to equation (3.254) gives:

$$\nabla_\mu J_T^\mu = (\nabla_\mu K_\nu) T^{\mu\nu} + K_\nu (\nabla_\mu T^{\mu\nu}) \tag{3.255}$$

The energy-momentum tensor is locally conserved by definition, meaning $\nabla_\mu T^{\mu\nu} = 0$, so the second term vanishes. For the first term, $\nabla_\mu K_\nu$ is antisymmetric (by Killing's equation) and $T^{\mu\nu}$ is symmetric. Their contraction is therefore zero:

$$(\nabla_\mu K_\nu)T^{\mu\nu} = 0 \tag{3.256}$$

Substituting these facts back into equation (3.255) yields the final result:

$$\nabla_\mu J^\mu_T = 0 \tag{3.257}$$

This proves that the current defined in equation (3.257) is indeed perfectly conserved.

Conserved Currents

Killing vectors allow us to define conserved quantities for fields, not just particles. Given a Killing vector K^μ and an energy-momentum tensor $T^{\mu\nu}$ that satisfies local conservation ($\nabla_\mu T^{\mu\nu} = 0$), we can construct a current:

$$J^\mu_T = K_\nu T^{\mu\nu}. \tag{3.258}$$

We verify its conservation by taking the divergence:

$$\nabla_\mu J^\mu_T = \nabla_\mu(K_\nu T^{\mu\nu}) = (\nabla_\mu K_\nu)T^{\mu\nu} + K_\nu(\nabla_\mu T^{\mu\nu}). \tag{3.259}$$

The second term is zero by conservation of $T^{\mu\nu}$. The first term vanishes because $T^{\mu\nu}$ is symmetric, while $\nabla_\mu K_\nu$ is antisymmetric (by Killing's equation).

$$\nabla_\mu J^\mu_T = 0. \tag{3.260}$$

If the spacetime admits a timelike Killing vector K^μ (time translation symmetry), we can define the total conserved energy of the system by integrating this current over a spacelike hypersurface Σ:

$$E_T = \int_\Sigma J^\mu_T n_\mu \sqrt{\gamma} d^3x. \tag{3.261}$$

Example: Symmetries of $\mathbb{R}^3$

Consider Euclidean 3-space with Cartesian coordinates and metric:

$$ds^2 = dx^2 + dy^2 + dz^2. \tag{3.262}$$

The metric is constant, so it is independent of all coordinates. **Translations:** There are three obvious Killing vectors corresponding to translations along the axes:

$$X = \partial_x = (1,0,0), \quad Y = \partial_y = (0,1,0), \quad Z = \partial_z = (0,0,1). \tag{3.263}$$

Rotations: There are also symmetries under rotation. Consider a rotation about the z-axis. In polar coordinates, the metric is $ds^2 = dr^2 + r^2 d\theta^2 + dz^2$, which is independent of θ. Thus $R = \partial_\theta$ is a Killing vector. To find its Cartesian components, we use the coordinate transformation $x = r\cos\theta, y = r\sin\theta$. The vector field is:

$$R_z = \frac{\partial x}{\partial\theta}\partial_x + \frac{\partial y}{\partial\theta}\partial_y = -y\partial_x + x\partial_y. \tag{3.264}$$

So the components are $R^\mu_z = (-y, x, 0)$. Let us verify Killing's equation explicitly for R_z. In Cartesian coordinates, $\Gamma = 0$, so $\nabla_\mu = \partial_\mu$. The covariant components are $K_\mu = \delta_{\mu\nu}R^\nu_z = (-y, x, 0)$. To find the Killing vector for rotation about the z-axis, we use the standard rotation matrix $M_z(\phi)$ for a counterclockwise rotation by an angle ϕ:

$$M_z(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{3.265}$$

Applying equation (3.265) to a position vector $\mathbf{x} = (x, y, z)^T$ gives the new coordinates $\mathbf{x}'(\phi)$:

$$\mathbf{x}'(\phi) = M_z(\phi)\mathbf{x} = \begin{pmatrix} x\cos\phi - y\sin\phi \\ x\sin\phi + y\cos\phi \\ z \end{pmatrix} \tag{3.266}$$

The Killing vector R^μ is the derivative of the transformation in equation (3.266) with respect to the parameter ϕ, evaluated at $\phi = 0$:

$$R^\mu = \left.\frac{d\mathbf{x}'}{d\phi}\right|_{\phi=0} = \left.\begin{pmatrix} -x\sin\phi - y\cos\phi \\ x\cos\phi - y\sin\phi \\ 0 \end{pmatrix}\right|_{\phi=0} \tag{3.267}$$

Evaluating equation (3.267) at $\phi = 0$ yields the components:

$$R^\mu = \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix} \tag{3.268}$$

This matches the components $R^\mu = (-y, x, 0)$ from the textbook. Similarly, the rotation matrix $M_x(\alpha)$ for an angle α about the x-axis is:

$$M_x(\alpha) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{pmatrix} \tag{3.269}$$

Applying equation (3.269) to the position vector yields:

$$\mathbf{x}'(\alpha) = M_x(\alpha)\mathbf{x} = \begin{pmatrix} x \\ y\cos\alpha - z\sin\alpha \\ y\sin\alpha + z\cos\alpha \end{pmatrix} \tag{3.270}$$

Taking the derivative of equation (3.270) with respect to α and evaluating at $\alpha = 0$ gives the Killing vector T^μ:

$$T^\mu = \left.\frac{d\mathbf{x}'}{d\alpha}\right|_{\alpha=0} = \left.\begin{pmatrix} 0 \\ -y\sin\alpha - z\cos\alpha \\ y\cos\alpha - z\sin\alpha \end{pmatrix}\right|_{\alpha=0} = \begin{pmatrix} 0 \\ -z \\ y \end{pmatrix} \tag{3.271}$$

Equation (3.271) provides the components $T^\mu = (0, -z, y)$ shown in equation (3.186). Finally, the rotation matrix $M_y(\beta)$ for an angle β about the y-axis is:

$$M_y(\beta) = \begin{pmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{pmatrix} \tag{3.272}$$

Applying equation (3.272) to the position vector results in:

$$\mathbf{x}'(\beta) = M_y(\beta)\mathbf{x} = \begin{pmatrix} x\cos\beta + z\sin\beta \\ y \\ -x\sin\beta + z\cos\beta \end{pmatrix} \tag{3.273}$$

Taking the derivative of equation (3.273) with respect to β and evaluating at $\beta = 0$ yields the Killing vector S^μ:

$$S^\mu = \left.\frac{d\mathbf{x}'}{d\beta}\right|_{\beta=0} = \left.\begin{pmatrix} -x\sin\beta + z\cos\beta \\ 0 \\ -x\cos\beta - z\sin\beta \end{pmatrix}\right|_{\beta=0} = \begin{pmatrix} z \\ 0 \\ -x \end{pmatrix} \tag{3.274}$$

Equation (3.274) exactly matches the components $S^\mu = (z, 0, -x)$

Example: Symmetries of S^2

Consider the 2-sphere with metric:

$$ds^2 = d\theta^2 + \sin^2\theta d\phi^2. \tag{3.275}$$

The metric is independent of ϕ, so we immediately identify the first Killing vector (rotation about the z-axis):

$$R = \partial_\phi. \tag{3.276}$$

The sphere behaves like the surface of a ball in $\mathbb{R}^3$, so it must have 3 rotational symmetries total. However, the other two rotations (about x and y) are not obvious in these coordinates because they change the θ coordinate. By transforming the Cartesian Killing vectors to spherical coordinates, we find:

$$S = \cos\phi\,\partial_\theta - \cot\theta\sin\phi\,\partial_\phi, \tag{3.277}$$

$$T = -\sin\phi\,\partial_\theta - \cot\theta\cos\phi\,\partial_\phi. \tag{3.278}$$

The non-zero metric components are:

$$g_{\theta\theta} = 1, \quad g_{\phi\phi} = \sin^2\theta \tag{3.279}$$

The non-zero Christoffel symbols for this metric are (standard results for a sphere):

$$\Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta \tag{3.280}$$

$$\Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \cot\theta \tag{3.281}$$

All others (like $\Gamma^\theta_{\theta\theta}$) are zero.

We are given the vector S in the basis $(\partial_\theta, \partial_\phi)$:

$$S = S^\theta\partial_\theta + S^\phi\partial_\phi = \cos\phi\,\partial_\theta - \cot\theta\sin\phi\,\partial_\phi \tag{3.282}$$

So the contravariant components are:

$$S^\theta = \cos\phi, \quad S^\phi = -\cot\theta\sin\phi \tag{3.283}$$

Killing's equation uses the covariant components $S_\mu = g_{\mu\nu}S^\nu$.

$$S_\theta = g_{\theta\theta}S^\theta = (1)(\cos\phi) = \cos\phi \tag{3.284}$$

$$\begin{aligned} S_\phi &= g_{\phi\phi}S^\phi = (\sin^2\theta)(-\cot\theta\sin\phi) \\ &= \sin^2\theta\left(-\frac{\cos\theta}{\sin\theta}\right)\sin\phi = -\sin\theta\cos\theta\sin\phi \end{aligned} \tag{3.285}$$

We must check $\nabla_\mu S_\nu + \nabla_\nu S_\mu = 0$ for all independent pairs of indices: (θ,θ), (ϕ,ϕ), and (θ,ϕ).
We check $2\nabla_\theta S_\theta = 2(\partial_\theta S_\theta - \Gamma^\lambda_{\theta\theta}S_\lambda)$.

- $\partial_\theta S_\theta = \partial_\theta(\cos\phi) = 0$
- $\Gamma^\lambda_{\theta\theta} = 0$ (all symbols with $\theta\theta$ lower are zero)

$$\nabla_\theta S_\theta = 0 - 0 = 0 \quad \checkmark \tag{3.286}$$

We check $2\nabla_\phi S_\phi = 2(\partial_\phi S_\phi - \Gamma^\lambda_{\phi\phi}S_\lambda)$.

- **Derivative term:**

$$\begin{aligned} \partial_\phi S_\phi &= \partial_\phi(-\sin\theta\cos\theta\sin\phi) \\ &= -\sin\theta\cos\theta\cos\phi \end{aligned}$$

- **Connection term:** Sum over $\lambda = \theta, \phi$. Only $\Gamma^{\theta}_{\phi\phi}$ is non-zero.

$$\begin{aligned}\Gamma^{\lambda}_{\phi\phi} S_\lambda &= \Gamma^{\theta}_{\phi\phi} S_\theta + \Gamma^{\phi}_{\phi\phi} S_\phi \\ &= (-\sin\theta\cos\theta)(S_\theta) + (0) \\ &= (-\sin\theta\cos\theta)(\cos\phi)\end{aligned}$$

Subtracting the connection term from the derivative term:

$$\nabla_\phi S_\phi = (-\sin\theta\cos\theta\cos\phi) - (-\sin\theta\cos\theta\cos\phi) = 0 \quad \checkmark \tag{3.287}$$

We check $\nabla_\theta S_\phi + \nabla_\phi S_\theta = 0$. Formula: $(\partial_\theta S_\phi - \Gamma^{\lambda}_{\theta\phi} S_\lambda) + (\partial_\phi S_\theta - \Gamma^{\lambda}_{\phi\theta} S_\lambda)$. Note that $\Gamma^{\lambda}_{\theta\phi} = \Gamma^{\lambda}_{\phi\theta}$.

- **Derivatives:**

$$\begin{aligned}\partial_\theta S_\phi &= \partial_\theta(-\sin\theta\cos\theta\sin\phi) \\ &= -\sin\phi[\cos^2\theta - \sin^2\theta] \quad \text{(Product Rule)} \\ \partial_\phi S_\theta &= \partial_\phi(\cos\phi) = -\sin\phi\end{aligned}$$

  Sum of derivatives:

$$-\sin\phi(\cos^2\theta - \sin^2\theta) - \sin\phi = -\sin\phi(\cos^2\theta - \sin^2\theta + 1)$$

  Using $1 - \sin^2\theta = \cos^2\theta$, the term in brackets becomes $2\cos^2\theta$.

$$\text{Sum} = -2\cos^2\theta\sin\phi \tag{3.288}$$

- **Connection Terms:** We need to subtract $2\Gamma^{\lambda}_{\theta\phi} S_\lambda$. The only non-zero symbol here is $\Gamma^{\phi}_{\theta\phi} = \cot\theta$.

$$\begin{aligned}2\Gamma^{\lambda}_{\theta\phi} S_\lambda &= 2\Gamma^{\phi}_{\theta\phi} S_\phi \\ &= 2(\cot\theta)(-\sin\theta\cos\theta\sin\phi) \\ &= 2\left(\frac{\cos\theta}{\sin\theta}\right)(-\sin\theta\cos\theta\sin\phi) \\ &= -2\cos^2\theta\sin\phi\end{aligned}$$

Finally, calculate the full covariant derivative sum:

$$\nabla_\theta S_\phi + \nabla_\phi S_\theta = (-2\cos^2\theta\sin\phi) - (-2\cos^2\theta\sin\phi) = 0 \quad \checkmark \tag{3.289}$$

Conclusion for S

Since $\nabla_{(\mu} S_{\nu)} = 0$ for all components, the vector field S is indeed a Killing vector.

Extension: Checking Vector T

We now apply the same procedure to explicitly verify the third rotational Killing vector:

$$T = -\sin\phi\,\partial_\theta - \cot\theta\cos\phi\,\partial_\phi \tag{3.290}$$

The contravariant components are:

$$T^\theta = -\sin\phi, \quad T^\phi = -\cot\theta\cos\phi \tag{3.291}$$

Lowering the indices with the metric components gives the covariant components:

$$T_\theta = g_{\theta\theta} T^\theta = (1)(-\sin\phi) = -\sin\phi \tag{3.292}$$

$$\begin{aligned} T_\phi &= g_{\phi\phi} T^\phi = (\sin^2\theta)(-\cot\theta\cos\phi) \\ &= \sin^2\theta\left(-\frac{\cos\theta}{\sin\theta}\right)\cos\phi = -\sin\theta\cos\theta\cos\phi \end{aligned} \tag{3.293}$$

We again check Killing's equation $\nabla_\mu T_\nu + \nabla_\nu T_\mu = 0$ for all independent pairs.
Check $2\nabla_\theta T_\theta$:

- $\partial_\theta T_\theta = \partial_\theta(-\sin\phi) = 0$
- $\Gamma^\lambda_{\theta\theta} = 0$

$$\nabla_\theta T_\theta = 0 - 0 = 0 \quad \checkmark \tag{3.294}$$

Check $2\nabla_\phi T_\phi$:

- **Derivative term:**

$$\begin{aligned} \partial_\phi T_\phi &= \partial_\phi(-\sin\theta\cos\theta\cos\phi) \\ &= \sin\theta\cos\theta\sin\phi \end{aligned}$$

- **Connection term:**

$$\begin{aligned} \Gamma^\lambda_{\phi\phi} T_\lambda &= \Gamma^\theta_{\phi\phi} T_\theta = (-\sin\theta\cos\theta)(-\sin\phi) \\ &= \sin\theta\cos\theta\sin\phi \end{aligned}$$

Subtracting the connection term from the derivative term:

$$\nabla_\phi T_\phi = (\sin\theta\cos\theta\sin\phi) - (\sin\theta\cos\theta\sin\phi) = 0 \quad \checkmark \tag{3.295}$$

Check $\nabla_\theta T_\phi + \nabla_\phi T_\theta$:

- **Derivatives:**

$$\begin{aligned} \partial_\theta T_\phi &= \partial_\theta(-\sin\theta\cos\theta\cos\phi) = -\cos\phi[\cos^2\theta - \sin^2\theta] \\ \partial_\phi T_\theta &= \partial_\phi(-\sin\phi) = -\cos\phi \end{aligned}$$

  Sum of derivatives:

$$-\cos\phi(\cos^2\theta - \sin^2\theta) - \cos\phi = -\cos\phi(\cos^2\theta - \sin^2\theta + 1)$$

  Using $1 - \sin^2\theta = \cos^2\theta$, this simplifies to:

$$\text{Sum} = -2\cos^2\theta\cos\phi \tag{3.296}$$

- **Connection Terms:**

$$\begin{aligned} 2\Gamma^\lambda_{\theta\phi} T_\lambda &= 2\Gamma^\phi_{\theta\phi} T_\phi \\ &= 2(\cot\theta)(-\sin\theta\cos\theta\cos\phi) \\ &= 2\left(\frac{\cos\theta}{\sin\theta}\right)(-\sin\theta\cos\theta\cos\phi) \\ &= -2\cos^2\theta\cos\phi \end{aligned}$$

Subtracting the connection terms from the derivative terms:

$$\nabla_\theta T_\phi + \nabla_\phi T_\theta = (-2\cos^2\theta\cos\phi) - (-2\cos^2\theta\cos\phi) = 0 \quad \checkmark \tag{3.297}$$

Conclusion for T

Since all symmetric covariant derivatives vanish, the vector field T perfectly satisfies Killing's equation and is a valid Killing vector for the 2-sphere.

3.10 Maximally Symmetric Spaces

Counting Isometries

We ask: what is the most symmetric a space can possibly be? To quantify this, we count the number of independent isometries (Killing vectors). In a flat Euclidean space $\mathbb{R}^n$, we have:

- **Translations:** n directions (moving any point to any other point).
- **Rotations:** Rotations about a point. A rotation is defined by a plane spanned by two axes. The number of independent planes in n dimensions is the number of pairs of axes: $\binom{n}{2} = \frac{1}{2}n(n-1)$.

The total number of independent symmetries is the sum: **Derivation of the Number of Independent Killing Vectors**

In a maximally symmetric space of dimension n, the total number of independent symmetries (Killing vectors) is the sum of the translational symmetries and the rotational symmetries.

1. Translational Symmetries
For any point p in an n-dimensional manifold, we can choose the origin of our coordinate system. Since there are n independent directions in the space (corresponding to the n axes), there are n independent translations.

$$N_{\text{trans}} = n \tag{3.298}$$

2. Rotational Symmetries
A rotation is defined by a plane spanned by two distinct axes. The number of independent rotations is the number of unique pairs of axes we can choose from the n available axes. Mathematically, this is "n choose 2":

$$N_{\text{rot}} = \binom{n}{2} = \frac{n(n-1)}{2} \tag{3.299}$$

3. Total Symmetries
The total number of independent Killing vectors N is the sum of translations and rotations:

$$N = N_{\text{trans}} + N_{\text{rot}} \tag{3.300}$$

Substituting the values from above:

$$N = n + \frac{1}{2}n(n-1) \tag{3.301}$$

4. Algebraic Simplification
To verify equation (3.189), we simplify the expression by finding a common denominator:

$$\begin{aligned} N &= \frac{2n}{2} + \frac{n(n-1)}{2} \\ &= \frac{2n + n^2 - n}{2} \\ &= \frac{n^2 + n}{2} \end{aligned}$$

Finally, factor out n from the numerator:

$$N = \frac{1}{2}n(n+1) \tag{3.302}$$

This confirms that a maximally symmetric space in n dimensions has $\frac{1}{2}n(n+1)$ independent Killing vectors.

A manifold that possesses this maximum number of Killing vectors ($\frac{1}{2}n(n+1)$) is called a **Maximally Symmetric Space**. Because there are isometries mapping any point to any other point (homogeneity) and any direction to any other direction (isotropy), the curvature scalar R must be constant everywhere, and the curvature tensor must look the same in all directions.

The Curvature Tensor of a Maximally Symmetric Space

We can derive the specific form of the Riemann tensor for such a space. In locally inertial coordinates at a point p, the metric is $g_{\mu\nu} = \eta_{\mu\nu}$. Since the space is isotropic, the Riemann tensor components must be invariant under Lorentz transformations (or rotations). The only rank-4 tensor we can construct from the metric that shares the symmetries of the Riemann tensor ($R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu} = -R_{\rho\sigma\nu\mu} = R_{\mu\nu\rho\sigma}$) is composed of products of the metric itself:

$$R_{\rho\sigma\mu\nu} \propto (g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu}). \tag{3.303}$$

A maximally symmetric space is both homogeneous (looks the same at every point) and isotropic (looks the same in every direction). This high degree of symmetry implies that the Riemann curvature tensor $R_{\rho\sigma\mu\nu}$ must be constructed solely from the metric tensor $g_{\mu\nu}$, as there are no other preferred vector or tensor fields to use. The only tensor of rank 4 that can be built from the metric and shares the symmetry properties of the Riemann tensor (antisymmetric in the first and second pair of indices, symmetric under exchange of pairs) is a linear combination of products of the metric. We write this as:

$$R_{\rho\sigma\mu\nu} = K(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu}) \tag{3.304}$$

where K is an unknown constant that determines the curvature scale. The Ricci tensor $R_{\sigma\nu}$ is defined as the contraction of the first and third indices of the Riemann tensor:

$$R_{\sigma\nu} = g^{\rho\mu}R_{\rho\sigma\mu\nu} \tag{3.305}$$

Substitute the Ansatz from equation (3.304) into this definition:

$$R_{\sigma\nu} = g^{\rho\mu}[K(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu})] \tag{3.306}$$

We distribute the inverse metric $g^{\rho\mu}$ across the terms:

$$R_{\sigma\nu} = K[(g^{\rho\mu}g_{\rho\mu})g_{\sigma\nu} - (g^{\rho\mu}g_{\rho\nu})g_{\sigma\mu}] \tag{3.307}$$

Now we apply two fundamental identities of n-dimensional geometry:

- The contraction of the metric with its inverse is the dimension of the manifold n:

$$g^{\rho\mu}g_{\rho\mu} = \delta^{\mu}_{\mu} = n$$

- The contraction of the metric with its inverse on mixed indices is the Kronecker delta:

$$g^{\rho\mu}g_{\rho\nu} = \delta^{\mu}_{\nu}$$

Substituting these back into the expression for $R_{\sigma\nu}$:

$$\begin{aligned} R_{\sigma\nu} &= K[ng_{\sigma\nu} - \delta^{\mu}_{\nu}g_{\sigma\mu}] \\ &= K[ng_{\sigma\nu} - g_{\sigma\nu}] \\ &= K(n-1)g_{\sigma\nu} \end{aligned}$$

The Ricci scalar R is the trace of the Ricci tensor:

$$R = g^{\sigma\nu} R_{\sigma\nu} \tag{3.308}$$

Substitute the result for $R_{\sigma\nu}$ we just derived:

$$R = g^{\sigma\nu}[K(n-1)g_{\sigma\nu}] \tag{3.309}$$

Factor out the constants:

$$R = K(n-1)(g^{\sigma\nu} g_{\sigma\nu}) \tag{3.310}$$

Once again, use the identity $g^{\sigma\nu} g_{\sigma\nu} = n$:

$$R = K(n-1)n \tag{3.311}$$

We now have a simple algebraic equation relating the scalar curvature R to our constant K:

$$R = n(n-1)K \tag{3.312}$$

Solving for K:

$$K = \frac{R}{n(n-1)} \tag{3.313}$$

Finally, we substitute this value of K back into our original Ansatz (Equation 3.304):

$$R_{\rho\sigma\mu\nu} = \frac{R}{n(n-1)}(g_{\rho\mu} g_{\sigma\nu} - g_{\rho\nu} g_{\sigma\mu}) \tag{3.314}$$

This confirms that for a maximally symmetric space, the Riemann tensor is entirely determined by the scalar curvature R and the dimension n.

Classification by Curvature

Since R is constant, we categorize these spaces by the sign of R:

- **Positive ($R > 0$):** The sphere S^n.
- **Zero ($R = 0$):** Flat Euclidean space $\mathbb{R}^n$.
- **Negative ($R < 0$):** The hyperboloid H^n.

Example: The Poincaré Half-Plane (H^2)

A concrete representation of the 2-dimensional hyperboloid H^2 is the Poincaré half-plane. **Coordinates:** (x, y) with $y > 0$. **Metric:**

$$ds^2 = \frac{a^2}{y^2}(dx^2 + dy^2). \tag{3.315}$$

Here a is a constant related to the radius of curvature.

Distance Measurements Let's measure the distance along a vertical line ($x = \text{const}$) from y_1 to y_2.

$$L = \int_{y_1}^{y_2} \sqrt{g_{yy}} dy = \int_{y_1}^{y_2} \sqrt{\frac{a^2}{y^2}} dy = a \int_{y_1}^{y_2} \frac{dy}{y} = a \ln\left(\frac{y_2}{y_1}\right). \tag{3.316}$$

Intuition: As $y \to 0$, the distance diverges. The boundary $y = 0$ is infinitely far away.

Christoffel Symbols The non-zero metric components are $g_{xx} = g_{yy} = a^2 y^{-2}$. The inverse components are $g^{xx} = g^{yy} = a^{-2} y^2$. Derivatives:

$$\partial_y g_{xx} = \partial_y (a^2 y^{-2}) = -2a^2 y^{-3}, \quad \partial_y g_{yy} = -2a^2 y^{-3}. \tag{3.317}$$

(Derivatives with respect to x vanish). Calculate Γ^x_{xy}:

$$\Gamma^x_{xy} = \frac{1}{2} g^{xx} (\partial_x g_{xy} + \partial_y g_{xx} - \partial_x g_{xy}) = \frac{1}{2}(a^{-2}y^2)(-2a^2 y^{-3}) = -y^{-1}. \tag{3.318}$$

Calculate Γ^y_{xx}:

$$\Gamma^y_{xx} = \frac{1}{2} g^{yy} (\partial_x g_{yx} + \partial_x g_{xy} - \partial_y g_{xx}) = \frac{1}{2}(a^{-2}y^2)(-(-2a^2 y^{-3})) = y^{-1}. \tag{3.319}$$

Calculate Γ^y_{yy}:

$$\Gamma^y_{yy} = \frac{1}{2} g^{yy} (\partial_y g_{yy}) = \frac{1}{2}(a^{-2}y^2)(-2a^2 y^{-3}) = -y^{-1}. \tag{3.320}$$

Summary of symbols:

$$\Gamma^x_{xy} = \Gamma^x_{yx} = -\frac{1}{y}, \quad \Gamma^y_{xx} = \frac{1}{y}, \quad \Gamma^y_{yy} = -\frac{1}{y}. \tag{3.321}$$

Geodesics The geodesic equations turn out to be:

$$\frac{d^2x}{d\tau^2} - \frac{2}{y}\frac{dx}{d\tau}\frac{dy}{d\tau} = 0, \quad \frac{d^2y}{d\tau^2} + \frac{1}{y}\left(\frac{dx}{d\tau}\right)^2 - \frac{1}{y}\left(\frac{dy}{d\tau}\right)^2 = 0. \tag{3.322}$$

It can be shown that the solutions satisfy $(x - x_0)^2 + y^2 = L^2$. Geometrically, the geodesics are **semicircles** centered on the x-axis (or vertical straight lines).

Curvature Calculation We compute one component of the Riemann tensor, $R^x{}_{yxy}$.

$$R^x{}_{yxy} = \partial_x \Gamma^x_{yy} - \partial_y \Gamma^x_{xy} + \Gamma^x_{x\lambda}\Gamma^\lambda_{yy} - \Gamma^x_{y\lambda}\Gamma^\lambda_{xy}. \tag{3.323}$$

Terms:
1. $\partial_x \Gamma^x_{yy} = \partial_x(-1/y) = 0$.

2. $\partial_y \Gamma^x_{xy} = \partial_y(-1/y) = +y^{-2}$. So the term is $-y^{-2}$.

3. $\Gamma^x_{x\lambda}\Gamma^\lambda_{yy}$: Only $\lambda = y$ contributes (since $\Gamma^y_{yy} \neq 0$).

$\Gamma^x_{xy}\Gamma^y_{yy} = (-1/y)(-1/y) = y^{-2}$.

4. $\Gamma^x_{y\lambda}\Gamma^\lambda_{xy}$: Sum over $\lambda = x, y$.

$\lambda = x : \Gamma^x_{yx}\Gamma^x_{xx} = (-1/y)(0) = 0$.

$\lambda = y : \Gamma^x_{yy}\Gamma^y_{xy} = (-1/y)(0) = 0$ (Wait, $\Gamma^y_{xy} = 0$).

Summing them up:

$$R^x{}_{yxy} = 0 - (y^{-2}) + y^{-2} - 0 = 0? \tag{3.324}$$

Correction: Let me re-evaluate term 4 carefully. $\Gamma^x_{y\lambda}\Gamma^\lambda_{xy}$.

$$\lambda = x \to \Gamma^x_{yx}\Gamma^x_{xx}$$

. We know Γ^x_{xx} was not calculated above. Let's check $\Gamma^x_{xx} = \frac{1}{2}g^{xx}(\partial_x g_{xx} + \dots) = 0$. Correct. $\lambda = y \to \Gamma^x_{yy}\Gamma^y_{xy}$. We check

$$\Gamma^y_{xy} = \frac{1}{2}g^{yy}(\partial_x g_{yy} + \partial_y g_{xy} - \partial_x g_{yy}) = 0$$

. Correct. Wait, let's re-evaluate Term 3: $\Gamma^x_{x\lambda}\Gamma^\lambda_{yy}$. Indices: Sum over λ. $\lambda = x$: $\Gamma^x_{xx}\Gamma^x_{yy} = 0$. $\lambda = y$: $\Gamma^x_{xy}\Gamma^y_{yy} = (-y^{-1})(-y^{-1}) = y^{-2}$. Let's re-evaluate Term 2: $-\partial_y\Gamma^x_{xy} = -\partial_y(-y^{-1}) = -(y^{-2}) = -y^{-2}$. So sum is $-y^{-2} + y^{-2} = 0$? This implies zero curvature, which is wrong for a hyperboloid. Let's check the definition of Riemann again: $R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}$. We want $R^x{}_{yxy}$. Indices: $\rho = x, \sigma = y, \mu = x, \nu = y$.

$$R^x{}_{yxy} = \partial_x\Gamma^x_{yy} - \partial_y\Gamma^x_{xy} + \Gamma^x_{x\lambda}\Gamma^\lambda_{yy} - \Gamma^x_{y\lambda}\Gamma^\lambda_{xy}. \tag{3.325}$$

1. $\partial_x\Gamma^x_{yy} = 0$.
2. $-\partial_y\Gamma^x_{xy} = -\partial_y(-y^{-1}) = -(y^{-2})$.
3. Sum λ: $\Gamma^x_{xx}\Gamma^x_{yy} + \Gamma^x_{xy}\Gamma^y_{yy} = (0) + (-y^{-1})(-y^{-1}) = +y^{-2}$.
4. Sum λ: $\Gamma^x_{yx}\Gamma^x_{xy} + \Gamma^x_{yy}\Gamma^y_{xy} = (-y^{-1})(-y^{-1}) + (0) = +y^{-2}$.

Result:

$$R^x{}_{yxy} = 0 - y^{-2} + y^{-2} - y^{-2} = -y^{-2}. \tag{3.326}$$

This is non-zero. Correct.

Lower the index to get R_{xyxy}:

$$R_{xyxy} = g_{xx}R^x{}_{yxy} = (a^2y^{-2})(-y^{-2}) = -a^2y^{-4}. \tag{3.327}$$

Calculate Ricci Tensor R_{yy}:

$$R_{yy} = R^x{}_{yxy} = -y^{-2}. \tag{3.328}$$

Calculate Ricci Tensor R_{xx}:

$$R_{xx} = R^y{}_{xyx} = g^{yy}R_{yxyx} = g^{yy}(-R_{xyxy}) = (a^{-2}y^2)(a^2y^{-4}) = y^{-2}\dots \tag{3.329}$$

Wait, symmetry $R_{xyxy} = -a^2y^{-4}$. $R_{xx} = g^{yy}R_{yxyx}$. Note $R_{yxyx} = -R_{xyyx} = R_{xyxy} = -a^2y^{-4}$. So $R_{xx} = (a^{-2}y^2)(-a^2y^{-4}) = -y^{-2}$. Ricci Scalar R:

$$R = g^{xx}R_{xx} + g^{yy}R_{yy} = (a^{-2}y^2)(-y^{-2}) + (a^{-2}y^2)(-y^{-2}) = -a^{-2} - a^{-2} = -\frac{2}{a^2}. \tag{3.330}$$

The curvature scalar is constant and negative. This confirms H^2 is a maximally symmetric space of negative curvature.

Global Topology vs Local Geometry

The curvature scalar R is a **local** property. It tells us about the intrinsic geometry in a small neighborhood of a point, such as whether parallel lines converge, diverge, or remain parallel. However, this local curvature does not uniquely determine the **global** topology—the overall "shape" or connectivity of the manifold. Consider the example of the Torus T^2. We can define a flat Euclidean metric on a square, $ds^2 = dx^2 + dy^2$, and identify the opposite edges such that $x \sim x + L$ and $y \sim y + L$. Locally, the metric is flat everywhere, so the Riemann tensor vanishes and $R = 0$. Globally, however, identifying the edges folds the square into a torus, which is topologically distinct from an infinite flat plane. This deep connection between local geometry and global topology is formalized by the **Gauss-Bonnet Theorem** for compact, two-dimensional Riemannian manifolds:

$$\chi(M) = \frac{1}{4\pi}\int_M R\sqrt{|g|}\, d^2x = 2(1-g). \tag{3.331}$$

Deriving the Euler Characteristic Formula: $\chi = 2(1-g)$
The topological invariant known as the Euler characteristic, denoted by χ, is defined for a triangulated surface as:

$$\chi = V - E + F \tag{3.332}$$

where V is the number of vertices, E is the number of edges, and F is the number of faces. To understand how genus (g) factors into this, we start with the simplest closed surface: a sphere ($g = 0$). If we map a basic polyhedron like a tetrahedron onto a sphere, we have 4 vertices, 6 edges, and 4 triangular faces. Calculating this gives $\chi_{\text{sphere}} = 4-6+4 = 2$. Now, consider how we topologically add a "hole" or a "handle" to create a surface of genus $g = 1$ (a torus). To attach a handle to our sphere, we must first cut out two separate triangular faces to create openings. Removing these two faces reduces our total face count F by 2, which reduces the Euler characteristic by 2. Next, we connect these two open boundaries with a cylindrical tube (the handle). A topological cylinder constructed from triangles can be shown to have an internal Euler characteristic of exactly 0 (the new vertices, edges, and faces required to build the tube perfectly cancel out: $\Delta V - \Delta E + \Delta F = 0$). Therefore, the net change in the Euler characteristic for attaching one handle is exactly -2. If we start with a sphere ($\chi = 2$) and add g handles to reach a genus g, we subtract 2 for every handle added. This logically yields the formula:

$$\chi = 2 - 2g = 2(1 - g) \tag{3.333}$$

Sketch of the Proof for the Global Gauss-Bonnet Theorem

To prove the global theorem, we rely on a process called triangulation, breaking the continuous curved manifold M down into a mesh of flat-looking triangles. First, we state the **Local Gauss-Bonnet Theorem** for a single triangle Δ with interior angles α_i on a curved surface:

$$\int_{\Delta} K dA + \int_{\partial\Delta} k_g ds = \sum_{i=1}^{3} \alpha_i - \pi \tag{3.334}$$

Here, K is the Gaussian Curvature, k_g is the geodesic curvature of the edges, dA is the area element, and ds is the line element along the boundary $\partial\Delta$. To find the global curvature, we sum equation (3.334) over all the faces F in our triangulation of the manifold M:

$$\sum_{\text{Faces}} \left(\int_{\Delta} K dA + \int_{\partial\Delta} k_g ds \right) = \sum_{\text{Faces}} \left(\sum \alpha_i - \pi \right) \tag{3.335}$$

Two massive cancellations occur in equation (3.335). First, every internal edge is shared by exactly two adjacent triangles. When we integrate the geodesic curvature k_g along the edges, we traverse the edge in one direction for the first triangle, and the opposite direction for the second. Thus, $\int k_g ds$ perfectly cancels out to zero for every shared edge. Second, the sum of all interior angles α_i around any single vertex V must complete a full circle, adding up to 2π. So, $\sum \alpha_i = 2\pi V$. Substituting these geometric facts:

$$\int_M K dA = 2\pi V - \pi F \tag{3.336}$$

We must now relate equation (3.336) to our edges E. In a strict triangulation, every face has 3 edges, and every edge is shared by exactly 2 faces. This gives the relation $3F = 2E$, or $2E - 3F = 0$. Let us algebraically manipulate the expression $2\pi(V - E + F)$:

$$2\pi(V - E + F) = 2\pi V - 2\pi E + 2\pi F \tag{3.337}$$

Substitute $2E = 3F$ into the middle term:

$$2\pi V - \pi(3F) + 2\pi F = 2\pi V - \pi F \tag{3.338}$$

Notice that equation (3.338) perfectly matches the right-hand side of equation (3.336). Using the definition of the Euler characteristic from equation (3.332), we conclude:

$$\int_M K dA = 2\pi(V - E + F) = 2\pi\chi \tag{3.339}$$

Finally, we map equation (3.339) to the specific notation used in your textbook. In exactly two dimensions, the Ricci scalar R is strictly twice the Gaussian curvature ($R = 2K$). The invariant area element dA is written in covariant tensor notation as $\sqrt{|g|}d^2x$. Substituting $K = R/2$ and $dA = \sqrt{|g|}d^2x$:

$$\int_M \left(\frac{R}{2}\right) \sqrt{|g|}d^2x = 2\pi\chi \tag{3.340}$$

Dividing both sides of equation (3.340) by 2π and substituting equation (3.333) yields the exact formula (1.320) from the text:

$$\frac{1}{4\pi}\int_M R\sqrt{|g|}d^2x = \chi = 2(1-g) \tag{3.341}$$

This completes the conceptual proof bridging local differential geometry with global topology. Here, $\sqrt{|g|}\,d^2x$ is the invariant volume (or area) element, $\chi(M)$ is the Euler characteristic, and g is the genus (intuitively, the number of "holes" in the surface).

- **Sphere ($g = 0$):** $\chi = 2$. The integral of the curvature must be strictly positive ($\int R > 0$). While the curvature can vary, a sphere cannot admit a globally flat metric; it requires positive curvature on average.
- **Torus ($g = 1$):** $\chi = 0$. The integral of the curvature vanishes ($\int R = 0$). This admits our example of the flat torus ($R = 0$ everywhere), but it also allows regions of positive and negative curvature that perfectly cancel out.
- **Higher genus ($g \geq 2$):** $\chi < 0$. The integral is strictly negative ($\int R < 0$). These surfaces necessarily have a net negative curvature.

In advanced theoretical frameworks like String Theory, physical processes involve summing over all possible worldsheet topologies. Physicists simplify calculations by choosing a "fiducial metric"—a maximally symmetric representative metric for each topological class: a round, positively curved metric for the sphere; a flat metric for the torus; and a uniformly negatively curved metric (hyperbolic) for higher-genus surfaces.

The Hyperbolic Plane: Proving the Equations

To understand spaces of constant negative curvature, we examine the Poincaré half-plane, which models a two-dimensional hyperbolic geometry. Based on the context provided, we define the metric on the upper half-plane ($y > 0$) as:

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu = a^2\frac{dx^2+dy^2}{y^2}. \tag{3.342}$$

Here, $g_{xx} = g_{yy} = a^2/y^2$, and the off-diagonal components are zero.

1. Proving Equation (3.343): Path Length Let us calculate the distance Δs along a vertical path from y_1 to y_2 at a constant x. Since x is constant, $dx = 0$. The line element reduces to $ds = \sqrt{g_{yy}}dy$.

$$\begin{aligned}\Delta s &= \int_{y_1}^{y_2}\sqrt{g_{\mu\nu}\frac{dx^\mu}{dy}\frac{dx^\nu}{dy}}\,dy \\ &= \int_{y_1}^{y_2}\sqrt{\frac{a^2}{y^2}}\,dy \\ &= a\int_{y_1}^{y_2}\frac{dy}{y} = a\ln\left(\frac{y_2}{y_1}\right).\end{aligned} \tag{3.343}$$

This result shows that the distance does not simply scale as $y_2 - y_1$ like it would in Euclidean space. Crucially, if we take the limit as $y_1 \to 0$, the path length diverges to infinity ($\Delta s \to \infty$). The line $y = 0$ acts as a boundary that is infinitely far away for any observer living on the hyperboloid.

2. Proving Equation (3.344): Christoffel Symbols We calculate the Christoffel symbols using the metric components $g_{xx} = g_{yy} = a^2y^{-2}$ and their inverses $g^{xx} = g^{yy} = a^{-2}y^2$. Note that the scale factor a will perfectly cancel out.

$$\Gamma^x_{xy} = \frac{1}{2}g^{xx}(\partial_x g_{xy} + \partial_y g_{xx} - \partial_x g_{xy}) = \frac{1}{2}\left(\frac{y^2}{a^2}\right)\partial_y\left(\frac{a^2}{y^2}\right) = \frac{1}{2}\left(\frac{y^2}{a^2}\right)\left(-2\frac{a^2}{y^3}\right) = -y^{-1}.$$
$$\Gamma^y_{xx} = \frac{1}{2}g^{yy}(\partial_x g_{xy} + \partial_x g_{yx} - \partial_y g_{xx}) = \frac{1}{2}\left(\frac{y^2}{a^2}\right)\left(-\partial_y\frac{a^2}{y^2}\right) = \frac{1}{2}\left(\frac{y^2}{a^2}\right)\left(2\frac{a^2}{y^3}\right) = y^{-1}.$$
$$\Gamma^y_{yy} = \frac{1}{2}g^{yy}(\partial_y g_{yy} + \partial_y g_{yy} - \partial_y g_{yy}) = \frac{1}{2}\left(\frac{y^2}{a^2}\right)\partial_y\left(\frac{a^2}{y^2}\right) = \frac{1}{2}\left(\frac{y^2}{a^2}\right)\left(-2\frac{a^2}{y^3}\right) = -y^{-1}.$$

Because the metric is symmetric and diagonal, all other symbols vanish, yielding the exact set from the text:

$$\Gamma^x_{xy} = \Gamma^x_{yx} = -y^{-1}, \quad \Gamma^y_{xx} = y^{-1}, \quad \Gamma^y_{yy} = -y^{-1}. \tag{3.344}$$

3. Proving Equation (3.351): Geodesics We want to show that geodesics satisfy $(x-x_0)^2+y^2 = l^2$. We start with the geodesic equation $\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0$. For the x-component (using prime for $d/d\lambda$):

$$x'' + 2\Gamma^x_{xy}x'y' = 0 \implies x'' - \frac{2}{y}x'y' = 0 \tag{3.345}$$

Dividing by x' gives $\frac{x''}{x'} = 2\frac{y'}{y}$. Integrating both sides with respect to λ yields $\ln(x') = 2\ln(y) + C$, which simplifies to $x' = ky^2$ for some constant k. Instead of solving the y-geodesic equation directly, we use the fact that the length of the tangent vector is constant along an affinely parameterized geodesic. Let's parameterize by arc length (s), so $g_{\mu\nu}x'^\mu x'^\nu = 1$:

$$\frac{a^2}{y^2}\left((x')^2 + (y')^2\right) = 1 \implies (x')^2 + (y')^2 = \frac{y^2}{a^2} \tag{3.346}$$

Substitute $x' = ky^2$ into this magnitude equation:

$$k^2y^4 + (y')^2 = \frac{y^2}{a^2} \implies (y')^2 = y^2\left(\frac{1}{a^2} - k^2y^2\right) \tag{3.347}$$

To find the shape of the path, we find dx/dy:

$$\frac{dx}{dy} = \frac{x'}{y'} = \frac{ky^2}{\pm y\sqrt{\frac{1}{a^2} - k^2y^2}} = \frac{ky}{\pm\sqrt{\frac{1}{a^2} - k^2y^2}} \tag{3.348}$$

Let us define a new constant $l = \frac{1}{ak}$. Substituting $k = \frac{1}{al}$ gives:

$$\frac{dx}{dy} = \frac{y/(al)}{\pm\sqrt{\frac{1}{a^2} - \frac{y^2}{a^2l^2}}} = \frac{y/l}{\pm\sqrt{1-(y/l)^2}} = \frac{y}{\pm\sqrt{l^2-y^2}} \tag{3.349}$$

Integrate with respect to y:

$$x = \int \frac{y}{\pm\sqrt{l^2-y^2}}\,dy = \mp\sqrt{l^2-y^2} + x_0 \tag{3.350}$$

Rearranging terms:

$$(x-x_0) = \mp\sqrt{l^2-y^2} \implies (x-x_0)^2 = l^2 - y^2 \implies (x-x_0)^2 + y^2 = l^2. \tag{3.351}$$

These curves describe semicircles with centers located on the x-axis, confirming the textbook's statement. In the limit as $x_0 \to \infty$ and $l \to \infty$ with $l - x_0$ fixed, we get a straight vertical line.

4. Proving Equation (3.354): Riemann Tensor Component Following the discussion of S^2, we calculate a representative component of the Riemann tensor. We compute $R^x{}_{yxy}$ directly from the definition of the Riemann tensor in terms of Christoffel symbols:

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma} \tag{3.352}$$

Assigning indices $(\rho, \sigma, \mu, \nu) = (x, y, x, y)$ to equation (3.352):

$$R^x{}_{yxy} = \partial_x \Gamma^x_{yy} - \partial_y \Gamma^x_{xy} + \Gamma^x_{x\lambda}\Gamma^\lambda_{yy} - \Gamma^x_{y\lambda}\Gamma^\lambda_{xy} \tag{3.353}$$

We evaluate term by term using equation (3.344):

- $\partial_x \Gamma^x_{yy} = \partial_x(0) = 0$
- $-\partial_y \Gamma^x_{xy} = -\partial_y(-y^{-1}) = -y^{-2}$
- $\Gamma^x_{x\lambda}\Gamma^\lambda_{yy} = \Gamma^x_{xx}\Gamma^x_{yy} + \Gamma^x_{xy}\Gamma^y_{yy} = (0)(0) + (-y^{-1})(-y^{-1}) = y^{-2}$
- $-\Gamma^x_{y\lambda}\Gamma^\lambda_{xy} = -(\Gamma^x_{yx}\Gamma^x_{xy} + \Gamma^x_{yy}\Gamma^y_{xy}) = -((-y^{-1})(-y^{-1}) + (0)(0)) = -y^{-2}$

Summing these contributions gives:

$$R^x{}_{yxy} = 0 - y^{-2} + y^{-2} - y^{-2} = -y^{-2}. \tag{3.354}$$

This perfectly matches equation (3.354) as defined in the text.

Lorentzian Signatures

The classification extends to spacetime (Lorentzian metric):

- $R = 0$: **Minkowski Space** (Flat).
- $R > 0$: **de Sitter Space (dS)** (Positive curvature, like a sphere).
- $R < 0$: **Anti-de Sitter Space (AdS)** (Negative curvature, like a hyperboloid).

3.11 Geodesic Deviation

The Question of Parallel Lines

In Euclidean geometry, the **Parallel Postulate** states that initially parallel lines remain parallel forever. In curved spacetime, this is false. Geodesics that start parallel can converge or diverge. For example, two longitude lines on Earth start parallel at the equator (both perpendicular to the equator) but converge at the pole.

This convergence or divergence is the most direct physical manifestation of curvature. In General Relativity, it corresponds to **tidal forces**. If you are falling into a black hole, the gravity at your feet is stronger than at your head, stretching you. This stretching is exactly the relative acceleration of the geodesics followed by your head and your feet.

The Setup: A Family of Geodesics

Consider a family of geodesics $\gamma_s(t)$.

- t is the affine parameter (e.g., proper time) along each geodesic.
- s is a parameter that labels *which* geodesic we are on.

This defines a 2D surface in the manifold mapped by coordinates (s, t) to points $x^\mu(s, t)$. We define two vector fields: 1. **Tangent Vector** T^μ: Points along the geodesic (velocity).

$$T^\mu = \frac{\partial x^\mu}{\partial t}. \tag{3.355}$$

2. **Deviation Vector** S^μ: Points from one geodesic to its neighbor (separation).

$$S^\mu = \frac{\partial x^\mu}{\partial s}. \tag{3.356}$$

The Commutator Property

Since T and S are coordinate basis vectors for the sub-manifold spanned by (s, t) (specifically $T = \partial_t$ and $S = \partial_s$), partial derivatives with respect to s and t commute.

$$\frac{\partial}{\partial s}\left(\frac{\partial x^\mu}{\partial t}\right) = \frac{\partial}{\partial t}\left(\frac{\partial x^\mu}{\partial s}\right) \implies \partial_s T^\mu = \partial_t S^\mu. \tag{3.357}$$

In covariant language, the Lie derivative vanishes $[S, T] = 0$. This implies a crucial identity relating their covariant derivatives:

$$T^\nu \nabla_\nu S^\mu = S^\nu \nabla_\nu T^\mu \quad (\text{or } \nabla_T S = \nabla_S T). \tag{3.358}$$

To prove the commutator property $[S, T]^\mu = 0$ in covariant form, we start with the definition of the Lie bracket of two vector fields evaluated using covariant derivatives:

$$[S, T]^\mu = S^\nu \nabla_\nu T^\mu - T^\nu \nabla_\nu S^\mu \tag{3.359}$$

We expand the covariant derivatives in equation (3.359) using the standard formula $\nabla_\nu V^\mu = \partial_\nu V^\mu + \Gamma^\mu_{\nu\lambda} V^\lambda$:

$$[S, T]^\mu = S^\nu (\partial_\nu T^\mu + \Gamma^\mu_{\nu\lambda} T^\lambda) - T^\nu (\partial_\nu S^\mu + \Gamma^\mu_{\nu\lambda} S^\lambda) \tag{3.360}$$

Distributing the vectors yields:

$$[S, T]^\mu = S^\nu \partial_\nu T^\mu + S^\nu \Gamma^\mu_{\nu\lambda} T^\lambda - T^\nu \partial_\nu S^\mu - T^\nu \Gamma^\mu_{\nu\lambda} S^\lambda \tag{3.361}$$

In the last term of equation (3.361), ν and λ are dummy summation indices, so we can swap their labels $(T^\lambda \Gamma^\mu_{\lambda\nu} S^\nu)$. We can then group the connection terms together:

$$S^\nu \Gamma^\mu_{\nu\lambda} T^\lambda - T^\lambda \Gamma^\mu_{\lambda\nu} S^\nu = S^\nu T^\lambda (\Gamma^\mu_{\nu\lambda} - \Gamma^\mu_{\lambda\nu}) \tag{3.362}$$

In General Relativity, we use the Levi-Civita connection, which is torsion-free. This means the Christoffel symbols are symmetric in their lower indices $(\Gamma^\mu_{\nu\lambda} = \Gamma^\mu_{\lambda\nu})$. Consequently, the term in equation (3.362) perfectly cancels out to zero. Equation (3.361) then simplifies to the standard partial derivative definition of the Lie bracket:

$$[S, T]^\mu = S^\nu \partial_\nu T^\mu - T^\nu \partial_\nu S^\mu \tag{3.363}$$

Next, we apply the definitions of the tangent and deviation vectors provided in your text: $T^\mu = \frac{\partial x^\mu}{\partial t}$ and $S^\nu = \frac{\partial x^\nu}{\partial s}$. The operator $S^\nu \partial_\nu$ represents the directional derivative along the s-parameterized curves, which is simply the partial derivative $\frac{\partial}{\partial s}$. Similarly, $T^\nu \partial_\nu = \frac{\partial}{\partial t}$. Applying these to equation (3.363):

$$S^\nu \partial_\nu T^\mu = \frac{\partial}{\partial s}\left(\frac{\partial x^\mu}{\partial t}\right) = \frac{\partial^2 x^\mu}{\partial s \partial t} \tag{3.364}$$

$$T^\nu \partial_\nu S^\mu = \frac{\partial}{\partial t}\left(\frac{\partial x^\mu}{\partial s}\right) = \frac{\partial^2 x^\mu}{\partial t \partial s} \tag{3.365}$$

Substituting equations (3.364) and (3.365) back into equation (3.363) gives:

$$[S, T]^\mu = \frac{\partial^2 x^\mu}{\partial s \partial t} - \frac{\partial^2 x^\mu}{\partial t \partial s} \tag{3.366}$$

By Schwarz's theorem, the mixed partial derivatives of the smooth coordinate functions $x^\mu(s, t)$ commute, meaning $\frac{\partial^2 x^\mu}{\partial s \partial t} = \frac{\partial^2 x^\mu}{\partial t \partial s}$. Therefore, equation (3.366) evaluates exactly to zero:

$$[S, T]^\mu = 0 \tag{3.367}$$

Since we established in equation (3.359) that $[S, T]^\mu = S^\nu \nabla_\nu T^\mu - T^\nu \nabla_\nu S^\mu$, setting it equal to zero directly yields the desired identity:

$$S^\nu \nabla_\nu T^\mu - T^\nu \nabla_\nu S^\mu = 0 \implies T^\nu \nabla_\nu S^\mu = S^\nu \nabla_\nu T^\mu \tag{3.368}$$

Relative Velocity and Acceleration

We want to see how the separation S^μ changes as we move along the geodesic (increment t). The **relative velocity** of the geodesics is the rate of change of S^μ:

$$V^\mu = \frac{DS^\mu}{dt} = T^\nu \nabla_\nu S^\mu. \tag{3.369}$$

The **relative acceleration** is the second rate of change:

$$A^\mu = \frac{D^2 S^\mu}{dt^2} = \frac{D}{dt}(V^\mu) = T^\rho \nabla_\rho (T^\nu \nabla_\nu S^\mu). \tag{3.370}$$

Derivation of the Geodesic Deviation Equation

We now manipulate the expression for A^μ to find a relation to curvature. Start with the definition:

$$A^\mu = T^\rho \nabla_\rho (T^\nu \nabla_\nu S^\mu). \tag{3.371}$$

Step 1: Use the commutator property (3.358) to swap T and S inside the parentheses ($\nabla_T S \to \nabla_S T$):

$$A^\mu = T^\rho \nabla_\rho (S^\nu \nabla_\nu T^\mu). \tag{3.372}$$

Step 2: Apply the Leibniz rule to expand the outer derivative:

$$A^\mu = (T^\rho \nabla_\rho S^\nu)(\nabla_\nu T^\mu) + S^\nu T^\rho \nabla_\rho \nabla_\nu T^\mu. \tag{3.373}$$

Step 3: In the second term, we want to swap the order of differentiation $\nabla_\rho \nabla_\nu$ to $\nabla_\nu \nabla_\rho$. The definition of the Riemann tensor tells us:

$$[\nabla_\rho, \nabla_\nu] T^\mu = R^\mu{}_{\sigma\rho\nu} T^\sigma \implies \nabla_\rho \nabla_\nu T^\mu = \nabla_\nu \nabla_\rho T^\mu + R^\mu{}_{\sigma\rho\nu} T^\sigma. \tag{3.374}$$

Substitute this into our expression for A^μ:

$$A^\mu = (T^\rho \nabla_\rho S^\nu)(\nabla_\nu T^\mu) + S^\nu T^\rho \left(\nabla_\nu \nabla_\rho T^\mu + R^\mu{}_{\sigma\rho\nu} T^\sigma \right). \tag{3.375}$$

Step 4: Distribute terms. The term with the Riemann tensor is:

$$S^\nu T^\rho R^\mu{}_{\sigma\rho\nu} T^\sigma = R^\mu{}_{\sigma\rho\nu} T^\sigma T^\rho S^\nu. \tag{3.376}$$

The remaining derivative term is $S^\nu T^\rho \nabla_\nu \nabla_\rho T^\mu$. We use Leibniz rule "in reverse" to pull S^ν outside:

$$S^\nu T^\rho \nabla_\nu \nabla_\rho T^\mu = S^\nu \nabla_\nu (T^\rho \nabla_\rho T^\mu) - S^\nu (\nabla_\nu T^\rho)(\nabla_\rho T^\mu). \tag{3.377}$$

Step 5: Apply the Geodesic Equation. Since T is a geodesic tangent vector, $T^\rho \nabla_\rho T^\mu = 0$. Therefore:

$$S^\nu \nabla_\nu (\underbrace{T^\rho \nabla_\rho T^\mu}_{0}) = 0. \tag{3.378}$$

So the second derivative term simplifies to just $-(S^\nu \nabla_\nu T^\rho)(\nabla_\rho T^\mu)$. Combining everything, we have:

$$A^\mu = \underbrace{(T^\rho \nabla_\rho S^\nu)(\nabla_\nu T^\mu)}_{\text{Term 1}} - \underbrace{(S^\nu \nabla_\nu T^\rho)(\nabla_\rho T^\mu)}_{\text{Term 2}} + R^\mu{}_{\sigma\rho\nu} T^\sigma T^\rho S^\nu. \tag{3.379}$$

Look closely at Term 1 and Term 2. Using the commutator property again ($T^\rho \nabla_\rho S^\nu = S^\rho \nabla_\rho T^\nu$): Term 1 becomes $(S^\rho \nabla_\rho T^\nu)(\nabla_\nu T^\mu)$. This cancels exactly with Term 2 (after relabeling dummy indices).

We are left with the **Geodesic Deviation Equation**:

$$\frac{D^2 S^\mu}{dt^2} = R^\mu{}_{\nu\rho\sigma} T^\nu T^\rho S^\sigma. \tag{3.380}$$

(Note: Indices on Riemann were relabeled to match standard form $R^\mu{}_{\nu\rho\sigma} T^\nu T^\rho S^\sigma$).

Physical Interpretation: Tidal Forces

This equation is the precise mathematical statement that **gravity is curvature**.

- **LHS:** The relative acceleration of two nearby particles falling freely. In Newtonian physics, this is the "tidal force" (difference in gravitational force).
- **RHS:** The Riemann tensor. This shows that tidal acceleration is proportional to curvature, the square of the velocity, and the separation distance.

Example: Falling towards Earth Consider two particles falling radially towards the Earth from space.

- **Math:** They move on radial geodesics. As they get closer to Earth, the geodesics converge toward the center of the Earth. The separation vector S^μ (pointing horizontally between them) shrinks. Thus, there is a relative acceleration $\frac{D^2 S}{dt^2}$ pointing inward (negative).
- **Physics:** The Newtonian tidal tensor is $\mathcal{E}_{ij} = \partial_i \partial_j \Phi$. The equation becomes roughly $\frac{d^2 S^i}{dt^2} \approx -\partial_i \partial_j \Phi S^j$.
- **Geometry:** This convergence proves spacetime is curved ($R \neq 0$). If spacetime were flat, parallel paths would stay parallel forever ($A^\mu = 0$).

This equation is frequently used to detect gravitational waves. As a wave passes, the curvature R oscillates, causing the distance S between test masses in an interferometer (like LIGO) to oscillate.

3.12 Supplement: Stokes's Theorem and Conservation Laws

The Generalized Stokes's Theorem

In calculus, you learned the Fundamental Theorem of Calculus: $\int_a^b \frac{df}{dx} dx = f(b) - f(a)$. This relates an integral over a volume (the interval) to values on the boundary (the endpoints). Stokes's Theorem generalizes this to manifolds of any dimension n. Let M be an n-dimensional manifold with boundary ∂M. Let ω be an $(n-1)$-form. The theorem states:

$$\int_M d\omega = \int_{\partial M} \omega. \tag{3.381}$$

This single elegant equation encompasses the divergence theorem, Green's theorem, and classical Stokes's theorem.

Intuition: The exterior derivative d measures how a form changes "locally." Integrating this change over the whole interior sums up to the net "flow" across the boundary. It is the ultimate statement that "what happens inside must cross the edge to get out."

Deriving the Vector Calculus Version

To make this useful for physics (like Electromagnetism), we need to translate it into vector notation using the metric and covariant derivatives.

Step 1: Relate the form to a vector. Let the $(n-1)$-form ω be the Hodge dual of a vector field V^μ:

$$\omega = *V. \tag{3.382}$$

In components, using the Levi-Civita tensor ϵ:

$$\omega_{\mu_1 \dots \mu_{n-1}} = \epsilon_{\nu \mu_1 \dots \mu_{n-1}} V^\nu. \tag{3.383}$$

Step 2: Calculate the exterior derivative. The exterior derivative $d\omega$ is an n-form. Any n-form is proportional to the volume element ϵ. We can find the proportionality factor by taking the dual again. Identity: For a vector V, the divergence is related to the duals by:

$$*d(*V) = (-1)^s \nabla_\mu V^\mu, \tag{3.384}$$

where s depends on the signature (-1 for Lorentzian, $+1$ for Euclidean). Therefore, the n-form $d\omega$ is:

$$d\omega = (\nabla_\mu V^\mu)\epsilon = (\nabla_\mu V^\mu)\sqrt{|g|}d^n x. \tag{3.385}$$

This handles the left-hand side of Stokes's theorem: it is the integral of the divergence.

Step 3: Handle the boundary integral. The boundary ∂M is an $(n-1)$-dimensional hypersurface. It has an induced metric γ_{ij} and an induced volume element $\hat{\epsilon}$. The restriction of our form ω to the boundary is determined by the unit normal vector n^μ to the boundary:

$$\omega|_{\partial M} = (n_\mu V^\mu)\hat{\epsilon} = (n_\mu V^\mu)\sqrt{|\gamma|}d^{n-1}y. \tag{3.386}$$

Orientation convention:

- If the boundary is **timelike**, n^μ points **inward**.
- If the boundary is **spacelike**, n^μ points **outward**.

Step 4: The Final Formula. Equating the two sides gives the Divergence Theorem on curved manifolds:

$$\int_M \nabla_\mu V^\mu \sqrt{|g|}d^n x = \int_{\partial M} n_\mu V^\mu \sqrt{|\gamma|}d^{n-1}y. \tag{3.387}$$

Intuition: The total "stuff" created inside a region (sum of divergence) equals the flux of "stuff" pushing out through the walls (normal component on boundary).

Application: Conservation of Charge

Assume we have a current vector J^μ that is locally conserved:

$$\nabla_\mu J^\mu = 0. \tag{3.388}$$

We define the total charge Q_Σ on a spacelike hypersurface Σ (like "all space at time t") as:

$$Q_\Sigma = -\int_\Sigma *J = -\int_\Sigma n_\mu J^\mu \sqrt{|\gamma|}d^{n-1}y. \tag{3.389}$$

(The minus sign is a convention to make charge positive when n^μ is future-pointing).

Does this charge change with time? Consider a 4D region R bounded by two time slices Σ_1 (past) and Σ_2 (future). Apply Stokes's theorem to the current J over region R:

$$\int_R \nabla_\mu J^\mu \sqrt{|g|}d^4 x = \int_{\partial R} n_\mu J^\mu \sqrt{|\gamma|}d^3 y. \tag{3.390}$$

The LHS is zero because the current is conserved. The RHS is the sum of integrals over the boundaries Σ_1 and Σ_2 (assuming fields vanish at spatial infinity). Note that on Σ_1 the normal points inward (future), and on Σ_2 it points outward (future). However, the boundary orientation convention in Stokes's theorem implies a relative minus sign.

$$0 = Q(\Sigma_2) - Q(\Sigma_1) \implies Q(\Sigma_1) = Q(\Sigma_2). \tag{3.391}$$

Thus, charge is globally conserved: the value is independent of the choice of time slice.

Application: Gauss's Law

We can calculate charge using only the field at the boundary (spatial infinity), without integrating over the interior volume. Maxwell's equations relate the field strength $F^{\mu\nu}$ to the current:

$$\nabla_\nu F^{\mu\nu} = J^\mu. \tag{3.392}$$

Substitute this into the charge definition:

$$Q = -\int_\Sigma n_\mu J^\mu \sqrt{|\gamma|} d^3y = -\int_\Sigma n_\mu (\nabla_\nu F^{\mu\nu}) \sqrt{|\gamma|} d^3y. \tag{3.393}$$

Now, apply Stokes's theorem *again*, but on the spatial hypersurface Σ itself. The boundary of the spatial volume Σ is the "surface at infinity" $\partial\Sigma$ (a 2-sphere).

$$Q = \int_{\partial\Sigma} n_\mu \sigma_\nu F^{\mu\nu} \sqrt{|\gamma^{(2)}|} d^2z. \tag{3.394}$$

Here:

- Σ is the 3D space.
- $\partial\Sigma$ is the 2D boundary sphere at infinity.
- n_μ is the normal to time (timelike).
- σ_ν is the normal to the sphere (radial spacelike).

Example: Point Charge in Minkowski Space Metric: $ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2$. Electric field: $E^r = \frac{q}{4\pi r^2}$ (radial). Field tensor: $F^{tr} = E^r$. Normals: $n_\mu = (1, 0, 0, 0)$ (time), $\sigma_\nu = (0, 1, 0, 0)$ (radial). Contract the normals with the tensor:

$$n_\mu \sigma_\nu F^{\mu\nu} = n_t \sigma_r F^{tr} = (-1)(1)E^r = -\frac{q}{4\pi r^2}. \tag{3.395}$$

(Note: n_μ is covariant, so $n_t = -1$ in $- + ++$ signature). Integrate over the sphere at infinity. Area element $\sqrt{|\gamma^{(2)}|} d^2z = r^2 \sin\theta d\theta d\phi$.

$$Q = \int_{S^2} \left(-\frac{q}{4\pi r^2}\right) (r^2 d\Omega) \times (-1 \text{ from formula sign}) = \frac{q}{4\pi} \int d\Omega = \frac{q}{4\pi}(4\pi) = q. \tag{3.396}$$

It works! We recovered the charge q solely by integrating the field flux at infinity.

Chapter 4

Gravitation

4.1 Physics in Curved Spacetime

Having established the mathematical framework of differential geometry, we are now prepared to examine the physics of gravitation as described by General Relativity. The subject fundamentally divides into two distinct but interconnected phenomena: how the gravitational field (spacetime curvature) influences the behavior of matter, and conversely, how matter determines the gravitational field.

In Newtonian gravity, these two elements are governed by two specific equations. First, the response of a body to a gravitational potential Φ is given by its acceleration:

$$\mathbf{a} = -\nabla\Phi \quad \Longrightarrow \quad \frac{d^2x^i}{dt^2} = -\partial_i\Phi. \tag{4.1}$$

Second, the generation of this potential by a matter density ρ is governed by Poisson's differential equation:

$$\nabla^2\Phi = 4\pi G\rho, \tag{4.2}$$

where G is Newton's gravitational constant.

In General Relativity, we must replace these statements with laws that describe how the curvature of spacetime acts on matter to manifest as gravity, and how energy and momentum influence spacetime to create curvature.

The Minimal-Coupling Principle

The foundation for generalizing laws of physics from flat Minkowski spacetime to curved spacetime rests upon the Einstein Equivalence Principle (EEP). The EEP asserts that in sufficiently small regions of spacetime, the laws of physics reduce to those of Special Relativity, making it impossible to locally detect a gravitational field. This universality of gravity suggests it is not a conventional force propagating through spacetime, but rather a fundamental feature of the background manifold itself.

This philosophy yields a direct operational recipe for translating physical laws into a curved-spacetime context, known as the **minimal-coupling principle**. The procedure consists of three steps:

1. Take a physical law valid in inertial coordinates in flat spacetime.
2. Write the law in a manifestly coordinate-invariant (tensorial) form.
3. Assert that this tensorial law remains true in curved spacetime.

Mathematically, this "Comma-Goes-to-Semicolon Rule" amounts to replacing the Minkowski metric $\eta_{\mu\nu}$ with the general metric $g_{\mu\nu}$, and replacing partial derivatives ∂_μ with covariant derivatives ∇_μ.

Derivation of the Geodesic Equation

Let us apply the minimal-coupling principle to the motion of a freely-falling particle. In flat spacetime, a free particle moves in a straight line, which is expressed by the vanishing of the second derivative of its parameterized path $x^\mu(\lambda)$:

$$\frac{d^2x^\mu}{d\lambda^2} = 0. \tag{4.3}$$

While $dx^\mu/d\lambda$ transforms as a valid vector (the tangent vector to the curve), the second derivative $d^2x^\mu/d\lambda^2$ does not transform as a tensor in general curvilinear coordinates. To apply our recipe, we must first rewrite equation (4.3) in a tensorial form using the directional derivative along the path.

Using the chain rule, the ordinary derivative with respect to λ can be written as an operator acting on the coordinates:

$$\frac{d}{d\lambda} = \frac{dx^\nu}{d\lambda}\partial_\nu. \tag{4.4}$$

Applying this to the tangent vector $\frac{dx^\mu}{d\lambda}$, equation (4.3) becomes:

$$\frac{dx^\nu}{d\lambda}\partial_\nu\left(\frac{dx^\mu}{d\lambda}\right) = 0. \tag{4.5}$$

We now apply the minimal-coupling principle by elevating the partial derivative to a covariant derivative ($\partial_\nu \to \nabla_\nu$):

$$\frac{dx^\nu}{d\lambda}\nabla_\nu\left(\frac{dx^\mu}{d\lambda}\right) = 0. \tag{4.6}$$

This is the condition that the tangent vector is parallel transported along its own path. We expand the covariant derivative acting on the vector $V^\mu = dx^\mu/d\lambda$:

$$\nabla_\nu V^\mu = \partial_\nu V^\mu + \Gamma^\mu_{\nu\rho}V^\rho. \tag{4.7}$$

Substituting this expansion back into our directional derivative:

$$\begin{aligned} \frac{dx^\nu}{d\lambda}\left[\partial_\nu\left(\frac{dx^\mu}{d\lambda}\right) + \Gamma^\mu_{\nu\rho}\left(\frac{dx^\rho}{d\lambda}\right)\right] &= 0 \\ \frac{dx^\nu}{d\lambda}\partial_\nu\left(\frac{dx^\mu}{d\lambda}\right) + \Gamma^\mu_{\nu\rho}\frac{dx^\nu}{d\lambda}\frac{dx^\rho}{d\lambda} &= 0. \end{aligned} \tag{4.8}$$

Recognizing that $\frac{dx^\nu}{d\lambda}\partial_\nu$ is simply the total derivative $\frac{d}{d\lambda}$ along the curve, we recover the **geodesic equation**:

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0, \tag{4.9}$$

where we have relabeled the dummy indices ν and ρ to ρ and σ respectively to match standard conventions. Thus, free particles in curved spacetime move along geodesics.

Intuition: The extra term involving the Christoffel symbols $\Gamma^\mu_{\rho\sigma}$ corrects for the apparent acceleration caused strictly by the curvature of the coordinate system. Geometrically, this equation demands that the curve is as "straight" as possible within the curved manifold.

Energy-Momentum Conservation

A straightforward second example is the conservation of energy and momentum. In flat spacetime, this is expressed as the vanishing divergence of the energy-momentum tensor:

$$\partial_\mu T^{\mu\nu} = 0. \tag{4.10}$$

Applying the minimal-coupling recipe, the appropriate generalization to curved spacetime immediately becomes:

$$\nabla_\mu T^{\mu\nu} = 0. \tag{4.11}$$

The Newtonian Limit

To prove that this geometric framework legitimately describes gravity, we must show that it reduces to Newtonian gravity in the appropriate limit. The **Newtonian limit** is defined by three strict requirements:

1. **Slow motion:** The particles move at velocities much less than the speed of light ($v \ll c$).
2. **Static field:** The gravitational field is unchanging with time.
3. **Weak field:** The gravitational field is a small perturbation on a flat spacetime background.

1. Simplifying the Geodesic Equation

Let us analyze the geodesic equation under these assumptions, using proper time τ as the affine parameter. The slow motion condition mathematically implies that the spatial components of the four-velocity are negligible compared to the time component:

$$\left|\frac{dx^i}{d\tau}\right| \ll \left|\frac{dt}{d\tau}\right|. \tag{4.12}$$

In the summation term of the geodesic equation, $\Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\tau}\frac{dx^\sigma}{d\tau}$, the dominant term by far will be the one where both $\rho = 0$ and $\sigma = 0$. We may therefore drop the spatial velocity terms, simplifying the equation to:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{00}\left(\frac{dt}{d\tau}\right)^2 = 0. \tag{4.13}$$

2. Evaluating the Christoffel Symbol

We must calculate the relevant connection coefficient, Γ^μ_{00}. By definition:

$$\Gamma^\mu_{00} = \frac{1}{2}g^{\mu\lambda}(\partial_0 g_{\lambda 0} + \partial_0 g_{0\lambda} - \partial_\lambda g_{00}). \tag{4.14}$$

The static field assumption mandates that all time derivatives of the metric vanish ($\partial_0 g_{\alpha\beta} = 0$). This eliminates the first two terms in the parentheses, yielding:

$$\Gamma^\mu_{00} = -\frac{1}{2}g^{\mu\lambda}\partial_\lambda g_{00}. \tag{4.15}$$

3. The Weak Field Approximation

The weak field assumption allows us to write the metric as the flat Minkowski metric $\eta_{\mu\nu}$ plus a small perturbation $h_{\mu\nu}$:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1. \tag{4.16}$$

We work in inertial coordinates with signature $(-+++)$, so $\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$. To compute equation (4.15), we need the inverse metric $g^{\mu\nu}$. We postulate that the inverse metric takes the form $g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}$, where indices on the perturbation are raised using the flat background metric ($h^{\mu\nu} = \eta^{\mu\rho}\eta^{\nu\sigma}h_{\rho\sigma}$). Let us verify this explicit expansion to linear order in h:

$$\begin{aligned}
g^{\mu\lambda}g_{\lambda\nu} &= (\eta^{\mu\lambda} - h^{\mu\lambda})(\eta_{\lambda\nu} + h_{\lambda\nu}) \\
&= \eta^{\mu\lambda}\eta_{\lambda\nu} + \eta^{\mu\lambda}h_{\lambda\nu} - h^{\mu\lambda}\eta_{\lambda\nu} - h^{\mu\lambda}h_{\lambda\nu} \\
&= \delta^\mu_\nu + \eta^{\mu\lambda}h_{\lambda\nu} - (\eta^{\mu\rho}\eta^{\lambda\sigma}h_{\rho\sigma})\eta_{\lambda\nu} + \mathcal{O}(h^2) \\
&= \delta^\mu_\nu + \eta^{\mu\lambda}h_{\lambda\nu} - \eta^{\mu\rho}\delta^\sigma_\nu h_{\rho\sigma} \\
&= \delta^\mu_\nu + \eta^{\mu\lambda}h_{\lambda\nu} - \eta^{\mu\rho}h_{\rho\nu}.
\end{aligned} \tag{4.17}$$

The linear terms perfectly cancel (by relabeling dummy index $\rho \to \lambda$), verifying that $g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}$ is the correct inverse to first order in the perturbation.

We now substitute this into our Christoffel symbol. Since $\partial_\lambda g_{00} = \partial_\lambda(\eta_{00} + h_{00}) = \partial_\lambda h_{00}$ is already of order $\mathcal{O}(h)$, the term $-h^{\mu\lambda}$ from the inverse metric would generate a negligible $\mathcal{O}(h^2)$ term. Thus, we may simply replace the inverse metric with the flat metric:

$$\Gamma^\mu_{00} \approx -\frac{1}{2}\eta^{\mu\lambda}\partial_\lambda h_{00}. \tag{4.18}$$

Substituting this back into the slow-motion geodesic equation (4.13):

$$\frac{d^2x^\mu}{d\tau^2} = \frac{1}{2}\eta^{\mu\lambda}\partial_\lambda h_{00}\left(\frac{dt}{d\tau}\right)^2. \tag{4.19}$$

4. Decomposing Time and Spatial Components

We examine the components of equation (4.19) separately. **Time Component ($\mu = 0$):**

$$\frac{d^2t}{d\tau^2} = \frac{1}{2}\eta^{0\lambda}\partial_\lambda h_{00}\left(\frac{dt}{d\tau}\right)^2 = \frac{1}{2}\eta^{00}\partial_0 h_{00}\left(\frac{dt}{d\tau}\right)^2. \tag{4.20}$$

Because the field is static, $\partial_0 h_{00} = 0$, leading to:

$$\frac{d^2t}{d\tau^2} = 0 \quad\Longrightarrow\quad \frac{dt}{d\tau} = \text{constant}. \tag{4.21}$$

This confirms that in the Newtonian limit, coordinate time t ticks at a constant rate relative to proper time τ.

Spatial Components ($\mu = i$): The spacelike components of the Minkowski inverse metric are simply the 3×3 Euclidean identity matrix, $\eta^{ij} = \delta^{ij}$.

$$\frac{d^2x^i}{d\tau^2} = \frac{1}{2}\eta^{ij}\partial_j h_{00}\left(\frac{dt}{d\tau}\right)^2 = \frac{1}{2}\delta^{ij}\partial_j h_{00}\left(\frac{dt}{d\tau}\right)^2 = \frac{1}{2}\partial_i h_{00}\left(\frac{dt}{d\tau}\right)^2. \tag{4.22}$$

We wish to write this equation purely in terms of coordinate time t, replacing the proper time derivatives. Using the chain rule rigorously:

$$\frac{d}{d\tau} = \frac{dt}{d\tau}\frac{d}{dt} \quad\Longrightarrow\quad \frac{d^2x^i}{d\tau^2} = \frac{d}{d\tau}\left(\frac{dt}{d\tau}\frac{dx^i}{dt}\right). \tag{4.23}$$

Applying the product rule gives:

$$\frac{d^2x^i}{d\tau^2} = \left(\frac{d^2t}{d\tau^2}\right)\frac{dx^i}{dt} + \frac{dt}{d\tau}\frac{d}{d\tau}\left(\frac{dx^i}{dt}\right). \tag{4.24}$$

We established that $d^2t/d\tau^2 = 0$. Re-applying the chain rule to the second term:

$$\frac{d^2x^i}{d\tau^2} = \left(\frac{dt}{d\tau}\right)^2\frac{d^2x^i}{dt^2}. \tag{4.25}$$

Substituting this back into our spatial equation allows the factors of $(dt/d\tau)^2$ to be canceled entirely:

$$\left(\frac{dt}{d\tau}\right)^2\frac{d^2x^i}{dt^2} = \frac{1}{2}\partial_i h_{00}\left(\frac{dt}{d\tau}\right)^2 \quad\Longrightarrow\quad \frac{d^2x^i}{dt^2} = \frac{1}{2}\partial_i h_{00}. \tag{4.26}$$

This is the equation of motion for a particle in a weak, static gravitational field.

5. Connection to the Newtonian Potential

We compare our geometric result with the classical Newtonian equation of motion from equation (4.1):

$$\frac{d^2x^i}{dt^2} = \frac{1}{2}\partial_i h_{00} \quad \text{vs} \quad \frac{d^2x^i}{dt^2} = -\partial_i\Phi. \tag{4.27}$$

For these two formalisms to describe the exact same physical reality, we must identify the metric perturbation with the Newtonian potential:

$$\frac{1}{2}\partial_i h_{00} = -\partial_i\Phi \quad\Longrightarrow\quad h_{00} = -2\Phi. \tag{4.28}$$

Since $g_{00} = \eta_{00} + h_{00}$ and we are using the signature where $\eta_{00} = -1$, the full time-time component of the metric is:

$$g_{00} = -(1 + 2\Phi). \tag{4.29}$$

This proves that the curvature of spacetime is sufficient to reproduce Newtonian gravity, provided the metric takes this specific form.

4.2 Einstein's Equation

Having established the geometrical framework and the response of test particles to curved spacetime, we turn to the second fundamental task of General Relativity: determining the field equations that govern how the metric $g_{\mu\nu}$ responds to the presence of energy and momentum.

Like Maxwell's equations in electrodynamics, the field equation of gravity cannot be rigorously derived from bedrock mathematical principles; it must be postulated and tested against experiment.

4.2.1 Motivating the Tensorial Equation

We seek a relativistic tensor equation that supersedes the classical Poisson equation for the Newtonian gravitational potential Φ:

$$\nabla^2\Phi = 4\pi G\rho, \tag{4.30}$$

where $\nabla^2 = \delta^{ij}\partial_i\partial_j$ is the spatial Laplacian, ρ is the mass density, and G is Newton's gravitational constant.

To generalize this into a coordinate-invariant tensorial form, we analyze its components:

- **The Source (Right-Hand Side):** The mass density ρ generalizes cleanly to the energy-momentum tensor $T_{\mu\nu}$.
- **The Field (Left-Hand Side):** We established previously that the Newtonian potential Φ is embedded within the metric tensor perturbation, $h_{00} = -2\Phi$.

Therefore, we expect the relativistic generalization to relate a symmetric $(0,2)$ tensor, constructed from second derivatives of the metric, to the energy-momentum tensor $T_{\mu\nu}$.

$$[\nabla^2 g]_{\mu\nu} \propto T_{\mu\nu}. \tag{4.31}$$

Our first impulse might be to apply the d'Alembertian operator $\Box = \nabla^\mu\nabla_\mu$ to the metric $g_{\mu\nu}$.

Fortunately, differential geometry provides a unique, non-trivial tensor constructed from up to second derivatives of the metric: the Riemann curvature tensor $R^\rho{}_{\sigma\mu\nu}$.

The Riemann tensor is a rank-4 tensor, whereas $T_{\mu\nu}$ is rank-2. We must contract indices to match ranks.

$$R_{\mu\nu} = \kappa T_{\mu\nu}, \tag{4.32}$$

where κ is an undetermined coupling constant.

4.2.2 The Conservation of Energy Problem

Equation (4.32) contains a fatal flaw regarding conservation laws. In a closed system, local energy and momentum must be conserved, which is expressed geometrically by the vanishing covariant divergence of the energy-momentum tensor:

$$\nabla^\mu T_{\mu\nu} = 0. \tag{4.33}$$

If our proposed field equation $R_{\mu\nu} = \kappa T_{\mu\nu}$ were true, taking the covariant divergence of both sides would require:

$$\nabla^\mu R_{\mu\nu} = 0. \tag{4.34}$$

However, this is not true for a general Riemannian manifold.

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2}\nabla_\nu R, \tag{4.35}$$

where $R = g^{\mu\nu}R_{\mu\nu}$ is the Ricci scalar.

Let us explicitly show the contradiction. If $R_{\mu\nu} = \kappa T_{\mu\nu}$, then taking the trace of both sides yields:

$$g^{\mu\nu}R_{\mu\nu} = \kappa g^{\mu\nu}T_{\mu\nu} \quad \Longrightarrow \quad R = \kappa T. \tag{4.36}$$

Substitute this into the Bianchi identity (4.35):

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2}\nabla_\nu(\kappa T). \tag{4.37}$$

But we assumed $\nabla^\mu R_{\mu\nu} = \kappa\nabla^\mu T_{\mu\nu} = 0$. Therefore, we are forced to conclude:

$$\frac{1}{2}\kappa\nabla_\nu T = 0 \quad\Longrightarrow\quad \nabla_\nu T = 0. \tag{4.38}$$

Since the covariant derivative of a scalar is simply its partial derivative ($\partial_\nu T = 0$), this implies that the trace of the energy-momentum tensor, T, must be a universal constant throughout all of spacetime.

4.2.3 The Einstein Tensor

To resolve this, we must replace the Ricci tensor on the left-hand side with a symmetric $(0, 2)$ tensor whose covariant divergence vanishes identically, irrespective of the specific metric.

We rearrange the contracted Bianchi identity (4.35) by moving all terms to one side:

$$\nabla^\mu R_{\mu\nu} - \frac{1}{2}\nabla_\nu R = 0. \tag{4.39}$$

We can rewrite $\nabla_\nu R$ using the metric tensor, since $\nabla^\mu g_{\mu\nu} = \delta^\mu_\nu$:

$$\nabla^\mu R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}\nabla^\mu R = 0 \quad\Longrightarrow\quad \nabla^\mu\left(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\right) = 0. \tag{4.40}$$

The quantity in the parentheses is the **Einstein Tensor**, $G_{\mu\nu}$:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}. \tag{4.41}$$

By mathematical necessity, the Einstein tensor is always divergence-free ($\nabla^\mu G_{\mu\nu} = 0$).

$$G_{\mu\nu} = \kappa T_{\mu\nu}. \tag{4.42}$$

This equation satisfies all our logical requirements: it equates a symmetric, divergence-free geometric tensor (containing up to second derivatives of the metric) to the symmetric, conserved energy-momentum tensor of matter.

4.2.4 The Trace-Reversed Form

To determine the coupling constant κ, it is highly advantageous to rewrite equation (4.42) by taking its trace. In a 4-dimensional spacetime, we contract both sides with $g^{\mu\nu}$:

$$\begin{aligned} g^{\mu\nu}G_{\mu\nu} &= \kappa g^{\mu\nu}T_{\mu\nu} \\ g^{\mu\nu}\left(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\right) &= \kappa T \\ R - \frac{1}{2}R(g^{\mu\nu}g_{\mu\nu}) &= \kappa T. \end{aligned} \tag{4.43}$$

Since the trace of the metric in 4 dimensions is $g^{\mu\nu}g_{\mu\nu} = \delta^\mu_\mu = 4$, we have:

$$R - \frac{1}{2}R(4) = \kappa T \quad\Longrightarrow\quad R - 2R = \kappa T \quad\Longrightarrow\quad R = -\kappa T. \tag{4.44}$$

We substitute this expression for the Ricci scalar back into the original field equation $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \kappa T_{\mu\nu}$:

$$R_{\mu\nu} - \frac{1}{2}(-\kappa T)g_{\mu\nu} = \kappa T_{\mu\nu}. \tag{4.45}$$

Rearranging to isolate the Ricci tensor yields the **trace-reversed form** of Einstein's Equation:

$$R_{\mu\nu} = \kappa\left(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu}\right). \tag{4.46}$$

Equations (4.42) and (4.46) are mathematically identical; one is simply an algebraic rearrangement of the other.

4.2.5 The Newtonian Limit and the Coupling Constant

To find κ, we evaluate the trace-reversed equation (4.46) in the Newtonian limit, which requires a weak, static gravitational field, and slowly moving particles.

Consider the source of gravity to be a perfect fluid:

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}, \tag{4.47}$$

where ρ is the rest-frame energy density, p is the pressure, and U^μ is the fluid four-velocity.

$$T_{\mu\nu} = \rho U_\mu U_\nu. \tag{4.48}$$

We evaluate this in the rest frame of the massive fluid (e.g., a star), so the spatial components of the four-velocity vanish: $U^\mu = (U^0, 0, 0, 0)$.

To find U^0, we enforce the normalization condition $g_{\mu\nu}U^\mu U^\nu = -1$.

$$g_{00}(U^0)^2 = -1 \quad \Longrightarrow \quad (-1 + h_{00})(U^0)^2 = -1 \quad \Longrightarrow \quad (U^0)^2 = (1 - h_{00})^{-1}. \tag{4.49}$$

Taylor expanding $(1 - h_{00})^{-1} \approx 1 + h_{00}$ to first order in the small perturbation h, we find:

$$U^0 \approx 1 + \frac{1}{2}h_{00}. \tag{4.50}$$

We must also find the covariant component $U_0 = g_{0\nu}U^\nu = g_{00}U^0$:

$$U_0 = (-1 + h_{00})\left(1 + \frac{1}{2}h_{00}\right) \approx -1 - \frac{1}{2}h_{00} + h_{00} = -1 + \frac{1}{2}h_{00}. \tag{4.51}$$

However, since the energy density ρ is already considered small in this nearly-flat limit, retaining the metric perturbation $\mathcal{O}(h)$ inside the velocity terms will only produce negligible second-order $\mathcal{O}(\rho h)$ corrections.

We substitute these into the 00-component of the energy-momentum tensor:

$$T_{00} = \rho(U_0)^2 = \rho(-1)^2 = \rho. \tag{4.52}$$

All other components T_{0i} and T_{ij} vanish to this order. The trace $T = g^{\mu\nu}T_{\mu\nu}$ is dominated entirely by the time component:

$$T = g^{00}T_{00} \approx (-1)\rho = -\rho. \tag{4.53}$$

We now plug $T_{00} = \rho$ and $T = -\rho$ into the 00-component of our trace-reversed field equation (4.46):

$$\begin{aligned} R_{00} &= \kappa\left(T_{00} - \frac{1}{2}Tg_{00}\right) \\ R_{00} &\approx \kappa\left[\rho - \frac{1}{2}(-\rho)(-1)\right] \\ R_{00} &\approx \kappa\left(\rho - \frac{1}{2}\rho\right) = \frac{1}{2}\kappa\rho. \end{aligned} \tag{4.54}$$

Evaluating the Ricci Component R_{00}

To link equation (4.46) to the Newtonian potential, we must explicitly calculate R_{00} in terms of the metric perturbation $h_{\mu\nu}$.

$$R_{00} = R^\lambda{}_{0\lambda 0} = R^0{}_{000} + R^i{}_{0i0}. \tag{4.55}$$

Because the Riemann tensor is antisymmetric in its first two indices ($R^0{}_{000} = -R^0{}_{000}$), the term $R^0{}_{000}$ vanishes identically.

$$R^i{}_{0j0} = \partial_j\Gamma^i_{00} - \partial_0\Gamma^i_{j0} + \Gamma^i_{j\lambda}\Gamma^\lambda_{00} - \Gamma^i_{0\lambda}\Gamma^\lambda_{j0}. \tag{4.56}$$

We apply our Newtonian limit conditions: 1. **Static Field:** All time derivatives vanish ($\partial_0 \to 0$), so $\partial_0 \Gamma^i_{j0} = 0$. 2. **Weak Field:** The Christoffel symbols Γ are strictly linear in the metric perturbation $h_{\mu\nu}$. Therefore, the quadratic products $(\Gamma)^2$ are of order $\mathcal{O}(h^2)$ and can be neglected.

This drastically simplifies the Riemann component to just the first term:

$$R^i{}_{0j0} \approx \partial_j \Gamma^i_{00}. \tag{4.57}$$

Substituting the static, weak-field Christoffel symbol $\Gamma^i_{00} \approx -\frac{1}{2}\delta^{ij}\partial_j h_{00}$ into our expression for the Ricci tensor gives:

$$\begin{aligned} R_{00} = R^i{}_{0i0} &= \partial_i(\Gamma^i_{00}) \\ &= \partial_i \left(-\frac{1}{2}\delta^{ij}\partial_j h_{00} \right) \\ &= -\frac{1}{2}\delta^{ij}\partial_i\partial_j h_{00} = -\frac{1}{2}\nabla^2 h_{00}. \end{aligned} \tag{4.58}$$

Extracting Newton's Constant

We equate our two derived expressions for R_{00}:

$$-\frac{1}{2}\nabla^2 h_{00} = \frac{1}{2}\kappa\rho \quad \Longrightarrow \quad \nabla^2 h_{00} = -\kappa\rho. \tag{4.59}$$

Recall our earlier identification linking the metric perturbation to the Newtonian gravitational potential: $h_{00} = -2\Phi$.

$$\nabla^2(-2\Phi) = -\kappa\rho \quad \Longrightarrow \quad 2\nabla^2\Phi = \kappa\rho \quad \Longrightarrow \quad \nabla^2\Phi = \frac{1}{2}\kappa\rho. \tag{4.60}$$

To perfectly match the classical Poisson equation, $\nabla^2\Phi = 4\pi G\rho$, the coupling constant must be set to $\kappa = 8\pi G$.

Substituting this value of κ back into our proposed equation (4.42) gives the final result—**Einstein's Field Equation**:

$$R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} = 8\pi G T_{\mu\nu}. \tag{4.61}$$

This profound equation tells us how the curvature of spacetime reacts to the presence of energy-momentum.

4.2.6 Vacuum Equations

We will often be interested in Einstein's equation in a vacuum, such as the region outside a star or planet. If $T_{\mu\nu} = 0$, the right-hand side of the trace-reversed equation (4.46) vanishes entirely.

$$R_{\mu\nu} = 0. \tag{4.62}$$

This is a considerably less formidable and highly useful form of the field equations for finding vacuum metrics.

Lagrangian Formulation in Curved Spacetime

1. The Physics: Why the Action Looks Like That

In classical mechanics, a particle takes a path that minimizes (or makes stationary) the action. In field theory, the "path" is the configuration of a field $\Phi(x)$ across all of spacetime.

- **The Flat Space Blueprint:** In flat space, the action is $S = \int \mathcal{L}\, d^n x$.

- **The Curved Space Reality:** In general relativity, the coordinate grid can stretch and warp. A coordinate volume $d^n x$ might represent a huge physical volume in one place and a tiny one in another.

To ensure our physics doesn't change just because our coordinate grid stretched, the action S must be a true, coordinate-independent number (a scalar).

1. We replace the flat-space Lagrangian with a true scalar $\hat{\mathcal{L}}$.
2. We replace regular partial derivatives ∂_μ with covariant derivatives ∇_μ because the field's rate of change must account for the curvature of the space it lives in.
3. We multiply by $\sqrt{-g}$ to convert the coordinate volume $d^n x$ into the invariant physical volume $\sqrt{-g}\, d^n x$.

2. The Math: Integration by Parts in Curved Space

To derive the equations of motion, we need to know how to move a covariant derivative from one term to another inside an integral.

Step A: The Product Rule Take two arbitrary objects, A and B, where one is a vector and the other a scalar (or they combine to form a vector). The covariant derivative obeys the standard Leibniz product rule:

$$\nabla_\mu (AB) = (\nabla_\mu A)B + A(\nabla_\mu B) \tag{4.63}$$

Step B: Rearranging We isolate the term we want to integrate by parts:

$$(\nabla_\mu A)B = \nabla_\mu (AB) - A(\nabla_\mu B) \tag{4.64}$$

Step C: Integrating and Stokes' Theorem Now, integrate all three terms over the physical spacetime volume:

$$\int (\nabla_\mu A)B\sqrt{-g}\, d^n x = \int \nabla_\mu (AB)\sqrt{-g}\, d^n x - \int A(\nabla_\mu B)\sqrt{-g}\, d^n x \tag{4.65}$$

The middle term is the integral of a total divergence. According to the generalized Stokes' theorem (the Divergence Theorem), the volume integral of a divergence turns into a surface integral over the boundary of that volume. Because we assume our field variations vanish at the boundaries of our universe (or at infinity), that boundary term is zero. We are left with the integration by parts formula:

$$\int (\nabla_\mu A)B\sqrt{-g}\, d^n x = -\int A(\nabla_\mu B)\sqrt{-g}\, d^n x \tag{4.66}$$

3. Deriving the Euler-Lagrange Equations

We vary the action $S = \int \hat{\mathcal{L}}\sqrt{-g}\, d^n x$ with respect to the field Φ by introducing a tiny perturbation: $\Phi \to \Phi + \delta\Phi$. We require the change in the action (δS) to be zero. Using the multivariable chain rule:

$$\delta S = \int \left[\frac{\partial \hat{\mathcal{L}}}{\partial \Phi}\delta\Phi + \frac{\partial \hat{\mathcal{L}}}{\partial(\nabla_\mu \Phi)}\delta(\nabla_\mu \Phi) \right] \sqrt{-g}\, d^n x = 0 \tag{4.67}$$

Because the variation δ and the derivative ∇_μ commute, we write $\delta(\nabla_\mu \Phi) = \nabla_\mu(\delta\Phi)$.

$$\delta S = \int \left[\frac{\partial \hat{\mathcal{L}}}{\partial \Phi}\delta\Phi + \frac{\partial \hat{\mathcal{L}}}{\partial(\nabla_\mu \Phi)}\nabla_\mu(\delta\Phi) \right] \sqrt{-g}\, d^n x = 0 \tag{4.68}$$

Applying our integration by parts formula to the second term (setting $A = \delta\Phi$ and $B = \frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\Phi)}$) gives:

$$\delta S = \int \left[\frac{\partial\hat{\mathcal{L}}}{\partial\Phi}\delta\Phi - \nabla_\mu \left(\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\Phi)} \right) \delta\Phi \right] \sqrt{-g}\, d^n x = 0 \tag{4.69}$$

Factoring out $\delta\Phi$:

$$\delta S = \int \left[\frac{\partial\hat{\mathcal{L}}}{\partial\Phi} - \nabla_\mu \left(\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\Phi)} \right) \right] \delta\Phi\sqrt{-g}\, d^n x = 0 \tag{4.70}$$

For this integral to vanish for *any* arbitrary variation $\delta\Phi$, the term inside the brackets must be identically zero everywhere. This yields the Euler-Lagrange equation.

4. Expanding the Scalar Field Example

The scalar Lagrangian is given by:

$$\hat{\mathcal{L}} = -\frac{1}{2}g^{\alpha\beta}(\nabla_\alpha\phi)(\nabla_\beta\phi) - V(\phi) \tag{4.71}$$

(Note: We use α and β as dummy indices to avoid confusion with the μ in the Euler-Lagrange equation.)

Step 1: Derivative with respect to ϕ The kinetic term doesn't contain ϕ directly, so we only differentiate the potential:

$$\frac{\partial\hat{\mathcal{L}}}{\partial\phi} = -\frac{dV}{d\phi} \tag{4.72}$$

Step 2: Derivative with respect to $\nabla_\mu\phi$ We apply the product rule to the kinetic term:

$$\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\phi)} = -\frac{1}{2}g^{\alpha\beta}\left[\frac{\partial(\nabla_\alpha\phi)}{\partial(\nabla_\mu\phi)}(\nabla_\beta\phi) + (\nabla_\alpha\phi)\frac{\partial(\nabla_\beta\phi)}{\partial(\nabla_\mu\phi)} \right] \tag{4.73}$$

The derivative $\frac{\partial(\nabla_\alpha\phi)}{\partial(\nabla_\mu\phi)}$ is the Kronecker delta δ^μ_α:

$$\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\phi)} = -\frac{1}{2}g^{\alpha\beta}\left[\delta^\mu_\alpha(\nabla_\beta\phi) + (\nabla_\alpha\phi)\delta^\mu_\beta \right] \tag{4.74}$$

The deltas force the indices to match μ:

$$\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\phi)} = -\frac{1}{2}\left[g^{\mu\beta}\nabla_\beta\phi + g^{\alpha\mu}\nabla_\alpha\phi \right] \tag{4.75}$$

Because the metric is symmetric ($g^{\alpha\mu} = g^{\mu\alpha}$), these two terms are identical and cancel the 1/2:

$$\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\phi)} = -g^{\mu\nu}\nabla_\nu\phi \tag{4.76}$$

Step 3: Plug into the Euler-Lagrange Equation Substitute our results from Step 1 and Step 2 into the master equation:

$$-\frac{dV}{d\phi} - \nabla_\mu\left(-g^{\mu\nu}\nabla_\nu\phi\right) = 0 \tag{4.77}$$

Because the metric is compatible with the covariant derivative ($\nabla_\mu g^{\mu\nu} = 0$), we can pull the metric outside the derivative:

$$-\frac{dV}{d\phi} + g^{\mu\nu}\nabla_\mu\nabla_\nu\phi = 0 \tag{4.78}$$

Recognizing $g^{\mu\nu}\nabla_\mu\nabla_\nu$ as the d'Alembertian operator $\Box$, we arrive at the final equation of motion:

$$\Box\phi - \frac{dV}{d\phi} = 0 \tag{4.79}$$

Summary Table

4.2.7 The Hilbert Action

We now turn to constructing an action for gravity itself.

Because the Equivalence Principle ensures that the metric can always be set to its canonical Minkowski form ($\eta_{\mu\nu}$) and its first derivatives can be set to zero ($\partial_\lambda g_{\mu\nu} = 0$) at any single point, any non-trivial scalar must involve at least *second* derivatives of the metric.

It can be rigorously proven that any non-trivial tensor made from products of the metric and its derivatives up to second order can be expressed entirely in terms of the metric and the Riemann tensor.

Recognizing this uniqueness, David Hilbert proposed that the Ricci scalar was the simplest and most natural choice for the gravitational Lagrangian.

$$S_H = \int \sqrt{-g} R \, d^n x. \tag{4.80}$$

4.2.8 Varying the Hilbert Action

To find the vacuum field equations, we must find the stationary points of the Hilbert action by applying the variational principle: $\delta S_H = 0$.

It is computationally more convenient to vary the action with respect to the *inverse* metric, $\delta g^{\mu\nu}$.

$$(\delta g^{\mu\lambda}) g_{\lambda\nu} + g^{\mu\lambda}(\delta g_{\lambda\nu}) = 0. \tag{4.81}$$

Multiplying by $g^{\nu\sigma}$ isolates $\delta g_{\mu\nu}$:

$$\delta g_{\mu\nu} = -g_{\mu\rho} g_{\nu\sigma} \delta g^{\rho\sigma}. \tag{4.82}$$

Thus, finding stationary points with respect to $\delta g^{\mu\nu}$ is entirely equivalent to finding them with respect to $\delta g_{\mu\nu}$.

We expand the Ricci scalar as $R = g^{\mu\nu} R_{\mu\nu}$.

$$\begin{aligned}
\delta S_H &= \delta \int d^n x \sqrt{-g}(g^{\mu\nu} R_{\mu\nu}) \\
&= \int d^n x \left[\sqrt{-g} g^{\mu\nu}(\delta R_{\mu\nu}) + \sqrt{-g} R_{\mu\nu}(\delta g^{\mu\nu}) + R(g^{\mu\nu} R_{\mu\nu})(\delta\sqrt{-g})\right] \\
&\equiv (\delta S)_1 + (\delta S)_2 + (\delta S)_3.
\end{aligned} \tag{4.83}$$

The middle term, $(\delta S)_2 = \int d^n x \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu}$, is already in the desired form (an expression multiplied by $\delta g^{\mu\nu}$).

Evaluating $(\delta S)_1$: Variation of the Ricci Tensor and the Boundary Term

The first term requires evaluating the variation of the Ricci tensor, $\delta R_{\mu\nu}$, which originates from the variation of the Riemann tensor. The Riemann tensor is defined entirely by the Levi-Civita connection (Christoffel symbols):

$$R^\rho{}_{\mu\lambda\nu} = \partial_\lambda \Gamma^\rho_{\nu\mu} - \partial_\nu \Gamma^\rho_{\lambda\mu} + \Gamma^\rho_{\lambda\sigma}\Gamma^\sigma_{\nu\mu} - \Gamma^\rho_{\nu\sigma}\Gamma^\sigma_{\lambda\mu}. \tag{4.84}$$

We evaluate the variation by treating the connection as the fundamental variable, shifting it by a small amount: $\Gamma^\rho_{\nu\mu} \to \Gamma^\rho_{\nu\mu} + \delta\Gamma^\rho_{\nu\mu}$. Crucially, while a single connection Γ is not a tensor, the *difference* between two connections, $\delta\Gamma^\rho_{\nu\mu}$, **is** a valid tensor. Because it is a tensor, we can take its covariant derivative.

The covariant derivative of this variation tensor is:

$$\nabla_\lambda(\delta\Gamma^\rho_{\nu\mu}) = \partial_\lambda(\delta\Gamma^\rho_{\nu\mu}) + \Gamma^\rho_{\lambda\sigma}\delta\Gamma^\sigma_{\nu\mu} - \Gamma^\sigma_{\lambda\nu}\delta\Gamma^\rho_{\sigma\mu} - \Gamma^\sigma_{\lambda\mu}\delta\Gamma^\rho_{\nu\sigma}. \tag{4.85}$$

By explicitly calculating the first-order variation of the Riemann tensor $\delta R^\rho{}_{\mu\lambda\nu}$ from its definition and comparing it to equation (4.98), the non-derivative terms perfectly cancel, yielding the **Palatini identity**:

$$\delta R^\rho{}_{\mu\lambda\nu} = \nabla_\lambda(\delta\Gamma^\rho_{\nu\mu}) - \nabla_\nu(\delta\Gamma^\rho_{\lambda\mu}). \tag{4.86}$$

Contracting the first and third indices ($\rho = \lambda$) gives the variation of the Ricci tensor:

$$\delta R_{\mu\nu} = \nabla_\rho(\delta\Gamma^\rho_{\nu\mu}) - \nabla_\nu(\delta\Gamma^\rho_{\rho\mu}). \tag{4.87}$$

We substitute this into the integral for $(\delta S)_1$:

$$(\delta S)_1 = \int d^n x\sqrt{-g} g^{\mu\nu}\left[\nabla_\rho(\delta\Gamma^\rho_{\nu\mu}) - \nabla_\nu(\delta\Gamma^\rho_{\rho\mu})\right]. \tag{4.88}$$

Because the Levi-Civita connection is metric-compatible ($\nabla_\rho g^{\mu\nu} = 0$), we can pull the inverse metric inside the covariant derivatives. Relabeling dummy indices allows us to factor the expression into a single covariant divergence of a vector field V^σ:

$$(\delta S)_1 = \int d^n x\sqrt{-g}\nabla_\sigma\left[g^{\mu\nu}(\delta\Gamma^\sigma_{\mu\nu}) - g^{\mu\sigma}(\delta\Gamma^\lambda_{\lambda\mu})\right] \equiv \int d^n x\sqrt{-g}\nabla_\sigma V^\sigma. \tag{4.89}$$

To rigorously prove that this term does not contribute to the equations of motion, we must express the vector V^σ purely in terms of the metric variation $\delta g_{\mu\nu}$. The variation of the Christoffel symbol in terms of the metric is:

$$\delta\Gamma^\sigma_{\mu\nu} = \frac{1}{2}g^{\sigma\rho}\left(\nabla_\mu\delta g_{\nu\rho} + \nabla_\nu\delta g_{\mu\rho} - \nabla_\rho\delta g_{\mu\nu}\right). \tag{4.90}$$

Substituting this into the two components of V^σ yields:

$$g^{\mu\nu}\delta\Gamma^\sigma_{\mu\nu} = \frac{1}{2}g^{\mu\nu}g^{\sigma\rho}\left(2\nabla_\mu\delta g_{\nu\rho} - \nabla_\rho\delta g_{\mu\nu}\right) = \nabla_\mu(\delta g^{\sigma\mu}) - \nabla^\sigma(\delta g^\mu_\mu), \tag{4.91}$$

$$g^{\mu\sigma}\delta\Gamma^\lambda_{\lambda\mu} = \frac{1}{2}g^{\mu\sigma}g^{\lambda\rho}\left(\nabla_\lambda\delta g_{\mu\rho} + \nabla_\mu\delta g_{\lambda\rho} - \nabla_\rho\delta g_{\lambda\mu}\right) = \frac{1}{2}\nabla^\sigma(\delta g^\lambda_\lambda) + \frac{1}{2}\nabla^\sigma(\delta g^\lambda_\lambda) - \frac{1}{2}\nabla^\sigma(\delta g^\lambda_\lambda) = \nabla^\sigma(\delta g^\mu_\mu). \tag{4.92}$$

Subtracting these results provides the explicit form of the boundary vector:

$$V^\sigma = \nabla_\mu(\delta g^{\sigma\mu}) - \nabla^\sigma(\delta g^\mu_\mu). \tag{4.93}$$

Thus, $(\delta S)_1$ is the spacetime integral of a pure covariant divergence. By the generalized Stokes's theorem, this volume integral converts perfectly into a surface integral over the boundary of the spacetime manifold $\partial\mathcal{M}$:

$$(\delta S)_1 = \int_{\mathcal{M}} d^n x\sqrt{-g}\nabla_\sigma V^\sigma = \int_{\partial\mathcal{M}} d^{n-1}y\sqrt{|h|}n_\sigma\left[\nabla_\mu(\delta g^{\sigma\mu}) - \nabla^\sigma(\delta g^\mu_\mu)\right], \tag{4.94}$$

where n_σ is the normal vector to the boundary and h is the determinant of the induced boundary metric.

In the standard formulation of the variational principle (Dirichlet boundary conditions), we demand that the metric variation and its first derivatives vanish on the boundary ($\delta g_{\mu\nu}|_{\partial\mathcal{M}} = 0$ and $\nabla_\rho\delta g_{\mu\nu}|_{\partial\mathcal{M}} = 0$). Under these strict boundary conditions, the entire surface integral evaluates to exactly zero. Therefore, the first term contributes nothing to the dynamical field equations:

$$(\delta S)_1 = 0. \tag{4.95}$$

4.3 The Palatini Formulation and Identity

4.3.1 Proof the Palatini Identity

To vary the action with respect to the connection, we must understand how the Riemann and Ricci tensors change under a small variation $\Gamma^\rho_{\mu\nu} \to \Gamma^\rho_{\mu\nu} + \delta\Gamma^\rho_{\mu\nu}$.

By definition, the Riemann tensor constructed from an arbitrary connection is:

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}. \tag{4.96}$$

Before varying, we must establish a crucial tensorial property. Under a general coordinate transformation $x \to x'$, a connection transforms as:

$$\Gamma^{\rho'}_{\mu'\nu'} = \frac{\partial x^{\rho'}}{\partial x^{\rho}} \frac{\partial x^{\mu}}{\partial x^{\mu'}} \frac{\partial x^{\nu}}{\partial x^{\nu'}} \Gamma^{\rho}_{\mu\nu} + \frac{\partial x^{\rho'}}{\partial x^{\lambda}} \frac{\partial^2 x^{\lambda}}{\partial x^{\mu'} \partial x^{\nu'}}. \tag{4.97}$$

Because of the second term involving the second derivative of the coordinates, the connection itself is *not* a tensor. However, if we take the difference of two valid connections defined on the same manifold, $\delta\Gamma^{\rho}_{\mu\nu} = \hat{\Gamma}^{\rho}_{\mu\nu} - \Gamma^{\rho}_{\mu\nu}$, the non-tensorial second-derivative terms perfectly subtract out. Thus, the variation $\delta\Gamma^{\rho}_{\mu\nu}$ **is a true tensor of rank (1,2)**.

Because $\delta\Gamma^{\rho}_{\mu\nu}$ is a tensor, we can meaningfully take its covariant derivative with respect to the background connection Γ. The expansion of this covariant derivative is:

$$\nabla_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) = \partial_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) + \Gamma^{\rho}_{\mu\lambda}\delta\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\lambda}_{\mu\nu}\delta\Gamma^{\rho}_{\lambda\sigma} - \Gamma^{\lambda}_{\mu\sigma}\delta\Gamma^{\rho}_{\nu\lambda}. \tag{4.98}$$

We now apply the variation directly to the definition of the Riemann tensor in equation (4.96). Dropping terms of order $\mathcal{O}(\delta\Gamma^2)$, we obtain the first-order variation:

$$\delta R^{\rho}{}_{\sigma\mu\nu} = \partial_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) - \partial_{\nu}(\delta\Gamma^{\rho}_{\mu\sigma}) + (\delta\Gamma^{\rho}_{\mu\lambda})\Gamma^{\lambda}_{\nu\sigma} + \Gamma^{\rho}_{\mu\lambda}(\delta\Gamma^{\lambda}_{\nu\sigma}) - (\delta\Gamma^{\rho}_{\nu\lambda})\Gamma^{\lambda}_{\mu\sigma} - \Gamma^{\rho}_{\nu\lambda}(\delta\Gamma^{\lambda}_{\mu\sigma}). \tag{4.99}$$

We aim to express this variation not in terms of partial derivatives, but in terms of the covariant derivatives we defined in equation (4.98). Let us explicitly calculate the difference of two covariant derivatives of our variation tensor:

$$\begin{aligned}\nabla_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) - \nabla_{\nu}(\delta\Gamma^{\rho}_{\mu\sigma}) = &\left[\partial_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) + \Gamma^{\rho}_{\mu\lambda}\delta\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\lambda}_{\mu\nu}\delta\Gamma^{\rho}_{\lambda\sigma} - \Gamma^{\lambda}_{\mu\sigma}\delta\Gamma^{\rho}_{\nu\lambda}\right] \\ &- \left[\partial_{\nu}(\delta\Gamma^{\rho}_{\mu\sigma}) + \Gamma^{\rho}_{\nu\lambda}\delta\Gamma^{\lambda}_{\mu\sigma} - \Gamma^{\lambda}_{\nu\mu}\delta\Gamma^{\rho}_{\lambda\sigma} - \Gamma^{\lambda}_{\nu\sigma}\delta\Gamma^{\rho}_{\mu\lambda}\right].\end{aligned} \tag{4.100}$$

We distribute the negative sign and group the terms:

$$\begin{aligned}\nabla_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) - \nabla_{\nu}(\delta\Gamma^{\rho}_{\mu\sigma}) =& \partial_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) - \partial_{\nu}(\delta\Gamma^{\rho}_{\mu\sigma}) \\ &+ \Gamma^{\rho}_{\mu\lambda}\delta\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\delta\Gamma^{\lambda}_{\mu\sigma} \\ &- \Gamma^{\lambda}_{\mu\nu}\delta\Gamma^{\rho}_{\lambda\sigma} + \Gamma^{\lambda}_{\nu\mu}\delta\Gamma^{\rho}_{\lambda\sigma} \\ &- \Gamma^{\lambda}_{\mu\sigma}\delta\Gamma^{\rho}_{\nu\lambda} + \Gamma^{\lambda}_{\nu\sigma}\delta\Gamma^{\rho}_{\mu\lambda}.\end{aligned} \tag{4.101}$$

Because we assumed the connection is torsion-free ($\Gamma^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\nu\mu}$), the two terms in the third line cancel each other exactly. Rearranging the remaining terms yields:

$$\nabla_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) - \nabla_{\nu}(\delta\Gamma^{\rho}_{\mu\sigma}) = \partial_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) - \partial_{\nu}(\delta\Gamma^{\rho}_{\mu\sigma}) + \Gamma^{\rho}_{\mu\lambda}\delta\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\delta\Gamma^{\lambda}_{\mu\sigma} + \Gamma^{\lambda}_{\nu\sigma}\delta\Gamma^{\rho}_{\mu\lambda} - \Gamma^{\lambda}_{\mu\sigma}\delta\Gamma^{\rho}_{\nu\lambda}. \tag{4.102}$$

Comparing this result term-by-term with equation (4.99), we find an exact match. This establishes the **Palatini identity** for the variation of the Riemann tensor:

$$\delta R^{\rho}{}_{\sigma\mu\nu} = \nabla_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) - \nabla_{\nu}(\delta\Gamma^{\rho}_{\mu\sigma}). \tag{4.103}$$

To find the variation of the Ricci tensor, we contract the first and third indices ($\rho = \mu$). Relabeling the remaining indices to conventional forms (letting $\sigma \to \mu$ and $\nu \to \nu$), we obtain the variation of the Ricci tensor:

$$\delta R_{\mu\nu} = \nabla_{\lambda}(\delta\Gamma^{\lambda}_{\mu\nu}) - \nabla_{\nu}(\delta\Gamma^{\lambda}_{\mu\lambda}). \tag{4.104}$$

This beautifully compact equation demonstrates that the variation of the Ricci tensor is simply the difference of two covariant divergences.

4.3.2 Variation of the Action and Integration by Parts

We now return to the Palatini action in equation (**??**). The total variation of the action δS must vanish for arbitrary variations of both $g^{\mu\nu}$ and $\Gamma^{\rho}_{\mu\nu}$.

Varying with respect to $g^{\mu\nu}$ trivially recovers the standard algebraic relation $R_{\mu\nu}(\Gamma) - \frac{1}{2}R(\Gamma)g_{\mu\nu} = 0$, exactly as in the Einstein-Hilbert derivation.

The profound physics emerges when we vary the action with respect to the connection Γ:

$$\delta_\Gamma S = \int d^4x \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu}. \tag{4.105}$$

We substitute the Palatini identity (4.104) into the integral:

$$\delta_\Gamma S = \int d^4x \sqrt{-g} g^{\mu\nu} \left[\nabla_\lambda (\delta \Gamma^\lambda_{\mu\nu}) - \nabla_\nu (\delta \Gamma^\lambda_{\mu\lambda}) \right]. \tag{4.106}$$

To isolate $\delta\Gamma$, we must integrate by parts. However, because Γ and g are assumed to be fully independent, we **cannot** assume that $\nabla_\lambda g^{\mu\nu} = 0$. We must treat the term $\sqrt{-g}g^{\mu\nu}$ as a general tensor density and rigorously apply the Leibniz rule for covariant derivatives.

Let us define the tensor density $\mathfrak{g}^{\mu\nu} = \sqrt{-g}g^{\mu\nu}$. The product rule dictates:

$$\nabla_\lambda (\mathfrak{g}^{\mu\nu} \delta \Gamma^\lambda_{\mu\nu}) = (\nabla_\lambda \mathfrak{g}^{\mu\nu}) \delta \Gamma^\lambda_{\mu\nu} + \mathfrak{g}^{\mu\nu} \nabla_\lambda (\delta \Gamma^\lambda_{\mu\nu}). \tag{4.107}$$

Rearranging this to solve for the term appearing in our action:

$$\mathfrak{g}^{\mu\nu} \nabla_\lambda (\delta \Gamma^\lambda_{\mu\nu}) = \nabla_\lambda (\mathfrak{g}^{\mu\nu} \delta \Gamma^\lambda_{\mu\nu}) - (\nabla_\lambda \mathfrak{g}^{\mu\nu}) \delta \Gamma^\lambda_{\mu\nu}. \tag{4.108}$$

A similar expansion applies to the second term in equation (4.106):

$$\mathfrak{g}^{\mu\nu} \nabla_\nu (\delta \Gamma^\lambda_{\mu\lambda}) = \nabla_\nu (\mathfrak{g}^{\mu\nu} \delta \Gamma^\lambda_{\mu\lambda}) - (\nabla_\nu \mathfrak{g}^{\mu\nu}) \delta \Gamma^\lambda_{\mu\lambda}. \tag{4.109}$$

Substituting these into the variation of the action yields:

$$\delta_\Gamma S = \int d^4x \left[\nabla_\lambda (\mathfrak{g}^{\mu\nu} \delta \Gamma^\lambda_{\mu\nu}) - \nabla_\nu (\mathfrak{g}^{\mu\nu} \delta \Gamma^\lambda_{\mu\lambda}) - (\nabla_\lambda \mathfrak{g}^{\mu\nu}) \delta \Gamma^\lambda_{\mu\nu} + (\nabla_\nu \mathfrak{g}^{\mu\nu}) \delta \Gamma^\lambda_{\mu\lambda} \right]. \tag{4.110}$$

The first two terms are total covariant divergences of vector densities. For a torsion-free connection, the covariant divergence of any vector density $\mathcal{V}^\lambda$ reduces exactly to an ordinary partial divergence: $\nabla_\lambda \mathcal{V}^\lambda = \partial_\lambda \mathcal{V}^\lambda$. By Stokes's theorem, the volume integral of a partial divergence transforms into a boundary integral at infinity. Assuming the variations $\delta\Gamma$ vanish at the boundary of spacetime, these total derivative terms evaluate strictly to zero.

We are left entirely with the terms where the covariant derivative acts on the metric density:

$$\delta_\Gamma S = \int d^4x \left[-(\nabla_\lambda \mathfrak{g}^{\mu\nu}) \delta \Gamma^\lambda_{\mu\nu} + (\nabla_\nu \mathfrak{g}^{\mu\nu}) \delta \Gamma^\lambda_{\mu\lambda} \right] = 0. \tag{4.111}$$

To factor out $\delta\Gamma^\lambda_{\mu\nu}$, we must align the dummy indices in the second term. We swap $\lambda \leftrightarrow \nu$, yielding $(\nabla_\lambda \mathfrak{g}^{\mu\lambda}) \delta \Gamma^\nu_{\mu\nu}$. Since $\delta \Gamma^\nu_{\mu\nu} = \delta^\nu_\lambda \delta \Gamma^\lambda_{\mu\nu}$, we can write:

$$\delta_\Gamma S = \int d^4x \left[-\nabla_\lambda \mathfrak{g}^{\mu\nu} + \delta^\nu_\lambda \nabla_\sigma \mathfrak{g}^{\mu\sigma} \right] \delta \Gamma^\lambda_{\mu\nu} = 0. \tag{4.112}$$

Because the connection is symmetric, the variation $\delta\Gamma^\lambda_{\mu\nu}$ is symmetric in its lower indices (μ, ν). Therefore, for the integral to vanish for arbitrary variations, the *symmetric part* of the bracketed expression must vanish identically:

$$-\nabla_\lambda \mathfrak{g}^{\mu\nu} + \frac{1}{2} \delta^\nu_\lambda \nabla_\sigma \mathfrak{g}^{\mu\sigma} + \frac{1}{2} \delta^\mu_\lambda \nabla_\sigma \mathfrak{g}^{\nu\sigma} = 0. \tag{4.113}$$

4.3.3 Recovering Metric Compatibility

Equation (4.113) acts as the dynamical equation of motion for the connection. We must solve it to find the explicit form of $\nabla_\lambda \mathfrak{g}^{\mu\nu}$.

We take the trace of equation (4.113) by contracting the indices λ and ν ($\lambda = \nu$). Remembering that in 4-dimensional spacetime, the trace of the Kronecker delta is $\delta^\nu_\nu = 4$:

$$\begin{aligned} -\nabla_\nu \mathfrak{g}^{\mu\nu} + \frac{1}{2}(4)\nabla_\sigma \mathfrak{g}^{\mu\sigma} + \frac{1}{2}\delta^\mu_\nu \nabla_\sigma \mathfrak{g}^{\nu\sigma} &= 0 \\ -\nabla_\sigma \mathfrak{g}^{\mu\sigma} + 2\nabla_\sigma \mathfrak{g}^{\mu\sigma} + \frac{1}{2}\nabla_\sigma \mathfrak{g}^{\mu\sigma} &= 0 \\ \frac{3}{2}\nabla_\sigma \mathfrak{g}^{\mu\sigma} &= 0. \end{aligned} \tag{4.114}$$

This proves that the vector density must be divergence-free: $\nabla_\sigma \mathfrak{g}^{\mu\sigma} = 0$.

We substitute this crucial result back into our primary constraint equation (4.113). Because $\nabla_\sigma \mathfrak{g}^{\mu\sigma} = 0$, the second and third terms in the equation vanish entirely, leaving the strict requirement:

$$\nabla_\lambda \mathfrak{g}^{\mu\nu} = 0 \quad \Longrightarrow \quad \nabla_\lambda(\sqrt{-g}g^{\mu\nu}) = 0. \tag{4.115}$$

To prove that this mandates full metric compatibility, we apply the product rule to the density:

$$\sqrt{-g}\nabla_\lambda g^{\mu\nu} + g^{\mu\nu}\nabla_\lambda \sqrt{-g} = 0. \tag{4.116}$$

We can find the covariant derivative of the determinant factor by investigating the determinant of the density $\mathfrak{g}^{\mu\nu}$ itself. The determinant of the matrix $\sqrt{-g}g^{\mu\nu}$ in 4 dimensions is:

$$\det(\mathfrak{g}^{\mu\nu}) = (\sqrt{-g})^4 \det(g^{\mu\nu}) = g^2(g^{-1}) = g. \tag{4.117}$$

Since $\nabla_\lambda \mathfrak{g}^{\mu\nu} = 0$, the covariant derivative of its matrix determinant must also vanish, meaning $\nabla_\lambda g = 0$. Since g is a constant under covariant differentiation, $\sqrt{-g}$ must also be constant:

$$\nabla_\lambda \sqrt{-g} = 0. \tag{4.118}$$

Substituting this back into equation (4.116) annihilates the second term, yielding the final, monumental result:

$$\sqrt{-g}\nabla_\lambda g^{\mu\nu} = 0 \quad \Longrightarrow \quad \nabla_\lambda g^{\mu\nu} = 0. \tag{4.119}$$

Evaluating $(\delta S)_3$: Variation of the Determinant

To evaluate the third term, we must find the variation of the metric determinant, δg.

$$\ln(\det M) = \mathrm{Tr}(\ln M). \tag{4.120}$$

Varying this identity with respect to M yields Jacobi's formula:

$$\frac{1}{\det M}\delta(\det M) = \mathrm{Tr}(M^{-1}\delta M). \tag{4.121}$$

Applying this to the metric tensor matrix ($M = g_{\mu\nu}$, $\det M = g$, $M^{-1} = g^{\mu\nu}$):

$$\frac{1}{g}\delta g = \mathrm{Tr}(g^{\mu\nu}\delta g_{\mu\nu}) = g^{\mu\nu}\delta g_{\mu\nu}. \tag{4.122}$$

We convert the variation to be with respect to the inverse metric using equation (4.82):

$$\delta g = g(g^{\mu\nu}\delta g_{\mu\nu}) = g(g^{\mu\nu}[-g_{\mu\rho}g_{\nu\sigma}\delta g^{\rho\sigma}]) = -g(\delta^\nu_\rho g_{\nu\sigma}\delta g^{\rho\sigma}) = -g(g_{\rho\sigma}\delta g^{\rho\sigma}). \tag{4.123}$$

Relabeling indices gives $\delta g = -g(g_{\mu\nu}\delta g^{\mu\nu})$.

$$\begin{aligned}\delta\sqrt{-g} &= \frac{1}{2\sqrt{-g}}(-\delta g) \\ &= \frac{1}{2\sqrt{-g}}(g(g_{\mu\nu}\delta g^{\mu\nu})) \\ &= -\frac{1}{2}\left(\frac{-g}{\sqrt{-g}}\right) g_{\mu\nu}\delta g^{\mu\nu} \\ &= -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}. \end{aligned}\tag{4.124}$$

We substitute this result back into the third term of the action variation:

$$(\delta S)_3 = \int d^n x R(\delta\sqrt{-g}) = \int d^n x\sqrt{-g}\left(-\frac{1}{2}Rg_{\mu\nu}\right)\delta g^{\mu\nu}. \tag{4.125}$$

4.3.4 Deriving the Vacuum Equations

We now assemble the full variation δS_H by combining the non-vanishing components $(\delta S)_2$ and $(\delta S)_3$:

$$\begin{aligned}\delta S_H &= \int d^n x\sqrt{-g}R_{\mu\nu}\delta g^{\mu\nu} + \int d^n x\sqrt{-g}\left(-\frac{1}{2}Rg_{\mu\nu}\right)\delta g^{\mu\nu} \\ \delta S_H &= \int d^n x\sqrt{-g}\left[R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\right]\delta g^{\mu\nu}. \end{aligned}\tag{4.126}$$

The general definition of a functional derivative for an action S with respect to a set of fields Φ^i is:

$$\delta S = \int \sum_i \left(\frac{\delta S}{\delta \Phi^i}\delta\Phi^i\right) d^n x. \tag{4.127}$$

Comparing this definition to our expression for δS_H, where our field is $g^{\mu\nu}$, we identify the functional derivative of the Hilbert action:

$$\frac{1}{\sqrt{-g}}\frac{\delta S_H}{\delta g^{\mu\nu}} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}. \tag{4.128}$$

The principle of least action demands that the action is stationary ($\delta S_H = 0$) for arbitrary variations $\delta g^{\mu\nu}$.

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 0. \tag{4.129}$$

The power of the Lagrangian approach is now evident.

4.3.5 Including Matter: The Energy-Momentum Tensor

To obtain the complete, non-vacuum Einstein equation, we must augment the gravitational Hilbert action with an action describing the matter fields, S_M.

$$S = \frac{1}{16\pi G}S_H + S_M, \tag{4.130}$$

where we have explicitly included the normalization constant $1/(16\pi G)$ on the gravitational term to ensure we recover the correct Newtonian limit.

We apply the variational principle to the total action with respect to the inverse metric:

$$\delta S = \frac{1}{16\pi G}\delta S_H + \delta S_M = 0. \tag{4.131}$$

Dividing by the volume measure $\sqrt{-g}$ and utilizing our previous result for δS_H:

$$\frac{1}{\sqrt{-g}}\frac{\delta S}{\delta g^{\mu\nu}} = \frac{1}{16\pi G}\left(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\right) + \frac{1}{\sqrt{-g}}\frac{\delta S_M}{\delta g^{\mu\nu}} = 0. \tag{4.132}$$

Rearranging this equation isolates the geometric terms on the left:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = -16\pi G \frac{1}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}}. \tag{4.133}$$

By comparing this to the known form of Einstein's equation ($G_{\mu\nu} = 8\pi G T_{\mu\nu}$), we are led to a profound, fundamental definition of the energy-momentum tensor $T_{\mu\nu}$ for any matter field:

$$T_{\mu\nu} \equiv \frac{-2}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}}. \tag{4.134}$$

Verification via the Scalar Field

To verify that equation (4.134) is a sensible physical definition, we apply it to the scalar field action defined in equation (**??**).

$$\begin{aligned}
\delta S_\phi &= \delta \int d^n x \left[\sqrt{-g} \left(-\frac{1}{2} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - V(\phi) \right) \right] \\
&= \int d^n x \left[\sqrt{-g}\,\delta \left(-\frac{1}{2} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - V(\phi) \right) + \left(-\frac{1}{2} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - V(\phi) \right) \delta\sqrt{-g} \right].
\end{aligned} \tag{4.135}$$

In the first term, the only explicit metric dependence is the inverse metric $g^{\rho\sigma}$, so its variation is $-\frac{1}{2}\delta g^{\mu\nu}(\nabla_\mu \phi \nabla_\nu \phi)$.

$$\delta S_\phi = \int d^n x \sqrt{-g}\,\delta g^{\mu\nu} \left[-\frac{1}{2} \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} \left(-\frac{1}{2} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - V(\phi) \right) \right]. \tag{4.136}$$

We extract the functional derivative $\delta S_\phi / \delta g^{\mu\nu}$ and apply our new definition (4.134):

$$\begin{aligned}
T^{(\phi)}_{\mu\nu} &= \frac{-2}{\sqrt{-g}} \frac{\delta S_\phi}{\delta g^{\mu\nu}} \\
&= -2 \left[-\frac{1}{2} \nabla_\mu \phi \nabla_\nu \phi + \frac{1}{4} g_{\mu\nu} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi + \frac{1}{2} g_{\mu\nu} V(\phi) \right] \\
T^{(\phi)}_{\mu\nu} &= \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - g_{\mu\nu} V(\phi).
\end{aligned} \tag{4.137}$$

In the flat spacetime limit ($g_{\mu\nu} \to \eta_{\mu\nu}$, $\nabla \to \partial$), this precisely recovers the standard energy-momentum tensor for a scalar field derived in classical field theory.

The Hilbert vs. Canonical Tensor

In flat spacetime field theory, Noether's theorem defines a "canonical" energy-momentum tensor $S^{\mu\nu}$ associated with symmetry under spacetime translations:

$$S^{\mu\nu} = \frac{\delta \mathcal{L}}{\delta(\partial_\mu \Phi^i)} \partial^\nu \Phi^i - \eta^{\mu\nu} \mathcal{L}. \tag{4.138}$$

While $S^{\mu\nu}$ is conserved ($\partial_\mu S^{\mu\nu} = 0$), it suffers from significant drawbacks: it is not always symmetric ($S^{\mu\nu} \neq S^{\nu\mu}$), it is not guaranteed to be gauge-invariant, and it cannot always be generalized to curved spacetime.

In stark contrast, the Hilbert definition (4.134) derived from the metric variation is vastly superior for General Relativity.

4.3.6 Properties of Einstein's Equation

We conclude by outlining the mathematical nature of Einstein's Equation.

While it appears to be a system of ten independent equations (since $G_{\mu\nu}$ and $T_{\mu\nu}$ are symmetric 4×4 matrices), the contracted Bianchi identity ($\nabla^\mu G_{\mu\nu} = 0$) imposes four differential constraints on the system.

This under-determination of four equations is not a mathematical flaw, but a physical necessity.

Finally, the equations are highly non-linear.

4.4 Properties of Einstein's Equation

Einstein's equation functions as a system of second-order partial differential equations determining the metric tensor field $g_{\mu\nu}$. To fully utilize this framework, we must rigorously analyze its mathematical properties, its degrees of freedom, its inherent non-linearity, and how it governs the physical motion of matter.

4.4.1 Degrees of Freedom and Diffeomorphism Invariance

Because both the Einstein tensor $G_{\mu\nu}$ and the energy-momentum tensor $T_{\mu\nu}$ are symmetric $(0,2)$ tensors, Einstein's equation appears to constitute ten independent differential equations. This seemingly provides a perfectly determined system for the ten independent component functions of the metric tensor.

However, the geometry of the manifold imposes constraints. The contracted Bianchi identity, $\nabla^\mu G_{\mu\nu} = 0$, enforces four differential constraints on the Ricci tensor and scalar. Consequently, there are only six truly independent dynamical equations governing the metric.

Mathematical and Physical Interpretation: This under-determination is not a flaw, but a mathematically necessary reflection of the theory's coordinate independence (diffeomorphism invariance). If a specific metric $g_{\mu\nu}(x)$ solves the field equations in a coordinate system x^μ, the tensor transformation law guarantees it will also be a solution in any smoothly transformed coordinate system $x^{\mu'}(x)$. The freedom to arbitrarily choose these four transformation functions $x^{\mu'}(x^\mu)$ perfectly accounts for the four "missing" constraining equations. General Relativity only determines the six physical, coordinate-independent degrees of freedom of the gravitational field.

4.4.2 The Non-Linearity of Gravity

A defining characteristic of General Relativity is its strict non-linearity. In Newtonian gravity, the gravitational potential of a composite system is the linear superposition of the individual potentials of its constituents. In General Relativity, the principle of superposition fails entirely.

Mathematically, the field equations are non-linear because the Ricci scalar and tensor are constructed from the Riemann tensor, which contains products of the Christoffel symbols $(\Gamma^\rho_{\mu\nu}\Gamma^\sigma_{\alpha\beta})$. These symbols inherently contain the inverse metric multiplied by first derivatives of the metric.

Physically, this non-linearity arises because the gravitational field couples to itself. The Equivalence Principle dictates that all forms of energy and momentum gravitate. Because the gravitational field itself contains energy, it must act as its own source.

Intuition Example (Perturbative Gravity): In quantum electrodynamics (a linear theory), the electromagnetic interaction between two electrons is mediated by the exchange of a virtual photon. Because photons carry no electric charge, they do not couple to one another. In contrast, the hypothetical quantum force carrier of gravity, the graviton, couples to energy-momentum. Therefore, gravitons interact with other gravitons. A massive body does not just pull on a test particle; the gravitational field of the body pulls on its own gravitational field, compounding the spacetime curvature. This back-reaction is what causes deviations from Newtonian orbits, such as the anomalous perihelion precession of Mercury.

4.4.3 Raychaudhuri's Equation and the Attractiveness of Gravity

To understand exactly how the energy-momentum distribution dictates the physical motion of matter, we consider the geometric evolution of a small, initially stationary ball of free test particles. The particles follow timelike geodesics with four-velocities U^μ. The fractional rate of change of the volume of this ball is quantified by the expansion scalar, $\theta = \nabla_\mu U^\mu$.

The purely geometric evolution of this expansion is governed by Raychaudhuri's equation:

$$\frac{d\theta}{d\tau} = 2\omega^2 - 2\sigma^2 - \frac{1}{3}\theta^2 - R_{\mu\nu}U^\mu U^\nu, \tag{4.139}$$

where τ is proper time, ω^2 measures the rotation (vorticity), and σ^2 measures the shear (volume-preserving distortion).

Let us evaluate this system at an initial moment where all particles in the ball are mutually at rest. The initial expansion, rotation, and shear strictly vanish: $\theta = \omega = \sigma = 0$. We construct a locally inertial coordinate system $x^{\hat{\mu}}$ at the center of the ball, aligning the time axis with the particles' rest frame so that $U^{\hat{\mu}} = (1, 0, 0, 0)$. Raychaudhuri's equation simplifies immensely to:

$$\frac{d\theta}{d\tau} = -R_{\hat{0}\hat{0}}. \tag{4.140}$$

We now introduce the physical source via the trace-reversed Einstein equation:

$$R_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right). \tag{4.141}$$

In our locally inertial rest frame, the stress-energy tensor is decomposed into the energy density $\rho = T_{\hat{0}\hat{0}}$ and the principal spatial pressures $p_k = T_{\hat{k}\hat{k}}$. The trace is:

$$T = \eta^{\hat{\alpha}\hat{\beta}} T_{\hat{\alpha}\hat{\beta}} = -T_{\hat{0}\hat{0}} + T_{\hat{1}\hat{1}} + T_{\hat{2}\hat{2}} + T_{\hat{3}\hat{3}} = -\rho + p_x + p_y + p_z. \tag{4.142}$$

Substituting this trace into the 00-component of Einstein's equation (noting $\eta_{\hat{0}\hat{0}} = -1$):

$$\begin{aligned} R_{\hat{0}\hat{0}} &= 8\pi G \left[\rho - \frac{1}{2}(-\rho + p_x + p_y + p_z)(-1) \right] \\ R_{\hat{0}\hat{0}} &= 8\pi G \left[\rho - \frac{1}{2}\rho + \frac{1}{2}(p_x + p_y + p_z) \right] \\ R_{\hat{0}\hat{0}} &= 4\pi G(\rho + p_x + p_y + p_z). \end{aligned} \tag{4.143}$$

Equating this to the kinematic expansion yields the physical equation of motion for the volume:

$$\frac{d\theta}{d\tau} = -4\pi G(\rho + p_x + p_y + p_z). \tag{4.144}$$

Geometric and Physical Interpretation: Equation (4.144) proves that gravity is universally attractive ($d\theta/d\tau < 0$, causing the volume to shrink) as long as the sum $\rho + \sum p_i$ is positive. Remarkably, pressure acts as a direct source of gravitational attraction. This is a severe departure from Newtonian gravity, where Poisson's equation ($\nabla^2\Phi = 4\pi G\rho$) depends exclusively on the mass density. While local pressure gradients exert physical forces that push matter apart, the uniform presence of pressure dynamically increases the strength of the attractive gravitational field.

4.4.4 Derivation of the Weyl Tensor Propagation Equation

The textbook concisely presents the divergence of the Weyl tensor and its relation to the energy-momentum tensor. However, deriving this propagation equation requires meticulously combining the second Bianchi identity with the full expansion of the Weyl tensor. We perform this explicit derivation here.

1. The Contracted Bianchi Identity

We begin with the second Bianchi identity for the Riemann curvature tensor:

$$\nabla_\lambda R_{\rho\sigma\mu\nu} + \nabla_\mu R_{\rho\sigma\nu\lambda} + \nabla_\nu R_{\rho\sigma\lambda\mu} = 0. \tag{4.145}$$

We wish to find the divergence of the Riemann tensor, $\nabla^\rho R_{\rho\sigma\mu\nu}$. We contract equation (4.145) with the inverse metric $g^{\rho\lambda}$ (setting $\lambda = \alpha, \rho = \alpha$ and summing):

$$\nabla^\rho R_{\rho\sigma\mu\nu} + \nabla_\mu R^\rho{}_{\sigma\nu\rho} + \nabla_\nu R^\rho{}_{\sigma\rho\mu} = 0. \tag{4.146}$$

Using the antisymmetry of the Riemann tensor ($R^{\rho}{}_{\sigma\nu\rho} = -R^{\rho}{}_{\sigma\rho\nu} = -R_{\sigma\nu}$), we evaluate the two traced terms:

$$\nabla_\mu R^{\rho}{}_{\sigma\nu\rho} = -\nabla_\mu R_{\sigma\nu} \tag{4.147}$$

$$\nabla_\nu R^{\rho}{}_{\sigma\rho\mu} = \nabla_\nu R_{\sigma\mu}. \tag{4.148}$$

Substituting these back yields the contracted Bianchi identity for the Riemann tensor:

$$\nabla^\rho R_{\rho\sigma\mu\nu} - \nabla_\mu R_{\sigma\nu} + \nabla_\nu R_{\sigma\mu} = 0 \quad \Longrightarrow \quad \nabla^\rho R_{\rho\sigma\mu\nu} = \nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}. \tag{4.149}$$

Using anti-symmetrization bracket notation $A_{[\mu\nu]} = \frac{1}{2}(A_{\mu\nu} - A_{\nu\mu})$, this is compactly written as:

$$\nabla^\rho R_{\rho\sigma\mu\nu} = 2\nabla_{[\mu} R_{\nu]\sigma}. \tag{4.150}$$

2. The Divergence of the Weyl Tensor

The full Weyl tensor $C_{\rho\sigma\mu\nu}$ in a 4-dimensional spacetime is defined as the trace-free part of the Riemann tensor:

$$C_{\rho\sigma\mu\nu} = R_{\rho\sigma\mu\nu} - \frac{1}{2}\left(g_{\rho\mu}R_{\sigma\nu} - g_{\rho\nu}R_{\sigma\mu} - g_{\sigma\mu}R_{\rho\nu} + g_{\sigma\nu}R_{\rho\mu}\right) + \frac{1}{6}R\left(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu}\right). \tag{4.151}$$

We take the covariant divergence ∇^ρ of this entire expression. Because the Levi-Civita connection is metric-compatible ($\nabla^\rho g_{\alpha\beta} = 0$), the metric tensors pass freely through the derivative:

$$\nabla^\rho C_{\rho\sigma\mu\nu} = \nabla^\rho R_{\rho\sigma\mu\nu} - \frac{1}{2}\left(\delta^\rho_\mu \nabla^\rho R_{\sigma\nu} - \delta^\rho_\nu \nabla^\rho R_{\sigma\mu} - g_{\sigma\mu}\nabla^\rho R_{\rho\nu} + g_{\sigma\nu}\nabla^\rho R_{\rho\mu}\right) + \frac{1}{6}\left(\delta^\rho_\mu g_{\sigma\nu}\nabla^\rho R - \delta^\rho_\nu g_{\sigma\mu}\nabla^\rho R\right). \tag{4.152}$$

We evaluate these contractions term by term:

- The first term is simply the contracted Bianchi identity (4.149): $\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}$.
- In the second group, the Kronecker deltas map $\rho \to \mu$ and $\rho \to \nu$: $\delta^\rho_\mu \nabla^\rho R_{\sigma\nu} = \nabla_\mu R_{\sigma\nu}$.
- Also in the second group, we apply the twice-contracted Bianchi identity ($\nabla^\rho R_{\rho\alpha} = \frac{1}{2}\nabla_\alpha R$) to the terms $g_{\sigma\mu}\nabla^\rho R_{\rho\nu} = \frac{1}{2} g_{\sigma\mu}\nabla_\nu R$.
- In the third group, the deltas again map ρ to the lower indices: $\delta^\rho_\mu g_{\sigma\nu}\nabla^\rho R = g_{\sigma\nu}\nabla_\mu R$.

Substituting all these evaluated terms back into the divergence gives:

$$\nabla^\rho C_{\rho\sigma\mu\nu} = (\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}) - \frac{1}{2}\left(\nabla_\mu R_{\sigma\nu} - \nabla_\nu R_{\sigma\mu} - \frac{1}{2}g_{\sigma\mu}\nabla_\nu R + \frac{1}{2}g_{\sigma\nu}\nabla_\mu R\right) + \frac{1}{6}\left(g_{\sigma\nu}\nabla_\mu R - g_{\sigma\mu}\nabla_\nu R\right). \tag{4.153}$$

Since the Ricci tensor is symmetric ($R_{\mu\nu} = R_{\nu\mu}$), we can combine the Ricci derivative terms:

$$\begin{aligned} \nabla^\rho C_{\rho\sigma\mu\nu} &= \left(1 - \frac{1}{2}\right)(\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}) + \left(\frac{1}{4} - \frac{1}{6}\right) g_{\sigma\mu}\nabla_\nu R - \left(\frac{1}{4} - \frac{1}{6}\right) g_{\sigma\nu}\nabla_\mu R \\ &= \frac{1}{2}(\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}) + \frac{1}{12}\left(g_{\sigma\mu}\nabla_\nu R - g_{\sigma\nu}\nabla_\mu R\right). \end{aligned} \tag{4.154}$$

We rewrite this result using the anti-symmetrization brackets:

$$\frac{1}{2}(\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}) = \nabla_{[\mu} R_{\nu]\sigma} \tag{4.155}$$

$$\frac{1}{12}\left(g_{\sigma\mu}\nabla_\nu R - g_{\sigma\nu}\nabla_\mu R\right) = \frac{1}{6}\left[\frac{1}{2}\left(g_{\sigma\mu}\nabla_\nu R - g_{\sigma\nu}\nabla_\mu R\right)\right] = \frac{1}{6} g_{\sigma[\mu}\nabla_{\nu]} R. \tag{4.156}$$

(Note in (4.156) that swapping μ and ν absorbs the negative sign, giving $+g_{\sigma[\mu}\nabla_{\nu]}R$). Summing these yields the geometric identity for the propagation of the Weyl tensor:

$$\nabla^\rho C_{\rho\sigma\mu\nu} = \nabla_{[\mu} R_{\nu]\sigma} + \frac{1}{6} g_{\sigma[\mu}\nabla_{\nu]} R. \tag{4.157}$$

3. Introducing the Matter Source

We now couple this purely geometric identity to the matter distribution using the trace-reversed Einstein field equation:

$$R_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right). \tag{4.158}$$

Taking the trace of this equation gives the Ricci scalar: $R = -8\pi G T$.

We substitute these into the two terms on the right-hand side of equation (4.157). For the first term, we take the covariant derivative and anti-symmetrize:

$$\begin{aligned} \nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma} &= 8\pi G \left[\nabla_\mu \left(T_{\nu\sigma} - \frac{1}{2} T g_{\nu\sigma} \right) - \nabla_\nu \left(T_{\mu\sigma} - \frac{1}{2} T g_{\mu\sigma} \right) \right] \\ &= 8\pi G \left[(\nabla_\mu T_{\nu\sigma} - \nabla_\nu T_{\mu\sigma}) - \frac{1}{2} (g_{\nu\sigma} \nabla_\mu T - g_{\mu\sigma} \nabla_\nu T) \right]. \end{aligned} \tag{4.159}$$

Converting this to bracket notation:

$$2\nabla_{[\mu} R_{\nu]\sigma} = 8\pi G \left(2\nabla_{[\mu} T_{\nu]\sigma} - g_{\sigma[\nu} \nabla_{\mu]} T \right). \tag{4.160}$$

Because $g_{\sigma[\nu} \nabla_{\mu]} T = -g_{\sigma[\mu} \nabla_{\nu]} T$, we divide by 2 to obtain:

$$\nabla_{[\mu} R_{\nu]\sigma} = 8\pi G \left(\nabla_{[\mu} T_{\nu]\sigma} + \frac{1}{2} g_{\sigma[\mu} \nabla_{\nu]} T \right). \tag{4.161}$$

For the second term in (4.157), we simply substitute $R = -8\pi G T$:

$$\frac{1}{6} g_{\sigma[\mu} \nabla_{\nu]} R = \frac{1}{6} g_{\sigma[\mu} \nabla_{\nu]} (-8\pi G T) = 8\pi G \left(-\frac{1}{6} g_{\sigma[\mu} \nabla_{\nu]} T \right). \tag{4.162}$$

Finally, we sum equations (4.161) and (4.162) to assemble the full Weyl propagation equation:

$$\begin{aligned} \nabla^\rho C_{\rho\sigma\mu\nu} &= 8\pi G \left[\nabla_{[\mu} T_{\nu]\sigma} + \left(\frac{1}{2} - \frac{1}{6} \right) g_{\sigma[\mu} \nabla_{\nu]} T \right] \\ \nabla^\rho C_{\rho\sigma\mu\nu} &= 8\pi G \left(\nabla_{[\mu} T_{\nu]\sigma} + \frac{1}{3} g_{\sigma[\mu} \nabla_{\nu]} T \right). \end{aligned} \tag{4.163}$$

This explicitly demonstrates how gradients in the local energy, momentum, and pressure precisely dictate the radiation of tidal curvature across empty spacetime.

4.4.5 Quantum Gravity and the Planck Scale

A significant limitation of General Relativity is that it is a classical field theory. Standard quantization procedures applied to the metric yield a non-renormalizable theory; when considering higher-order quantum loops, ultraviolet divergences appear that cannot be absorbed by a finite number of parameters.

The breakdown of classical relativity is strictly isolated to extreme physical regimes determined by the **Planck Scale**, constructed from G, c, and $\hbar$:

$$\text{Planck Mass:} \quad m_P = \left(\frac{\hbar c}{G} \right)^{1/2} \approx 1.22 \times 10^{19}\,\text{GeV} \tag{4.164}$$

$$\text{Planck Length:} \quad l_P = \left(\frac{\hbar G}{c^3} \right)^{1/2} \approx 1.62 \times 10^{-33}\,\text{cm} \tag{4.165}$$

$$\text{Planck Time:} \quad t_P = \left(\frac{\hbar G}{c^5} \right)^{1/2} \approx 5.39 \times 10^{-44}\,\text{sec} \tag{4.166}$$

$$\text{Planck Energy:} \quad E_P = \left(\frac{\hbar c^5}{G} \right)^{1/2} \approx 1.22 \times 10^{19}\,\text{GeV} \tag{4.167}$$

At scales approaching l_P or E_P, the smooth manifold structure of spacetime is expected to be fundamentally altered by quantum fluctuations, requiring a complete theory of quantum gravity, such as String Theory.

To rigorously demonstrate that these specific combinations yield the correct dimensions, we can perform a dimensional analysis. Let the fundamental dimensions be denoted as Mass $[M]$, Length $[L]$, and Time $[T]$. The dimensions of the fundamental constants are:

$$[c] = LT^{-1} \tag{4.168}$$

$$[\hbar] = ML^2T^{-1} \tag{4.169}$$

$$[G] = M^{-1}L^3T^{-2} \tag{4.170}$$

Substituting these base dimensions into the equations for the Planck units proves their consistency:

$$[m_P] = \left[\frac{\hbar c}{G}\right]^{1/2} = \left(\frac{(ML^2T^{-1})(LT^{-1})}{M^{-1}L^3T^{-2}}\right)^{1/2} = \left(\frac{ML^3T^{-2}}{M^{-1}L^3T^{-2}}\right)^{1/2} = (M^2)^{1/2} = M \tag{4.171}$$

$$[l_P] = \left[\frac{\hbar G}{c^3}\right]^{1/2} = \left(\frac{(ML^2T^{-1})(M^{-1}L^3T^{-2})}{(LT^{-1})^3}\right)^{1/2} = \left(\frac{L^5T^{-3}}{L^3T^{-3}}\right)^{1/2} = (L^2)^{1/2} = L \tag{4.172}$$

$$[t_P] = \left[\frac{\hbar G}{c^5}\right]^{1/2} = \left(\frac{(ML^2T^{-1})(M^{-1}L^3T^{-2})}{(LT^{-1})^5}\right)^{1/2} = \left(\frac{L^5T^{-3}}{L^5T^{-5}}\right)^{1/2} = (T^2)^{1/2} = T \tag{4.173}$$

$$[E_P] = \left[\frac{\hbar c^5}{G}\right]^{1/2} = \left(\frac{(ML^2T^{-1})(L^5T^{-5})}{M^{-1}L^3T^{-2}}\right)^{1/2} = \left(\frac{ML^7T^{-6}}{M^{-1}L^3T^{-2}}\right)^{1/2} = (M^2L^4T^{-4})^{1/2} = ML^2T^{-2} \tag{4.174}$$

This matches the standard macroscopic dimensions for Energy ($[E] = [\text{Force} \times \text{Distance}] = ML^2T^{-2}$).

4.5 The Cosmological Constant

A fundamental and distinguishing characteristic of General Relativity is that the gravitational field couples to the entire energy-momentum tensor. In classical Newtonian mechanics and standard quantum field theory, the absolute zero-point of energy is physically arbitrary; only relative changes in energy between states dictate the dynamical equations of motion. Consequently, one can freely add a constant potential V_0 to the system without altering the physics. In gravitation, however, the absolute magnitude of the energy fundamentally matters. The total, un-normalized energy density acts as a direct source of spacetime curvature.

4.5.1 Vacuum Energy and Lorentz Invariance

This behavior inherently opens up the physical possibility of vacuum energy: an intrinsic, non-vanishing energy density characteristic of entirely empty space. If the vacuum is to be truly empty, it must not possess any preferred frame of reference or distinguished spatial direction. Therefore, if a non-zero energy density exists, its associated energy-momentum tensor must remain strictly Lorentz invariant in any locally inertial coordinate system.

In a locally inertial frame, the only $(0, 2)$ tensor that is invariant under the full Lorentz group is the Minkowski metric $\eta_{\hat{\mu}\hat{\nu}}$. Thus, the vacuum stress-energy tensor must be proportional to the metric:

$$T^{(vac)}_{\hat{\mu}\hat{\nu}} = -\rho_{vac}\eta_{\hat{\mu}\hat{\nu}}, \tag{4.175}$$

where ρ_{vac} is a constant scalar representing the vacuum energy density. By the principle of general covariance, we can elevate this locally inertial expression to a general coordinate-invariant tensor equation valid in any reference frame:

$$T^{(vac)}_{\mu\nu} = -\rho_{vac}g_{\mu\nu}. \tag{4.176}$$

Furthermore, the energy density ρ_{vac} must be a universal constant throughout spacetime, because any non-zero gradient ($\nabla_\lambda \rho_{vac} \neq 0$) would establish a preferred direction, breaking the assumed Lorentz invariance of the vacuum.

4.5.2 Equation of State for the Vacuum

We can deduce the physical properties of this vacuum energy by comparing equation (4.176) to the standard stress-energy tensor for a perfect fluid:

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}. \tag{4.177}$$

For the fluid tensor to strictly match the vacuum tensor $-\rho_{vac}g_{\mu\nu}$, the term proportional to the four-velocities $U_\mu U_\nu$ must identically vanish. This enforces the condition $\rho_{vac}+p_{vac} = 0$, which immediately yields the vacuum equation of state:

$$p_{vac} = -\rho_{vac}. \tag{4.178}$$

This establishes that a positive vacuum energy density necessitates a strictly isotropic negative pressure of equal magnitude.

4.5.3 Modifying Einstein's Equation

We can decompose the total energy-momentum of the universe into a standard matter/radiation piece $T^{(M)}_{\mu\nu}$ and the vacuum piece $T^{(vac)}_{\mu\nu}$. Substituting this decomposition into the standard field equation gives:

$$\begin{aligned} R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} &= 8\pi G\left(T^{(M)}_{\mu\nu} + T^{(vac)}_{\mu\nu}\right) \\ R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} &= 8\pi G\left(T^{(M)}_{\mu\nu} - \rho_{vac}g_{\mu\nu}\right). \end{aligned} \tag{4.179}$$

Rearranging the terms to place all geometric components on the left-hand side yields:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + 8\pi G\rho_{vac}g_{\mu\nu} = 8\pi G T^{(M)}_{\mu\nu}. \tag{4.180}$$

Historically, Einstein introduced a new term, the **cosmological constant** Λ, to construct a static cosmological model. The modified field equation takes the form:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \tag{4.181}$$

By directly comparing equation (4.180) with equation (4.181), it is evident that the cosmological constant is mathematically and physically equivalent to a uniform vacuum energy density:

$$\rho_{vac} = \frac{\Lambda}{8\pi G}. \tag{4.182}$$

In modern theoretical physics, the terms ”cosmological constant” and ”vacuum energy” are utilized interchangeably.

4.5.4 Lagrangian Derivation

This modification can be derived rigorously from the principle of least action. The standard Hilbert Lagrangian density for gravity is $\hat{\mathcal{L}}_H = R$, which is the simplest scalar constructable from the metric. However, an even simpler scalar is a fundamental constant. We can augment the gravitational action by including a constant term -2Λ:

$$S = \int d^4x\sqrt{-g}\left[\frac{1}{16\pi G}(R - 2\Lambda) + \hat{\mathcal{L}}_M\right]. \tag{4.183}$$

To verify this produces the correct field equation, we compute the variation of the new term with respect to the inverse metric $\delta g^{\mu\nu}$. Recalling the variation of the determinant $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}$:

$$\begin{aligned} \delta S_\Lambda &= \delta\int d^4x\sqrt{-g}\left(\frac{-2\Lambda}{16\pi G}\right) \\ &= \int d^4x\left(\frac{-2\Lambda}{16\pi G}\right)\left(-\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}\right) \\ &= \frac{1}{16\pi G}\int d^4x\sqrt{-g}\left(\Lambda g_{\mu\nu}\right)\delta g^{\mu\nu}. \end{aligned} \tag{4.184}$$

When combined with the variation of the Ricci scalar (which generates the Einstein tensor $G_{\mu\nu}$), this variation rigorously yields the $+\Lambda g_{\mu\nu}$ term on the left-hand side of Einstein's equation.

4.5.5 The Cosmological Constant Problem

While vacuum energy is easily accommodated within the mathematics of General Relativity, calculating its expected magnitude is highly problematic. In quantum field theory, the vacuum state possesses zero-point energy due to quantum fluctuations. Consider a simple harmonic oscillator with classical potential $V(x) = \frac{1}{2}\omega^2 x^2$. The Heisenberg uncertainty principle forbids the particle from resting exactly at the minimum with zero momentum, shifting the ground state energy to $E_0 = \frac{1}{2}\hbar\omega$.

A free quantum field can be decomposed via Fourier transform into an infinite collection of harmonic oscillators in momentum space. The frequency of each mode is $\omega = \sqrt{m^2 + k^2}$, where k is the wave vector magnitude. The total vacuum energy density is the integral over all momentum modes:

$$\rho_{vac} = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2} \hbar \sqrt{m^2 + k^2}. \tag{4.185}$$

This integral diverges to infinity. If we impose an ultraviolet momentum cutoff k_{max} (representing the scale at which our field theory breaks down), dimensional analysis indicates the vacuum energy density scales as:

$$\rho_{vac} \sim \hbar k_{max}^4. \tag{4.186}$$

If we assume classical field theory is valid up to the reduced Planck scale $\overline{m}_P \sim 10^{18}$ GeV, we calculate an immense vacuum energy density:

$$\rho_{vac}^{(theory)} \sim (10^{18}\ \mathrm{GeV})^4 \sim 10^{112}\ \mathrm{erg/cm}^3. \tag{4.187}$$

Conversely, cosmological observations of the universe's large-scale structure and expansion constrain the physical vacuum energy to be extremely small:

$$|\rho_\Lambda^{(obs)}| \leq (10^{-12}\ \mathrm{GeV})^4 \sim 10^{-8}\ \mathrm{erg/cm}^3. \tag{4.188}$$

The ratio between the theoretical expectation in equation (4.187) and the observational bound in equation (4.188) reveals a discrepancy of 120 orders of magnitude. Because no known symmetry naturally cancels this massive vacuum energy to such a precise, near-zero value, this massive theoretical failure is known as the "cosmological constant problem".

4.6 Energy Conditions

Einstein's field equation, $G_{\mu\nu} = 8\pi G T_{\mu\nu}$, is fundamentally a system of differential equations determining the metric $g_{\mu\nu}$ for a given matter distribution $T_{\mu\nu}$. However, because the Einstein tensor satisfies the contracted Bianchi identity identically ($\nabla^\mu G_{\mu\nu} = 0$), one can technically postulate any arbitrary smooth metric $g_{\mu\nu}$, compute its corresponding Einstein tensor, and simply define that tensor to be $8\pi G T_{\mu\nu}$. Because of the Bianchi identity, the resulting energy-momentum tensor is guaranteed to be conserved.

This implies that any arbitrary spacetime geometry—including unphysical ones such as traversable wormholes or warp drives—is a mathematical solution to Einstein's equation for *some* hypothetical matter distribution. To filter out these unphysical spacetimes, we must impose coordinate-invariant restrictions on the energy-momentum tensor. These restrictions are known as **energy conditions**. They limit the arbitrariness of $T_{\mu\nu}$ by contracting it with physically meaningful vectors (timelike or null).

To develop physical intuition, we evaluate each energy condition for the case of a perfect fluid:

$$T_{\mu\nu} = (\rho + p) U_\mu U_\nu + p g_{\mu\nu}, \tag{4.189}$$

where U^μ is the fluid four-velocity, ρ is the rest-frame energy density, and p is the isotropic pressure.

4.6.1 The Weak Energy Condition (WEC)

The Weak Energy Condition mandates that the local energy density measured by any physical observer must be non-negative. Since an observer's worldline is defined by a timelike tangent vector t^μ, the WEC states:

$$T_{\mu\nu}t^\mu t^\nu \geq 0 \quad \text{for all timelike vectors } t^\mu. \tag{4.190}$$

To translate this into conditions on ρ and p, we evaluate this contraction in a locally inertial reference frame where the fluid is at rest, meaning $U^\mu = (1, 0, 0, 0)$. In this frame, the metric is $g_{\mu\nu} = \eta_{\mu\nu}$, and the stress-energy tensor is diagonal: $T_{\mu\nu} = \mathrm{diag}(\rho, p, p, p)$.

Let $t^\mu = (t^0, t^i)$ be an arbitrary timelike vector, which satisfies $t_\mu t^\mu < 0$. In our local frame:

$$-(t^0)^2 + (t^i)^2 < 0 \implies (t^i)^2 < (t^0)^2, \tag{4.191}$$

where $(t^i)^2 \equiv \delta_{ij}t^i t^j$. We can define a squared velocity parameter $v^2 = (t^i)^2/(t^0)^2$, where $0 \leq v^2 < 1$.

The WEC contraction is:

$$\begin{aligned} T_{\mu\nu}t^\mu t^\nu &= T_{00}(t^0)^2 + T_{ij}t^i t^j \\ &= \rho(t^0)^2 + p(t^i)^2 \\ &= (t^0)^2 \left[\rho + p\frac{(t^i)^2}{(t^0)^2}\right] \\ &= (t^0)^2(\rho + pv^2). \end{aligned} \tag{4.192}$$

For this expression to remain non-negative for *all* valid timelike observers (all $v^2 \in [0, 1)$), we must check the boundaries of the interval:

1. At $v = 0$ (an observer comoving with the fluid), we require $\rho \geq 0$.
2. In the limit $v \to 1$ (an observer moving at nearly the speed of light), we require $\rho + p \geq 0$.

Thus, the WEC dictates two physical constraints:

$$\rho \geq 0 \quad \text{and} \quad \rho + p \geq 0. \tag{4.193}$$

4.6.2 The Null Energy Condition (NEC)

The Null Energy Condition is a weaker version of the WEC, applied only to massless particles (photons). It states:

$$T_{\mu\nu}l^\mu l^\nu \geq 0 \quad \text{for all null vectors } l^\mu. \tag{4.194}$$

Using the same perfect fluid setup, a null vector satisfies $l_\mu l^\mu = 0$, meaning $(l^0)^2 = (l^i)^2$. The contraction becomes:

$$\begin{aligned} T_{\mu\nu}l^\mu l^\nu &= \rho(l^0)^2 + p(l^i)^2 \\ &= \rho(l^0)^2 + p(l^0)^2 \\ &= (l^0)^2(\rho + p). \end{aligned} \tag{4.195}$$

Since $(l^0)^2 > 0$, the NEC imposes a single condition:

$$\rho + p \geq 0. \tag{4.196}$$

Notably, the NEC allows for a negative energy density ($\rho < 0$), provided there is a sufficiently large positive pressure to compensate.

4.6.3 The Dominant Energy Condition (DEC)

The Dominant Energy Condition incorporates the WEC, but adds the requirement that the flow of energy-momentum cannot exceed the speed of light. Physically, the four-momentum density measured by an observer t^μ is $P^\mu = -T^\mu_{\ \nu} t^\nu$. The DEC requires that P^μ is a causal (non-spacelike) vector pointing in the same time direction as t^μ:

$$T_{\mu\nu} t^\mu t^\nu \geq 0 \quad \text{and} \quad P_\mu P^\mu \leq 0 \quad \text{for all timelike } t^\mu. \tag{4.197}$$

Let us evaluate the causal constraint $P_\mu P^\mu \leq 0$ for a perfect fluid in its rest frame. The mixed tensor $T^\mu_{\ \nu}$ is:

$$T^0_{\ 0} = g^{00} T_{00} = (-1)(\rho) = -\rho, \quad T^i_{\ j} = g^{ik} T_{kj} = \delta^{ik}(p\delta_{kj}) = p\delta^i_j. \tag{4.198}$$

The components of the momentum density P^μ are:

$$P^0 = -T^0_{\ \nu} t^\nu = -T^0_{\ 0} t^0 = -(-\rho) t^0 = \rho t^0 \tag{4.199}$$

$$P^i = -T^i_{\ \nu} t^\nu = -T^i_{\ j} t^j = -p t^i. \tag{4.200}$$

We calculate the invariant norm:

$$\begin{aligned} P_\mu P^\mu &= -(P^0)^2 + (P^i)^2 \\ &= -(\rho t^0)^2 + (-p t^i)^2 \\ &= -\rho^2 (t^0)^2 + p^2 (t^i)^2. \end{aligned} \tag{4.201}$$

For P^μ to be non-spacelike, we require $P_\mu P^\mu \leq 0$:

$$p^2 (t^i)^2 \leq \rho^2 (t^0)^2 \implies \frac{p^2}{\rho^2} \leq \frac{(t^0)^2}{(t^i)^2}. \tag{4.202}$$

Since t^μ is timelike, the ratio $(t^0)^2/(t^i)^2$ is strictly greater than 1. For the inequality to hold for *all* timelike vectors (even as $(t^i)^2 \to (t^0)^2$), we must satisfy the strictest limit:

$$\frac{p^2}{\rho^2} \leq 1 \implies p^2 \leq \rho^2. \tag{4.203}$$

Combining this with the WEC requirement ($\rho \geq 0$), the DEC is mathematically equivalent to:

$$\rho \geq |p|. \tag{4.204}$$

This ensures the energy density is non-negative and strictly bounds the magnitude of the pressure.

4.6.4 The Null Dominant Energy Condition (NDEC)

The Null Dominant Energy Condition enforces the DEC, but exclusively for null vectors l^μ:

$$T_{\mu\nu} l^\mu l^\nu \geq 0 \quad \text{and} \quad P_\mu P^\mu \leq 0 \quad \text{where } P^\mu = -T^\mu_{\ \nu} l^\nu. \tag{4.205}$$

Re-evaluating the invariant norm for a null vector ($(l^0)^2 = (l^i)^2$):

$$\begin{aligned} P_\mu P^\mu &= -\rho^2 (l^0)^2 + p^2 (l^i)^2 \\ &= (l^0)^2 (p^2 - \rho^2). \end{aligned} \tag{4.206}$$

For $P_\mu P^\mu \leq 0$, we require $p^2 \leq \rho^2$, which yields $|\rho| \geq |p|$. Coupled with the NEC ($\rho + p \geq 0$), the NDEC permits negative energy densities ($\rho < 0$) under one highly constrained circumstance: if ρ is negative, the condition $|\rho| \geq |p|$ implies $-\rho \geq p$. However, the NEC requires $p \geq -\rho$. The only way both can be true is if $p = -\rho$.

Thus, the NDEC perfectly excludes all sources forbidden by the DEC, with the singular exception of a negative cosmological constant (negative vacuum energy).

4.6.5 The Strong Energy Condition (SEC)

The Strong Energy Condition differs structurally from the others by involving the trace of the energy-momentum tensor, $T = T^\lambda{}_\lambda$. It dictates:

$$T_{\mu\nu}t^\mu t^\nu \geq \frac{1}{2}Tt^\sigma t_\sigma \quad \text{for all timelike vectors } t^\mu. \tag{4.207}$$

For a perfect fluid, the trace is $T = -\rho + 3p$. The invariant length of the timelike vector is $t^\sigma t_\sigma = -(t^0)^2 + (t^i)^2 = -(t^0)^2(1 - v^2)$. Substituting these into the SEC inequality:

$$\begin{aligned}(t^0)^2(\rho + pv^2) &\geq \frac{1}{2}(-\rho + 3p)\left[-(t^0)^2(1 - v^2)\right] \\ \rho + pv^2 &\geq \frac{1}{2}(\rho - 3p)(1 - v^2).\end{aligned} \tag{4.208}$$

Evaluating this at the velocity limits:

1. At $v = 0$: $\rho \geq \frac{1}{2}(\rho - 3p) \implies 2\rho \geq \rho - 3p \implies \rho + 3p \geq 0$.
2. As $v \to 1$: $\rho + p \geq 0$.

The Strong Energy Condition does *not* imply the WEC, because it does not strictly require $\rho \geq 0$. It does, however, forbid large negative pressures.

The SEC and Gravitational Attraction

The mathematical form of the SEC has a profound geometric origin rooted in Raychaudhuri's equation. Recall that the purely geometric expansion of a family of geodesics is driven by the contraction of the Ricci tensor with the four-velocity: $d\theta/d\tau \approx -R_{\mu\nu}U^\mu U^\nu$.

Einstein's equation is $R_{\mu\nu} = 8\pi G(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu})$. Contracting this with arbitrary timelike vectors t^μ yields:

$$\begin{aligned}R_{\mu\nu}t^\mu t^\nu &= 8\pi G\left(T_{\mu\nu}t^\mu t^\nu - \frac{1}{2}Tg_{\mu\nu}t^\mu t^\nu\right) \\ &= 8\pi G\left(T_{\mu\nu}t^\mu t^\nu - \frac{1}{2}Tt_\sigma t^\sigma\right).\end{aligned} \tag{4.209}$$

For gravity to be universally attractive ($d\theta/d\tau \leq 0$), the quantity $R_{\mu\nu}t^\mu t^\nu$ must be non-negative. We see immediately that this requires:

$$T_{\mu\nu}t^\mu t^\nu - \frac{1}{2}Tt_\sigma t^\sigma \geq 0 \implies T_{\mu\nu}t^\mu t^\nu \geq \frac{1}{2}Tt_\sigma t^\sigma. \tag{4.210}$$

This is the exact definition of the Strong Energy Condition. Therefore, the SEC is the precise mathematical requirement for gravity to act as an attractive force on all possible timelike trajectories.

Notice that vacuum energy (where $p = -\rho$) violently violates the SEC, because $\rho + 3p = \rho - 3\rho = -2\rho$, which is negative for positive vacuum energy. This SEC violation is the mathematical reason why a positive cosmological constant generates repulsive gravity, leading to the accelerated expansion of the universe.

4.7 The Equivalence Principle Revisited

The Einstein Equivalence Principle (EEP) served as the foundational heuristic for generalizing the laws of Special Relativity to the curved spacetime of General Relativity. By positing that the laws of physics reduce to their Special Relativistic forms in sufficiently small, locally inertial reference frames, we deduced the minimal-coupling procedure: replacing the Minkowski metric $\eta_{\mu\nu}$ with the dynamical metric $g_{\mu\nu}$, and replacing partial derivatives ∂_μ with covariant derivatives ∇_μ.

However, the Equivalence Principle is not a fundamental, mathematically rigorous axiom of the final theory. Rather, it is a phenomenological guide. In practice, the EEP is typically invoked to justify four distinct propositions regarding the formulation of gravitational physics:

1. **General Covariance:** The laws of physics must be expressible in a manifestly coordinate-invariant, tensorial form.
2. **Metric Theory of Gravity:** There exists a symmetric metric tensor $g_{\mu\nu}$ defined on the spacetime manifold, and the curvature of this metric is the sole geometric manifestation of gravity.
3. **Uniqueness of the Gravitational Field:** There do not exist any other long-range fields (e.g., secondary scalar or vector fields) that masquerade as gravity or mediate a "fifth force."
4. **Minimal Coupling:** Matter fields interact with the gravitational field exclusively through the metric and its first derivatives (via the connection). There are no direct algebraic couplings between matter fields and the Riemann curvature tensor.

These four propositions possess different levels of fundamental validity. The first is a mathematical tautology; any physical law can be written in a generally covariant form given sufficient auxiliary fields. The second is the defining hallmark of General Relativity and appears robust. The third is a testable empirical hypothesis. The fourth, however, is merely a low-energy approximation.

4.7.1 The Approximation of Minimal Coupling

From the perspective of modern effective field theory, the minimal-coupling rule is not an absolute law but the leading-order term in a low-energy expansion. Consider a scalar field ϕ with mass m. The minimally coupled action is:

$$S_{\text{min}} = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} m^2 \phi^2 \right]. \tag{4.211}$$

However, the requirements of general covariance and dimensional analysis do not forbid non-minimal couplings. One could easily construct a perfectly valid, generally covariant term that couples the scalar field directly to the Ricci scalar R:

$$S_{\text{non-min}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} \xi R \phi^2 \right], \tag{4.212}$$

where ξ is a dimensionless coupling constant. Such a term vanishes in flat spacetime (where $R = 0$), meaning it satisfies the strict formulation of the Equivalence Principle (that physics reduces to Special Relativity in locally flat regions), yet it violently violates the minimal-coupling heuristic.

In fact, quantum field theory in curved spacetime demonstrates that even if ξ is set to zero classically, loop corrections (vacuum polarization) will dynamically generate this non-minimal coupling term. At macroscopic scales and low energies, the curvature R is incredibly small, rendering these terms physically negligible. Thus, minimal coupling is highly successful in astrophysical applications, but it cannot be viewed as a fundamental geometric prohibition.

4.8 Alternative Theories of Gravity

Given that General Relativity is a classical field theory, and that our motivations for minimal coupling are approximate, it is scientifically imperative to consider alternative formulations of gravity. These modifications are typically motivated by the quest for a quantum theory of gravity, or by the cosmological mysteries of dark matter and dark energy.

4.8.1 Higher-Order Actions: $f(R)$ Gravity

The Einstein-Hilbert action assumes that the gravitational Lagrangian is simply the Ricci scalar R. The most direct generalization is to replace R with an arbitrary analytic function $f(R)$:

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{16\pi G} f(R) + \mathcal{L}_M \right]. \tag{4.213}$$

To determine the modified field equations, we must perform a rigorous variational calculus derivation. We vary the action with respect to the inverse metric $g^{\mu\nu}$:

$$\delta S_{f(R)} = \frac{1}{16\pi G} \int d^4x \left[\delta(\sqrt{-g}) f(R) + \sqrt{-g}\,\delta(f(R))\right]. \tag{4.214}$$

Using the established identity for the variation of the determinant, $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g} g_{\mu\nu}\delta g^{\mu\nu}$, and applying the chain rule to the function $f(R)$, we obtain:

$$\delta S_{f(R)} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[-\frac{1}{2} f(R) g_{\mu\nu}\delta g^{\mu\nu} + f'(R)\delta R\right], \tag{4.215}$$

where $f'(R) \equiv df/dR$. We must now expand the variation of the Ricci scalar, $\delta R = \delta(g^{\mu\nu}R_{\mu\nu})$:

$$\delta R = R_{\mu\nu}\delta g^{\mu\nu} + g^{\mu\nu}\delta R_{\mu\nu}. \tag{4.216}$$

Recall the Palatini identity for the variation of the Ricci tensor:

$$\delta R_{\mu\nu} = \nabla_\lambda(\delta\Gamma^\lambda_{\mu\nu}) - \nabla_\nu(\delta\Gamma^\lambda_{\mu\lambda}). \tag{4.217}$$

By substituting the variation of the Levi-Civita connection in terms of the metric variation, one finds that the contraction $g^{\mu\nu}\delta R_{\mu\nu}$ can be expressed as a total derivative involving covariant d'Alembertians:

$$g^{\mu\nu}\delta R_{\mu\nu} = \nabla_\mu\nabla_\nu(\delta g^{\mu\nu}) - \Box(g_{\mu\nu}\delta g^{\mu\nu}), \tag{4.218}$$

where $\Box = \nabla^\sigma\nabla_\sigma$. We substitute this into our action variation:

$$\delta S_{f(R)} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[f'(R)R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu}\right] \delta g^{\mu\nu} + \frac{1}{16\pi G} \int d^4x \sqrt{-g} f'(R) \left[\nabla_\mu\nabla_\nu - g_{\mu\nu}\Box\right] \delta g^{\mu\nu}. \tag{4.219}$$

The second integral contains derivatives acting on $\delta g^{\mu\nu}$. We must integrate by parts twice to shift the derivatives onto $f'(R)$. Assuming the boundary terms vanish at infinity, the integration by parts yields:

$$\int d^4x \sqrt{-g} f'(R) \nabla_\mu\nabla_\nu(\delta g^{\mu\nu}) = \int d^4x \sqrt{-g} \left[\nabla_\mu\nabla_\nu f'(R)\right] \delta g^{\mu\nu}, \tag{4.220}$$

and similarly for the d'Alembertian term. Gathering all terms proportional to $\delta g^{\mu\nu}$:

$$\delta S_{f(R)} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[f'(R)R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + \nabla_\mu\nabla_\nu f'(R) - g_{\mu\nu}\Box f'(R)\right] \delta g^{\mu\nu}. \tag{4.221}$$

Adding the variation of the matter action, defined as $\delta S_M = -\frac{1}{2}\int d^4x \sqrt{-g} T_{\mu\nu}\delta g^{\mu\nu}$, and demanding that the total variation vanishes yields the exact **modified field equations for $f(R)$ gravity**:

$$f'(R)R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + \left(\nabla_\mu\nabla_\nu - g_{\mu\nu}\Box\right) f'(R) = 8\pi G T_{\mu\nu}. \tag{4.222}$$

Physical Interpretation: If $f(R) = R$, then $f'(R) = 1$. The derivative terms $\nabla_\mu\nabla_\nu(1)$ trivially vanish, and equation (4.222) perfectly recovers standard General Relativity. However, for non-linear functions (e.g., $f(R) = R + \alpha R^2$), the field equations become fourth-order differential equations for the metric, due to the $\Box f'(R)$ term containing second derivatives of the Ricci scalar (which itself contains second derivatives of the metric). Such theories introduce a new, propagating scalar degree of freedom embedded within the geometry.

4.8.2 Scalar-Tensor Theories

Another prominent class of alternatives abandons the idea that geometry is described solely by a tensor field. In Scalar-Tensor theories, such as Brans-Dicke theory, the gravitational coupling constant G is promoted to a dynamical scalar field $\phi(x)$.

The action for a generalized scalar-tensor theory takes the form:

$$S = \int d^4x\sqrt{-g}\left[\phi R - \frac{\omega(\phi)}{\phi}g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi - V(\phi) + \mathcal{L}_M\right], \tag{4.223}$$

where $\omega(\phi)$ is a coupling parameter and $V(\phi)$ is a potential. Unlike the non-minimal coupling in equation (4.212), here the scalar field dictates the absolute strength of gravity. Varying this action with respect to $g^{\mu\nu}$ generates a modified Einstein equation where the scalar field acts as a geometric source term. Varying the action with respect to ϕ generates a generalized Klein-Gordon equation, demonstrating that the matter distribution $T_{\mu\nu}$ acts as a direct source for the scalar field. This violates the Strong Equivalence Principle, as the local gravitational "constant" becomes dependent on the surrounding cosmic matter distribution.

4.8.3 Torsion and Non-Christoffel Connections

General Relativity rests upon the Fundamental Theorem of Riemannian Geometry, which assumes the connection $\Gamma^\rho_{\mu\nu}$ is both metric-compatible ($\nabla_\lambda g_{\mu\nu} = 0$) and symmetric in its lower indices ($\Gamma^\rho_{\mu\nu} = \Gamma^\rho_{\nu\mu}$).

If we relax the symmetry requirement, the antisymmetric part of the connection defines a new valid tensor, the **torsion tensor**:

$$T^\rho{}_{\mu\nu} = \Gamma^\rho_{\mu\nu} - \Gamma^\rho_{\nu\mu}. \tag{4.224}$$

In Einstein-Cartan theory, spacetime is permitted to have non-vanishing torsion. While mass and energy source spacetime curvature, intrinsic quantum mechanical spin is postulated to source spacetime torsion. Because macroscopic objects have randomly oriented intrinsic spins that average to zero, torsion is utterly negligible in the Solar System. However, at extreme densities (such as inside a collapsing neutron star), torsion would manifest as a repulsive spin-spin interaction, potentially halting the formation of a physical singularity.

4.8.4 Additional Tensor Fields and Geometric Generalizations

When exploring alternatives to General Relativity, one might consider relaxing the fundamental geometric axioms of the manifold, such as metric compatibility. The Fundamental Theorem of Riemannian Geometry guarantees a unique Levi-Civita connection $\mathring{\Gamma}^\rho_{\mu\nu}$ that is both symmetric (torsion-free) and metric-compatible ($\mathring{\nabla}_\lambda g_{\mu\nu} = 0$).

If we drop the requirement of metric compatibility, the covariant derivative of the metric no longer vanishes ($\nabla_\lambda g_{\mu\nu} = Q_{\lambda\mu\nu} \neq 0$), defining the **non-metricity tensor** $Q_{\lambda\mu\nu}$. However, any arbitrary affine connection $\Gamma^\rho_{\mu\nu}$ can be strictly decomposed into the unique Levi-Civita connection plus a tensorial correction $C^\rho_{\mu\nu}$ (often related to the contortion or defect tensor):

$$\Gamma^\rho_{\mu\nu} = \mathring{\Gamma}^\rho_{\mu\nu} + C^\rho_{\mu\nu}. \tag{4.225}$$

Because the difference between any two connections is a true tensor, $C^\rho_{\mu\nu}$ transforms covariantly. Consequently, substituting this decomposed connection into the Riemann curvature tensor effectively splits the curvature into a purely Riemannian part (calculated solely from $g_{\mu\nu}$) plus covariant terms involving $C^\rho_{\mu\nu}$ and its derivatives.

Physical Interpretation: From the perspective of the action principle, a theory of gravity formulated with a non-metric-compatible connection (or a connection with torsion) is mathematically isomorphic to standard General Relativity (using the Levi-Civita connection) coupled to additional, independent tensor fields. Because these extra degrees of freedom do not fundamentally alter the underlying Riemannian structure, theories involving non-metricity or torsion are often viewed not as radical replacements for General Relativity, but simply as General Relativity augmented by exotic matter fields.

4.9 Variational Derivations of Advanced Field Equations

To solidify the Lagrangian formulation of General Relativity, we present explicitly expanded derivations of two critical results: the derivation of the electromagnetic energy-momentum tensor via metric variation, and the Palatini formulation of gravity, which dynamically generates metric compatibility rather than assuming it a priori.

4.9.1 The Electromagnetic Stress-Energy Tensor

Consider the action for the electromagnetic field in a curved spacetime background. The Maxwell Lagrangian density is constructed from the antisymmetric Faraday tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, where A_μ is the four-potential. The action is:

$$S_{\text{EM}} = \int d^4x \sqrt{-g} \left(-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + A_\mu J^\mu \right), \tag{4.226}$$

where J^μ is the conserved charge current. To derive the stress-energy tensor $T_{\mu\nu}$ using the Hilbert definition, we must vary this action with respect to the inverse metric $g^{\rho\sigma}$.

First, we rewrite the scalar contraction $F^{\mu\nu}F_{\mu\nu}$ explicitly in terms of the inverse metric to expose all metric dependencies:

$$F^{\mu\nu} F_{\mu\nu} = g^{\mu\alpha} g^{\nu\beta} F_{\mu\nu} F_{\alpha\beta}. \tag{4.227}$$

Crucially, the Faraday tensor $F_{\mu\nu}$ is defined purely by partial derivatives of the potential A_μ, which are entirely independent of the metric. Thus, $\delta F_{\mu\nu} = 0$.

We vary the action S_{EM} with respect to $g^{\rho\sigma}$:

$$\begin{aligned} \delta S_{\text{EM}} &= \delta \int d^4x \left[\sqrt{-g} \left(-\frac{1}{4} g^{\mu\alpha} g^{\nu\beta} F_{\mu\nu} F_{\alpha\beta} \right) \right] \\ &= \int d^4x \left[(\delta\sqrt{-g}) \left(-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} \right) - \frac{1}{4} \sqrt{-g} \delta(g^{\mu\alpha} g^{\nu\beta}) F_{\mu\nu} F_{\alpha\beta} \right]. \end{aligned} \tag{4.228}$$

Recall the identity for the variation of the metric determinant: $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g} g_{\rho\sigma} \delta g^{\rho\sigma}$.

Next, we expand the variation of the metric product using the product rule:

$$\delta(g^{\mu\alpha} g^{\nu\beta}) = (\delta g^{\mu\alpha}) g^{\nu\beta} + g^{\mu\alpha} (\delta g^{\nu\beta}). \tag{4.229}$$

Substituting this into the second term of the integral:

$$-\frac{1}{4}\sqrt{-g} \left[(\delta g^{\mu\alpha}) g^{\nu\beta} F_{\mu\nu} F_{\alpha\beta} + g^{\mu\alpha} (\delta g^{\nu\beta}) F_{\mu\nu} F_{\alpha\beta} \right]. \tag{4.230}$$

We relabel dummy indices to factor out $\delta g^{\rho\sigma}$. In the first term, let $\mu \to \rho$ and $\alpha \to \sigma$. In the second term, let $\nu \to \rho$ and $\beta \to \sigma$:

$$-\frac{1}{4}\sqrt{-g} \left[\delta g^{\rho\sigma} (g^{\nu\beta} F_{\rho\nu} F_{\sigma\beta}) + \delta g^{\rho\sigma} (g^{\mu\alpha} F_{\mu\rho} F_{\alpha\sigma}) \right]. \tag{4.231}$$

Using the antisymmetry of the Faraday tensor ($F_{\mu\rho} = -F_{\rho\mu}$ and $F_{\alpha\sigma} = -F_{\sigma\alpha}$), the second term becomes identical to the first:

$$-\frac{1}{4}\sqrt{-g} \left[2\delta g^{\rho\sigma} F_\rho{}^\beta F_{\sigma\beta} \right] = -\frac{1}{2}\sqrt{-g} F_{\rho\lambda} F_\sigma{}^\lambda \delta g^{\rho\sigma}. \tag{4.232}$$

Combining both parts of the variation:

$$\delta S_{\text{EM}} = \int d^4x \sqrt{-g} \left[\frac{1}{8} g_{\rho\sigma} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} F_{\rho\lambda} F_\sigma{}^\lambda \right] \delta g^{\rho\sigma}. \tag{4.233}$$

Using the Hilbert definition of the stress-energy tensor, $T_{\rho\sigma} = \frac{-2}{\sqrt{-g}} \frac{\delta S_{\text{EM}}}{\delta g^{\rho\sigma}}$, we multiply the bracketed term by -2 to obtain the final result:

$$T_{\rho\sigma} = F_{\rho\lambda} F_\sigma{}^\lambda - \frac{1}{4} g_{\rho\sigma} F_{\mu\nu} F^{\mu\nu}. \tag{4.234}$$

Properties: By contracting with the inverse metric ($T = g^{\rho\sigma}T_{\rho\sigma}$), we find $T = F^{\sigma\lambda}F_{\sigma\lambda} - \frac{1}{4}(4)F_{\mu\nu}F^{\mu\nu} = 0$. The electromagnetic stress-energy tensor is strictly traceless in four dimensions, reflecting the conformal invariance of Maxwell's equations and the masslessness of the photon.

4.9.2 The Palatini Formulation of General Relativity

In the standard derivation of Einstein's field equations from the Einstein-Hilbert action, we assumed from the outset that the connection $\Gamma^\lambda_{\mu\nu}$ was the Levi-Civita connection, fully determined by the metric. The Palatini formulation takes a more fundamental approach: it treats the metric $g_{\mu\nu}$ and the connection $\Gamma^\lambda_{\mu\nu}$ as entirely independent dynamical variables.

The action is formally identical, but the Ricci tensor is now considered a function solely of the connection, $R_{\mu\nu}(\Gamma)$:

$$S[g,\Gamma] = \int d^4x\sqrt{-g}g^{\mu\nu}R_{\mu\nu}(\Gamma). \tag{4.235}$$

Varying the action with respect to the metric $g^{\mu\nu}$ proceeds exactly as before, trivially yielding $R_{\mu\nu}(\Gamma) - \frac{1}{2}R(\Gamma)g_{\mu\nu} = 0$.

The profoundly new physics emerges when we vary the action with respect to the connection. Because $\delta\Gamma^\lambda_{\mu\nu}$ is a tensor, the variation of the Ricci tensor is given by the Palatini identity:

$$\delta R_{\mu\nu} = \nabla_\lambda(\delta\Gamma^\lambda_{\mu\nu}) - \nabla_\nu(\delta\Gamma^\lambda_{\mu\lambda}). \tag{4.236}$$

We insert this into the action variation:

$$\delta_\Gamma S = \int d^4x\sqrt{-g}g^{\mu\nu}\left[\nabla_\lambda(\delta\Gamma^\lambda_{\mu\nu}) - \nabla_\nu(\delta\Gamma^\lambda_{\mu\lambda})\right] = 0. \tag{4.237}$$

Because the connection is an independent variable, we *cannot* assume that $\nabla_\lambda g^{\mu\nu} = 0$. Therefore, we cannot pull $g^{\mu\nu}$ inside the covariant derivatives. Instead, we must integrate by parts. The generalized integration by parts rule in curved space (derived from the divergence theorem) states that for a vector V^λ and scalar f:

$$\int d^4x\sqrt{-g}f\nabla_\lambda V^\lambda = -\int d^4x\sqrt{-g}V^\lambda\nabla_\lambda f + \text{boundary terms}. \tag{4.238}$$

However, the object we are varying against is a tensor density, $\sqrt{-g}g^{\mu\nu}$. Applying the chain rule in reverse (integration by parts) to our variation yields:

$$\delta_\Gamma S = \int d^4x\left[-\nabla_\lambda(\sqrt{-g}g^{\mu\nu})\delta\Gamma^\lambda_{\mu\nu} + \nabla_\nu(\sqrt{-g}g^{\mu\nu})\delta\Gamma^\lambda_{\mu\lambda}\right]. \tag{4.239}$$

We relabel dummy indices in the second term to factor out $\delta\Gamma^\lambda_{\mu\nu}$. Let $\nu \to \lambda$ and $\lambda \to \nu$:

$$\delta_\Gamma S = \int d^4x\left[-\nabla_\lambda(\sqrt{-g}g^{\mu\nu}) + \delta^\nu_\lambda\nabla_\sigma(\sqrt{-g}g^{\mu\sigma})\right]\delta\Gamma^\lambda_{\mu\nu} = 0. \tag{4.240}$$

For this action to be stationary under arbitrary variations $\delta\Gamma^\lambda_{\mu\nu}$, the quantity in the brackets must vanish identically:

$$\nabla_\lambda(\sqrt{-g}g^{\mu\nu}) - \delta^\nu_\lambda\nabla_\sigma(\sqrt{-g}g^{\mu\sigma}) = 0. \tag{4.241}$$

We can solve this differential equation by contracting indices. Taking the trace by setting $\nu = \lambda$ yields:

$$\nabla_\sigma(\sqrt{-g}g^{\mu\sigma}) - 4\nabla_\sigma(\sqrt{-g}g^{\mu\sigma}) = 0 \implies -3\nabla_\sigma(\sqrt{-g}g^{\mu\sigma}) = 0. \tag{4.242}$$

This requires that $\nabla_\sigma(\sqrt{-g}g^{\mu\sigma}) = 0$. Substituting this back into equation (4.241) eliminates the second term entirely, leaving the strict requirement:

$$\nabla_\lambda(\sqrt{-g}g^{\mu\nu}) = 0. \tag{4.243}$$

Expanding the covariant derivative of the scalar density $\sqrt{-g}$:

$$\sqrt{-g}\nabla_\lambda g^{\mu\nu} + g^{\mu\nu}\nabla_\lambda\sqrt{-g} = 0. \tag{4.244}$$

By utilizing the identity $\nabla_\lambda\sqrt{-g} = \frac{1}{2}\sqrt{-g}g^{\alpha\beta}\nabla_\lambda g_{\alpha\beta}$, one can algebraically show that this implies the covariant derivative of the metric itself must vanish:

$$\nabla_\lambda g^{\mu\nu} = 0. \tag{4.245}$$

Geometric Conclusion: The Palatini formulation is extraordinarily profound. By treating the metric and the connection as completely independent geometric objects, the principle of least action dynamically forces the connection to be metric-compatible. We do not have to assume Riemannian geometry a priori; the dynamics of the gravitational field independently demand it.

Chapter 5

The Schwarzschild Solution

5.1 The Schwarzschild Solution

5.1.1 Metric Ansatz and Symmetry Justification

We aim to describe the exterior gravitational field of a non-rotating, unevolving, spherically symmetric massive body. We begin by considering the metric of flat Minkowski space in spherical coordinates $x^\mu = (t, r, \theta, \phi)$, which inherently possesses spherical symmetry:

$$ds^2_{\text{Minkowski}} = -dt^2 + dr^2 + r^2 d\Omega^2, \tag{5.1}$$

where $d\Omega^2$ represents the line element of a unit two-sphere:

$$d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2. \tag{5.2}$$

To describe a curved spacetime that maintains these physical symmetries, we must impose constraints on the general metric tensor $g_{\mu\nu}$.

First, we require the spacetime to be static. The rigorous condition for a static metric dictates that all metric components are independent of the time coordinate t, and the metric is invariant under time reversal, $t \to -t$. Let us examine a generic metric containing off-diagonal time-space components:

$$ds^2 = g_{tt}dt^2 + 2g_{ti}dtdx^i + g_{ij}dx^i dx^j. \tag{5.3}$$

Applying the time reversal transformation $t \to -t$, the differential transforms as $dt \to -dt$. Substituting this into the metric yields:

$$ds^2 = g_{tt}(-dt)^2 + 2g_{ti}(-dt)dx^i + g_{ij}dx^i dx^j, \tag{5.4}$$

$$ds^2 = g_{tt}dt^2 - 2g_{ti}dtdx^i + g_{ij}dx^i dx^j. \tag{5.5}$$

For the geometry to remain invariant under this transformation, the cross terms must equate to their negative counterparts:

$$2g_{ti}dtdx^i = -2g_{ti}dtdx^i. \tag{5.6}$$

This condition mandates that $g_{ti} = 0$. Consequently, a static spacetime admits no cross terms between the temporal and spatial coordinates.

Second, we impose spherical symmetry. The metric must preserve the $SO(3)$ symmetry of the spheres, implying that the angular dependence must remain strictly proportional to $d\Omega^2$. The remaining metric components cannot depend on the angular coordinates (θ, ϕ) and are strictly functions of the radial coordinate r. To ensure the signature of the metric remains Lorentzian $(-, +, +, +)$ throughout the spacetime, we express the unknown metric coefficients as exponential functions. The most general static, spherically symmetric metric ansatz is therefore:

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + e^{2\gamma(r)}r^2 d\Omega^2. \tag{5.7}$$

In general relativity, coordinates hold no intrinsic physical meaning a priori, granting the freedom to perform diffeomorphisms to simplify the metric. We perform a coordinate transformation to absorb the arbitrary function $e^{2\gamma(r)}$ into the definition of the radial coordinate. Let us define a new radial coordinate $\bar{r}$ as:

$$\bar{r} = e^{\gamma(r)}r. \tag{5.8}$$

To express the metric in terms of $\overline{r}$, we compute its total differential. Applying the product and chain rules with respect to r:

$$d\overline{r} = \frac{d}{dr}\left(e^{\gamma(r)}r\right)dr, \tag{5.9}$$

$$d\overline{r} = \left(\frac{d}{dr}\left(e^{\gamma(r)}\right)r + e^{\gamma(r)}\frac{d}{dr}(r)\right)dr, \tag{5.10}$$

$$d\overline{r} = \left(e^{\gamma(r)}\frac{d\gamma(r)}{dr}r + e^{\gamma(r)}\right)dr. \tag{5.11}$$

Factoring out $e^{\gamma(r)}$ yields:

$$d\overline{r} = e^{\gamma(r)}\left(1 + r\frac{d\gamma(r)}{dr}\right)dr. \tag{5.12}$$

We isolate dr to substitute back into the metric:

$$dr = e^{-\gamma(r)}\left(1 + r\frac{d\gamma(r)}{dr}\right)^{-1} d\overline{r}. \tag{5.13}$$

Squaring this differential, we obtain:

$$dr^2 = e^{-2\gamma(r)}\left(1 + r\frac{d\gamma(r)}{dr}\right)^{-2} d\overline{r}^2. \tag{5.14}$$

We now substitute dr^2 and $e^{2\gamma(r)}r^2 = \overline{r}^2$ into the general ansatz:

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}\left[e^{-2\gamma(r)}\left(1 + r\frac{d\gamma(r)}{dr}\right)^{-2} d\overline{r}^2\right] + \overline{r}^2 d\Omega^2, \tag{5.15}$$

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)-2\gamma(r)}\left(1 + r\frac{d\gamma(r)}{dr}\right)^{-2} d\overline{r}^2 + \overline{r}^2 d\Omega^2. \tag{5.16}$$

Since r is a function of $\overline{r}$, the functions $\alpha(r)$ and $\beta(r)$ are implicitly functions of $\overline{r}$. We can group the factors modulating $d\overline{r}^2$ into a single new exponential function. Let us define:

$$e^{2\overline{\beta}(\overline{r})} = e^{2\beta(r)-2\gamma(r)}\left(1 + r\frac{d\gamma(r)}{dr}\right)^{-2}, \tag{5.17}$$

and assign $\overline{\alpha}(\overline{r}) = \alpha(r)$. The transformed metric takes the form:

$$ds^2 = -e^{2\overline{\alpha}(\overline{r})}dt^2 + e^{2\overline{\beta}(\overline{r})}d\overline{r}^2 + \overline{r}^2 d\Omega^2. \tag{5.18}$$

Because coordinates are mere labels, we are free to drop the overbars and formally replace $\overline{r} \to r$, recovering the familiar radial coordinate label. The resulting metric is precisely as general as the original ansatz but exhibits a significantly simpler algebraic structure:

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2 d\Omega^2. \tag{5.19}$$

5.1.2 Inverse Metric and Determinant

Covariant Metric Tensor Components

We establish the covariant components of the metric tensor, denoted by $g_{\mu\nu}$, arising from the static, spherically symmetric line element derived in Part 1. We adopt the coordinate system $x^\mu = (x^0, x^1, x^2, x^3) = (t, r, \theta, \phi)$. The invariant interval is given by:

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta \, d\phi^2. \tag{5.20}$$

By direct comparison with the general quadratic form $g_{\mu\nu}dx^\mu dx^\nu$, we extract the non-vanishing covariant components of the metric tensor:

$$g_{tt} = -e^{2\alpha(r)}, \tag{5.21}$$

$$g_{rr} = e^{2\beta(r)}, \tag{5.22}$$

$$g_{\theta\theta} = r^2, \tag{5.23}$$

$$g_{\phi\phi} = r^2 \sin^2\theta. \tag{5.24}$$

All off-diagonal components (where $\mu \neq \nu$) are strictly zero:

$$g_{\mu\nu} = 0 \quad \text{for} \quad \mu \neq \nu. \tag{5.25}$$

We can represent the covariant metric tensor explicitly in matrix form as:

$$g_{\mu\nu} = \begin{pmatrix} -e^{2\alpha(r)} & 0 & 0 & 0 \\ 0 & e^{2\beta(r)} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2\sin^2\theta \end{pmatrix}. \tag{5.26}$$

Computation of the Inverse Metric Tensor

The contravariant metric tensor (or inverse metric), denoted by $g^{\mu\nu}$, is defined by the fundamental identity:

$$g^{\mu\lambda} g_{\lambda\nu} = \delta^\mu_\nu, \tag{5.27}$$

where δ^μ_ν is the Kronecker delta, taking the value 1 when $\mu = \nu$ and 0 when $\mu \neq \nu$, and we implicitly sum over the dummy index $\lambda \in \{t, r, \theta, \phi\}$.

We compute the diagonal elements by setting $\mu = \nu$. Let us explicitly expand the summation for the temporal component $\mu = \nu = t$:

$$\sum_\lambda g^{t\lambda} g_{\lambda t} = \delta^t_t. \tag{5.28}$$

$$g^{tt} g_{tt} + g^{tr} g_{rt} + g^{t\theta} g_{\theta t} + g^{t\phi} g_{\phi t} = 1. \tag{5.29}$$

Substituting the known values $g_{rt} = 0$, $g_{\theta t} = 0$, and $g_{\phi t} = 0$, the equation simplifies to:

$$g^{tt} g_{tt} = 1. \tag{5.30}$$

Substituting the explicit function for g_{tt}:

$$g^{tt}\left(-e^{2\alpha(r)}\right) = 1. \tag{5.31}$$

Solving for the contravariant component g^{tt}:

$$g^{tt} = \frac{1}{-e^{2\alpha(r)}}. \tag{5.32}$$

$$g^{tt} = -e^{-2\alpha(r)}. \tag{5.33}$$

Applying the identical summation expansion to the remaining diagonal components $\mu = \nu = r$, $\mu = \nu = \theta$, and $\mu = \nu = \phi$ yields:

$$g^{rr} g_{rr} = 1, \tag{5.34}$$

$$g^{\theta\theta} g_{\theta\theta} = 1, \tag{5.35}$$

$$g^{\phi\phi} g_{\phi\phi} = 1. \tag{5.36}$$

Substituting the respective covariant components and solving:

$$g^{rr} = \frac{1}{g_{rr}} = \frac{1}{e^{2\beta(r)}} = e^{-2\beta(r)}, \tag{5.37}$$

$$g^{\theta\theta} = \frac{1}{g_{\theta\theta}} = \frac{1}{r^2}, \tag{5.38}$$

$$g^{\phi\phi} = \frac{1}{g_{\phi\phi}} = \frac{1}{r^2 \sin^2\theta}. \tag{5.39}$$

To verify that the off-diagonal components of the inverse metric vanish, we set $\mu \neq \nu$. Let us take $\mu = t$ and $\nu = r$ as an example, expanding the sum over λ:

$$\sum_\lambda g^{t\lambda} g_{\lambda r} = \delta^t_r. \tag{5.40}$$

$$g^{tt} g_{tr} + g^{tr} g_{rr} + g^{t\theta} g_{\theta r} + g^{t\phi} g_{\phi r} = 0. \tag{5.41}$$

Substituting $g_{tr} = 0$, $g_{\theta r} = 0$, and $g_{\phi r} = 0$:

$$g^{tr} g_{rr} = 0. \tag{5.42}$$

Since $g_{rr} = e^{2\beta(r)} \neq 0$, it must follow that:

$$g^{tr} = 0. \tag{5.43}$$

By symmetry and identical algebraic expansion for all $\mu \neq \nu$, we conclude that all off-diagonal components $g^{\mu\nu}$ vanish. The explicit matrix representation of the inverse metric is:

$$g^{\mu\nu} = \begin{pmatrix} -e^{-2\alpha(r)} & 0 & 0 & 0 \\ 0 & e^{-2\beta(r)} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2\sin^2\theta} \end{pmatrix}. \tag{5.44}$$

Verification of the Inverse Metric via Matrix Multiplication

We rigorously verify the inverse relation by performing the explicit matrix multiplication $g^{\mu\lambda} g_{\lambda\nu}$, which must yield the identity matrix I (equivalent to the components of δ^μ_ν):

$$g^{\mu\lambda} g_{\lambda\nu} = \begin{pmatrix} -e^{-2\alpha(r)} & 0 & 0 & 0 \\ 0 & e^{-2\beta(r)} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2\sin^2\theta} \end{pmatrix} \begin{pmatrix} -e^{2\alpha(r)} & 0 & 0 & 0 \\ 0 & e^{2\beta(r)} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2\sin^2\theta \end{pmatrix}. \tag{5.45}$$

Performing the row-by-column multiplications for the diagonal elements:

$$(g \cdot g)_{tt} = \left(-e^{-2\alpha(r)}\right)\left(-e^{2\alpha(r)}\right) + 0 + 0 + 0, \tag{5.46}$$

$$(g \cdot g)_{rr} = 0 + \left(e^{-2\beta(r)}\right)\left(e^{2\beta(r)}\right) + 0 + 0, \tag{5.47}$$

$$(g \cdot g)_{\theta\theta} = 0 + 0 + \left(\frac{1}{r^2}\right)\left(r^2\right) + 0, \tag{5.48}$$

$$(g \cdot g)_{\phi\phi} = 0 + 0 + 0 + \left(\frac{1}{r^2\sin^2\theta}\right)\left(r^2\sin^2\theta\right). \tag{5.49}$$

Evaluating the products of the exponential and algebraic terms:

$$(g \cdot g)_{tt} = e^{-2\alpha(r)+2\alpha(r)} = e^0 = 1, \tag{5.50}$$

$$(g \cdot g)_{rr} = e^{-2\beta(r)+2\beta(r)} = e^0 = 1, \tag{5.51}$$

$$(g \cdot g)_{\theta\theta} = 1, \tag{5.52}$$

$$(g \cdot g)_{\phi\phi} = 1. \tag{5.53}$$

Since the product of two diagonal matrices yields another diagonal matrix, the off-diagonal terms are trivially zero. Thus, we reconstruct the Kronecker delta matrix:

$$g^{\mu\lambda}g_{\lambda\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \delta^{\mu}_{\nu}. \tag{5.54}$$

This completes the verification.

Computation of the Metric Determinant

Let g denote the determinant of the covariant metric tensor, defined as $g = \det(g_{\mu\nu})$. Because the matrix representation of $g_{\mu\nu}$ is purely diagonal, its determinant is exactly the product of its diagonal elements:

$$g = g_{tt}g_{rr}g_{\theta\theta}g_{\phi\phi}. \tag{5.55}$$

Substituting the explicit covariant components into this product:

$$g = \left(-e^{2\alpha(r)}\right)\left(e^{2\beta(r)}\right)\left(r^2\right)\left(r^2\sin^2\theta\right). \tag{5.56}$$

We group the constant sign, the exponential terms, and the polynomial/trigonometric terms:

$$g = -\left(e^{2\alpha(r)}e^{2\beta(r)}\right)\left(r^2 \cdot r^2\right)\left(\sin^2\theta\right). \tag{5.57}$$

Applying the exponent addition rule $e^A e^B = e^{A+B}$:

$$g = -e^{2\alpha(r)+2\beta(r)}r^4\sin^2\theta. \tag{5.58}$$

Factoring the coefficient of 2 in the exponential term yields the final expression for the determinant:

$$g = -e^{2(\alpha(r)+\beta(r))}r^4\sin^2\theta. \tag{5.59}$$

Computation of the Invariant Volume Element Factor

In tensorial integration and the derivation of covariant divergence, the invariant volume element involves the factor $\sqrt{-g}$. We compute this explicitly. First, we determine $-g$:

$$-g = -\left(-e^{2(\alpha(r)+\beta(r))}r^4\sin^2\theta\right). \tag{5.60}$$

$$-g = e^{2(\alpha(r)+\beta(r))}r^4\sin^2\theta. \tag{5.61}$$

Taking the principal square root of both sides:

$$\sqrt{-g} = \sqrt{e^{2(\alpha(r)+\beta(r))}r^4\sin^2\theta}. \tag{5.62}$$

We distribute the square root (which is equivalent to raising to the power of $\frac{1}{2}$) across the multiplicative factors:

$$\sqrt{-g} = \left(e^{2(\alpha(r)+\beta(r))}\right)^{\frac{1}{2}}\left(r^4\right)^{\frac{1}{2}}\left(\sin^2\theta\right)^{\frac{1}{2}}. \tag{5.63}$$

Applying the power rule $(x^a)^b = x^{ab}$ to each term:

$$\left(e^{2(\alpha(r)+\beta(r))}\right)^{\frac{1}{2}} = e^{\alpha(r)+\beta(r)}, \tag{5.64}$$

$$\left(r^4\right)^{\frac{1}{2}} = r^2, \tag{5.65}$$

$$\left(\sin^2\theta\right)^{\frac{1}{2}} = \sin\theta. \tag{5.66}$$

(Note: we assume the standard spherical coordinate domain $0 \leq \theta \leq \pi$, wherein $\sin\theta \geq 0$, so the principal root is simply $\sin\theta$). Multiplying these simplified factors yields the final explicit form:

$$\sqrt{-g} = e^{\alpha(r)+\beta(r)}r^2\sin\theta. \tag{5.67}$$

Deriving Christoffel Symbols via the Lagrangian Method

We begin by defining the Lagrangian $\mathcal{L}$ for the geodesics of a test particle moving in the spacetime. The Lagrangian is given by the kinetic term associated with the metric:

$$\mathcal{L} = g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu \tag{5.68}$$

where $\dot{x}^\mu = \frac{dx^\mu}{d\lambda}$ denotes the derivative with respect to an affine parameter λ. Using the time-dependent spherically symmetric metric, $ds^2 = -e^{2\alpha(t,r)}dt^2 + e^{2\beta(t,r)}dr^2 + r^2 d\Omega^2$, we explicitly expand the Lagrangian in the coordinates (t, r, θ, ϕ):

$$\mathcal{L} = -e^{2\alpha(t,r)}\dot{t}^2 + e^{2\beta(t,r)}\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2 \tag{5.69}$$

The motion of a particle traversing a geodesic satisfies the Euler-Lagrange equations for each coordinate x^μ:

$$\frac{d}{d\lambda}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu}\right) - \frac{\partial \mathcal{L}}{\partial x^\mu} = 0 \tag{5.70}$$

We will evaluate this equation for each coordinate independently, manipulate the result into the canonical geodesic form, and extract the corresponding Christoffel symbols by matching the coefficients term-by-term with the general geodesic equation:

$$\ddot{x}^\mu + \Gamma^\mu_{\rho\sigma}\dot{x}^\rho\dot{x}^\sigma = 0 \tag{5.71}$$

1. The t-equation

We evaluate the Euler-Lagrange equation (5.70) for $x^\mu = t$. The derivative with respect to the velocity $\dot{t}$ is:

$$\frac{\partial \mathcal{L}}{\partial \dot{t}} = -2e^{2\alpha(t,r)}\dot{t} \tag{5.72}$$

Taking the total derivative with respect to λ requires applying the chain rule to the exponential factor, keeping in mind that α depends on both t and r:

$$\begin{aligned}
\frac{d}{d\lambda}\left(\frac{\partial \mathcal{L}}{\partial \dot{t}}\right) &= -2\frac{d}{d\lambda}\left(e^{2\alpha(t,r)}\right)\dot{t} - 2e^{2\alpha(t,r)}\ddot{t} \\
&= -2\left(e^{2\alpha(t,r)} \cdot 2\left[\frac{\partial \alpha}{\partial t}\dot{t} + \frac{\partial \alpha}{\partial r}\dot{r}\right]\right)\dot{t} - 2e^{2\alpha(t,r)}\ddot{t} \\
&= -4e^{2\alpha(t,r)}\partial_t\alpha\,\dot{t}^2 - 4e^{2\alpha(t,r)}\partial_r\alpha\,\dot{t}\dot{r} - 2e^{2\alpha(t,r)}\ddot{t}
\end{aligned} \tag{5.73}$$

The derivative of the Lagrangian with respect to the coordinate t is:

$$\frac{\partial \mathcal{L}}{\partial t} = -2\dot{t}^2 e^{2\alpha(t,r)}\partial_t\alpha + 2\dot{r}^2 e^{2\beta(t,r)}\partial_t\beta \tag{5.74}$$

Substituting these into the Euler-Lagrange equation:

$$-2e^{2\alpha(t,r)}\ddot{t} - 4e^{2\alpha(t,r)}\partial_t\alpha\,\dot{t}^2 - 4e^{2\alpha(t,r)}\partial_r\alpha\,\dot{t}\dot{r} - \left(-2e^{2\alpha(t,r)}\partial_t\alpha\,\dot{t}^2 + 2e^{2\beta(t,r)}\partial_t\beta\,\dot{r}^2\right) = 0 \tag{5.75}$$

We divide the entire equation by $-2e^{2\alpha(t,r)}$ to isolate $\ddot{t}$:

$$\ddot{t} + 2\partial_t\alpha\,\dot{t}^2 + 2\partial_r\alpha\,\dot{t}\dot{r} - \partial_t\alpha\,\dot{t}^2 + e^{2(\beta-\alpha)}\partial_t\beta\,\dot{r}^2 = 0 \tag{5.76}$$

Simplifying the algebraically similar $\dot{t}^2$ terms:

$$\ddot{t} + (\partial_t\alpha)\dot{t}^2 + e^{2(\beta-\alpha)}(\partial_t\beta)\dot{r}^2 + 2(\partial_r\alpha)\dot{t}\dot{r} = 0 \tag{5.77}$$

Comparing equation (5.77) directly to the canonical t-geodesic equation:

$$\ddot{t} + \Gamma^t_{tt}\dot{t}^2 + \Gamma^t_{rr}\dot{r}^2 + \Gamma^t_{\theta\theta}\dot{\theta}^2 + \Gamma^t_{\phi\phi}\dot{\phi}^2 + 2\Gamma^t_{tr}\dot{t}\dot{r} + \cdots = 0 \tag{5.78}$$

We immediately extract the following non-zero Christoffel symbols by coefficient matching:

$$\Gamma^t_{tt} = \partial_t\alpha \tag{5.79}$$
$$\Gamma^t_{rr} = e^{2(\beta-\alpha)}\partial_t\beta \tag{5.80}$$
$$\Gamma^t_{tr} = \partial_r\alpha \tag{5.81}$$

2. The r-equation

We evaluate the Euler-Lagrange equation for $x^\mu = r$. The derivative with respect to $\dot{r}$ is:

$$\frac{\partial \mathcal{L}}{\partial \dot{r}} = 2e^{2\beta(t,r)}\dot{r} \tag{5.82}$$

Taking the total derivative with respect to λ:

$$\begin{aligned} \frac{d}{d\lambda}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}}\right) &= 2\left(e^{2\beta(t,r)} \cdot 2\left[\frac{\partial \beta}{\partial t}\dot{t} + \frac{\partial \beta}{\partial r}\dot{r}\right]\right)\dot{r} + 2e^{2\beta(t,r)}\ddot{r} \\ &= 4e^{2\beta(t,r)}\partial_t\beta\,\dot{t}\dot{r} + 4e^{2\beta(t,r)}\partial_r\beta\,\dot{r}^2 + 2e^{2\beta(t,r)}\ddot{r} \end{aligned} \tag{5.83}$$

The derivative of the Lagrangian with respect to the coordinate r is:

$$\frac{\partial \mathcal{L}}{\partial r} = -2\dot{t}^2 e^{2\alpha(t,r)}\partial_r\alpha + 2\dot{r}^2 e^{2\beta(t,r)}\partial_r\beta + 2r\dot{\theta}^2 + 2r\sin^2\theta\dot{\phi}^2 \tag{5.84}$$

Substituting these into the Euler-Lagrange equation:

$$2e^{2\beta}\ddot{r} + 4e^{2\beta}\partial_t\beta\,\dot{t}\dot{r} + 4e^{2\beta}\partial_r\beta\,\dot{r}^2 - \left(-2e^{2\alpha}\partial_r\alpha\,\dot{t}^2 + 2e^{2\beta}\partial_r\beta\,\dot{r}^2 + 2r\dot{\theta}^2 + 2r\sin^2\theta\dot{\phi}^2\right) = 0 \tag{5.85}$$

Divide the entire equation by $2e^{2\beta(t,r)}$ to isolate $\ddot{r}$:

$$\ddot{r} + 2\partial_t\beta\,\dot{t}\dot{r} + 2\partial_r\beta\,\dot{r}^2 + e^{2(\alpha-\beta)}\partial_r\alpha\,\dot{t}^2 - \partial_r\beta\,\dot{r}^2 - re^{-2\beta}\dot{\theta}^2 - r\sin^2\theta e^{-2\beta}\dot{\phi}^2 = 0 \tag{5.86}$$

Simplifying the algebraically similar $\dot{r}^2$ terms:

$$\ddot{r} + e^{2(\alpha-\beta)}(\partial_r\alpha)\dot{t}^2 + (\partial_r\beta)\dot{r}^2 - re^{-2\beta}\dot{\theta}^2 - r\sin^2\theta e^{-2\beta}\dot{\phi}^2 + 2(\partial_t\beta)\dot{t}\dot{r} = 0 \tag{5.87}$$

Comparing equation (5.87) with the canonical r-geodesic equation:

$$\ddot{r} + \Gamma^r_{tt}\dot{t}^2 + \Gamma^r_{rr}\dot{r}^2 + \Gamma^r_{\theta\theta}\dot{\theta}^2 + \Gamma^r_{\phi\phi}\dot{\phi}^2 + 2\Gamma^r_{tr}\dot{t}\dot{r} + \cdots = 0 \tag{5.88}$$

We extract the following non-zero Christoffel symbols:

$$\Gamma^r_{tt} = e^{2(\alpha-\beta)}\partial_r\alpha \tag{5.89}$$
$$\Gamma^r_{rr} = \partial_r\beta \tag{5.90}$$
$$\Gamma^r_{\theta\theta} = -re^{-2\beta} \tag{5.91}$$
$$\Gamma^r_{\phi\phi} = -re^{-2\beta}\sin^2\theta \tag{5.92}$$
$$\Gamma^r_{tr} = \partial_t\beta \tag{5.93}$$

3. The θ-equation

We evaluate the Euler-Lagrange equation for $x^\mu = \theta$. The derivative with respect to $\dot{\theta}$ is:

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = 2r^2\dot{\theta} \tag{5.94}$$

Taking the total derivative with respect to λ:

$$\frac{d}{d\lambda}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) = 4r\dot{r}\dot{\theta} + 2r^2\ddot{\theta} \tag{5.95}$$

The derivative of the Lagrangian with respect to the coordinate θ is:

$$\frac{\partial \mathcal{L}}{\partial \theta} = 2r^2\sin\theta\cos\theta\,\dot{\phi}^2 \tag{5.96}$$

Substituting these into the Euler-Lagrange equation:

$$2r^2\ddot{\theta} + 4r\dot{r}\dot{\theta} - 2r^2 \sin\theta\cos\theta\,\dot{\phi}^2 = 0 \tag{5.97}$$

Divide the entire equation by $2r^2$:

$$\ddot{\theta} + (-\sin\theta\cos\theta)\dot{\phi}^2 + 2\left(\frac{1}{r}\right)\dot{r}\dot{\theta} = 0 \tag{5.98}$$

Comparing equation (5.98) to the canonical θ-geodesic equation:

$$\ddot{\theta} + \Gamma^{\theta}_{\phi\phi}\dot{\phi}^2 + 2\Gamma^{\theta}_{r\theta}\dot{r}\dot{\theta} + \cdots = 0 \tag{5.99}$$

We extract the non-zero Christoffel symbols:

$$\Gamma^{\theta}_{\phi\phi} = -\sin\theta\cos\theta \tag{5.100}$$

$$\Gamma^{\theta}_{r\theta} = \frac{1}{r} \tag{5.101}$$

4. The ϕ-equation

We evaluate the Euler-Lagrange equation for $x^\mu = \phi$. The derivative with respect to $\dot{\phi}$ is:

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 2r^2\sin^2\theta\,\dot{\phi} \tag{5.102}$$

Taking the total derivative with respect to λ:

$$\begin{aligned}\frac{d}{d\lambda}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right) &= 2\frac{d}{d\lambda}(r^2\sin^2\theta)\dot{\phi} + 2r^2\sin^2\theta\,\ddot{\phi} \\ &= 2\left(2r\dot{r}\sin^2\theta + r^2(2\sin\theta\cos\theta)\dot{\theta}\right)\dot{\phi} + 2r^2\sin^2\theta\,\ddot{\phi} \\ &= 4r\sin^2\theta\,\dot{r}\dot{\phi} + 4r^2\sin\theta\cos\theta\,\dot{\theta}\dot{\phi} + 2r^2\sin^2\theta\,\ddot{\phi}\end{aligned} \tag{5.103}$$

The derivative of the Lagrangian with respect to the coordinate ϕ is strictly zero:

$$\frac{\partial \mathcal{L}}{\partial \phi} = 0 \tag{5.104}$$

Substituting these into the Euler-Lagrange equation:

$$2r^2\sin^2\theta\,\ddot{\phi} + 4r\sin^2\theta\,\dot{r}\dot{\phi} + 4r^2\sin\theta\cos\theta\,\dot{\theta}\dot{\phi} = 0 \tag{5.105}$$

Divide the entire equation by $2r^2\sin^2\theta$ to isolate $\ddot{\phi}$:

$$\ddot{\phi} + \frac{2}{r}\dot{r}\dot{\phi} + 2\frac{\cos\theta}{\sin\theta}\dot{\theta}\dot{\phi} = 0 \tag{5.106}$$

Simplifying utilizing $\cot\theta = \frac{\cos\theta}{\sin\theta}$:

$$\ddot{\phi} + 2\left(\frac{1}{r}\right)\dot{r}\dot{\phi} + 2(\cot\theta)\dot{\theta}\dot{\phi} = 0 \tag{5.107}$$

Comparing equation (5.107) to the canonical ϕ-geodesic equation:

$$\ddot{\phi} + 2\Gamma^{\phi}_{r\phi}\dot{r}\dot{\phi} + 2\Gamma^{\phi}_{\theta\phi}\dot{\theta}\dot{\phi} + \cdots = 0 \tag{5.108}$$

We extract the non-zero Christoffel symbols:

$$\Gamma^{\phi}_{r\phi} = \frac{1}{r} \tag{5.109}$$

$$\Gamma^{\phi}_{\theta\phi} = \cot\theta \tag{5.110}$$

5. Summary and Verification of Symmetry

The complete list of explicitly derived non-zero Christoffel symbols obtained from the Lagrangian action method is:

$$\Gamma^t_{tt} = \partial_t \alpha \tag{5.111}$$
$$\Gamma^t_{rr} = e^{2(\beta-\alpha)} \partial_t \beta \tag{5.112}$$
$$\Gamma^t_{tr} = \partial_r \alpha \tag{5.113}$$
$$\Gamma^r_{tt} = e^{2(\alpha-\beta)} \partial_r \alpha \tag{5.114}$$
$$\Gamma^r_{rr} = \partial_r \beta \tag{5.115}$$
$$\Gamma^r_{tr} = \partial_t \beta \tag{5.116}$$
$$\Gamma^r_{\theta\theta} = -re^{-2\beta} \tag{5.117}$$
$$\Gamma^r_{\phi\phi} = -re^{-2\beta} \sin^2\theta \tag{5.118}$$
$$\Gamma^\theta_{r\theta} = \frac{1}{r} \tag{5.119}$$
$$\Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta \tag{5.120}$$
$$\Gamma^\phi_{r\phi} = \frac{1}{r} \tag{5.121}$$
$$\Gamma^\phi_{\theta\phi} = \cot\theta \tag{5.122}$$

To verify the mathematical symmetry required by the torsion-free Levi-Civita connection ($\Gamma^\mu_{\rho\sigma} = \Gamma^\mu_{\sigma\rho}$), we observe the coefficient matching for the mixed derivative terms carefully. For instance, in equation (5.77), the term $2\partial_r\alpha\dot{t}\dot{r}$ directly matches the general geodesic summation over ρ and σ, which expands to $\Gamma^t_{tr}\dot{t}\dot{r} + \Gamma^t_{rt}\dot{r}\dot{t}$. Because $\dot{t}\dot{r} = \dot{r}\dot{t}$, this sum is $2\Gamma^t_{tr}\dot{t}\dot{r}$, perfectly satisfying $\Gamma^t_{tr} = \Gamma^t_{rt} = \partial_r\alpha$. This exact factor of 2 appears identically in equations (5.87), (5.98), and (5.107) for all respective cross-terms ($\dot{t}\dot{r}, \dot{r}\dot{\theta}, \dot{r}\dot{\phi}, \dot{\theta}\dot{\phi}$), fully guaranteeing the structural lower-index symmetry of the derived Christoffel symbols without any a priori assumptions.

5.1.3 Derivation of the Ricci Tensor

The Contracted Christoffel Symbol Identity

The Ricci tensor $R_{\mu\nu}$ is defined as the contraction of the first and third indices of the Riemann tensor:

$$R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu} = \partial_\lambda \Gamma^\lambda_{\mu\nu} - \partial_\nu \Gamma^\lambda_{\mu\lambda} + \Gamma^\lambda_{\sigma\lambda}\Gamma^\sigma_{\mu\nu} - \Gamma^\lambda_{\sigma\nu}\Gamma^\sigma_{\mu\lambda}. \tag{5.123}$$

To simplify the computation, we utilize the identity for the contracted Christoffel symbol, which relates the trace of the connection to the determinant of the metric g:

$$\Gamma^\lambda_{\mu\lambda} = \partial_\mu \ln\sqrt{-g}. \tag{5.124}$$

We derived $\sqrt{-g}$ for the Schwarzschild metric ansatz in Part 2:

$$\sqrt{-g} = e^{\alpha(r)+\beta(r)} r^2 \sin\theta. \tag{5.125}$$

Taking the natural logarithm yields:

$$\ln\sqrt{-g} = \alpha(r) + \beta(r) + 2\ln r + \ln(\sin\theta). \tag{5.126}$$

We evaluate the partial derivatives of this expression to find the non-zero components of the contracted Christoffel symbols. For the radial coordinate ($\mu = r$):

$$\Gamma^\lambda_{r\lambda} = \partial_r \left[\alpha(r) + \beta(r) + 2\ln r + \ln(\sin\theta)\right]. \tag{5.127}$$

$$\Gamma^\lambda_{r\lambda} = \partial_r\alpha + \partial_r\beta + \frac{2}{r}. \tag{5.128}$$

For the polar angle ($\mu = \theta$):

$$\Gamma^{\lambda}_{\theta\lambda} = \partial_\theta \left[\alpha(r) + \beta(r) + 2\ln r + \ln(\sin\theta)\right]. \tag{5.129}$$

$$\Gamma^{\lambda}_{\theta\lambda} = \frac{1}{\sin\theta}\cos\theta = \cot\theta. \tag{5.130}$$

The temporal and azimuthal derivatives vanish ($\Gamma^{\lambda}_{t\lambda} = 0$ and $\Gamma^{\lambda}_{\phi\lambda} = 0$). We will substitute these identities directly into the Ricci tensor formula.

Temporal Component: R_{tt}

Expanding the definition of the Ricci tensor for $\mu = t, \nu = t$:

$$R_{tt} = \partial_\lambda \Gamma^{\lambda}_{tt} - \partial_t \Gamma^{\lambda}_{t\lambda} + \Gamma^{\lambda}_{\sigma\lambda}\Gamma^{\sigma}_{tt} - \Gamma^{\lambda}_{\sigma t}\Gamma^{\sigma}_{t\lambda}. \tag{5.131}$$

Since $\Gamma^{\lambda}_{t\lambda} = 0$, the second term $\partial_t \Gamma^{\lambda}_{t\lambda}$ vanishes. We expand the summation over λ in the first term. The only non-zero component of the form Γ^{λ}_{tt} is Γ^{r}_{tt}:

$$\partial_\lambda \Gamma^{\lambda}_{tt} = \partial_r \Gamma^{r}_{tt}. \tag{5.132}$$

Substituting $\Gamma^{r}_{tt} = e^{2(\alpha-\beta)}\partial_r\alpha$:

$$\partial_r \Gamma^{r}_{tt} = \partial_r \left(e^{2(\alpha-\beta)}\partial_r\alpha\right). \tag{5.133}$$

Applying the product and chain rules:

$$\partial_r \Gamma^{r}_{tt} = 2(\partial_r\alpha - \partial_r\beta)e^{2(\alpha-\beta)}\partial_r\alpha + e^{2(\alpha-\beta)}\partial_r^2\alpha. \tag{5.134}$$

$$\partial_r \Gamma^{r}_{tt} = e^{2(\alpha-\beta)}\left[2(\partial_r\alpha)^2 - 2\partial_r\alpha\partial_r\beta + \partial_r^2\alpha\right]. \tag{5.135}$$

Next, we expand the third term $\Gamma^{\lambda}_{\sigma\lambda}\Gamma^{\sigma}_{tt}$. Summing over σ, the only non-zero Γ^{σ}_{tt} is for $\sigma = r$:

$$\Gamma^{\lambda}_{\sigma\lambda}\Gamma^{\sigma}_{tt} = \Gamma^{\lambda}_{r\lambda}\Gamma^{r}_{tt}. \tag{5.136}$$

Substituting the derived identity for $\Gamma^{\lambda}_{r\lambda}$:

$$\Gamma^{\lambda}_{r\lambda}\Gamma^{r}_{tt} = \left(\partial_r\alpha + \partial_r\beta + \frac{2}{r}\right)\left(e^{2(\alpha-\beta)}\partial_r\alpha\right). \tag{5.137}$$

$$\Gamma^{\lambda}_{r\lambda}\Gamma^{r}_{tt} = e^{2(\alpha-\beta)}\left[(\partial_r\alpha)^2 + \partial_r\alpha\partial_r\beta + \frac{2}{r}\partial_r\alpha\right]. \tag{5.138}$$

Finally, we expand the fourth term $-\Gamma^{\lambda}_{\sigma t}\Gamma^{\sigma}_{t\lambda}$ by summing over all valid index pairs (λ, σ) that produce non-zero Christoffel symbols. The non-zero terms require $\{\lambda = t, \sigma = r\}$ and $\{\lambda = r, \sigma = t\}$:

$$-\Gamma^{\lambda}_{\sigma t}\Gamma^{\sigma}_{t\lambda} = -\left(\Gamma^{t}_{rt}\Gamma^{r}_{tt} + \Gamma^{r}_{tt}\Gamma^{t}_{tr}\right). \tag{5.139}$$

Using symmetry $\Gamma^{t}_{rt} = \Gamma^{t}_{tr}$:

$$-\Gamma^{\lambda}_{\sigma t}\Gamma^{\sigma}_{t\lambda} = -2\Gamma^{t}_{tr}\Gamma^{r}_{tt}. \tag{5.140}$$

Substituting the known symbols:

$$-\Gamma^{\lambda}_{\sigma t}\Gamma^{\sigma}_{t\lambda} = -2(\partial_r\alpha)\left(e^{2(\alpha-\beta)}\partial_r\alpha\right) = e^{2(\alpha-\beta)}\left[-2(\partial_r\alpha)^2\right]. \tag{5.141}$$

Assembling all terms for R_{tt}:

$$\begin{aligned} R_{tt} = {} & e^{2(\alpha-\beta)}\left[2(\partial_r\alpha)^2 - 2\partial_r\alpha\partial_r\beta + \partial_r^2\alpha\right] \\ & + e^{2(\alpha-\beta)}\left[(\partial_r\alpha)^2 + \partial_r\alpha\partial_r\beta + \frac{2}{r}\partial_r\alpha\right] \\ & + e^{2(\alpha-\beta)}\left[-2(\partial_r\alpha)^2\right]. \end{aligned} \tag{5.142}$$

Factoring out $e^{2(\alpha-\beta)}$ and canceling algebraically:

$$R_{tt} = e^{2(\alpha-\beta)}\left[\partial_r^2\alpha + (\partial_r\alpha)^2 - \partial_r\alpha\partial_r\beta + \frac{2}{r}\partial_r\alpha\right]. \tag{5.143}$$

Radial Component: R_{rr}

Expanding the definition for $\mu = r, \nu = r$:

$$R_{rr} = \partial_\lambda \Gamma^\lambda_{rr} - \partial_r \Gamma^\lambda_{r\lambda} + \Gamma^\lambda_{\sigma\lambda}\Gamma^\sigma_{rr} - \Gamma^\lambda_{\sigma r}\Gamma^\sigma_{r\lambda}. \tag{5.144}$$

Summing over λ for the first term (Γ^r_{rr} is the only non-zero component):

$$\partial_\lambda \Gamma^\lambda_{rr} = \partial_r \Gamma^r_{rr} = \partial_r(\partial_r \beta) = \partial_r^2 \beta. \tag{5.145}$$

For the second term, we differentiate our log-determinant identity:

$$-\partial_r \Gamma^\lambda_{r\lambda} = -\partial_r \left(\partial_r \alpha + \partial_r \beta + \frac{2}{r} \right) = -\partial_r^2 \alpha - \partial_r^2 \beta + \frac{2}{r^2}. \tag{5.146}$$

For the third term, the sum over σ isolates $\sigma = r$:

$$\Gamma^\lambda_{\sigma\lambda}\Gamma^\sigma_{rr} = \Gamma^\lambda_{r\lambda}\Gamma^r_{rr}. \tag{5.147}$$

Substituting the identity:

$$\Gamma^\lambda_{r\lambda}\Gamma^r_{rr} = \left(\partial_r \alpha + \partial_r \beta + \frac{2}{r} \right)(\partial_r \beta) = \partial_r \alpha \partial_r \beta + (\partial_r \beta)^2 + \frac{2}{r}\partial_r \beta. \tag{5.148}$$

For the fourth term, $-\Gamma^\lambda_{\sigma r}\Gamma^\sigma_{r\lambda}$, we must sum over all non-zero diagonal pairings where $\lambda = \sigma$, because $\Gamma^\sigma_{r\lambda}$ is only non-zero when the upper index matches the non-radial lower index:

$$-\Gamma^\lambda_{\sigma r}\Gamma^\sigma_{r\lambda} = -\left(\Gamma^t_{tr}\Gamma^t_{rt} + \Gamma^r_{rr}\Gamma^r_{rr} + \Gamma^\theta_{\theta r}\Gamma^\theta_{r\theta} + \Gamma^\phi_{\phi r}\Gamma^\phi_{r\phi} \right). \tag{5.149}$$

Substituting the individual Christoffel symbols:

$$-\Gamma^\lambda_{\sigma r}\Gamma^\sigma_{r\lambda} = -\left[(\partial_r \alpha)^2 + (\partial_r \beta)^2 + \left(\frac{1}{r}\right)^2 + \left(\frac{1}{r}\right)^2 \right]. \tag{5.150}$$

$$-\Gamma^\lambda_{\sigma r}\Gamma^\sigma_{r\lambda} = -(\partial_r \alpha)^2 - (\partial_r \beta)^2 - \frac{2}{r^2}. \tag{5.151}$$

Assembling all terms for R_{rr}:

$$\begin{aligned} R_{rr} = {} & \partial_r^2 \beta - \partial_r^2 \alpha - \partial_r^2 \beta + \frac{2}{r^2} \\ & + \partial_r \alpha \partial_r \beta + (\partial_r \beta)^2 + \frac{2}{r}\partial_r \beta \\ & - (\partial_r \alpha)^2 - (\partial_r \beta)^2 - \frac{2}{r^2}. \end{aligned} \tag{5.152}$$

Canceling strictly opposite terms ($\partial_r^2 \beta$, $\frac{2}{r^2}$, and $(\partial_r \beta)^2$):

$$R_{rr} = -\partial_r^2 \alpha - (\partial_r \alpha)^2 + \partial_r \alpha \partial_r \beta + \frac{2}{r}\partial_r \beta. \tag{5.153}$$

Polar Component: $R_{\theta\theta}$

Expanding the definition for $\mu = \theta, \nu = \theta$:

$$R_{\theta\theta} = \partial_\lambda \Gamma^\lambda_{\theta\theta} - \partial_\theta \Gamma^\lambda_{\theta\lambda} + \Gamma^\lambda_{\sigma\lambda}\Gamma^\sigma_{\theta\theta} - \Gamma^\lambda_{\sigma\theta}\Gamma^\sigma_{\theta\lambda}. \tag{5.154}$$

Summing over λ for the first term:

$$\partial_\lambda \Gamma^\lambda_{\theta\theta} = \partial_r \Gamma^r_{\theta\theta} = \partial_r \left(-re^{-2\beta} \right). \tag{5.155}$$

Applying the product rule:

$$\partial_r \Gamma^r_{\theta\theta} = -e^{-2\beta} - r\left(-2\partial_r\beta e^{-2\beta}\right) = e^{-2\beta}\left(2r\partial_r\beta - 1\right). \tag{5.156}$$

For the second term, we differentiate the angular part of the log-determinant identity:

$$-\partial_\theta \Gamma^\lambda_{\theta\lambda} = -\partial_\theta(\cot\theta) = \csc^2\theta. \tag{5.157}$$

For the third term, the sum over σ isolates $\sigma = r$:

$$\Gamma^\lambda_{\sigma\lambda}\Gamma^\sigma_{\theta\theta} = \Gamma^\lambda_{r\lambda}\Gamma^r_{\theta\theta}. \tag{5.158}$$

Substituting the components:

$$\Gamma^\lambda_{r\lambda}\Gamma^r_{\theta\theta} = \left(\partial_r\alpha + \partial_r\beta + \frac{2}{r}\right)\left(-re^{-2\beta}\right) = e^{-2\beta}\left(-r\partial_r\alpha - r\partial_r\beta - 2\right). \tag{5.159}$$

For the fourth term, we sum $-\Gamma^\lambda_{\sigma\theta}\Gamma^\sigma_{\theta\lambda}$ over non-zero valid pairs. The required pairs are $\{\lambda = \phi, \sigma = \phi\}$, $\{\lambda = r, \sigma = \theta\}$, and $\{\lambda = \theta, \sigma = r\}$:

$$-\Gamma^\lambda_{\sigma\theta}\Gamma^\sigma_{\theta\lambda} = -\left(\Gamma^\phi_{\phi\theta}\Gamma^\phi_{\theta\phi} + \Gamma^r_{\theta\theta}\Gamma^\theta_{\theta r} + \Gamma^\theta_{r\theta}\Gamma^r_{\theta\theta}\right). \tag{5.160}$$

$$-\Gamma^\lambda_{\sigma\theta}\Gamma^\sigma_{\theta\lambda} = -\left[(\cot\theta)^2 + \left(-re^{-2\beta}\right)\left(\frac{1}{r}\right) + \left(\frac{1}{r}\right)\left(-re^{-2\beta}\right)\right]. \tag{5.161}$$

$$-\Gamma^\lambda_{\sigma\theta}\Gamma^\sigma_{\theta\lambda} = -\cot^2\theta + 2e^{-2\beta}. \tag{5.162}$$

Assembling all terms for $R_{\theta\theta}$:

$$R_{\theta\theta} = e^{-2\beta}\left(2r\partial_r\beta - 1\right) + \csc^2\theta + e^{-2\beta}\left(-r\partial_r\alpha - r\partial_r\beta - 2\right) - \cot^2\theta + 2e^{-2\beta}. \tag{5.163}$$

We group the trigonometric terms and use the Pythagorean identity $\csc^2\theta - \cot^2\theta = 1$:

$$R_{\theta\theta} = e^{-2\beta}\left[2r\partial_r\beta - 1 - r\partial_r\alpha - r\partial_r\beta - 2 + 2\right] + 1. \tag{5.164}$$

$$R_{\theta\theta} = e^{-2\beta}\left[r\partial_r\beta - r\partial_r\alpha - 1\right] + 1. \tag{5.165}$$

Azimuthal Component: $R_{\phi\phi}$

Expanding the definition for $\mu = \phi, \nu = \phi$:

$$R_{\phi\phi} = \partial_\lambda\Gamma^\lambda_{\phi\phi} - \partial_\phi\Gamma^\lambda_{\phi\lambda} + \Gamma^\lambda_{\sigma\lambda}\Gamma^\sigma_{\phi\phi} - \Gamma^\lambda_{\sigma\phi}\Gamma^\sigma_{\phi\lambda}. \tag{5.166}$$

Summing over λ for the first term requires both $\lambda = r$ and $\lambda = \theta$:

$$\partial_\lambda\Gamma^\lambda_{\phi\phi} = \partial_r\Gamma^r_{\phi\phi} + \partial_\theta\Gamma^\theta_{\phi\phi}. \tag{5.167}$$

We compute each derivative separately:

$$\partial_r\left(-re^{-2\beta}\sin^2\theta\right) = \sin^2\theta\left(-e^{-2\beta} + 2r\partial_r\beta e^{-2\beta}\right). \tag{5.168}$$

$$\partial_\theta\left(-\sin\theta\cos\theta\right) = -\cos^2\theta + \sin^2\theta. \tag{5.169}$$

The second term vanishes since $\Gamma^\lambda_{\phi\lambda} = 0$:

$$-\partial_\phi\Gamma^\lambda_{\phi\lambda} = 0. \tag{5.170}$$

For the third term, summing over σ requires $\sigma = r$ and $\sigma = \theta$:

$$\Gamma^\lambda_{\sigma\lambda}\Gamma^\sigma_{\phi\phi} = \Gamma^\lambda_{r\lambda}\Gamma^r_{\phi\phi} + \Gamma^\lambda_{\theta\lambda}\Gamma^\theta_{\phi\phi}. \tag{5.171}$$

Substituting the respective identities and symbols:

$$\Gamma^{\lambda}_{\sigma\lambda}\Gamma^{\sigma}_{\phi\phi} = \left(\partial_r\alpha + \partial_r\beta + \frac{2}{r}\right)\left(-re^{-2\beta}\sin^2\theta\right) + (\cot\theta)(-\sin\theta\cos\theta). \tag{5.172}$$

$$\Gamma^{\lambda}_{\sigma\lambda}\Gamma^{\sigma}_{\phi\phi} = \sin^2\theta\, e^{-2\beta}\left(-r\partial_r\alpha - r\partial_r\beta - 2\right) - \cos^2\theta. \tag{5.173}$$

For the fourth term, we sum $-\Gamma^{\lambda}_{\sigma\phi}\Gamma^{\sigma}_{\phi\lambda}$ over valid pairs. The non-zero combinations are $\{\lambda = r, \sigma = \phi\}$, $\{\lambda = \phi, \sigma = r\}$, $\{\lambda = \theta, \sigma = \phi\}$, and $\{\lambda = \phi, \sigma = \theta\}$:

$$-\Gamma^{\lambda}_{\sigma\phi}\Gamma^{\sigma}_{\phi\lambda} = -\left(\Gamma^{r}_{\phi\phi}\Gamma^{\phi}_{\phi r} + \Gamma^{\phi}_{r\phi}\Gamma^{r}_{\phi\phi} + \Gamma^{\theta}_{\phi\phi}\Gamma^{\phi}_{\phi\theta} + \Gamma^{\phi}_{\theta\phi}\Gamma^{\theta}_{\phi\phi}\right). \tag{5.174}$$

$$-\Gamma^{\lambda}_{\sigma\phi}\Gamma^{\sigma}_{\phi\lambda} = -2\Gamma^{r}_{\phi\phi}\Gamma^{\phi}_{\phi r} - 2\Gamma^{\theta}_{\phi\phi}\Gamma^{\phi}_{\phi\theta}. \tag{5.175}$$

$$-\Gamma^{\lambda}_{\sigma\phi}\Gamma^{\sigma}_{\phi\lambda} = -2\left(-re^{-2\beta}\sin^2\theta\right)\left(\frac{1}{r}\right) - 2\left(-\sin\theta\cos\theta\right)\left(\frac{\cos\theta}{\sin\theta}\right). \tag{5.176}$$

$$-\Gamma^{\lambda}_{\sigma\phi}\Gamma^{\sigma}_{\phi\lambda} = 2e^{-2\beta}\sin^2\theta + 2\cos^2\theta. \tag{5.177}$$

Assembling all terms for $R_{\phi\phi}$:

$$\begin{aligned} R_{\phi\phi} = {} & \sin^2\theta\left(-e^{-2\beta} + 2r\partial_r\beta e^{-2\beta}\right) - \cos^2\theta + \sin^2\theta \\ & + \sin^2\theta\, e^{-2\beta}\left(-r\partial_r\alpha - r\partial_r\beta - 2\right) - \cos^2\theta \\ & + 2e^{-2\beta}\sin^2\theta + 2\cos^2\theta. \end{aligned} \tag{5.178}$$

Notice that the purely angular terms completely cancel: $-\cos^2\theta - \cos^2\theta + 2\cos^2\theta = 0$. We factor out $\sin^2\theta$ from the remaining terms:

$$R_{\phi\phi} = \sin^2\theta\left[e^{-2\beta}\left(-1 + 2r\partial_r\beta - r\partial_r\alpha - r\partial_r\beta - 2 + 2\right) + 1\right]. \tag{5.179}$$

$$R_{\phi\phi} = \sin^2\theta\left[e^{-2\beta}\left(r\partial_r\beta - r\partial_r\alpha - 1\right) + 1\right]. \tag{5.180}$$

This confirms the structural symmetry condition $R_{\phi\phi} = \sin^2\theta\, R_{\theta\theta}$.

5.1.4 *Einstein Equations and Solution*

The Vacuum Condition

The exterior gravitational field of a massive body is situated in a region devoid of matter and energy. Therefore, the energy-momentum tensor vanishes, $T_{\mu\nu} = 0$. By the Einstein field equations, this mandates that the Ricci tensor must vanish identically everywhere in the exterior spacetime:

$$R_{\mu\nu} = 0. \tag{5.181}$$

This establishes a system of non-linear ordinary differential equations for the unknown metric functions $\alpha(r)$ and $\beta(r)$. From the derivation of the Ricci tensor components, the non-trivial vacuum equations are:

$$R_{tt} = e^{2(\alpha-\beta)}\left[\partial_r^2\alpha + (\partial_r\alpha)^2 - \partial_r\alpha\partial_r\beta + \frac{2}{r}\partial_r\alpha\right] = 0, \tag{5.182}$$

$$R_{rr} = -\partial_r^2\alpha - (\partial_r\alpha)^2 + \partial_r\alpha\partial_r\beta + \frac{2}{r}\partial_r\beta = 0, \tag{5.183}$$

$$R_{\theta\theta} = e^{-2\beta}\left[r(\partial_r\beta - \partial_r\alpha) - 1\right] + 1 = 0. \tag{5.184}$$

(The equation $R_{\phi\phi} = \sin^2\theta R_{\theta\theta} = 0$ provides no additional constraints beyond $R_{\theta\theta} = 0$.)

Derivation of the Constraint $\alpha = -\beta$

To decouple the differential equations, we construct a linear combination of the temporal and radial equations designed to eliminate the second derivatives of the metric functions. We multiply R_{tt} by the factor $e^{2(\beta-\alpha)}$ and add the result to R_{rr}:

$$e^{2(\beta-\alpha)} R_{tt} + R_{rr} = 0. \tag{5.185}$$

Substituting the explicit forms of the Ricci tensor components:

$$\begin{aligned} e^{2(\beta-\alpha)} R_{tt} + R_{rr} = &\left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{2}{r}\partial_r \alpha\right] \\ &+ \left[-\partial_r^2 \alpha - (\partial_r \alpha)^2 + \partial_r \alpha \partial_r \beta + \frac{2}{r}\partial_r \beta\right]. \end{aligned} \tag{5.186}$$

The second-order derivatives $\partial_r^2 \alpha$ and the non-linear first-order terms strictly cancel:

$$e^{2(\beta-\alpha)} R_{tt} + R_{rr} = \frac{2}{r}\partial_r \alpha + \frac{2}{r}\partial_r \beta = 0. \tag{5.187}$$

Factoring the algebraic multiplier yields:

$$\frac{2}{r}\left(\partial_r \alpha + \partial_r \beta\right) = 0. \tag{5.188}$$

Since $r \neq 0$ in the exterior region, the term within the parenthesis must vanish identically. By linearity of the derivative operator, this becomes:

$$\partial_r \left(\alpha + \beta\right) = 0. \tag{5.189}$$

Integrating with respect to the radial coordinate r:

$$\alpha(r) + \beta(r) = c, \tag{5.190}$$

where c is a constant of integration. We resolve this constant by exploiting coordinate freedom. A rescaling of the time coordinate $t \to \tilde{t} = e^{-c} t$ transforms the temporal metric component as $g_{\tilde{t}\tilde{t}} = e^{2c} g_{tt} = -e^{2(\alpha+c)}$. We can formally absorb the integration constant c into the definition of the time coordinate without altering the geometry. Imposing this choice effectively sets $c = 0$, strictly yielding:

$$\alpha(r) = -\beta(r). \tag{5.191}$$

Explicit Solution of the Differential Equations

With the constraint $\beta(r) = -\alpha(r)$, we now address the angular vacuum equation $R_{\theta\theta} = 0$:

$$e^{-2\beta}\left[r(\partial_r \beta - \partial_r \alpha) - 1\right] + 1 = 0. \tag{5.192}$$

Substituting $\beta = -\alpha$ and consequently $\partial_r \beta = -\partial_r \alpha$:

$$e^{2\alpha}\left[r(-\partial_r \alpha - \partial_r \alpha) - 1\right] + 1 = 0. \tag{5.193}$$

$$e^{2\alpha}\left(-2r\partial_r \alpha - 1\right) + 1 = 0. \tag{5.194}$$

Rearranging terms by isolating the exponential function:

$$e^{2\alpha}\left(2r\partial_r \alpha + 1\right) = 1. \tag{5.195}$$

We recognize the left-hand side as the result of the product rule applied to the function $re^{2\alpha}$. We compute the total radial derivative explicitly to verify:

$$\partial_r \left(re^{2\alpha}\right) = \left(\partial_r r\right) e^{2\alpha} + r\left(\partial_r e^{2\alpha}\right). \tag{5.196}$$

$$\partial_r \left(re^{2\alpha}\right) = e^{2\alpha} + r\left(2\partial_r\alpha\, e^{2\alpha}\right). \tag{5.197}$$

$$\partial_r \left(re^{2\alpha}\right) = e^{2\alpha}\left(1 + 2r\partial_r\alpha\right). \tag{5.198}$$

Consequently, the differential equation $R_{\theta\theta} = 0$ simplifies to exactly:

$$\partial_r \left(re^{2\alpha}\right) = 1. \tag{5.199}$$

Integrating both sides with respect to r:

$$\int \partial_r \left(re^{2\alpha}\right) dr = \int 1\, dr. \tag{5.200}$$

$$re^{2\alpha(r)} = r + C_1, \tag{5.201}$$

where C_1 is a new constant of integration. Dividing by the radial coordinate r:

$$e^{2\alpha(r)} = 1 + \frac{C_1}{r}. \tag{5.202}$$

We define the integration constant as $C_1 = -R_S$, where R_S will be physically identified as the Schwarzschild radius. The final form for the temporal metric component coefficient is:

$$e^{2\alpha(r)} = 1 - \frac{R_S}{r}. \tag{5.203}$$

Using the symmetry constraint $\beta(r) = -\alpha(r)$, the radial metric component coefficient is:

$$e^{2\beta(r)} = e^{-2\alpha(r)} = \left(1 - \frac{R_S}{r}\right)^{-1}. \tag{5.204}$$

These explicit forms for $\alpha(r)$ and $\beta(r)$ identically satisfy the remaining vacuum equation $R_{rr} = 0$ (and consequently $R_{tt} = 0$), concluding the integration of the Einstein equations for this geometric ansatz.

5.1.5 Final Schwarzschild Metric and Interpretation

Substitution of the Metric Functions

We established the general static, spherically symmetric metric ansatz as:

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2 d\Omega^2, \tag{5.205}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2$. By solving the vacuum Einstein field equations $R_{\mu\nu} = 0$, we derived the explicit forms for the exponential metric functions:

$$e^{2\alpha(r)} = 1 - \frac{R_S}{r}, \tag{5.206}$$

$$e^{2\beta(r)} = \left(1 - \frac{R_S}{r}\right)^{-1}, \tag{5.207}$$

where R_S is a constant of integration. Substituting these functions directly into the metric ansatz yields the spacetime geometry in terms of the undetermined constant R_S:

$$ds^2 = -\left(1 - \frac{R_S}{r}\right)dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1}dr^2 + r^2 d\Omega^2. \tag{5.208}$$

The Weak-Field Limit and the Schwarzschild Radius

To bestow physical meaning upon the integration constant R_S, we mandate that the exact solution must asymptotically reduce to the known weak-field Newtonian limit at large spatial distances ($r \gg R_S$). In the weak-field, stationary approximation, the temporal component of the metric tensor relates to the Newtonian gravitational potential Φ via:

$$g_{tt} \approx -(1 + 2\Phi). \tag{5.209}$$

For a spherically symmetric body of mass M, the classical Newtonian gravitational potential is given by:

$$\Phi = -\frac{GM}{r}. \tag{5.210}$$

Substituting this potential into the weak-field metric relation yields the expected asymptotic form of the temporal component:

$$g_{tt} \approx -\left[1 + 2\left(-\frac{GM}{r}\right)\right], \tag{5.211}$$

$$g_{tt} \approx -\left(1 - \frac{2GM}{r}\right). \tag{5.212}$$

We now isolate the exact temporal component g_{tt} from our derived vacuum solution:

$$g_{tt} = -\left(1 - \frac{R_S}{r}\right). \tag{5.213}$$

Equating the exact component to the weak-field limit approximation dictates:

$$-\left(1 - \frac{R_S}{r}\right) = -\left(1 - \frac{2GM}{r}\right). \tag{5.214}$$

Multiplying both sides by -1 and subtracting 1 yields:

$$-\frac{R_S}{r} = -\frac{2GM}{r}. \tag{5.215}$$

Multiplying by $-r$ (valid for $r \neq 0$) strictly determines the constant of integration:

$$R_S = 2GM. \tag{5.216}$$

This physically significant distance parameter R_S is defined as the Schwarzschild radius.

The Final Schwarzschild Metric

Having rigorously determined the integration constant in terms of the mass M of the gravitating body and the gravitational constant G, we substitute $R_S = 2GM$ back into the derived line element. The complete, exact Schwarzschild metric describing the exterior vacuum spacetime of a spherically symmetric mass is:

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \tag{5.217}$$

Expanding the angular differential element $d\Omega^2$ for absolute completeness, we obtain the final explicit form:

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta \, d\phi^2. \tag{5.218}$$

5.2 Brikhoff's Theorem

5.2.1 General Spherically Symmetric Metric

We begin by constructing the metric on a general spherically symmetric spacetime in the arbitrary coordinates (a, b, θ, ϕ). By the assumption of spherical symmetry, there are no cross terms between the angular directions along the 2-spheres and the transverse directions. Specifically, a basis vector such as ∂_a must be orthogonal to the angular basis vector ∂_θ, ensuring the absence of terms like $da\, d\theta$. The line element is therefore written as:

$$ds^2 = g_{aa}(a,b)da^2 + g_{ab}(a,b)(dadb + dbda) + g_{bb}(a,b)db^2 + r^2(a,b)d\Omega^2 \tag{5.219}$$

Here, $r(a, b)$ acts as an arbitrary function representing the areal radius, and $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$ is the standard line element of a unit sphere. Since the tensor product of coordinate differentials in a line element is symmetric ($dadb = dbda$), we can explicitly combine the cross terms:

$$ds^2 = g_{aa}(a,b)da^2 + 2g_{ab}(a,b)dadb + g_{bb}(a,b)db^2 + r^2(a,b)d\Omega^2 \tag{5.220}$$

To simplify the geometry, we change the coordinate basis from (a, b) to (a, r) by inverting the function $r(a, b)$ to solve for $b(a, r)$. This transformation is valid provided r is not a function of a alone. We compute the exact differential of the new function $b(a, r)$ using the chain rule:

$$db = \frac{\partial b}{\partial a}da + \frac{\partial b}{\partial r}dr \tag{5.221}$$

We must substitute this expression into the (a, b) metric. First, we expand the squared differential db^2:

$$\begin{aligned} db^2 &= \left(\frac{\partial b}{\partial a}da + \frac{\partial b}{\partial r}dr\right)^2 \\ &= \left(\frac{\partial b}{\partial a}\right)^2 da^2 + 2\left(\frac{\partial b}{\partial a}\right)\left(\frac{\partial b}{\partial r}\right) dadr + \left(\frac{\partial b}{\partial r}\right)^2 dr^2 \end{aligned} \tag{5.222}$$

Next, we expand the cross term $2g_{ab}dadb$:

$$\begin{aligned} 2g_{ab}dadb &= 2g_{ab}da\left(\frac{\partial b}{\partial a}da + \frac{\partial b}{\partial r}dr\right) \\ &= 2g_{ab}\frac{\partial b}{\partial a}da^2 + 2g_{ab}\frac{\partial b}{\partial r}dadr \end{aligned} \tag{5.223}$$

Substituting these expanded terms back into the metric yields:

$$\begin{aligned} ds^2 = g_{aa}da^2 &+ \left[2g_{ab}\frac{\partial b}{\partial a}da^2 + 2g_{ab}\frac{\partial b}{\partial r}dadr\right] \\ &+ g_{bb}\left[\left(\frac{\partial b}{\partial a}\right)^2 da^2 + 2\left(\frac{\partial b}{\partial a}\right)\left(\frac{\partial b}{\partial r}\right) dadr + \left(\frac{\partial b}{\partial r}\right)^2 dr^2\right] + r^2 d\Omega^2 \end{aligned} \tag{5.224}$$

We now group the terms by the fundamental differentials da^2, $dadr$, and dr^2:

$$\begin{aligned} ds^2 = &\left[g_{aa} + 2g_{ab}\frac{\partial b}{\partial a} + g_{bb}\left(\frac{\partial b}{\partial a}\right)^2\right] da^2 \\ &+ 2\left[g_{ab}\frac{\partial b}{\partial r} + g_{bb}\left(\frac{\partial b}{\partial a}\right)\left(\frac{\partial b}{\partial r}\right)\right] dadr \\ &+ \left[g_{bb}\left(\frac{\partial b}{\partial r}\right)^2\right] dr^2 + r^2 d\Omega^2 \end{aligned} \tag{5.225}$$

We define the quantities inside the square brackets as our new effective metric components in the (a, r) coordinate system, denoted as $\tilde{g}_{aa}(a, r)$, $\tilde{g}_{ar}(a, r)$, and $\tilde{g}_{rr}(a, r)$:

$$\tilde{g}_{aa}(a, r) = g_{aa} + 2g_{ab}\frac{\partial b}{\partial a} + g_{bb}\left(\frac{\partial b}{\partial a}\right)^2 \tag{5.226}$$

$$\tilde{g}_{ar}(a, r) = g_{ab}\frac{\partial b}{\partial r} + g_{bb}\frac{\partial b}{\partial a}\frac{\partial b}{\partial r} \tag{5.227}$$

$$\tilde{g}_{rr}(a, r) = g_{bb}\left(\frac{\partial b}{\partial r}\right)^2 \tag{5.228}$$

Dropping the tildes for simplicity, the metric takes the form:

$$ds^2 = g_{aa}(a, r)da^2 + g_{ar}(a, r)(dadr + drda) + g_{rr}(a, r)dr^2 + r^2 d\Omega^2 \tag{5.229}$$

where we have explicitly split $2g_{ar}dadr$ back into $g_{ar}(dadr + drda)$ to match the required canonical form.

Our next objective is to eliminate the off-diagonal $dadr$ cross term entirely by introducing a new time coordinate $t(a, r)$. The exact differential of $t(a, r)$ is given by:

$$dt = \frac{\partial t}{\partial a}da + \frac{\partial t}{\partial r}dr \tag{5.230}$$

Squaring this differential produces:

$$dt^2 = \left(\frac{\partial t}{\partial a}\right)^2 da^2 + 2\left(\frac{\partial t}{\partial a}\right)\left(\frac{\partial t}{\partial r}\right) dadr + \left(\frac{\partial t}{\partial r}\right)^2 dr^2 \tag{5.231}$$

We demand that the metric be reducible to the strictly diagonal form:

$$ds^2 = m(t, r)dt^2 + n(t, r)dr^2 + r^2 d\Omega^2 \tag{5.232}$$

To establish the equivalence, we substitute our expanded expression for dt^2 into this target diagonal form:

$$\begin{aligned} m(t, r)dt^2 + n(t, r)dr^2 &= m(t, r)\left[\left(\frac{\partial t}{\partial a}\right)^2 da^2 + 2\left(\frac{\partial t}{\partial a}\right)\left(\frac{\partial t}{\partial r}\right) dadr + \left(\frac{\partial t}{\partial r}\right)^2 dr^2\right] + n(t, r)dr^2 \\ &= \left[m(t, r)\left(\frac{\partial t}{\partial a}\right)^2\right] da^2 + 2\left[m(t, r)\left(\frac{\partial t}{\partial a}\right)\left(\frac{\partial t}{\partial r}\right)\right] dadr \\ &\quad + \left[m(t, r)\left(\frac{\partial t}{\partial r}\right)^2 + n(t, r)\right] dr^2 \end{aligned} \tag{5.233}$$

For this expanded target metric to identically match our (a, r) metric, the coefficients of da^2, $dadr$, and dr^2 must strictly equal g_{aa}, g_{ar}, and g_{rr} respectively. This equivalence yields a system of three coupled partial differential equations:

$$g_{aa} = m\left(\frac{\partial t}{\partial a}\right)^2 \tag{5.234}$$

$$g_{ar} = m\left(\frac{\partial t}{\partial a}\right)\left(\frac{\partial t}{\partial r}\right) \tag{5.235}$$

$$g_{rr} = m\left(\frac{\partial t}{\partial r}\right)^2 + n \tag{5.236}$$

These represent three distinct equations for the three unknown functions $t(a, r)$, $m(t, r)$, and $n(t, r)$. Because the system is exactly constrained (up to initial conditions for t), we are mathematically guaranteed to find a solution that diagonalizes the metric. The line element is successfully brought to the form:

$$ds^2 = m(t,r)dt^2 + n(t,r)dr^2 + r^2 d\Omega^2 \tag{5.237}$$

Finally, we must enforce the geometric requirement that spacetime possesses a Lorentzian signature. By analogy with flat Minkowski spacetime where the metric is $ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2$, we identify t as the temporal coordinate and deduce that the coefficient $m(t,r)$ must be strictly negative. To intrinsically guarantee that m remains negative and n remains positive, we introduce two arbitrary real-valued functions $\alpha(t,r)$ and $\beta(t,r)$:

$$m(t,r) = -e^{2\alpha(t,r)} \tag{5.238}$$

$$n(t,r) = e^{2\beta(t,r)} \tag{5.239}$$

Substituting these definitions into the diagonalized metric, we arrive at the standard general form for a spherically symmetric Lorentzian spacetime:

$$ds^2 = -e^{2\alpha(t,r)}dt^2 + e^{2\beta(t,r)}dr^2 + r^2 d\Omega^2 \tag{5.240}$$

5.2.2 Inverse Metric and Determinant

The Covariant Metric Tensor

From the general spherically symmetric line element derived in Part 1,

$$ds^2 = -e^{2\alpha(t,r)}dt^2 + e^{2\beta(t,r)}dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta\, d\phi^2, \tag{5.241}$$

we can read off the non-zero components of the covariant metric tensor $g_{\mu\nu}$. Since there are no cross terms (e.g., $dt\,dr$, $d\theta\,d\phi$), the metric is strictly diagonal. In the coordinate basis $x^\mu = (t, r, \theta, \phi)$, the metric components are:

$$g_{tt} = -e^{2\alpha(t,r)} \tag{5.242}$$

$$g_{rr} = e^{2\beta(t,r)} \tag{5.243}$$

$$g_{\theta\theta} = r^2 \tag{5.244}$$

$$g_{\phi\phi} = r^2 \sin^2\theta \tag{5.245}$$

All off-diagonal components vanish identically: $g_{\mu\nu} = 0$ for $\mu \neq \nu$.

The Contravariant (Inverse) Metric Tensor

The inverse metric tensor $g^{\mu\nu}$ is defined by the algebraic relation:

$$g^{\mu\lambda} g_{\lambda\nu} = \delta^\mu_\nu \tag{5.246}$$

where δ^μ_ν is the Kronecker delta. Because the covariant metric tensor is purely diagonal, its matrix representation is also diagonal, and the inverse matrix is simply obtained by taking the algebraic reciprocal of each individual main diagonal element. Therefore, the non-zero components of the contravariant metric tensor $g^{\mu\nu}$ are:

$$g^{tt} = \frac{1}{g_{tt}} = \frac{1}{-e^{2\alpha(t,r)}} = -e^{-2\alpha(t,r)} \tag{5.247}$$

$$g^{rr} = \frac{1}{g_{rr}} = \frac{1}{e^{2\beta(t,r)}} = e^{-2\beta(t,r)} \tag{5.248}$$

$$g^{\theta\theta} = \frac{1}{g_{\theta\theta}} = \frac{1}{r^2} = r^{-2} \tag{5.249}$$

$$g^{\phi\phi} = \frac{1}{g_{\phi\phi}} = \frac{1}{r^2 \sin^2\theta} = r^{-2} \csc^2\theta \tag{5.250}$$

Similarly, all off-diagonal components of the inverse metric vanish: $g^{\mu\nu} = 0$ for $\mu \neq \nu$.

Verification of the Inverse Metric

To maintain full mathematical rigor, we explicitly verify the inverse condition $g^{\mu\lambda}g_{\lambda\nu} = \delta^\mu_\nu$ by expanding the summation over the dummy index $\lambda \in \{t, r, \theta, \phi\}$.

For the diagonal temporal component ($\mu = \nu = t$):

$$\begin{aligned} g^{t\lambda}g_{\lambda t} &= g^{tt}g_{tt} + \underbrace{g^{tr}g_{rt}}_{0} + \underbrace{g^{t\theta}g_{\theta t}}_{0} + \underbrace{g^{t\phi}g_{\phi t}}_{0} \\ &= \left(-e^{-2\alpha(t,r)}\right)\left(-e^{2\alpha(t,r)}\right) \\ &= e^{-2\alpha(t,r)+2\alpha(t,r)} = e^0 = 1 = \delta^t_t \end{aligned} \tag{5.251}$$

For the diagonal radial component ($\mu = \nu = r$):

$$\begin{aligned} g^{r\lambda}g_{\lambda r} &= \underbrace{g^{rt}g_{tr}}_{0} + g^{rr}g_{rr} + \underbrace{g^{r\theta}g_{\theta r}}_{0} + \underbrace{g^{r\phi}g_{\phi r}}_{0} \\ &= \left(e^{-2\beta(t,r)}\right)\left(e^{2\beta(t,r)}\right) \\ &= e^{-2\beta(t,r)+2\beta(t,r)} = e^0 = 1 = \delta^r_r \end{aligned} \tag{5.252}$$

For the diagonal polar component ($\mu = \nu = \theta$):

$$\begin{aligned} g^{\theta\lambda}g_{\lambda\theta} &= \underbrace{g^{\theta t}g_{t\theta}}_{0} + \underbrace{g^{\theta r}g_{r\theta}}_{0} + g^{\theta\theta}g_{\theta\theta} + \underbrace{g^{\theta\phi}g_{\phi\theta}}_{0} \\ &= \left(r^{-2}\right)\left(r^2\right) \\ &= r^{-2+2} = r^0 = 1 = \delta^\theta_\theta \end{aligned} \tag{5.253}$$

For the diagonal azimuthal component ($\mu = \nu = \phi$):

$$\begin{aligned} g^{\phi\lambda}g_{\lambda\phi} &= \underbrace{g^{\phi t}g_{t\phi}}_{0} + \underbrace{g^{\phi r}g_{r\phi}}_{0} + \underbrace{g^{\phi\theta}g_{\theta\phi}}_{0} + g^{\phi\phi}g_{\phi\phi} \\ &= \left(r^{-2}\csc^2\theta\right)\left(r^2\sin^2\theta\right) \\ &= \left(\frac{1}{r^2\sin^2\theta}\right)\left(r^2\sin^2\theta\right) = 1 = \delta^\phi_\phi \end{aligned} \tag{5.254}$$

For the off-diagonal components ($\mu \neq \nu$), we examine $\mu = t, \nu = r$ as a representative example. Since $g^{\mu\nu}$ and $g_{\mu\nu}$ are strictly diagonal, at least one term in each product within the summation will be identically zero:

$$\begin{aligned} g^{t\lambda}g_{\lambda r} &= g^{tt}\underbrace{g_{tr}}_{0} + \underbrace{g^{tr}}_{0}g_{rr} + \underbrace{g^{t\theta}}_{0}\underbrace{g_{\theta r}}_{0} + \underbrace{g^{t\phi}}_{0}\underbrace{g_{\phi r}}_{0} \\ &= \left(-e^{-2\alpha(t,r)}\right)(0) + (0)\left(e^{2\beta(t,r)}\right) + 0 + 0 \\ &= 0 = \delta^t_r \end{aligned} \tag{5.255}$$

By the diagonal symmetry of the matrices, this result holds trivially for all off-diagonal contractions ($\mu \neq \nu$). Thus, the inverse relationship is rigorously verified.

Computation of the Metric Determinant

The determinant of the covariant metric tensor, conventionally denoted as $g \equiv \det(g_{\mu\nu})$, is the determinant of the 4×4 metric matrix. For any purely diagonal matrix, the determinant simplifies to the direct product of its main diagonal elements:

$$g = \det(g_{\mu\nu}) = \prod_\mu g_{\mu\mu} = g_{tt}g_{rr}g_{\theta\theta}g_{\phi\phi} \tag{5.256}$$

Substituting the explicit, non-zero components into this product:

$$g = \left(-e^{2\alpha(t,r)}\right)\left(e^{2\beta(t,r)}\right)\left(r^2\right)\left(r^2\sin^2\theta\right) \tag{5.257}$$

We mathematically consolidate the terms step-by-step. First, we group the exponential terms together and the polynomial r terms together:

$$\begin{aligned} g &= -\left(e^{2\alpha(t,r)}e^{2\beta(t,r)}\right)\left(r^2\cdot r^2\right)\sin^2\theta \\ &= -e^{2\alpha(t,r)+2\beta(t,r)}\left(r^4\right)\sin^2\theta \end{aligned} \tag{5.258}$$

Factoring out the common factor of 2 in the exponential argument, we arrive at the final closed-form expression for the metric determinant:

$$g = -e^{2(\alpha(t,r)+\beta(t,r))}r^4\sin^2\theta \tag{5.259}$$

Computation of the Invariant Volume Measure

In General Relativity, integration over the spacetime manifold utilizes the invariant volume measure element, $\sqrt{-g}$. We first isolate the positive quantity $-g$ by multiplying the previous result by -1:

$$-g = e^{2(\alpha(t,r)+\beta(t,r))}r^4\sin^2\theta \tag{5.260}$$

Next, we apply the square root operator to both sides of the equation:

$$\sqrt{-g} = \sqrt{e^{2(\alpha(t,r)+\beta(t,r))}r^4\sin^2\theta} \tag{5.261}$$

Since the factors inside the square root are products of strictly positive real numbers (assuming the standard coordinate ranges where $r > 0$ and $0 < \theta < \pi$), we can safely distribute the square root over each multiplicative factor:

$$\begin{aligned} \sqrt{-g} &= \sqrt{e^{2(\alpha(t,r)+\beta(t,r))}}\cdot\sqrt{r^4}\cdot\sqrt{\sin^2\theta} \\ &= \left(e^{2(\alpha(t,r)+\beta(t,r))}\right)^{1/2}\left(r^4\right)^{1/2}(\sin\theta) \\ &= e^{\alpha(t,r)+\beta(t,r)}r^2\sin\theta \end{aligned} \tag{5.262}$$

This yields the final rigorous form for the scalar density used in the spacetime measure element:

$$\sqrt{-g} = e^{\alpha(t,r)+\beta(t,r)}r^2\sin\theta \tag{5.263}$$

5.2.3 Christoffel Symbols (Time-Dependent Case)

The General Christoffel Symbol Formula

The Christoffel symbols of the second kind, which encode the connection coefficients for the metric, are defined by the equation:

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2}\sum_{\sigma\in\{t,r,\theta,\phi\}} g^{\lambda\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}) \tag{5.264}$$

Because our coordinate basis (t, r, θ, ϕ) yields a strictly diagonal metric and inverse metric, all off-diagonal components vanish ($g^{\lambda\sigma} = 0$ for $\lambda \neq \sigma$). Consequently, the summation over the dummy index σ explicitly collapses to a single non-zero term where $\sigma = \lambda$:

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2}g^{\lambda\lambda}(\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}) \tag{5.265}$$

(Note: The repeated index λ in equation (5.265) is fixed by the left-hand side and does not imply summation). We will use this reduced formula to compute all non-zero components, maintaining both time (t) and radial (r) dependencies for full generality.

1. The Temporal Components ($\lambda = t$)

For $\lambda = t$, the relevant inverse metric component is $g^{tt} = -e^{-2\alpha(t,r)}$. Equation (5.265) becomes:

$$\Gamma^t_{\mu\nu} = \frac{1}{2} g^{tt} (\partial_\mu g_{\nu t} + \partial_\nu g_{\mu t} - \partial_t g_{\mu\nu}) \tag{5.266}$$

Case $\mu = t, \nu = t$:

$$\begin{aligned}
\Gamma^t_{tt} &= \frac{1}{2} g^{tt} (\partial_t g_{tt} + \partial_t g_{tt} - \partial_t g_{tt}) \\
&= \frac{1}{2} g^{tt} \partial_t g_{tt} \\
&= \frac{1}{2} \left(-e^{-2\alpha(t,r)}\right) \partial_t \left(-e^{2\alpha(t,r)}\right) \\
&= \frac{1}{2} \left(-e^{-2\alpha(t,r)}\right) \left(-2e^{2\alpha(t,r)} \partial_t \alpha(t,r)\right) \\
&= \left(\frac{2}{2}\right) \left(e^{-2\alpha(t,r)} e^{2\alpha(t,r)}\right) \partial_t \alpha(t,r) \\
&= \partial_t \alpha(t,r)
\end{aligned} \tag{5.267}$$

Case $\mu = t, \nu = r$: Since the metric is diagonal, $g_{rt} = 0$.

$$\begin{aligned}
\Gamma^t_{tr} &= \frac{1}{2} g^{tt} (\partial_t g_{rt} + \partial_r g_{tt} - \partial_t g_{tr}) \\
&= \frac{1}{2} g^{tt} (0 + \partial_r g_{tt} - 0) \\
&= \frac{1}{2} g^{tt} \partial_r g_{tt} \\
&= \frac{1}{2} \left(-e^{-2\alpha(t,r)}\right) \partial_r \left(-e^{2\alpha(t,r)}\right) \\
&= \frac{1}{2} \left(-e^{-2\alpha(t,r)}\right) \left(-2e^{2\alpha(t,r)} \partial_r \alpha(t,r)\right) \\
&= \partial_r \alpha(t,r)
\end{aligned} \tag{5.268}$$

By the fundamental symmetry of the Christoffel symbols in their lower indices ($\Gamma^\lambda_{\mu\nu} = \Gamma^\lambda_{\nu\mu}$), it immediately follows that $\Gamma^t_{rt} = \partial_r \alpha(t,r)$.

Case $\mu = r, \nu = r$:

$$\begin{aligned}
\Gamma^t_{rr} &= \frac{1}{2} g^{tt} (\partial_r g_{rt} + \partial_r g_{rt} - \partial_t g_{rr}) \\
&= \frac{1}{2} g^{tt} (0 + 0 - \partial_t g_{rr}) \\
&= -\frac{1}{2} g^{tt} \partial_t g_{rr} \\
&= -\frac{1}{2} \left(-e^{-2\alpha(t,r)}\right) \partial_t \left(e^{2\beta(t,r)}\right) \\
&= \frac{1}{2} e^{-2\alpha(t,r)} \left(2e^{2\beta(t,r)} \partial_t \beta(t,r)\right) \\
&= e^{2(\beta(t,r) - \alpha(t,r))} \partial_t \beta(t,r)
\end{aligned} \tag{5.269}$$

For the angular components ($\mu, \nu \in \{\theta, \phi\}$), the required derivatives are $\partial_t g_{\theta\theta} = \partial_t(r^2) = 0$ and $\partial_t g_{\phi\phi} = \partial_t(r^2 \sin^2\theta) = 0$. Consequently, $\Gamma^t_{\theta\theta} = 0$ and $\Gamma^t_{\phi\phi} = 0$.

2. The Radial Components ($\lambda = r$)

For $\lambda = r$, the relevant inverse metric component is $g^{rr} = e^{-2\beta(t,r)}$. Equation (5.265) becomes:

$$\Gamma^r_{\mu\nu} = \frac{1}{2} g^{rr} (\partial_\mu g_{\nu r} + \partial_\nu g_{\mu r} - \partial_r g_{\mu\nu}) \tag{5.270}$$

Case $\mu = t, \nu = t$: Since $g_{tr} = 0$:

$$\begin{aligned}
\Gamma^r_{tt} &= \frac{1}{2} g^{rr} (\partial_t g_{tr} + \partial_t g_{tr} - \partial_r g_{tt}) \\
&= -\frac{1}{2} g^{rr} \partial_r g_{tt} \\
&= -\frac{1}{2} \left(e^{-2\beta(t,r)}\right) \partial_r \left(-e^{2\alpha(t,r)}\right) \\
&= -\frac{1}{2} \left(e^{-2\beta(t,r)}\right) \left(-2e^{2\alpha(t,r)} \partial_r \alpha(t,r)\right) \\
&= e^{2(\alpha(t,r)-\beta(t,r))} \partial_r \alpha(t,r)
\end{aligned} \tag{5.271}$$

Case $\mu = t, \nu = r$:

$$\begin{aligned}
\Gamma^r_{tr} &= \frac{1}{2} g^{rr} (\partial_t g_{rr} + \partial_r g_{tr} - \partial_r g_{tr}) \\
&= \frac{1}{2} g^{rr} \partial_t g_{rr} \\
&= \frac{1}{2} \left(e^{-2\beta(t,r)}\right) \partial_t \left(e^{2\beta(t,r)}\right) \\
&= \frac{1}{2} \left(e^{-2\beta(t,r)}\right) \left(2e^{2\beta(t,r)} \partial_t \beta(t,r)\right) \\
&= \partial_t \beta(t,r)
\end{aligned} \tag{5.272}$$

By symmetry, $\Gamma^r_{rt} = \partial_t \beta(t,r)$.

Case $\mu = r, \nu = r$:

$$\begin{aligned}
\Gamma^r_{rr} &= \frac{1}{2} g^{rr} (\partial_r g_{rr} + \partial_r g_{rr} - \partial_r g_{rr}) \\
&= \frac{1}{2} g^{rr} \partial_r g_{rr} \\
&= \frac{1}{2} \left(e^{-2\beta(t,r)}\right) \partial_r \left(e^{2\beta(t,r)}\right) \\
&= \frac{1}{2} \left(e^{-2\beta(t,r)}\right) \left(2e^{2\beta(t,r)} \partial_r \beta(t,r)\right) \\
&= \partial_r \beta(t,r)
\end{aligned} \tag{5.273}$$

Case $\mu = \theta, \nu = \theta$:

$$\begin{aligned}
\Gamma^r_{\theta\theta} &= \frac{1}{2} g^{rr} (\partial_\theta g_{\theta r} + \partial_\theta g_{r\theta} - \partial_r g_{\theta\theta}) \\
&= -\frac{1}{2} g^{rr} \partial_r g_{\theta\theta} \\
&= -\frac{1}{2} \left(e^{-2\beta(t,r)}\right) \partial_r \left(r^2\right) \\
&= -\frac{1}{2} e^{-2\beta(t,r)} (2r) \\
&= -r e^{-2\beta(t,r)}
\end{aligned} \tag{5.274}$$

Case $\mu = \phi, \nu = \phi$:

$$\begin{aligned}
\Gamma^r_{\phi\phi} &= \frac{1}{2} g^{rr} (\partial_\phi g_{\phi r} + \partial_\phi g_{r\phi} - \partial_r g_{\phi\phi}) \\
&= -\frac{1}{2} g^{rr} \partial_r g_{\phi\phi} \\
&= -\frac{1}{2} \left(e^{-2\beta(t,r)}\right) \partial_r \left(r^2 \sin^2\theta\right) \\
&= -\frac{1}{2} e^{-2\beta(t,r)} \left(2r \sin^2\theta\right) \\
&= -r e^{-2\beta(t,r)} \sin^2\theta
\end{aligned} \tag{5.275}$$

3. The Polar Components ($\lambda = \theta$)

For $\lambda = \theta$, the relevant inverse metric component is $g^{\theta\theta} = r^{-2}$. Equation (5.265) becomes:

$$\Gamma^{\theta}_{\mu\nu} = \frac{1}{2} g^{\theta\theta} (\partial_\mu g_{\nu\theta} + \partial_\nu g_{\mu\theta} - \partial_\theta g_{\mu\nu}) \tag{5.276}$$

Case $\mu = r, \nu = \theta$:

$$\begin{aligned}
\Gamma^{\theta}_{r\theta} &= \frac{1}{2} g^{\theta\theta} (\partial_r g_{\theta\theta} + \partial_\theta g_{r\theta} - \partial_\theta g_{r\theta}) \\
&= \frac{1}{2} g^{\theta\theta} \partial_r g_{\theta\theta} \\
&= \frac{1}{2} \left(\frac{1}{r^2}\right) \partial_r \left(r^2\right) \\
&= \frac{1}{2} \left(\frac{1}{r^2}\right) (2r) \\
&= \frac{1}{r}
\end{aligned} \tag{5.277}$$

By symmetry, $\Gamma^{\theta}_{\theta r} = \frac{1}{r}$.
Case $\mu = \phi, \nu = \phi$:

$$\begin{aligned}
\Gamma^{\theta}_{\phi\phi} &= \frac{1}{2} g^{\theta\theta} (\partial_\phi g_{\phi\theta} + \partial_\phi g_{\theta\phi} - \partial_\theta g_{\phi\phi}) \\
&= -\frac{1}{2} g^{\theta\theta} \partial_\theta g_{\phi\phi} \\
&= -\frac{1}{2} \left(\frac{1}{r^2}\right) \partial_\theta \left(r^2 \sin^2\theta\right) \\
&= -\frac{1}{2} \left(\frac{1}{r^2}\right) \left(r^2 \cdot 2\sin\theta\cos\theta\right) \\
&= -\sin\theta\cos\theta
\end{aligned} \tag{5.278}$$

4. The Azimuthal Components ($\lambda = \phi$)

For $\lambda = \phi$, the relevant inverse metric component is $g^{\phi\phi} = \frac{1}{r^2\sin^2\theta}$. Equation (5.265) becomes:

$$\Gamma^{\phi}_{\mu\nu} = \frac{1}{2} g^{\phi\phi} (\partial_\mu g_{\nu\phi} + \partial_\nu g_{\mu\phi} - \partial_\phi g_{\mu\nu}) \tag{5.279}$$

Case $\mu = r, \nu = \phi$:

$$\begin{aligned}
\Gamma^{\phi}_{r\phi} &= \frac{1}{2} g^{\phi\phi} (\partial_r g_{\phi\phi} + \partial_\phi g_{r\phi} - \partial_\phi g_{r\phi}) \\
&= \frac{1}{2} g^{\phi\phi} \partial_r g_{\phi\phi} \\
&= \frac{1}{2} \left(\frac{1}{r^2\sin^2\theta}\right) \partial_r \left(r^2 \sin^2\theta\right) \\
&= \frac{1}{2} \left(\frac{1}{r^2\sin^2\theta}\right) \left(2r\sin^2\theta\right) \\
&= \frac{1}{r}
\end{aligned} \tag{5.280}$$

By symmetry, $\Gamma^{\phi}_{\phi r} = \frac{1}{r}$.

Case $\mu = \theta, \nu = \phi$:

$$\begin{aligned}
\Gamma^{\phi}_{\theta\phi} &= \frac{1}{2} g^{\phi\phi}(\partial_\theta g_{\phi\phi} + \partial_\phi g_{\theta\phi} - \partial_\phi g_{\theta\phi}) \\
&= \frac{1}{2} g^{\phi\phi} \partial_\theta g_{\phi\phi} \\
&= \frac{1}{2} \left(\frac{1}{r^2 \sin^2\theta} \right) \partial_\theta \left(r^2 \sin^2\theta \right) \\
&= \frac{1}{2} \left(\frac{1}{r^2 \sin^2\theta} \right) \left(r^2 \cdot 2 \sin\theta \cos\theta \right) \\
&= \frac{\sin\theta\cos\theta}{\sin^2\theta} \\
&= \frac{\cos\theta}{\sin\theta} \\
&= \cot\theta
\end{aligned} \tag{5.281}$$

By symmetry, $\Gamma^{\phi}_{\phi\theta} = \cot\theta$. All other components identically vanish due to the diagonal structure of the metric and the independence of the coordinates from ϕ.

5.2.4 Riemann Curvature Tensor

The Riemann curvature tensor measures the non-commutativity of covariant derivatives acting on vector fields. It is intrinsically related to the tidal forces experienced in the spacetime geometry. The tensor is defined entirely in terms of the Christoffel symbols and their partial derivatives[cite: 1, 2]:

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma} \tag{5.282}$$

Because the Riemann tensor has 256 components in 4-dimensional spacetime, we will explicitly construct a set of representative non-zero components to rigorously demonstrate the derivation process, keeping both time and radial dependencies active[cite: 2].

1. Derivation of $R^{t}{}_{rtr}$

We assign the indices $(\rho, \sigma, \mu, \nu) = (t, r, t, r)$. Substituting these into the definition of the Riemann tensor yields:

$$R^{t}{}_{rtr} = \partial_t \Gamma^{t}_{rr} - \partial_r \Gamma^{t}_{tr} + \Gamma^{t}_{t\lambda}\Gamma^{\lambda}_{rr} - \Gamma^{t}_{r\lambda}\Gamma^{\lambda}_{tr} \tag{5.283}$$

We explicitly expand the summations over the dummy index $\lambda \in \{t, r, \theta, \phi\}$:

$$\Gamma^{t}_{t\lambda}\Gamma^{\lambda}_{rr} = \Gamma^{t}_{tt}\Gamma^{t}_{rr} + \Gamma^{t}_{tr}\Gamma^{r}_{rr} + \Gamma^{t}_{t\theta}\Gamma^{\theta}_{rr} + \Gamma^{t}_{t\phi}\Gamma^{\phi}_{rr} \tag{5.284}$$

$$\Gamma^{t}_{r\lambda}\Gamma^{\lambda}_{tr} = \Gamma^{t}_{rt}\Gamma^{t}_{tr} + \Gamma^{t}_{rr}\Gamma^{r}_{tr} + \Gamma^{t}_{r\theta}\Gamma^{\theta}_{tr} + \Gamma^{t}_{r\phi}\Gamma^{\phi}_{tr} \tag{5.285}$$

From the Christoffel symbols derived in Part 3, we know that all components with mixed angular and non-angular lower indices vanish (e.g., $\Gamma^{\theta}_{rr} = \Gamma^{\phi}_{rr} = \Gamma^{\theta}_{tr} = \Gamma^{\phi}_{tr} = 0$ and $\Gamma^{t}_{t\theta} = \Gamma^{t}_{t\phi} = \Gamma^{t}_{r\theta} = \Gamma^{t}_{r\phi} = 0$). Equations (5.284) and (5.285) thus reduce to:

$$\Gamma^{t}_{t\lambda}\Gamma^{\lambda}_{rr} = \Gamma^{t}_{tt}\Gamma^{t}_{rr} + \Gamma^{t}_{tr}\Gamma^{r}_{rr} \tag{5.286}$$

$$\Gamma^{t}_{r\lambda}\Gamma^{\lambda}_{tr} = \Gamma^{t}_{rt}\Gamma^{t}_{tr} + \Gamma^{t}_{rr}\Gamma^{r}_{tr} \tag{5.287}$$

We now evaluate the four fundamental terms of $R^{t}{}_{rtr}$ individually, substituting the explicit non-zero Christoffel symbols: **Term 1:**

$$\begin{aligned}
\partial_t \Gamma^{t}_{rr} &= \partial_t \left(e^{2(\beta(t,r)-\alpha(t,r))} \partial_t \beta(t,r) \right) \\
&= \left(\partial_t e^{2(\beta(t,r)-\alpha(t,r))} \right) \partial_t \beta(t,r) + e^{2(\beta(t,r)-\alpha(t,r))} \partial_t^2 \beta(t,r) \\
&= 2(\partial_t \beta(t,r) - \partial_t \alpha(t,r)) e^{2(\beta(t,r)-\alpha(t,r))} \partial_t \beta(t,r) + e^{2(\beta(t,r)-\alpha(t,r))} \partial_t^2 \beta(t,r)
\end{aligned} \tag{5.288}$$

Term 2:

$$-\partial_r \Gamma^t_{tr} = -\partial_r \left(\partial_r \alpha(t,r)\right) = -\partial_r^2 \alpha(t,r) \tag{5.289}$$

Term 3:

$$\begin{aligned}\Gamma^t_{tt}\Gamma^t_{rr} + \Gamma^t_{tr}\Gamma^r_{rr} &= (\partial_t \alpha(t,r)) \left(e^{2(\beta(t,r)-\alpha(t,r))}\partial_t \beta(t,r)\right) + (\partial_r \alpha(t,r))(\partial_r \beta(t,r)) \\ &= e^{2(\beta(t,r)-\alpha(t,r))}\partial_t \alpha(t,r)\partial_t \beta(t,r) + \partial_r \alpha(t,r)\partial_r \beta(t,r)\end{aligned} \tag{5.290}$$

Term 4:

$$\begin{aligned}-\left(\Gamma^t_{rt}\Gamma^t_{tr} + \Gamma^t_{rr}\Gamma^r_{tr}\right) &= -\left[(\partial_r \alpha(t,r))(\partial_r \alpha(t,r)) + \left(e^{2(\beta(t,r)-\alpha(t,r))}\partial_t \beta(t,r)\right)(\partial_t \beta(t,r))\right] \\ &= -(\partial_r \alpha(t,r))^2 - e^{2(\beta(t,r)-\alpha(t,r))}(\partial_t \beta(t,r))^2\end{aligned} \tag{5.291}$$

Summing these four terms and organizing them by the exponential prefactor $e^{2(\beta(t,r)-\alpha(t,r))}$ and standard derivative terms[cite: 2]:

$$\begin{aligned}R^t{}_{rtr} = e^{2(\beta-\alpha)} &\left[2(\partial_t \beta)^2 - 2\partial_t \alpha \partial_t \beta + \partial_t^2 \beta + \partial_t \alpha \partial_t \beta - (\partial_t \beta)^2\right] \\ &+ \left[-\partial_r^2 \alpha + \partial_r \alpha \partial_r \beta - (\partial_r \alpha)^2\right]\end{aligned} \tag{5.292}$$

Combining algebraically similar terms yields the final expression:

$$R^t{}_{rtr} = e^{2(\beta-\alpha)} \left[\partial_t^2 \beta + (\partial_t \beta)^2 - \partial_t \alpha \partial_t \beta\right] - \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta\right] \tag{5.293}$$

2. Derivation of $R^\theta{}_{\phi\theta\phi}$

We assign the indices $(\rho, \sigma, \mu, \nu) = (\theta, \phi, \theta, \phi)$. The Riemann definition is:

$$R^\theta{}_{\phi\theta\phi} = \partial_\theta \Gamma^\theta_{\phi\phi} - \partial_\phi \Gamma^\theta_{\theta\phi} + \Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{\phi\phi} - \Gamma^\theta_{\phi\lambda}\Gamma^\lambda_{\theta\phi} \tag{5.294}$$

We observe from Part 3 that $\Gamma^\theta_{\theta\phi} = 0$, causing the term $\partial_\phi \Gamma^\theta_{\theta\phi}$ to vanish entirely. We expand the summations over λ for the remaining terms:

$$\Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{\phi\phi} = \Gamma^\theta_{\theta t}\Gamma^t_{\phi\phi} + \Gamma^\theta_{\theta r}\Gamma^r_{\phi\phi} + \Gamma^\theta_{\theta\theta}\Gamma^\theta_{\phi\phi} + \Gamma^\theta_{\theta\phi}\Gamma^\phi_{\phi\phi} \tag{5.295}$$

$$\Gamma^\theta_{\phi\lambda}\Gamma^\lambda_{\theta\phi} = \Gamma^\theta_{\phi t}\Gamma^t_{\theta\phi} + \Gamma^\theta_{\phi r}\Gamma^r_{\theta\phi} + \Gamma^\theta_{\phi\theta}\Gamma^\theta_{\theta\phi} + \Gamma^\theta_{\phi\phi}\Gamma^\phi_{\theta\phi} \tag{5.296}$$

Applying the identically zero Christoffel symbols simplifies these to:

$$\Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{\phi\phi} = \Gamma^\theta_{\theta r}\Gamma^r_{\phi\phi} \tag{5.297}$$

$$\Gamma^\theta_{\phi\lambda}\Gamma^\lambda_{\theta\phi} = \Gamma^\theta_{\phi\phi}\Gamma^\phi_{\theta\phi} \tag{5.298}$$

We evaluate the terms sequentially: **Term 1:**

$$\begin{aligned}\partial_\theta \Gamma^\theta_{\phi\phi} &= \partial_\theta(-\sin\theta\cos\theta) \\ &= -(\cos\theta\cos\theta - \sin\theta\sin\theta) \\ &= \sin^2\theta - \cos^2\theta\end{aligned} \tag{5.299}$$

Term 2:

$$\begin{aligned}\Gamma^\theta_{\theta r}\Gamma^r_{\phi\phi} &= \left(\frac{1}{r}\right)\left(-re^{-2\beta(t,r)}\sin^2\theta\right) \\ &= -e^{-2\beta(t,r)}\sin^2\theta\end{aligned} \tag{5.300}$$

Term 3:

$$\begin{aligned}-\Gamma^\theta_{\phi\phi}\Gamma^\phi_{\theta\phi} &= -\left(-\sin\theta\cos\theta\right)(\cot\theta) \\ &= \sin\theta\cos\theta\left(\frac{\cos\theta}{\sin\theta}\right) \\ &= \cos^2\theta\end{aligned} \tag{5.301}$$

Summing the evaluated terms[cite: 2]:

$$\begin{aligned} R^{\theta}{}_{\phi\theta\phi} &= (\sin^2\theta - \cos^2\theta) - e^{-2\beta(t,r)}\sin^2\theta + \cos^2\theta \\ &= \sin^2\theta - e^{-2\beta(t,r)}\sin^2\theta \\ &= (1 - e^{-2\beta(t,r)})\sin^2\theta \end{aligned} \tag{5.302}$$

3. Derivation of $R^{r}{}_{\theta r\theta}$

We assign the indices $(\rho, \sigma, \mu, \nu) = (r, \theta, r, \theta)$. The definition reads:

$$R^{r}{}_{\theta r\theta} = \partial_r\Gamma^r_{\theta\theta} - \partial_\theta\Gamma^r_{r\theta} + \Gamma^r_{r\lambda}\Gamma^\lambda_{\theta\theta} - \Gamma^r_{\theta\lambda}\Gamma^\lambda_{r\theta} \tag{5.303}$$

Since $\Gamma^r_{r\theta} = 0$, the partial derivative $\partial_\theta\Gamma^r_{r\theta} = 0$. Expanding the summations over λ:

$$\Gamma^r_{r\lambda}\Gamma^\lambda_{\theta\theta} = \Gamma^r_{rt}\Gamma^t_{\theta\theta} + \Gamma^r_{rr}\Gamma^r_{\theta\theta} + \Gamma^r_{r\theta}\Gamma^\theta_{\theta\theta} + \Gamma^r_{r\phi}\Gamma^\phi_{\theta\theta} = \Gamma^r_{rr}\Gamma^r_{\theta\theta} \tag{5.304}$$

$$\Gamma^r_{\theta\lambda}\Gamma^\lambda_{r\theta} = \Gamma^r_{\theta t}\Gamma^t_{r\theta} + \Gamma^r_{\theta r}\Gamma^r_{r\theta} + \Gamma^r_{\theta\theta}\Gamma^\theta_{r\theta} + \Gamma^r_{\theta\phi}\Gamma^\phi_{r\theta} = \Gamma^r_{\theta\theta}\Gamma^\theta_{r\theta} \tag{5.305}$$

Evaluating the non-zero terms: **Term 1:**

$$\begin{aligned} \partial_r\Gamma^r_{\theta\theta} &= \partial_r\left(-re^{-2\beta(t,r)}\right) \\ &= -e^{-2\beta(t,r)} - r\left(-2\partial_r\beta(t,r)e^{-2\beta(t,r)}\right) \\ &= -e^{-2\beta(t,r)} + 2re^{-2\beta(t,r)}\partial_r\beta(t,r) \end{aligned} \tag{5.306}$$

Term 2:

$$\begin{aligned} \Gamma^r_{rr}\Gamma^r_{\theta\theta} &= (\partial_r\beta(t,r))\left(-re^{-2\beta(t,r)}\right) \\ &= -re^{-2\beta(t,r)}\partial_r\beta(t,r) \end{aligned} \tag{5.307}$$

Term 3:

$$\begin{aligned} -\Gamma^r_{\theta\theta}\Gamma^\theta_{r\theta} &= -\left(-re^{-2\beta(t,r)}\right)\left(\frac{1}{r}\right) \\ &= e^{-2\beta(t,r)} \end{aligned} \tag{5.308}$$

Summing these evaluated terms[cite: 2]:

$$\begin{aligned} R^{r}{}_{\theta r\theta} &= \left(-e^{-2\beta(t,r)} + 2re^{-2\beta(t,r)}\partial_r\beta(t,r)\right) - re^{-2\beta(t,r)}\partial_r\beta(t,r) + e^{-2\beta(t,r)} \\ &= 2re^{-2\beta(t,r)}\partial_r\beta(t,r) - re^{-2\beta(t,r)}\partial_r\beta(t,r) \\ &= re^{-2\beta(t,r)}\partial_r\beta(t,r) \end{aligned} \tag{5.309}$$

This demonstrates the rigorous explicit expansion of the curvature tensor components directly from the connection coefficients.

5.2.5 Ricci Tensor

The Ricci tensor is defined by the contraction of the first and third indices of the Riemann curvature tensor[cite: 1, 2]:

$$R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu} \tag{5.310}$$

Expanding the summation over the dummy index $\lambda \in \{t, r, \theta, \phi\}$ gives the explicit formula for any component:

$$R_{\mu\nu} = R^{t}{}_{\mu t\nu} + R^{r}{}_{\mu r\nu} + R^{\theta}{}_{\mu\theta\nu} + R^{\phi}{}_{\mu\phi\nu} \tag{5.311}$$

We will compute the non-vanishing components of the Ricci tensor $(R_{tt}, R_{rr}, R_{tr}, R_{\theta\theta}, R_{\phi\phi})$ explicitly by expanding the corresponding Riemann tensor components[cite: 2]. The definition of the Riemann tensor is:

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma} \tag{5.312}$$

1. Derivation of R_{tt}

Applying equation (5.311) for $\mu = t, \nu = t$:

$$R_{tt} = R^t{}_{ttt} + R^r{}_{trt} + R^\theta{}_{t\theta t} + R^\phi{}_{t\phi t} \tag{5.313}$$

Component $R^t{}_{ttt}$: By the antisymmetry of the Riemann tensor in its last two indices ($R^\rho{}_{\sigma\mu\nu} = -R^\rho{}_{\sigma\nu\mu}$), any component with identical last indices vanishes identically:

$$R^t{}_{ttt} = 0 \tag{5.314}$$

Component $R^r{}_{trt}$: Using equation (5.312) with $(\rho, \sigma, \mu, \nu) = (r, t, r, t)$:

$$R^r{}_{trt} = \partial_r \Gamma^r_{tt} - \partial_t \Gamma^r_{rt} + \Gamma^r_{r\lambda}\Gamma^\lambda_{tt} - \Gamma^r_{t\lambda}\Gamma^\lambda_{rt} \tag{5.315}$$

We expand the summations over λ, keeping only non-zero terms:

$$\Gamma^r_{r\lambda}\Gamma^\lambda_{tt} = \Gamma^r_{rt}\Gamma^t_{tt} + \Gamma^r_{rr}\Gamma^r_{tt} \tag{5.316}$$
$$\Gamma^r_{t\lambda}\Gamma^\lambda_{rt} = \Gamma^r_{tt}\Gamma^t_{rt} + \Gamma^r_{tr}\Gamma^r_{rt} \tag{5.317}$$

Substitute the Christoffel symbols derived in Part 3:

$$\partial_r \Gamma^r_{tt} = \partial_r \left(e^{2(\alpha-\beta)}\partial_r\alpha\right) = 2(\partial_r\alpha - \partial_r\beta)e^{2(\alpha-\beta)}\partial_r\alpha + e^{2(\alpha-\beta)}\partial_r^2\alpha \tag{5.318}$$
$$-\partial_t \Gamma^r_{rt} = -\partial_t(\partial_t\beta) = -\partial_t^2\beta \tag{5.319}$$
$$\Gamma^r_{rt}\Gamma^t_{tt} + \Gamma^r_{rr}\Gamma^r_{tt} = (\partial_t\beta)(\partial_t\alpha) + (\partial_r\beta)\left(e^{2(\alpha-\beta)}\partial_r\alpha\right) \tag{5.320}$$
$$-\left(\Gamma^r_{tt}\Gamma^t_{rt} + \Gamma^r_{tr}\Gamma^r_{rt}\right) = -\left[\left(e^{2(\alpha-\beta)}\partial_r\alpha\right)(\partial_r\alpha) + (\partial_t\beta)(\partial_t\beta)\right] = -e^{2(\alpha-\beta)}(\partial_r\alpha)^2 - (\partial_t\beta)^2 \tag{5.321}$$

Summing these evaluated terms and grouping by spatial and temporal derivatives:

$$\begin{aligned} R^r{}_{trt} &= e^{2(\alpha-\beta)}\left[2(\partial_r\alpha)^2 - 2\partial_r\alpha\partial_r\beta + \partial_r^2\alpha + \partial_r\alpha\partial_r\beta - (\partial_r\alpha)^2\right] \\ &\quad - \partial_t^2\beta + \partial_t\alpha\partial_t\beta - (\partial_t\beta)^2 \\ &= e^{2(\alpha-\beta)}\left[\partial_r^2\alpha + (\partial_r\alpha)^2 - \partial_r\alpha\partial_r\beta\right] - \left[\partial_t^2\beta + (\partial_t\beta)^2 - \partial_t\alpha\partial_t\beta\right] \end{aligned} \tag{5.322}$$

Component $R^\theta{}_{t\theta t}$:

$$R^\theta{}_{t\theta t} = \partial_t \Gamma^\theta_{\theta t} - \partial_\theta \Gamma^\theta_{tt} + \Gamma^\theta_{t\lambda}\Gamma^\lambda_{\theta t} - \Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{tt} \tag{5.323}$$

Since $\Gamma^\theta_{\theta t} = 0$ and $\Gamma^\theta_{tt} = 0$, the partial derivatives vanish. Expanding the summations:

$$\Gamma^\theta_{t\lambda}\Gamma^\lambda_{\theta t} = \Gamma^\theta_{tr}\Gamma^r_{\theta t} = 0 \tag{5.324}$$
$$-\Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{tt} = -\Gamma^\theta_{\theta r}\Gamma^r_{tt} = -\left(\frac{1}{r}\right)\left(e^{2(\alpha-\beta)}\partial_r\alpha\right) = -\frac{1}{r}e^{2(\alpha-\beta)}\partial_r\alpha \tag{5.325}$$

Wait, the definition of Riemann is $R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \dots$. For $(\rho, \sigma, \mu, \nu) = (\theta, t, \theta, t)$:

$$R^\theta{}_{t\theta t} = \partial_\theta \Gamma^\theta_{tt} - \partial_t \Gamma^\theta_{\theta t} + \Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{tt} - \Gamma^\theta_{t\lambda}\Gamma^\lambda_{\theta t} \tag{5.326}$$

The partial derivatives are still zero. The sum evaluates to:

$$\Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{tt} = \Gamma^\theta_{\theta r}\Gamma^r_{tt} = \left(\frac{1}{r}\right)\left(e^{2(\alpha-\beta)}\partial_r\alpha\right) = \frac{1}{r}e^{2(\alpha-\beta)}\partial_r\alpha \tag{5.327}$$

Component $R^\phi{}_{t\phi t}$: By identical symmetry in the ϕ coordinate:

$$R^\phi{}_{t\phi t} = \Gamma^\phi_{\phi r}\Gamma^r_{tt} = \left(\frac{1}{r}\right)\left(e^{2(\alpha-\beta)}\partial_r\alpha\right) = \frac{1}{r}e^{2(\alpha-\beta)}\partial_r\alpha \tag{5.328}$$

Total R_{tt}: Summing the components:

$$R_{tt} = -\left[\partial_t^2\beta + (\partial_t\beta)^2 - \partial_t\alpha\partial_t\beta\right] + e^{2(\alpha-\beta)}\left[\partial_r^2\alpha + (\partial_r\alpha)^2 - \partial_r\alpha\partial_r\beta + \frac{2}{r}\partial_r\alpha\right] \tag{5.329}$$

2. Derivation of R_{rr}

Applying equation (5.311) for $\mu = r, \nu = r$:

$$R_{rr} = R^t{}_{rtr} + R^r{}_{rrr} + R^\theta{}_{r\theta r} + R^\phi{}_{r\phi r} \tag{5.330}$$

Component $R^t{}_{rtr}$: This was fully derived in Part 4:

$$R^t{}_{rtr} = e^{2(\beta-\alpha)}\left[\partial_t^2\beta + (\partial_t\beta)^2 - \partial_t\alpha\partial_t\beta\right] - \left[\partial_r^2\alpha + (\partial_r\alpha)^2 - \partial_r\alpha\partial_r\beta\right] \tag{5.331}$$

(Note: $R^r{}_{rrr} = 0$ by antisymmetry).

Component $R^\theta{}_{r\theta r}$:

$$R^\theta{}_{r\theta r} = \partial_\theta\Gamma^\theta_{rr} - \partial_r\Gamma^\theta_{\theta r} + \Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{rr} - \Gamma^\theta_{r\lambda}\Gamma^\lambda_{\theta r} \tag{5.332}$$

$$\partial_\theta\Gamma^\theta_{rr} = 0 \tag{5.333}$$

$$-\partial_r\Gamma^\theta_{\theta r} = -\partial_r\left(\frac{1}{r}\right) = \frac{1}{r^2} \tag{5.334}$$

$$\Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{rr} = \Gamma^\theta_{\theta r}\Gamma^r_{rr} = \left(\frac{1}{r}\right)(\partial_r\beta) = \frac{1}{r}\partial_r\beta \tag{5.335}$$

$$-\Gamma^\theta_{r\lambda}\Gamma^\lambda_{\theta r} = -\Gamma^\theta_{r\theta}\Gamma^\theta_{\theta r} = -\left(\frac{1}{r}\right)\left(\frac{1}{r}\right) = -\frac{1}{r^2} \tag{5.336}$$

Summing these gives:

$$R^\theta{}_{r\theta r} = \frac{1}{r^2} + \frac{1}{r}\partial_r\beta - \frac{1}{r^2} = \frac{1}{r}\partial_r\beta \tag{5.337}$$

Component $R^\phi{}_{r\phi r}$: By spherical symmetry, this matches the θ component exactly:

$$R^\phi{}_{r\phi r} = \frac{1}{r}\partial_r\beta \tag{5.338}$$

Total R_{rr}: Summing the components[cite: 2]:

$$R_{rr} = e^{2(\beta-\alpha)}\left[\partial_t^2\beta + (\partial_t\beta)^2 - \partial_t\alpha\partial_t\beta\right] - \left[\partial_r^2\alpha + (\partial_r\alpha)^2 - \partial_r\alpha\partial_r\beta - \frac{2}{r}\partial_r\beta\right] \tag{5.339}$$

3. Derivation of R_{tr}

Applying equation (5.311) for $\mu = t, \nu = r$:

$$R_{tr} = R^t{}_{ttr} + R^r{}_{trr} + R^\theta{}_{t\theta r} + R^\phi{}_{t\phi r} \tag{5.340}$$

Components $R^t{}_{ttr}$ and $R^r{}_{trr}$:

$$\begin{aligned} R^t{}_{ttr} &= \partial_t\Gamma^t_{rt} - \partial_r\Gamma^t_{tt} + \Gamma^t_{t\lambda}\Gamma^\lambda_{rt} - \Gamma^t_{r\lambda}\Gamma^\lambda_{tt} \\ &= \partial_t(\partial_r\alpha) - \partial_r(\partial_t\alpha) + \left(\Gamma^t_{tt}\Gamma^t_{rt} + \Gamma^t_{tr}\Gamma^r_{rt}\right) - \left(\Gamma^t_{rt}\Gamma^t_{tt} + \Gamma^t_{rr}\Gamma^r_{tt}\right) \\ &= 0 + (\partial_t\alpha)(\partial_r\alpha) + (\partial_r\alpha)(\partial_t\beta) - (\partial_r\alpha)(\partial_t\alpha) - \left(e^{2(\beta-\alpha)}\partial_t\beta\right)\left(e^{2(\alpha-\beta)}\partial_r\alpha\right) \\ &= \partial_r\alpha\partial_t\beta - \partial_t\beta\partial_r\alpha = 0 \end{aligned} \tag{5.341}$$

By identical summation expansion, $R^r{}_{trr} = 0$.

Component $R^\theta{}_{t\theta r}$:

$$R^\theta{}_{t\theta r} = \partial_\theta\Gamma^\theta_{rt} - \partial_r\Gamma^\theta_{t\theta} + \Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{rt} - \Gamma^\theta_{r\lambda}\Gamma^\lambda_{t\theta} \tag{5.342}$$

The connection coefficients Γ^θ_{rt} and $\Gamma^\theta_{t\theta}$ are zero, so the derivatives vanish.

$$\Gamma^\theta_{\theta\lambda}\Gamma^\lambda_{rt} = \Gamma^\theta_{\theta r}\Gamma^r_{rt} = \left(\frac{1}{r}\right)(\partial_t\beta) = \frac{1}{r}\partial_t\beta \tag{5.343}$$

$$-\Gamma^\theta_{r\lambda}\Gamma^\lambda_{t\theta} = 0 \tag{5.344}$$

Thus, $R^\theta{}_{t\theta r} = \frac{1}{r}\partial_t\beta$.

Component $R^\phi{}_{t\phi r}$: By spherical symmetry, $R^\phi{}_{t\phi r} = \frac{1}{r}\partial_t\beta$.

Total R_{tr}: Summing the components[cite: 2]:

$$R_{tr} = \frac{2}{r}\partial_t\beta \tag{5.345}$$

4. Derivation of $R_{\theta\theta}$

Applying equation (5.311) for $\mu = \theta, \nu = \theta$:

$$R_{\theta\theta} = R^t{}_{\theta t\theta} + R^r{}_{\theta r\theta} + R^\theta{}_{\theta\theta\theta} + R^\phi{}_{\theta\phi\theta} \tag{5.346}$$

(Note: $R^\theta{}_{\theta\theta\theta} = 0$).

Component $R^t{}_{\theta t\theta}$:

$$\begin{aligned} R^t{}_{\theta t\theta} &= \partial_\theta \Gamma^t_{\theta t} - \partial_t \Gamma^t_{\theta\theta} + \Gamma^t_{\theta\lambda}\Gamma^\lambda_{\theta t} - \Gamma^t_{t\lambda}\Gamma^\lambda_{\theta\theta} \\ &= 0 - 0 + 0 - \Gamma^t_{tr}\Gamma^r_{\theta\theta} \\ &= -(\partial_r\alpha)\left(-re^{-2\beta}\right) = re^{-2\beta}\partial_r\alpha \end{aligned} \tag{5.347}$$

Wait, verifying index ordering: $R^\rho{}_{\sigma\mu\nu} \implies (\rho,\sigma,\mu,\nu) = (t,t,\theta,\theta)$? No, the trace is $R^\lambda{}_{\mu\lambda\nu}$. Thus, for $\mu = \theta, \nu = \theta$, we need $R^t{}_{\theta t\theta}$.

$$R^t{}_{\theta t\theta} = \partial_t \Gamma^t_{\theta\theta} - \partial_\theta \Gamma^t_{t\theta} + \Gamma^t_{t\lambda}\Gamma^\lambda_{\theta\theta} - \Gamma^t_{\theta\lambda}\Gamma^\lambda_{t\theta} = 0 - 0 + \Gamma^t_{tr}\Gamma^r_{\theta\theta} - 0 = -re^{-2\beta}\partial_r\alpha \tag{5.348}$$

Component $R^r{}_{\theta r\theta}$: This was fully derived in Part 4:

$$R^r{}_{\theta r\theta} = re^{-2\beta}\partial_r\beta \tag{5.349}$$

Component $R^\phi{}_{\theta\phi\theta}$:

$$R^\phi{}_{\theta\phi\theta} = \partial_\phi \Gamma^\phi_{\theta\theta} - \partial_\theta \Gamma^\phi_{\phi\theta} + \Gamma^\phi_{\phi\lambda}\Gamma^\lambda_{\theta\theta} - \Gamma^\phi_{\theta\lambda}\Gamma^\lambda_{\phi\theta} \tag{5.350}$$

$$-\partial_\theta \Gamma^\phi_{\phi\theta} = -\partial_\theta(\cot\theta) = \csc^2\theta \tag{5.351}$$

$$\Gamma^\phi_{\phi\lambda}\Gamma^\lambda_{\theta\theta} = \Gamma^\phi_{\phi r}\Gamma^r_{\theta\theta} = \left(\frac{1}{r}\right)\left(-re^{-2\beta}\right) = -e^{-2\beta} \tag{5.352}$$

$$-\Gamma^\phi_{\theta\lambda}\Gamma^\lambda_{\phi\theta} = -\Gamma^\phi_{\theta\phi}\Gamma^\phi_{\phi\theta} = -(\cot\theta)(\cot\theta) = -\cot^2\theta \tag{5.353}$$

Using the identity $\csc^2\theta - \cot^2\theta = 1$, we find:

$$R^\phi{}_{\theta\phi\theta} = \csc^2\theta - \cot^2\theta - e^{-2\beta} = 1 - e^{-2\beta} \tag{5.354}$$

Total $R_{\theta\theta}$: Summing the components[cite: 2]:

$$R_{\theta\theta} = -re^{-2\beta}\partial_r\alpha + re^{-2\beta}\partial_r\beta + 1 - e^{-2\beta} = e^{-2\beta}\left[r(\partial_r\beta - \partial_r\alpha) - 1\right] + 1 \tag{5.355}$$

5. Derivation of $R_{\phi\phi}$

Applying equation (5.311) for $\mu = \phi, \nu = \phi$:

$$R_{\phi\phi} = R^t{}_{\phi t\phi} + R^r{}_{\phi r\phi} + R^\theta{}_{\phi\theta\phi} + R^\phi{}_{\phi\phi\phi} \tag{5.356}$$

Evaluating the components analogously:

$$R^t{}_{\phi t\phi} = \Gamma^t_{tr}\Gamma^r_{\phi\phi} = (\partial_r\alpha)\left(-re^{-2\beta}\sin^2\theta\right) = -re^{-2\beta}\sin^2\theta\partial_r\alpha \tag{5.357}$$

$$\begin{aligned} R^r{}_{\phi r\phi} &= \partial_r \Gamma^r_{\phi\phi} + \Gamma^r_{rr}\Gamma^r_{\phi\phi} - \Gamma^r_{\phi\phi}\Gamma^\phi_{r\phi} \\ &= \partial_r\left(-re^{-2\beta}\sin^2\theta\right) + (\partial_r\beta)\left(-re^{-2\beta}\sin^2\theta\right) - \left(-re^{-2\beta}\sin^2\theta\right)\left(\frac{1}{r}\right) \\ &= \left(-e^{-2\beta}\sin^2\theta + 2re^{-2\beta}\sin^2\theta\partial_r\beta\right) - re^{-2\beta}\sin^2\theta\partial_r\beta + e^{-2\beta}\sin^2\theta \\ &= re^{-2\beta}\sin^2\theta\partial_r\beta \end{aligned} \tag{5.358}$$

$$R^\theta{}_{\phi\theta\phi} = (1 - e^{-2\beta})\sin^2\theta \quad \text{(Derived in Part 4)} \tag{5.359}$$

Total $R_{\phi\phi}$: Summing the components[cite: 2]:

$$\begin{aligned} R_{\phi\phi} &= \left(-re^{-2\beta}\partial_r\alpha + re^{-2\beta}\partial_r\beta + 1 - e^{-2\beta}\right)\sin^2\theta \\ &= R_{\theta\theta}\sin^2\theta \end{aligned} \tag{5.360}$$

5.2.6 Einstein Equations in Vacuum

The Vacuum Field Equations

In the absence of matter or energy sources, the spacetime geometry is governed by the vacuum Einstein field equations. This condition requires that the Ricci curvature tensor vanishes identically:

$$R_{\mu\nu} = 0 \tag{5.361}$$

Our objective is to solve this system of partial differential equations using the non-zero components of $R_{\mu\nu}$ derived in Part 5. We will systematically analyze these components to constrain the unknown metric functions $\alpha(t,r)$ and $\beta(t,r)$[cite: 2].

1. The R_{tr} Equation and the Constraint on $\beta(t,r)$

We begin with the off-diagonal component R_{tr}[cite: 2]. From our previous derivation, we have:

$$R_{tr} = \frac{2}{r}\partial_t\beta(t,r) \tag{5.362}$$

Applying the vacuum condition (5.361):

$$\frac{2}{r}\partial_t\beta(t,r) = 0 \tag{5.363}$$

For any physical point in the spacetime away from the singular origin ($r \neq 0$), we can multiply both sides of equation (5.363) by $\frac{r}{2}$:

$$\partial_t\beta(t,r) = 0 \tag{5.364}$$

Equation (5.364) mathematically guarantees that the metric function β has no dependence on the temporal coordinate t[cite: 2]. Therefore, we can express β as a function of the radial coordinate r alone[cite: 2]:

$$\beta(t,r) = \beta(r) \tag{5.365}$$

2. The $R_{\theta\theta}$ Equation and the Constraint on $\alpha(t,r)$

We next examine the angular component $R_{\theta\theta}$. Substituting the explicit result from Part 5 and applying the vacuum condition yields:

$$R_{\theta\theta} = e^{-2\beta(r)}\left[r(\partial_r\beta(r) - \partial_r\alpha(t,r)) - 1\right] + 1 = 0 \tag{5.366}$$

Notice that we have explicitly substituted $\beta(t,r) = \beta(r)$ into the expression based on equation (5.365).

To determine the temporal behavior of $\alpha(t,r)$, we take the partial derivative of equation (5.366) with respect to the time coordinate t[cite: 2]:

$$\partial_t\left(e^{-2\beta(r)}\left[r(\partial_r\beta(r) - \partial_r\alpha(t,r)) - 1\right] + 1\right) = \partial_t(0) \tag{5.367}$$

We expand this derivative systematically using the sum and product rules. Since $e^{-2\beta(r)}$ and r are strictly independent of t, they act as constants under the operator ∂_t:

$$e^{-2\beta(r)}\partial_t\left(r\partial_r\beta(r) - r\partial_r\alpha(t,r) - 1\right) + \partial_t(1) = 0 \tag{5.368}$$

Evaluating the inner temporal derivatives:

$$e^{-2\beta(r)}\left(r\underbrace{\partial_t\partial_r\beta(r)}_{0} - r\partial_t\partial_r\alpha(t,r) - \underbrace{\partial_t(1)}_{0}\right) + 0 = 0 \tag{5.369}$$

This simplifies to:

$$-re^{-2\beta(r)}\partial_t\partial_r\alpha(t,r) = 0 \tag{5.370}$$

Since neither the radial coordinate r (for $r \neq 0$) nor the exponential function $e^{-2\beta(r)}$ can equal zero, we must conclude[cite: 2]:

$$\partial_t \partial_r \alpha(t, r) = 0 \tag{5.371}$$

Assuming sufficient smoothness to commute the partial derivatives (Schwarz's theorem), we have:

$$\partial_r(\partial_t \alpha(t, r)) = 0 \tag{5.372}$$

This implies that the temporal derivative of α must be a function of t alone, meaning there is no mixing between t and r dependencies. Integrating with respect to r yields an additive separation of variables[cite: 2]:

$$\alpha(t, r) = f(r) + g(t) \tag{5.373}$$

where $f(r)$ is an arbitrary function of the radial coordinate and $g(t)$ is an arbitrary function of the time coordinate[cite: 2].

3. Vanishing of Time Derivatives in R_{tt} and R_{rr}

We now substitute our constraints ($\partial_t \beta = 0$, $\partial_t^2 \beta = 0$) into the remaining diagonal Ricci components, R_{tt} and R_{rr}, to verify that their governing dynamics are strictly spatial.

Starting with the R_{tt} component derived in Part 5:

$$R_{tt} = -\left[\partial_t^2 \beta + (\partial_t \beta)^2 - \partial_t \alpha \partial_t \beta\right] + e^{2(\alpha-\beta)}\left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{2}{r}\partial_r \alpha\right] = 0 \tag{5.374}$$

Applying $\partial_t \beta = 0$ causes the entire first bracket to vanish exactly:

$$R_{tt} = -\left[0 + 0^2 - (\partial_t \alpha)(0)\right] + e^{2(\alpha-\beta)}\left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{2}{r}\partial_r \alpha\right] = 0 \tag{5.375}$$

Furthermore, since $\alpha(t, r) = f(r)+g(t)$, all spatial derivatives of α isolate the function $f(r)$: $\partial_r \alpha(t, r) = f'(r)$ and $\partial_r^2 \alpha(t, r) = f''(r)$. The temporal part $g(t)$ is completely annihilated by the spatial derivatives. Because $e^{2(\alpha-\beta)} \neq 0$, we can divide it out to obtain the purely spatial differential equation:

$$f''(r) + (f'(r))^2 - f'(r)\beta'(r) + \frac{2}{r}f'(r) = 0 \tag{5.376}$$

Next, we evaluate the R_{rr} component:

$$R_{rr} = e^{2(\beta-\alpha)}\left[\partial_t^2 \beta + (\partial_t \beta)^2 - \partial_t \alpha \partial_t \beta\right] - \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta - \frac{2}{r}\partial_r \beta\right] = 0 \tag{5.377}$$

Again, the application of $\partial_t \beta = 0$ eliminates the entire temporal bracket:

$$R_{rr} = e^{2(\beta-\alpha)}\left[0 + 0^2 - (\partial_t \alpha)(0)\right] - \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta - \frac{2}{r}\partial_r \beta\right] = 0 \tag{5.378}$$

Multiplying by -1 and replacing the spatial derivatives of $\alpha(t, r)$ with derivatives of $f(r)$, we obtain the second purely spatial differential equation:

$$f''(r) + (f'(r))^2 - f'(r)\beta'(r) - \frac{2}{r}\beta'(r) = 0 \tag{5.379}$$

By systematically applying the $R_{tr} = 0$ and $R_{\theta\theta} = 0$ constraints, we have rigorously demonstrated that the fundamental structure of the spherically symmetric vacuum equations is entirely independent of temporal evolution, confirming that any spherically symmetric vacuum solution must ultimately be static.

Elimination of Time Dependence (Coordinate Redefinition)

1. Separation of Variables for $\alpha(t,r)$

In Part 6, we established from the $R_{tr} = 0$ Einstein equation that $\partial_t \beta(t,r) = 0$, which immediately restricts the function β to depend solely on the radial coordinate[cite: 2]:

$$\beta(t,r) = \beta(r) \tag{5.380}$$

Subsequently, substituting equation (5.380) into the $R_{\theta\theta} = 0$ equation led to the strict mixed-derivative constraint[cite: 2]:

$$\partial_t \partial_r \alpha(t,r) = 0 \tag{5.381}$$

We rigorously integrate this constraint to find the explicit functional form of $\alpha(t,r)$. First, we integrate with respect to the radial coordinate r:

$$\int \left(\partial_r \left(\partial_t \alpha(t,r)\right)\right) dr = \int 0 \, dr \tag{5.382}$$

$$\partial_t \alpha(t,r) = h(t) \tag{5.383}$$

Here, $h(t)$ is an arbitrary integration function that depends exclusively on the time coordinate t, arising because the integration was performed with respect to r.

Next, we integrate this result with respect to the temporal coordinate t:

$$\int \left(\partial_t \alpha(t,r)\right) dt = \int h(t) \, dt \tag{5.384}$$

$$\alpha(t,r) = \int h(t) \, dt + f(r) \tag{5.385}$$

The term $f(r)$ emerges as the constant of integration with respect to t, meaning it is an arbitrary function depending solely on r. By defining a new purely time-dependent function $g(t) = \int h(t) \, dt$, we arrive at the exact additive separation of variables[cite: 2]:

$$\alpha(t,r) = f(r) + g(t) \tag{5.386}$$

2. Substitution into the Line Element

We substitute the separated functions from equations (5.380) and (5.386) back into the general spherically symmetric Lorentzian line element derived at the end of Part 1:

$$ds^2 = -e^{2\alpha(t,r)} dt^2 + e^{2\beta(t,r)} dr^2 + r^2 d\Omega^2 \tag{5.387}$$

$$ds^2 = -e^{2(f(r)+g(t))} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2 \tag{5.388}$$

Using the algebraic properties of exponentials ($e^{A+B} = e^A e^B$), we factor the temporal metric component:

$$ds^2 = -e^{2f(r)} e^{2g(t)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2 \tag{5.389}$$

3. Coordinate Transformation to Absorb $g(t)$

Equation (5.389) shows that the time dependence of the entire metric is isolated within a single multiplicative factor, $e^{2g(t)}$, which acts solely on the dt^2 differential. This structure mathematically invites a coordinate transformation to completely eliminate this temporal dependence[cite: 2].

We introduce a new temporal coordinate, which we will temporarily denote as $\tilde{t}$, designed specifically to absorb the $e^{2g(t)} dt^2$ term. We define the total differential $d\tilde{t}$ such that its square identically matches the time-dependent portion of the metric[cite: 2]:

$$d\tilde{t}^2 = e^{2g(t)} dt^2 \tag{5.390}$$

Taking the positive square root of both sides (to preserve the forward direction of time), we obtain the differential relationship:

$$d\tilde{t} = e^{g(t)}dt \tag{5.391}$$

Because $g(t)$ is a function of t alone, $e^{g(t)}dt$ forms an exact differential. We can thus globally define the new time coordinate $\tilde{t}$ by direct integration:

$$\tilde{t}(t) = \int e^{g(t)}\,dt \tag{5.392}$$

We now substitute equation (5.390) directly back into our factorized metric (5.389):

$$ds^2 = -e^{2f(r)}\left(e^{2g(t)}dt^2\right) + e^{2\beta(r)}dr^2 + r^2 d\Omega^2 \tag{5.393}$$

$$ds^2 = -e^{2f(r)}d\tilde{t}^2 + e^{2\beta(r)}dr^2 + r^2 d\Omega^2 \tag{5.394}$$

4. Final Static Metric Form

In the new coordinate system $(\tilde{t}, r, \theta, \phi)$, there is absolutely no explicit dependence on the temporal coordinate $\tilde{t}$. The geometry is therefore rigorously proven to be static[cite: 2].

To conform to standard notation, we simply drop the tilde from our new time coordinate ($\tilde{t} \to t$) and relabel the arbitrary radial function $f(r)$ as our new, strictly radial $\alpha(r)$ function ($\alpha(r) \equiv f(r)$)[cite: 2]. Applying these relabelings to equation (5.394) yields the final static metric:

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2 d\Omega^2 \tag{5.395}$$

where it is now definitively established that:

$$\alpha = \alpha(r) \tag{5.396}$$

$$\beta = \beta(r) \tag{5.397}$$

This complete elimination of time dependence from a purely spherically symmetric ansatz constitutes the first half of Birkhoff's Theorem: any spherically symmetric vacuum solution to the Einstein field equations must necessarily be static[cite: 2].

Final Reduction to Schwarzschild Form

1. The Reduced Static Metric

In the preceding subsections, we rigorously analyzed the vacuum Einstein field equations ($R_{\mu\nu} = 0$) for a general spherically symmetric spacetime. We determined that the radial metric function β must be strictly time-independent ($\partial_t \beta = 0$), implying $\beta(t, r) = \beta(r)$[cite: 2]. Furthermore, the temporal metric function was shown to be additively separable: $\alpha(t, r) = f(r) + g(t)$[cite: 2].

By executing a coordinate transformation to a new temporal parameter, we absorbed the function $g(t)$ entirely[cite: 2]. This allowed us to redefine the temporal coefficient strictly as a function of the radial coordinate: $\alpha(t, r) \to \alpha(r)$[cite: 2].

Substituting these purely radial functions back into our diagonalized line element, the spacetime metric rigorously reduces to:

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2 d\Omega^2 \tag{5.398}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2$ represents the standard metric on the 2-sphere.

2. Emergence of the Timelike Killing Vector

Equation (5.398) reveals that all metric components are now manifestly independent of the temporal coordinate t[cite: 2]. Mathematically, this implies the existence of a continuous symmetry generated by a vector field. Specifically, we have proven a crucial geometrical result: any spherically symmetric vacuum metric necessarily possesses a timelike Killing vector field[cite: 2].

In our chosen coordinates, this Killing vector is simply the coordinate basis vector $K = \partial_t$. A metric that possesses a Killing vector that is timelike near infinity is classified as *stationary*[cite: 2]. However, our metric exhibits a more restrictive property. Because we successfully eliminated all time-space cross terms (such as $dt\,dr$ or $dt\,d\phi$) in Part 1, the timelike Killing vector is strictly orthogonal to the family of spatial hypersurfaces defined by $t = \text{constant}$[cite: 2]. A stationary metric satisfying this hypersurface-orthogonality condition is defined as *static*[cite: 2].

Thus, we conclude that the geometry of a spherically symmetric vacuum is inherently static, precluding any dynamical radial pulsations or time-dependent gravitational radiation from a purely spherically symmetric source[cite: 2].

3. Birkhoff's Theorem and the Schwarzschild Solution

Notice that the static, spherically symmetric metric derived in equation (5.398) is precisely the same mathematical ansatz originally utilized to derive the Schwarzschild solution[cite: 2].

By evaluating the spatial components of the vacuum Einstein equations ($R_{rr} = 0$ and $R_{\theta\theta} = 0$) for this static metric, one determines the specific functional forms of $\alpha(r)$ and $\beta(r)$. The standard derivation yields $\alpha(r) = -\beta(r)$ and subsequently $e^{2\alpha(r)} = e^{-2\beta(r)} = 1 - \frac{2GM}{r}$.

Because our derivation began from the most general possible spherically symmetric metric without assuming time-independence a priori, we have therefore proven Birkhoff's theorem: the unique spherically symmetric vacuum solution to Einstein's equations is the Schwarzschild metric[cite: 2].

The final, fully reduced geometric form of the spacetime is:

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 \left(d\theta^2 + \sin^2\theta\, d\phi^2\right) \tag{5.399}$$

This completes the rigorous reduction from a completely arbitrary, time-dependent, spherically symmetric coordinate basis to the exact, static Schwarzschild geometry.

5.3 Coordinate vs. Curvature Singularities

Inspection of the metric coefficients in equation (**??**) reveals two alarming locations where the components diverge or degenerate:

1. As $r \to 0$, the temporal component $g_{tt} \to \infty$ and the radial component $g_{rr} \to 0$.
2. As $r \to 2GM$, the radial component $g_{rr} \to \infty$ and the temporal component $g_{tt} \to 0$.

We must determine whether these represent true breakdowns of the spacetime geometry (curvature singularities) or merely failures of our specific coordinate chart (coordinate singularities).

A coordinate singularity occurs when the coordinate system itself becomes degenerate, even though the underlying manifold is perfectly smooth. A classical example is the origin of a flat two-dimensional plane described by polar coordinates:

$$ds^2 = dr^2 + r^2 d\theta^2. \tag{5.400}$$

The inverse metric component is $g^{\theta\theta} = 1/r^2$, which diverges at $r = 0$. However, the intrinsic geometry is flawlessly flat ($R = 0$).

To establish the existence of a true geometric singularity, we must evaluate coordinate-invariant scalar quantities constructed from the Riemann curvature tensor $R^\rho{}_{\sigma\mu\nu}$. Because the Schwarzschild metric is a vacuum solution, the Ricci scalar trivially vanishes ($R = g^{\mu\nu} R_{\mu\nu} = 0$), as does the contraction of the Ricci tensor ($R^{\mu\nu}R_{\mu\nu} = 0$). To detect a divergence in the full curvature, we must compute the completely contracted square of the Riemann tensor, known as the **Kretschmann scalar**:

$$K = R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}. \tag{5.401}$$

5.3.1 Explicit Calculation of the Kretschmann Scalar

To calculate K without being overwhelmed by the metric coefficients of the inverse metric $g^{\mu\nu}$, it is highly advantageous to project the Riemann tensor onto a local orthonormal tetrad (vielbein) basis $e_{\hat{\mu}}$. In this basis, the metric components are simply those of flat Minkowski space, $g_{\hat{\mu}\hat{\nu}} = \eta_{\hat{\mu}\hat{\nu}}$.

For the Schwarzschild metric, the natural non-coordinate orthonormal basis one-forms are:

$$\hat{\omega}^{\hat{t}} = \left(1 - \frac{2GM}{r}\right)^{1/2} dt \tag{5.402}$$

$$\hat{\omega}^{\hat{r}} = \left(1 - \frac{2GM}{r}\right)^{-1/2} dr \tag{5.403}$$

$$\hat{\omega}^{\hat{\theta}} = r d\theta \tag{5.404}$$

$$\hat{\omega}^{\hat{\phi}} = r \sin\theta d\phi \tag{5.405}$$

By solving Cartan's structural equations ($d\hat{\omega}^{\hat{\mu}} + \hat{\omega}^{\hat{\mu}}_{\hat{\nu}} \wedge \hat{\omega}^{\hat{\nu}} = 0$ and $\hat{\Theta}^{\hat{\mu}}{}_{\hat{\nu}} = d\hat{\omega}^{\hat{\mu}}_{\hat{\nu}} + \hat{\omega}^{\hat{\mu}}_{\hat{\lambda}} \wedge \hat{\omega}^{\hat{\lambda}}_{\hat{\nu}}$), one finds the non-vanishing, independent components of the Riemann curvature two-form, which yield the tensor components in the orthonormal frame. The non-zero independent components are:

$$R_{\hat{t}\hat{r}\hat{t}\hat{r}} = -\frac{2GM}{r^3} \tag{5.406}$$

$$R_{\hat{t}\hat{\theta}\hat{t}\hat{\theta}} = R_{\hat{t}\hat{\phi}\hat{t}\hat{\phi}} = \frac{GM}{r^3} \tag{5.407}$$

$$R_{\hat{r}\hat{\theta}\hat{r}\hat{\theta}} = R_{\hat{r}\hat{\phi}\hat{r}\hat{\phi}} = -\frac{GM}{r^3} \tag{5.408}$$

$$R_{\hat{\theta}\hat{\phi}\hat{\theta}\hat{\phi}} = \frac{2GM}{r^3} \tag{5.409}$$

Because we are in an orthonormal frame with signature $(- + ++)$, raising indices requires multiplying by -1 for each time index $\hat{t}$ and $+1$ for each spatial index. Therefore, squaring an individual component $R_{\hat{\alpha}\hat{\beta}\hat{\gamma}\hat{\delta}}$ yields a positive contribution if it contains two time indices (since $(-1)(-1) = +1$), and a positive contribution if it contains zero time indices.

The Kretschmann scalar is the full sum over all indices:

$$K = \sum_{\mu,\nu,\rho,\sigma} R^{\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}} R_{\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}}. \tag{5.410}$$

Due to the symmetries of the Riemann tensor ($R_{abcd} = -R_{bacd} = -R_{abdc} = R_{cdab}$), each strictly independent component with distinct indices generates 4 identical terms in the summation (e.g., $R_{0101}, R_{1001}, R_{0110}, R_{1010}$ all square to the same value).

Let us sum the contributions explicitly:

$$\begin{aligned} K = {} & 4 \times (R_{\hat{t}\hat{r}\hat{t}\hat{r}})^2 \\ & + 4 \times (R_{\hat{t}\hat{\theta}\hat{t}\hat{\theta}})^2 + 4 \times (R_{\hat{t}\hat{\phi}\hat{t}\hat{\phi}})^2 \\ & + 4 \times (R_{\hat{r}\hat{\theta}\hat{r}\hat{\theta}})^2 + 4 \times (R_{\hat{r}\hat{\phi}\hat{r}\hat{\phi}})^2 \\ & + 4 \times (R_{\hat{\theta}\hat{\phi}\hat{\theta}\hat{\phi}})^2. \end{aligned} \tag{5.411}$$

Substituting the physical values from our tetrad basis:

$$
\begin{aligned}
K &= 4\left(-\frac{2GM}{r^3}\right)^2 + 4\left(\frac{GM}{r^3}\right)^2 + 4\left(\frac{GM}{r^3}\right)^2 + 4\left(-\frac{GM}{r^3}\right)^2 + 4\left(-\frac{GM}{r^3}\right)^2 + 4\left(\frac{2GM}{r^3}\right)^2 \\
&= 4\left(\frac{4G^2M^2}{r^6}\right) + 4\left(\frac{G^2M^2}{r^6}\right) + 4\left(\frac{G^2M^2}{r^6}\right) + 4\left(\frac{G^2M^2}{r^6}\right) + 4\left(\frac{G^2M^2}{r^6}\right) + 4\left(\frac{4G^2M^2}{r^6}\right) \\
&= \frac{G^2M^2}{r^6}\left[16+4+4+4+4+16\right] \\
K &= \frac{48G^2M^2}{r^6}.
\end{aligned}
\tag{5.412}
$$

5.3.2 Physical Implications of the Singularities

The exact calculation of the Kretschmann scalar provides definitive answers regarding the nature of the divergences in the Schwarzschild metric.

5.3.3 The Central Singularity ($r = 0$)

As $r \to 0$, the Kretschmann scalar diverges to infinity ($K \to \infty$). Because K is a coordinate-invariant scalar, this proves unequivocally that $r = 0$ is a true, physical curvature singularity. At this locus, tidal forces become infinite, and the smooth pseudo-Riemannian manifold structure of General Relativity fundamentally breaks down.

5.3.4 The Schwarzschild Radius ($r = 2GM$)

Conversely, let us evaluate the Kretschmann scalar at the Schwarzschild radius, $r = 2GM$:

$$
K(r = 2GM) = \frac{48G^2M^2}{(2GM)^6} = \frac{48G^2M^2}{64G^6M^6} = \frac{3}{4G^4M^4}. \tag{5.413}
$$

The curvature here is strictly finite. Therefore, $r = 2GM$ is definitively a **coordinate singularity**. The apparent divergence in g_{rr} is an artifact of the specific choice of Schwarzschild time t and radial distance r. The geometry itself is perfectly regular at this surface. (In subsequent sections, we will introduce Eddington-Finkelstein or Kruskal-Szekeres coordinates to smoothly cross this surface, revealing it to be the event horizon of a black hole).

5.3.5 Astrophysical Relevance

It is crucial to recognize that the Schwarzschild metric is exclusively a *vacuum* solution ($T_{\mu\nu} = 0$). It is rigorously valid only in the empty space exterior to a spherically symmetric mass distribution.

For ordinary astrophysical bodies, the Schwarzschild radius is minuscule compared to the physical radius of the object. For example, the physical radius of the Sun is $R_\odot \approx 7 \times 10^5$ km, while its Schwarzschild radius is merely:

$$
R_{S(\odot)} = \frac{2GM_\odot}{c^2} \approx 3 \text{ km}. \tag{5.414}
$$

Because 3 km $\ll 7 \times 10^5$ km, the surface $r = 2GM$ lies deep within the stellar interior, where the vacuum metric does not apply. Inside the star, the geometry is described by the interior Schwarzschild metric (which depends on the internal pressure and density profiles of the matter) and is perfectly smooth at $r = 0$. The pathological features of $r = 2GM$ and $r = 0$ are only realized in nature if a massive star undergoes complete gravitational collapse, forming a black hole.

5.4 Geodesics of the Schwarzschild Geometry

Having established the Schwarzschild metric as the unique, spherically symmetric vacuum solution to Einstein's field equations, our next task is to explore the physical dynamics it dictates. We accomplish this by analyzing the trajectories of freely falling test particles and photons, which are governed by the geodesic equation.

The Schwarzschild metric in standard coordinates (t, r, θ, ϕ) is given by:

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2). \tag{5.415}$$

While one could compute the non-vanishing Christoffel symbols and substitute them directly into the standard geodesic equation ($\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0$), it is mathematically more elegant and computationally efficient to exploit the symmetries of the spacetime using the Lagrangian formulation.

5.4.1 Symmetries and Constants of Motion

The trajectories of free particles are the extrema of the proper time functional. For an affine parameter λ, the paths are stationary points of the Lagrangian:

$$\mathcal{L} = \frac{1}{2} g_{\mu\nu}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda}. \tag{5.416}$$

Substituting the Schwarzschild metric into this Lagrangian yields:

$$\mathcal{L} = \frac{1}{2}\left[-\left(1 - \frac{2GM}{r}\right)\dot{t}^2 + \left(1 - \frac{2GM}{r}\right)^{-1}\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2\right], \tag{5.417}$$

where dots denote differentiation with respect to the affine parameter λ.

We immediately identify two cyclic coordinates in the Lagrangian: t and ϕ. Because $\partial\mathcal{L}/\partial t = 0$ and $\partial\mathcal{L}/\partial\phi = 0$, the corresponding canonical momenta are strictly conserved along the geodesics.

These conserved quantities correspond geometrically to the existence of continuous isometries, generated by Killing vector fields. The spacetime is static, possessing a timelike Killing vector $K = \partial_t$, and spherically symmetric, possessing an azimuthal spacelike Killing vector $R = \partial_\phi$.

The conserved quantity associated with time translation invariance is the energy (per unit rest mass) of the particle, E:

$$-E \equiv p_t = \frac{\partial\mathcal{L}}{\partial\dot{t}} = -\left(1 - \frac{2GM}{r}\right)\dot{t} \quad \Longrightarrow \quad E = \left(1 - \frac{2GM}{r}\right)\frac{dt}{d\lambda}. \tag{5.418}$$

(Note: the negative sign is chosen conventionally so that $E > 0$ for regular particles). The conserved quantity associated with azimuthal rotational invariance is the angular momentum (per unit rest mass), L:

$$L \equiv p_\phi = \frac{\partial\mathcal{L}}{\partial\dot{\phi}} = r^2\sin^2\theta\frac{d\phi}{d\lambda}. \tag{5.419}$$

5.4.2 The Orbital Plane

Because the Schwarzschild spacetime is perfectly spherically symmetric, the conservation of angular momentum implies that the motion of any single particle must be confined to a two-dimensional plane. We can rigorously prove this by examining the Euler-Lagrange equation for the polar angle θ:

$$\frac{d}{d\lambda}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}}\right) - \frac{\partial\mathcal{L}}{\partial\theta} = 0. \tag{5.420}$$

Applying this to our Lagrangian (5.417):

$$\frac{d}{d\lambda}(r^2\dot{\theta}) - r^2\sin\theta\cos\theta\dot{\phi}^2 = 0. \tag{5.421}$$

We are free to orient our coordinate system such that the particle's initial position and velocity lie in the equatorial plane, meaning $\theta(0) = \pi/2$ and $\dot{\theta}(0) = 0$. If we evaluate the equation of motion at this initial instant, the second term vanishes because $\cos(\pi/2) = 0$. The equation reduces to $d(r^2\dot{\theta})/d\lambda = 0$. Since $r \neq 0$ and $\dot{\theta}$ is initially zero, $\dot{\theta}$ must remain identically zero for all λ.

Therefore, the orbit is permanently confined to the equatorial plane, and we may rigorously set:

$$\theta = \frac{\pi}{2}, \quad \sin\theta = 1, \quad \dot{\theta} = 0. \tag{5.422}$$

Under this condition, the expression for the conserved angular momentum (5.419) simplifies to:

$$L = r^2 \frac{d\phi}{d\lambda}. \tag{5.423}$$

5.4.3 The Radial Equation and Effective Potential

Rather than solving the second-order differential equation for $r(\lambda)$, we can utilize a fundamental first-order integral of motion: the normalization of the four-velocity. For any geodesic parameterized by an affine parameter, the quantity $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ is strictly constant. We define a constant ϵ such that:

$$g_{\mu\nu}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = -\epsilon. \tag{5.424}$$

Here, $\epsilon = 1$ for massive particles (timelike geodesics, where λ is the proper time τ), and $\epsilon = 0$ for massless particles (null geodesics, representing photons).

Expanding this normalization condition using the metric in the equatorial plane:

$$-\left(1 - \frac{2GM}{r}\right)\left(\frac{dt}{d\lambda}\right)^2 + \left(1 - \frac{2GM}{r}\right)^{-1}\left(\frac{dr}{d\lambda}\right)^2 + r^2\left(\frac{d\phi}{d\lambda}\right)^2 = -\epsilon. \tag{5.425}$$

We now eliminate the coordinate velocities $\dot{t}$ and $\dot{\phi}$ by substituting the conserved quantities E and L:

$$\frac{dt}{d\lambda} = E\left(1 - \frac{2GM}{r}\right)^{-1}, \tag{5.426}$$

$$\frac{d\phi}{d\lambda} = \frac{L}{r^2}. \tag{5.427}$$

Substituting these into equation (5.425) yields:

$$-\left(1 - \frac{2GM}{r}\right)\left[E\left(1 - \frac{2GM}{r}\right)^{-1}\right]^2 + \left(1 - \frac{2GM}{r}\right)^{-1}\left(\frac{dr}{d\lambda}\right)^2 + r^2\left(\frac{L}{r^2}\right)^2 = -\epsilon. \tag{5.428}$$

Simplifying the terms:

$$-E^2\left(1 - \frac{2GM}{r}\right)^{-1} + \left(1 - \frac{2GM}{r}\right)^{-1}\left(\frac{dr}{d\lambda}\right)^2 + \frac{L^2}{r^2} = -\epsilon. \tag{5.429}$$

To isolate the kinetic term $(\dot{r})^2$, we multiply the entire equation by the metric factor $\left(1 - \frac{2GM}{r}\right)$:

$$-E^2 + \left(\frac{dr}{d\lambda}\right)^2 + \left(1 - \frac{2GM}{r}\right)\frac{L^2}{r^2} = -\epsilon\left(1 - \frac{2GM}{r}\right). \tag{5.430}$$

Rearranging the terms and dividing by 2 places this equation into the exact mathematical form of a classical one-dimensional energy conservation equation:

$$\frac{1}{2}\left(\frac{dr}{d\lambda}\right)^2 + V_{\text{eff}}(r) = \mathcal{E}, \tag{5.431}$$

where the total effective energy is defined as $\mathcal{E} \equiv \frac{1}{2}E^2$, and the **effective potential** $V_{\text{eff}}(r)$ is:

$$V_{\text{eff}}(r) = \frac{1}{2}\epsilon - \epsilon\frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{r^3}. \tag{5.432}$$

Physical Interpretation of the Effective Potential: The effective potential governs the radial motion of the particle. The terms can be understood as follows: 1. $\frac{1}{2}\epsilon$: A constant rest-mass energy term. 2. $-\epsilon\frac{GM}{r}$: The standard, attractive Newtonian gravitational potential. 3. $+\frac{L^2}{2r^2}$: The repulsive classical centrifugal barrier. 4. $-\frac{GML^2}{r^3}$: A purely relativistic, highly attractive correction.

In Newtonian mechanics, the centrifugal barrier dominates at small r, preventing a particle with non-zero angular momentum from ever reaching the origin. In General Relativity, the $-1/r^3$ term dominates as $r \to 0$. This implies that, unlike in classical mechanics, particles with angular momentum can plunge directly into the central singularity if they cross the centrifugal barrier.

5.4.4 Null Geodesics: The Photon Sphere

Let us first analyze the trajectories of massless particles (photons), for which $\epsilon = 0$. The effective potential drastically simplifies to:

$$V_{\text{eff}}(r) = \frac{L^2}{2r^2} - \frac{GML^2}{r^3}. \tag{5.433}$$

Circular orbits occur at the extrema of the effective potential, where the radial force vanishes ($dV_{\text{eff}}/dr = 0$). Taking the derivative:

$$\frac{dV_{\text{eff}}}{dr} = -\frac{L^2}{r^3} + \frac{3GML^2}{r^4}. \tag{5.434}$$

Setting this to zero to find the critical radius r_c:

$$\frac{L^2}{r^3} = \frac{3GML^2}{r^4} \quad \implies \quad r_c = 3GM. \tag{5.435}$$

This remarkable result shows that there exists a circular orbit for photons at $r = 3GM$, regardless of their angular momentum (so long as $L \neq 0$). This surface is known as the **photon sphere**.

To determine the stability of this orbit, we evaluate the second derivative:

$$\frac{d^2V_{\text{eff}}}{dr^2} = \frac{3L^2}{r^4} - \frac{12GML^2}{r^5}. \tag{5.436}$$

Evaluating this at the photon sphere $r = 3GM$:

$$\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=3GM} = \frac{3L^2}{(3GM)^4} - \frac{12GML^2}{(3GM)^5} = \frac{3L^2}{81G^4M^4} - \frac{12L^2}{243G^4M^4} = \frac{L^2}{G^4M^4}\left(\frac{1}{27} - \frac{4}{81}\right) = -\frac{L^2}{81G^4M^4}. \tag{5.437}$$

Since the second derivative is strictly negative, the extremum is a local maximum. The circular orbit at $r = 3GM$ is therefore **highly unstable**. Any infinitesimal radial perturbation will cause the photon to either spiral inexorably into the black hole or escape to infinity.

5.4.5 Timelike Geodesics and the Innermost Stable Circular Orbit

We now turn to massive particles, for which $\epsilon = 1$. The effective potential is:

$$V_{\text{eff}}(r) = \frac{1}{2} - \frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{r^3}. \tag{5.438}$$

We again seek circular orbits by setting the first derivative to zero:

$$\frac{dV_{\text{eff}}}{dr} = \frac{GM}{r^2} - \frac{L^2}{r^3} + \frac{3GML^2}{r^4} = 0. \tag{5.439}$$

To find the roots, we multiply the entire equation by r^4/GM:

$$r^2 - \left(\frac{L^2}{GM}\right)r + 3L^2 = 0. \tag{5.440}$$

This is a simple quadratic equation in r, the roots of which yield the radii of the circular orbits:

$$r_c = \frac{\frac{L^2}{GM} \pm \sqrt{\left(\frac{L^2}{GM}\right)^2 - 12L^2}}{2} = \frac{L^2 \pm \sqrt{L^4 - 12G^2M^2L^2}}{2GM}. \tag{5.441}$$

The physical behavior of the orbits depends intimately on the discriminant of this equation, $\Delta = L^4 - 12G^2M^2L^2 = L^2(L^2 - 12G^2M^2)$.

Case 1: Large Angular Momentum ($L^2 > 12G^2M^2$) If the particle has sufficient angular momentum, there are two distinct real roots. The positive sign (+) corresponds to a local minimum of the effective potential, representing a **stable circular orbit**. The negative sign (−) corresponds to a local maximum, representing an **unstable circular orbit** closer to the central mass. In the Newtonian limit ($L \to \infty$), the stable orbit approaches $r \approx L^2/GM$ (the classical Keplerian result), and the unstable orbit approaches $r \approx 3GM$ (the photon sphere limit).

Case 2: The Critical Limit ($L^2 = 12G^2M^2$) As the angular momentum is decreased, the stable and unstable orbits migrate toward each other. They merge at an inflection point when the discriminant vanishes identically ($\Delta = 0$). This defines a critical minimum angular momentum required to maintain any circular orbit:

$$L_c = \sqrt{12}GM. \tag{5.442}$$

Substituting this critical value back into equation (5.441) gives the single, unique radius where the roots converge:

$$r_{\text{ISCO}} = \frac{12G^2M^2}{2GM} = 6GM. \tag{5.443}$$

This is a monumental result in astrophysics: the **Innermost Stable Circular Orbit (ISCO)**. In Newtonian mechanics, a stable circular orbit can exist arbitrarily close to a point mass. In General Relativity, no stable circular orbit can exist at a radius smaller than $r = 6GM$.

Case 3: Small Angular Momentum ($L^2 < 12G^2M^2$) If the angular momentum falls below the critical threshold, the discriminant is negative, meaning there are no real roots. The effective potential possesses no extrema and becomes monotonically attractive. A particle with such angular momentum possesses no centrifugal barrier capable of halting its radial descent; regardless of its energy, it will plunge directly into the central mass.

5.5 Experimental Tests of General Relativity

Having established the theoretical framework of the Schwarzschild geometry and the corresponding geodesic equations, we now turn to the observable consequences of these phenomena. Historically, Einstein proposed three classical tests to verify General Relativity: the deflection of light, the anomalous precession of planetary perihelia, and the gravitational redshift. In this section, we rigorously derive the latter two directly from the Schwarzschild metric.

5.5.1 The Precession of Perihelia

In Newtonian mechanics, the two-body problem yields a perfectly closed elliptical orbit (for bound states) due to the strict $1/r^2$ nature of the gravitational force. The fact that the orbit closes upon itself after exactly 2π radians is a unique property of the $1/r$ potential (a consequence of Bertrand's theorem). In General Relativity, the relativistic correction to the effective potential alters this periodicity, causing the major axis of the ellipse to slowly precess over time.

To calculate this precession rate, we must determine the orbital shape $r(\phi)$. We begin with the radial energy equation for a massive particle ($\epsilon = 1$) derived in the previous section:

$$\frac{1}{2}\left(\frac{dr}{d\lambda}\right)^2 + \frac{1}{2} - \frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{r^3} = \frac{1}{2}E^2. \tag{5.444}$$

We wish to eliminate the affine parameter λ in favor of the azimuthal angle ϕ. Using the chain rule and the conservation of angular momentum $L = r^2(d\phi/d\lambda)$, we have:

$$\frac{dr}{d\lambda} = \frac{dr}{d\phi}\frac{d\phi}{d\lambda} = \frac{L}{r^2}\frac{dr}{d\phi}. \tag{5.445}$$

Substitute this into equation (5.444) and multiply the entire equation by 2:

$$\frac{L^2}{r^4}\left(\frac{dr}{d\phi}\right)^2 + 1 - \frac{2GM}{r} + \frac{L^2}{r^2} - \frac{2GML^2}{r^3} = E^2. \tag{5.446}$$

To simplify this non-linear differential equation, we perform a standard change of variables. Let $u(\phi) = 1/r(\phi)$. The derivative transforms as:

$$\frac{dr}{d\phi} = \frac{d}{d\phi}\left(u^{-1}\right) = -\frac{1}{u^2}\frac{du}{d\phi} = -r^2\frac{du}{d\phi}. \tag{5.447}$$

Squaring this gives $(dr/d\phi)^2 = r^4(du/d\phi)^2$. Substituting this back into the orbital equation yields:

$$L^2\left(\frac{du}{d\phi}\right)^2 + 1 - 2GMu + L^2u^2 - 2GML^2u^3 = E^2. \tag{5.448}$$

To reduce the order of the powers, we differentiate the entire equation with respect to ϕ:

$$2L^2\left(\frac{du}{d\phi}\right)\frac{d^2u}{d\phi^2} - 2GM\frac{du}{d\phi} + 2L^2u\frac{du}{d\phi} - 6GML^2u^2\frac{du}{d\phi} = 0. \tag{5.449}$$

Assuming the orbit is not a perfect circle (so $du/d\phi \neq 0$), we may divide through by $2L^2(du/d\phi)$ to obtain a second-order linear differential equation with a non-linear source term:

$$\frac{d^2u}{d\phi^2} + u = \frac{GM}{L^2} + 3GMu^2. \tag{5.450}$$

For mathematical convenience, we introduce a dimensionless radial variable x, scaled to the Newtonian circular orbit radius:

$$x = \frac{L^2}{GM}u = \frac{L^2}{GMr}. \tag{5.451}$$

Since $u = (GM/L^2)x$, substituting this into our differential equation and multiplying by (L^2/GM) yields the central equation for perihelion precession:

$$\frac{d^2x}{d\phi^2} - 1 + x = \frac{3G^2M^2}{L^2}x^2. \tag{5.452}$$

Perturbative Solution

In Newtonian gravity, the non-linear term on the right-hand side vanishes completely, leaving a simple harmonic oscillator equation $\frac{d^2x}{d\phi^2} + x = 1$. In General Relativity, the term $\alpha \equiv 3G^2M^2/L^2$ is extremely small for planetary orbits, allowing us to solve equation (5.452) using perturbation theory.

We expand the solution into a zeroth-order Newtonian piece and a first-order relativistic correction:

$$x(\phi) = x_0(\phi) + x_1(\phi). \tag{5.453}$$

Substituting this into equation (5.452) and separating orders yields two equations:

$$\text{Zeroth order:}\quad \frac{d^2x_0}{d\phi^2} + x_0 = 1, \tag{5.454}$$

$$\text{First order:}\quad \frac{d^2x_1}{d\phi^2} + x_1 = \alpha x_0^2. \tag{5.455}$$

The general solution to the zeroth-order equation (5.454) describes a classic Newtonian ellipse:

$$x_0(\phi) = 1 + e\cos\phi, \tag{5.456}$$

where e is the eccentricity of the orbit. (We have chosen the phase such that the perihelion, the point of closest approach where x is maximized and r is minimized, occurs at $\phi = 0$).

We substitute x_0 into the first-order equation (5.455):

$$\frac{d^2x_1}{d\phi^2} + x_1 = \alpha(1 + e\cos\phi)^2 = \alpha(1 + 2e\cos\phi + e^2\cos^2\phi). \tag{5.457}$$

Using the trigonometric identity $\cos^2\phi = \frac{1}{2}(1+\cos 2\phi)$, the source term expands to:

$$\frac{d^2x_1}{d\phi^2} + x_1 = \alpha\left[\left(1 + \frac{1}{2}e^2\right) + 2e\cos\phi + \frac{1}{2}e^2\cos 2\phi\right]. \tag{5.458}$$

This is a driven harmonic oscillator. We must find the particular solutions for each forcing term:

1. The constant term $\alpha(1 + e^2/2)$ generates a constant shift: $x_1^{(a)} = \alpha(1 + e^2/2)$.
2. The double-frequency term $\frac{1}{2}\alpha e^2\cos 2\phi$ generates an oscillation. Guessing a form $A\cos 2\phi$, we find $(-4A + A)\cos 2\phi = \frac{1}{2}\alpha e^2\cos 2\phi$, yielding $x_1^{(b)} = -\frac{1}{6}\alpha e^2\cos 2\phi$.
3. The resonant term $2\alpha e\cos\phi$ matches the natural frequency of the homogenous equation. The particular solution must be of the form $D\phi\sin\phi$. Differentiating gives:

$$\frac{d}{d\phi}(D\phi\sin\phi) = D\sin\phi + D\phi\cos\phi, \tag{5.459}$$

$$\frac{d^2}{d\phi^2}(D\phi\sin\phi) = 2D\cos\phi - D\phi\sin\phi. \tag{5.460}$$

   Adding x_1 back in gives $(2D\cos\phi - D\phi\sin\phi) + D\phi\sin\phi = 2D\cos\phi$. Matching coefficients requires $2D = 2\alpha e$, so $D = \alpha e$. Thus, $x_1^{(c)} = \alpha e\phi\sin\phi$.

The full first-order correction is:

$$x_1(\phi) = \alpha\left[\left(1 + \frac{1}{2}e^2\right) + e\phi\sin\phi - \frac{1}{6}e^2\cos 2\phi\right]. \tag{5.461}$$

The constant and double-frequency terms represent unobservable, microscopic adjustments to the shape of the ellipse. However, the secular term $\alpha e\phi\sin\phi$ grows continuously with every revolution. This term encapsulates the perihelion precession.

Calculating the Precession Angle

Combining the zeroth and the dominant first-order secular term yields the approximate trajectory:

$$x(\phi) \approx 1 + e\cos\phi + \alpha e\phi\sin\phi. \tag{5.462}$$

Because α is infinitesimally small, we can rewrite this expression using the Taylor expansion of cosine, $\cos(\phi - \delta) \approx \cos\phi + \delta\sin\phi$:

$$x(\phi) \approx 1 + e\cos(\phi - \alpha\phi) = 1 + e\cos[(1-\alpha)\phi]. \tag{5.463}$$

The particle completes a full radial orbit and returns to perihelion when the argument of the cosine reaches 2π. Let this occur at angle ϕ_p. We set:

$$(1-\alpha)\phi_p = 2\pi \quad \Longrightarrow \quad \phi_p = \frac{2\pi}{1-\alpha}. \tag{5.464}$$

Using the binomial expansion $(1-\alpha)^{-1} \approx 1+\alpha$, the angular position of the perihelion is:

$$\phi_p \approx 2\pi(1+\alpha) = 2\pi + 2\pi\alpha. \tag{5.465}$$

Therefore, instead of returning to exactly 2π, the perihelion shifts forward by an angle $\Delta\phi$ per orbit:

$$\Delta\phi = 2\pi\alpha = \frac{6\pi G^2 M^2}{L^2}. \tag{5.466}$$

To compare this prediction with astronomical observations, we express the angular momentum L in terms of the semi-major axis a and eccentricity e of the Newtonian ellipse. From classical orbital mechanics, $L^2 \approx GMa(1-e^2)$. Substituting this into our result and restoring explicit factors of the speed of light c yields Einstein's precise formula for the precession:

$$\Delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)} \text{ radians per orbit.} \tag{5.467}$$

For Mercury, this geometric correction evaluates to precisely 43.0 arcseconds per century, flawlessly resolving the historical anomaly that had plagued Newtonian celestial mechanics.

5.5.2 Gravitational Redshift

The second major observable consequence of the Schwarzschild geometry is the gravitational redshift. While we previously motivated this effect using the Equivalence Principle, we can now derive it rigorously and exactly from the full metric, ensuring its validity beyond the weak-field approximation.

Consider an observer who is strictly stationary at a fixed radial coordinate r outside a spherical mass. The four-velocity $U^\mu = (U^t, 0, 0, 0)$ of this observer must satisfy the normalization condition for massive observers, $g_{\mu\nu}U^\mu U^\nu = -1$.

$$g_{tt}(U^t)^2 = -1 \quad\implies\quad -\left(1 - \frac{2GM}{r}\right)(U^t)^2 = -1. \tag{5.468}$$

Solving for the time component yields the observer's four-velocity:

$$U^\mu = \left(\left(1 - \frac{2GM}{r}\right)^{-1/2}, 0, 0, 0\right). \tag{5.469}$$

Now, consider a photon traveling radially outward from the mass to the observer. The photon's trajectory is a null geodesic parameterized by an affine parameter λ, possessing a four-momentum vector $p^\mu = dx^\mu/d\lambda$.

The angular frequency (energy) of the photon, strictly as measured by the stationary observer, is given by the invariant inner product of the photon's four-momentum and the observer's four-velocity:

$$\omega = -g_{\mu\nu}p^\mu U^\nu. \tag{5.470}$$

Because the observer has only a time component, this reduces to:

$$\omega = -g_{tt}p^t U^t = -\left[-\left(1 - \frac{2GM}{r}\right)\right]\left(\frac{dt}{d\lambda}\right)\left(1 - \frac{2GM}{r}\right)^{-1/2}. \tag{5.471}$$

We know from our geodesic analysis that the covariant energy $E = -K_\mu p^\mu$ corresponding to the timelike Killing vector is strictly conserved along the photon's path:

$$E = \left(1 - \frac{2GM}{r}\right)\frac{dt}{d\lambda} \quad\implies\quad \frac{dt}{d\lambda} = E\left(1 - \frac{2GM}{r}\right)^{-1}. \tag{5.472}$$

Substitute this definition of $dt/d\lambda$ into the measured frequency equation:

$$\begin{aligned}\omega &= \left(1 - \frac{2GM}{r}\right)\left[E\left(1 - \frac{2GM}{r}\right)^{-1}\right]\left(1 - \frac{2GM}{r}\right)^{-1/2} \\ \omega &= E\left(1 - \frac{2GM}{r}\right)^{-1/2}.\end{aligned} \tag{5.473}$$

Because the value E is a global constant of motion along the entire photon trajectory, we can relate the frequency measured by an observer emitting the photon at r_1 to an observer receiving it at a higher radius r_2. The ratio of the observed frequencies is:

$$\frac{\omega_2}{\omega_1} = \frac{E\left(1 - \frac{2GM}{r_2}\right)^{-1/2}}{E\left(1 - \frac{2GM}{r_1}\right)^{-1/2}} = \left(\frac{1 - \frac{2GM}{r_1}}{1 - \frac{2GM}{r_2}}\right)^{1/2}. \tag{5.474}$$

This is the exact gravitational frequency shift formula for the Schwarzschild metric.

The Weak-Field Limit

To connect this exact geometric result to familiar Newtonian concepts, we evaluate the ratio in the weak-field limit, where $2GM/r \ll 1$. We Taylor expand the square roots using $(1-x)^n \approx 1 - nx$:

$$\begin{aligned}\frac{\omega_2}{\omega_1} &= \left(1 - \frac{2GM}{r_1}\right)^{1/2}\left(1 - \frac{2GM}{r_2}\right)^{-1/2} \\ &\approx \left(1 - \frac{GM}{r_1}\right)\left(1 + \frac{GM}{r_2}\right) \\ &\approx 1 - \frac{GM}{r_1} + \frac{GM}{r_2} + \mathcal{O}\left(\frac{G^2M^2}{r^2}\right).\end{aligned} \tag{5.475}$$

Recognizing the classical Newtonian gravitational potential $\Phi(r) = -GM/r$, the shift reduces precisely to:

$$\frac{\omega_2}{\omega_1} \approx 1 + \Phi(r_1) - \Phi(r_2) = 1 - \Delta\Phi. \tag{5.476}$$

If the photon is climbing out of the gravitational well ($r_2 > r_1$), the potential difference is positive, and the ratio $\omega_2/\omega_1 < 1$. The frequency decreases, creating a **gravitational redshift**. This demonstrates how time fundamentally flows at different rates depending on the depth within a gravitational potential, a prediction verified flawlessly by the Pound-Rebka experiment.

5.6 Schwarzschild Black Holes and Causal Structure

Having demonstrated that the apparent singularities at $r = 2GM$ are merely artifacts of the Schwarzschild coordinate chart, we are compelled to investigate the physical nature of the spacetime geometry at and within this radius. We define objects whose physical extent is strictly bounded by this critical radius as **black holes**. To understand the dynamics of particles and fields in this extreme gravitational regime, we must rigorously analyze the causal structure of the manifold.

5.6.1 Causal Structure in Schwarzschild Coordinates

The causal structure of spacetime is entirely dictated by the behavior of the light cones, which are defined by the trajectories of null geodesics ($ds^2 = 0$). To analyze the radial propagation of light, we restrict our attention to radial null curves by setting $d\theta = 0$ and $d\phi = 0$. The Schwarzschild line element then reduces to:

$$0 = -\left(1 - \frac{2GM}{r}\right)dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2. \tag{5.477}$$

Rearranging this equation yields the coordinate velocity of a radial photon:

$$\frac{dt}{dr} = \pm\left(1 - \frac{2GM}{r}\right)^{-1}, \tag{5.478}$$

where the positive sign corresponds to outgoing light rays (r increasing with t) and the negative sign corresponds to ingoing light rays (r decreasing with t).

Asymptotic Behavior: In the asymptotic flat-space limit ($r \to \infty$), equation (5.478) reduces to $dt/dr = \pm 1$, cleanly recovering the standard 45° light cones of Minkowski spacetime. However, as the light ray approaches the Schwarzschild radius ($r \to 2GM$), the term $(1 - 2GM/r)$ approaches zero. Consequently, the derivative diverges:

$$\lim_{r \to 2GM} \frac{dt}{dr} = \pm\infty. \tag{5.479}$$

Plotted on a standard spacetime diagram with t on the vertical axis and r on the horizontal axis, the slopes of the light cones become vertical. The light cones "close up."

This mathematical divergence implies that, according to the coordinate time t, an infalling photon (or massive particle) requires an infinite amount of time to reach the surface $r = 2GM$. To a distant stationary observer, the infalling object appears to slow down asymptotically, its emitted light becoming infinitely gravitationally redshifted, but it never appears to cross the horizon. However, this is an optical and coordinate illusion; calculation of the proper time τ along the infalling geodesic reveals that the observer reaches and crosses $r = 2GM$ in a finite amount of their own experienced time. The failure lies strictly in the t coordinate, which ceases to be a valid global time parameter.

5.6.2 The Tortoise Coordinate

To resolve this coordinate pathology, our first strategy is to define a new spatial coordinate that pushes the problematic region to coordinate infinity, thereby "opening up" the light cones. We define the **tortoise coordinate**, r^*, by integrating equation (5.478) directly:

$$dt = \pm\frac{dr}{1 - 2GM/r} \quad \Longrightarrow \quad \int dt = \pm\int \frac{r}{r - 2GM} dr. \tag{5.480}$$

Performing the integration yields:

$$\pm t = r + 2GM \ln\left|\frac{r}{2GM} - 1\right| + \text{constant}. \tag{5.481}$$

We formally define the tortoise coordinate as:

$$r^* \equiv r + 2GM \ln\left|\frac{r}{2GM} - 1\right|. \tag{5.482}$$

Differentiating this definition confirms its utility:

$$dr^* = \left(1 + \frac{2GM}{r - 2GM}\right) dr = \left(\frac{r}{r - 2GM}\right) dr = \left(1 - \frac{2GM}{r}\right)^{-1} dr. \tag{5.483}$$

We can now substitute dr^* into the standard Schwarzschild metric. The radial kinetic term transforms as:

$$\left(1 - \frac{2GM}{r}\right)^{-1} dr^2 = \left(1 - \frac{2GM}{r}\right)(dr^*)^2. \tag{5.484}$$

The line element in the new (t, r^*, θ, ϕ) coordinates becomes:

$$ds^2 = \left(1 - \frac{2GM}{r}\right)\left[-dt^2 + (dr^*)^2\right] + r^2 d\Omega^2, \tag{5.485}$$

where r is now implicitly understood to be a function of r^*.

In this coordinate system, radial null geodesics satisfy $-dt^2 + (dr^*)^2 = 0$, meaning $dt/dr^* = \pm 1$. The light cones remain rigidly at 45° everywhere. We have successfully prevented the light cones from degenerating. However, the cost of this transformation is that the surface $r = 2GM$ has been mapped to $r^* \to -\infty$. While the light cones are well-behaved, we still cannot trace trajectories *through* the Schwarzschild radius in a finite coordinate span.

5.6.3 Eddington-Finkelstein Coordinates

To construct a coordinate system capable of penetrating the boundary at $r = 2GM$, we must abandon the notion of orthogonal time and space slicing. We seek a coordinate system adapted directly to the paths of the infalling photons.

Using the tortoise coordinate, we define the **advanced Eddington-Finkelstein time coordinate**, v, which is constant along ingoing radial null geodesics:

$$v \equiv t + r^*. \tag{5.486}$$

The differential is $dv = dt + dr^*$. From our derivation of the tortoise coordinate, we know $dr^* = (1 - 2GM/r)^{-1} dr$. Therefore, the differential of the original Schwarzschild time t is:

$$dt = dv - dr^* = dv - \left(1 - \frac{2GM}{r}\right)^{-1} dr. \tag{5.487}$$

We rigorously substitute this expression for dt back into the original Schwarzschild metric (5.415):

$$\begin{aligned} ds^2 &= -\left(1 - \frac{2GM}{r}\right)\left[dv - \left(1 - \frac{2GM}{r}\right)^{-1} dr\right]^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \\ &= -\left(1 - \frac{2GM}{r}\right)\left[dv^2 - 2\left(1 - \frac{2GM}{r}\right)^{-1} dvdr + \left(1 - \frac{2GM}{r}\right)^{-2} dr^2\right] \\ &\quad + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \end{aligned} \tag{5.488}$$

Expanding the bracket:

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dv^2 + 2dvdr - \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \tag{5.489}$$

The singular dr^2 terms cancel precisely and identically. The resulting metric in **advanced Eddington-Finkelstein coordinates** (v, r, θ, ϕ) is:

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dv^2 + 2dvdr + r^2 d\Omega^2. \tag{5.490}$$

Proof of Geometric Regularity

The metric component $g_{vv} = -(1 - 2GM/r)$ still vanishes at $r = 2GM$, but this no longer implies a geometric degeneracy. The metric is expressed as a matrix:

$$g_{\mu\nu} = \begin{pmatrix} -\left(1 - \frac{2GM}{r}\right) & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2\theta \end{pmatrix} \tag{5.491}$$

The determinant of this metric tensor is computed explicitly:

$$g = \det(g_{\mu\nu}) = (-(0) - (1)(1))(r^2)(r^2 \sin^2\theta) = -r^4 \sin^2\theta. \tag{5.492}$$

The determinant is entirely regular, non-zero, and negative everywhere except at the true singularity $r = 0$ and the coordinate poles $\theta = 0, \pi$. Because the determinant is non-degenerate, the metric is mathematically invertible at $r = 2GM$. This constitutes absolute geometric proof that the surface $r = 2GM$ is merely a coordinate singularity in the original Schwarzschild chart.

5.6.4 The Event Horizon and the Tipping of Light Cones

We now re-examine the causal structure in the fully extended Eddington-Finkelstein coordinates. Setting $ds^2 = 0$ for radial null curves in metric (5.490) gives:

$$0 = -\left(1 - \frac{2GM}{r}\right) dv^2 + 2dvdr = dv\left[-\left(1 - \frac{2GM}{r}\right) dv + 2dr\right]. \tag{5.493}$$

This yields two distinct solutions for the radial light rays:

$$\text{Ingoing null geodesics:} \quad \frac{dv}{dr} = 0 \quad \Longrightarrow \quad v = \text{constant}, \tag{5.494}$$

$$\text{Outgoing null geodesics:} \quad \frac{dv}{dr} = 2\left(1 - \frac{2GM}{r}\right)^{-1}. \tag{5.495}$$

Let us analyze the "outgoing" geodesics at different radii: 1. **Exterior ($r > 2GM$):** Here, $(1 - 2GM/r) > 0$. Therefore, $dv/dr > 0$. As v increases (moving forward in time), r increases. The light ray successfully escapes to spatial infinity. 2. **The Boundary ($r = 2GM$):** Here, $(1-2GM/r) = 0$. The derivative $dv/dr \to \infty$, meaning $dr/dv = 0$. The light ray is completely "frozen" at a constant radial coordinate. It propagates forward in time but fails to move outward in space. 3. **Interior ($r < 2GM$):** Here, $(1 - 2GM/r) < 0$. Therefore, $dv/dr < 0$. Because v must increase along any future-directed trajectory, r must strictly *decrease*.

Physical Interpretation: The Event Horizon. In the region $r < 2GM$, the entire future light cone tilts entirely toward the central singularity at $r = 0$. Because massive particles are restricted to move along timelike trajectories (strictly within the interior of the local light cone), and massless particles move along null trajectories (on the boundary of the light cone), **all physical objects in the interior region must move toward** $r = 0$.

The surface $r = 2GM$ acts as a unidirectional causal membrane. It is not a localized spatial obstruction, but a fundamentally temporal one. Once crossed, the radial coordinate r acts identically to a time coordinate—it becomes dynamically inevitable. Just as an observer outside the black hole is powerless to stop the advancement of tomorrow, an observer inside the black hole is powerless to stop their descent to $r = 0$.

This null hypersurface, separating the causally connected exterior from the causally disconnected interior, is the **event horizon**.

5.6.5 The Misleading Newtonian Analogy

It is a curious historical coincidence that one can calculate a "black hole radius" using strictly Newtonian mechanics that matches the exact general relativistic event horizon.

In classical mechanics, the escape velocity v_{esc} of a mass m from a central body of mass M is derived by equating kinetic energy to the gravitational binding energy:

$$\frac{1}{2} m v_{\text{esc}}^2 = \frac{GMm}{r} \quad \Longrightarrow \quad v_{\text{esc}} = \sqrt{\frac{2GM}{r}}. \tag{5.496}$$

If one naively demands that the escape velocity exceeds the speed of light ($v_{\text{esc}} = c = 1$), one obtains $r = 2GM$. John Michell and Pierre-Simon Laplace hypothesized "dark stars" using precisely this logic in the 18th century.

However, this analogy is conceptually flawed and physically misleading. The classical escape velocity represents the initial velocity required for an unpowered, purely ballistic trajectory to reach spatial infinity. In Newtonian mechanics, even if $v_{\text{esc}} > c$, a particle could still escape if it were subjected to continuous outward acceleration (e.g., an extremely powerful rocket traveling at a constant $v = c/2$).

In General Relativity, the event horizon represents a fundamental deformation of causal structure. Because the light cones have completely tilted inwards, there are no outward-pointing timelike or null vectors whatsoever. Continuous acceleration within the horizon does not lead to escape; it merely alters the proper time required to collide with the inescapable singularity at $r = 0$. The Newtonian "dark star" simply has a deep potential well; the relativistic black hole represents the end of space and time.